

普通高等学校规划教材

Ordinary Differential Equations (常微分方程)

主 编 张慧星 杨晓波
副主编 孙永征 邵 虎
吴彦强

中国矿业大学出版社

• 徐州 •

内容简介

本书是常微分方程课程的英文教材,由作者结合多年的全英文教学经验编写而成。主要内容包括常微分方程的初等积分法、线性微分方程组理论和常系数线性微分方程组的求解方法、高阶线性微分方程理论和常系数高阶线性微分方程的求解方法、解的存在唯一性理论、微分方程的定性理论以及常微分方程的数值求解方法等。为了读者更方便地运用 Maple 软件解决常微分方程的应用问题,本书给出了一些实际范例和 Maple 程序。

本书可作为高等学校数学与应用数学专业常微分方程课程的双语、全英文教材,也可供教师和科学技术人员参考。

图书在版编目(CIP)数据

常微分方程 = Ordinary Differential Equations :

英文 / 张慧星, 杨晓波主编. — 徐州 : 中国矿业大学出版社, 2022.10

ISBN 978 - 7 - 5646 - 5564 - 8

I. ①常… II. ①张… ②杨… III. ①常微分方程—高等学校—教材—英文 IV. ①O175.1

中国版本图书馆 CIP 数据核字(2022)第 175294 号

书 名 Ordinary Differential Equations (常微分方程)
主 编 张慧星 杨晓波
责任编辑 黄本斌
出版发行 中国矿业大学出版社有限责任公司
(江苏省徐州市解放南路 邮编 221008)
营销热线 (0516)83884103 83884995
出版服务 (0516)83995789 83884920
网 址 <http://www.cumtp.com> E-mail: cumtpvip@cumtp.com
印 刷 徐州中矿大印发科技有限公司
开 本 787 mm×1092 mm 1/16 印张 14.5 字数 470 千字
版次印次 2022 年 10 月第 1 版 2022 年 10 月第 1 次印刷
定 价 36.00 元

(图书出现印装质量问题,本社负责调换)

前言

常微分方程是研究自然界和社会中事物与现象运动变化规律的数学理论,在实际中有着广泛的应用。物理、化学、生物、控制科学、经济等领域中与变化率有关的问题几乎都可以用微分方程模型来研究,而认识实际系统随时间变化的规律,需要对微分方程的解的性态进行深入分析。了解微分方程的解的性态,通常有三种途径:① 求微分方程的解析解;② 求微分方程的数值解;③ 对解的性态进行定性分析。这三种途径各有特点和局限性,在常微分方程的研究中又相互补充,共同发展。

全书共分六章。

第一章,讲述常微分方程的一些基本概念,重点介绍求解一阶微分方程的初等积分法。在系统讲解各种典型方程解法的同时,特别强调解方程的灵活性和一些常用技巧。

第二章,讲述线性微分方程组的一般理论。首先,针对这类方程的初值问题,证明了解的存在唯一性;其次,给出了齐次线性微分方程组和非齐次线性微分方程组的通解基本结果;然后,介绍了常系数线性微分方程组的求解方法;最后,讨论了周期线性系统的解的性态。

第三章,讲述高阶线性微分方程的一般理论。主要介绍了高阶线性微分方程初值问题的解存在唯一性、高阶线性微分方程理论、常系数高阶线性微分方程的求解方法,在此基础上,还讨论了线性微分方程的幂级数解法和拉普拉斯变换求解方法。

第四章,讲述常微分方程的一般理论。主要内容包括皮亚诺存在定理、皮卡存在唯一性定理、解的延伸定理、解对初值与参数的连续相依性和可微性。这部分内容是常微分方程最重要的理论基础。

第五章,讲述在理论和应用上都很重要的定性理论。主要内容包括解的稳定性的概念、线性系统的零解稳定性、李雅普诺夫第二方法、平面自治系统的基本概念和有关性质、平面自治线性系统的奇点的类型与判定、平面自治系统的极限环等。

第六章,讲述微分方程的数值解法。主要内容包括数值方法的基本知识、欧拉方法、改进的欧拉方法和龙格-库塔法。结合实际例子,在求出微分方程解析解的基础上,运用 Maple 软件求得数值解,并在相应的地方都给出了 Maple

程序。

在附录中,本书引入了软件 Maple2020 的简单介绍,给出了它的基本运算功能,展示了如何应用 Maple2020 来解决微分方程的求解以及定性分析中的问题。

在编写时我们力求反映以下特点:首先,本书的内容体系和基本框架按照由易到难、从特殊到一般的认识规律,先讲述线性方程的基本理论,然后再介绍非线性方程的一般理论。比如在第二章中,用逐步逼近法先证明了线性方程初值问题的解存在唯一性。有了这个基础,在第四章中再用这个方法处理非线性方程时,对定理的条件和证明都能很好理解。其次,在内容的处理上,结合实际范例,注重启发式叙述,重视基本概念的引入和数学思想的介绍,以使读者更全面快速地掌握基本理论和方法,提高读者的数学素养和培养读者应用理论知识分析问题和解决问题的能力。再次,注重运用图形进行表述,使读者建立一个直观认识,加深对抽象理论的理解和掌握。最后,实现了基本理论、方法的应用和数值计算的有机结合,尝试应用 Maple 软件解决微分方程的求解和定性分析中的问题,并且全部给出了 Maple 程序,以使读者加强定性分析思想和数值计算的深度融合,取得更好的学习效果。

本书可以作为普通高等学校数学、应用数学等专业的本科生教材或参考书,也可以供相关教师和科学技术人员参考,特别是为从事数学的应用有关的科学技术研究人员提供必要的数学基础知识。

本书的出版得到了中国矿业大学数学学院、教务部和中国矿业大学出版社的大力支持,也得到中国矿业大学“全英语课程”建设项目(2018YY11)的资助。研究生高妍、任硕、成珍、李枫柳、王璞和赵磊为本书出版做了大量的工作,同时参与了对习题的求解和验证。在此谨向中国矿业大学数学学院等单位和在本书编写及出版过程中做出贡献的教师和学生表示衷心的感谢。

鉴于作者的水平所限,书中不可避免存在各种疏漏和不足之处,敬请读者和同行批评指正,作者将不胜感激。

作者
2022 年 4 月

Contents

Chapter 1 Elementary Integration Method 1

1.1 Fundamental Concepts 1

Exercise 1.1 5

1.2 Separable Equations 6

Exercise 1.2 9

1.3 Homogeneous Equations 10

Exercise 1.3 14

1.4 First-Order Linear Differential Equations 15

Exercise 1.4 18

1.5 Exact Differential Equations 20

Exercise 1.5 23

1.6 Integrating Factor Method 24

Exercise 1.6 28

1.7 Flexibility in Problem Solving 30

Exercise 1.7 35

1.8 Implicit First-Order Differential Equations 36

Exercise 1.8 42

1.9 High-Order Differential Equations 43

Exercise 1.9 46

Chapter 2 Linear Systems of Differential Equations 48

2.1 First-Order System of Differential Equations 48

2.2 First-Order Linear System of Differential Equations 51

2.3 Existence and Uniqueness for First-Order System 52

Exercise 2.3 55

2.4 General Theory of Linear Homogeneous Systems 56

Exercise 2.4 61

2.5 General Theory of Nonhomogeneous Linear System 63

Exercise 2.5 66

2.6 Linear Differential System with Constant Coefficients 67

Exercise 2.6 78

2.7	Periodic Linear Systems	80
	Exercise 2.7	84
Chapter 3 High-Order Linear Differential Equations		86
3.1	Introduction	86
	Exercise 3.1	88
3.2	Existence and Uniqueness for High-Order Equations	89
	Exercise 3.2	90
3.3	General Solutions of High-Order Homogeneous Linear Equations	90
	Exercise 3.3	94
3.4	Nonhomogeneous High-Order Linear Differential Equations	95
	Exercise 3.4	98
3.5	High-Order Homogeneous Linear Equations with Constant Coefficients	99
	Exercise 3.5	103
3.6	High-Order Nonhomogeneous Linear Equations with Constant Coefficients	104
	Exercise 3.6	108
3.7	Power Series Methods	109
	Exercise 3.7	113
3.8	Laplace Transform Methods	114
	Exercise 3.8	125
Chapter 4 General Theory		127
4.1	Peano’s Existence Theorem	127
	Exercise 4.1	132
4.2	Existence and Uniqueness of Solutions	132
	Exercise 4.2	139
4.3	Extension Theorem	141
	Exercise 4.3	145
4.4	Continuous Dependence on Initial Values and Parameters	146
	Exercise 4.4	149
Chapter 5 Qualitative Theory		150
5.1	Stability	150
	Exercise 5.1	152
5.2	Stability of Linear Systems	153
	Exercise 5.2	155
5.3	The Method of Lyapunov	156
	Exercise 5.3	160

5.4	Basic Concepts of Two-Dimensional Autonomous Systems	161
	Exercise 5.4	164
5.5	Critical Points of Two-Dimensional Linear Autonomous Systems	165
	Exercise 5.5	170
5.6	Limit Cycles of Two-Dimensional Autonomous Systems	172
	Exercise 5.6	177
Chapter 6	Numerical Methods for Ordinary Differential Equations	179
6.1	Introduction to Numerical Methods	179
6.2	Numerical Approximation; Euler’s Method	180
6.3	An Improvement in Euler’s Method	192
6.4	The Runge-Kutta Method	198
Appendix	Application of Maple2020 for Ordinary Differential Equations	207
A1	Introduction to Maple2020	207
A2	Basic Operations for Maple2020	209
A3	Maple2020 for Solving Ordinary Differential Equations	212
References		222

Chapter 1 Elementary Integration Method

Ordinary differential equations serve as mathematical models for many problems, not only in science and technology, but also in such diverse fields as sociology, economics, defense and demography.

The theory of differential equations and its applications to almost every branch of knowledge have resulted in a continued interest for students in many disciplines. This chapter introduces some examples and fundamental concepts of differential equations. These concepts are very important for studying the contents in the following chapters.

1.1 Fundamental Concepts

The most interesting natural phenomena involve change and are described by equations that relate to changing quantities. Because the derivative of the function is the changing rate with respect to the independent variable, it is natural that equations involving derivatives are frequently used to illustrate the changing universe. A **differential equation** is an equation relating an unknown function and one or more of its derivatives.

Example 1.1.1 Newton's law of cooling can be described in this way. The time rate of change of the temperature $T(t)$ of a body with respect to time t is proportional to the difference between T and the temperature S of the surrounding medium.

That is,

$$\frac{dT}{dt} = -k(T - S), \quad (1.1.1)$$

where k is a positive constant. Note that if $T > S$, then $\frac{dT}{dt} < 0$, so the temperature is a decreasing function of t and the body is cooling. But if $T < S$, then $\frac{dT}{dt} > 0$, so that T is an increasing function of t and the body is warming.

Thus the physical law is translated into a differential equation. If we can find an explicit formula for $T(t)$ with given values of k and S , then we may predict the future temperature of the body with the aid of this formula.

Example 1.1.2 The motion of a body along a straight line is described by its position function

$$s = s(t), \quad (1.1.2)$$

which gives the distance that the body (or more exactly, its center of mass) has traveled up to time t . The instantaneous velocity of the body is defined to be

$$v(t) = \frac{ds}{dt}. \quad (1.1.3)$$

Its acceleration $a(t)$ is

$$a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}. \quad (1.1.4)$$

According to Newton's second law of motion, the position function $s(t)$ satisfies the following differential equation

$$m \frac{d^2s}{dt^2} = F(t), \text{ that is, } F = ma, \quad (1.1.5)$$

where m is the mass of the body, $F(t)$ is a force which acts on the body and is directed along its line of motion.

If the force F is known, then the differential equation (1.1.5) can be integrated twice to obtain the position function $s(t)$ in terms of two constants of integration.

Example 1.1.3 Let $P(t)$ be a population with constant birth and death rates. In many simple cases, the time rate of change of $P(t)$ is proportional to the size of the population. That is,

$$\frac{dP}{dt} = kP, \quad (1.1.6)$$

where k is the constant of proportionality.

If a function together with its derivatives satisfy a differential equation, then we call this function be a **solution** of the differential equation.

Example 1.1.4 The equation

$$\frac{dy}{dx} + 2xy = 0 \quad (1.1.7)$$

is a differential equation. Here x is the independent variable and y is the unknown function. It is easy to check that the function $y = 2e^{-x^2}$ is a solution of the equation (1.1.7), that is,

$$\frac{d}{dx}(2e^{-x^2}) + 2x(2e^{-x^2}) \equiv 0, x \in (-\infty, +\infty).$$

Indeed, we will see later that all solutions of the equation (1.1.7) can be written in the form $y = ce^{-x^2}$, where c is an arbitrary real number.

Example 1.1.5 Verify that the function $x(t) = c_1 \sin 3t + c_2 \cos 3t$ satisfies the differential equation

$$x''(t) + 9x(t) = 0, -\infty < t < +\infty, \quad (1.1.8)$$

where c_1 and c_2 are two arbitrary constants.

Solution Differentiating $x(t)$ with respect to t , we have

$$\begin{aligned} x' &= 3c_1 \cos 3t - 3c_2 \sin 3t, \\ x'' &= -9c_1 \sin 3t - 9c_2 \cos 3t. \end{aligned}$$

Hence,

$$\begin{aligned} x'' + 9x(t) &= (-9c_1 \sin 3t - 9c_2 \cos 3t) + 9(c_1 \sin 3t + c_2 \cos 3t) \\ &= (-9c_1 + 9c_1) \sin 3t + (-9c_2 + 9c_2) \cos 3t \equiv 0. \end{aligned}$$

So $x(t) = c_1 \sin 3t + c_2 \cos 3t$ is a solution of the differential equation.

The **order** of a differential equation is the order of the highest derivative of the unknown function that appears in it.

The differential equation (1.1.1) is of first order, the ones in Example 1.1.2 and Example 1.1.5 are second-order equations, and

$$x^{(4)} + tx'' + x = e^t$$

is a fourth-order equation. The general form of an n th-order differential equation with independent variable t and unknown function $x = x(t)$ is

$$F(t, x, x', x'', \dots, x^{(n)}) = 0, \quad (1.1.9)$$

where F is a specific real-valued function of $n + 2$ variables. We will ordinarily assume that any differential equation under study can be solved explicitly for the highest derivative, that is, the equation can be written in the so-called normal form

$$x^{(n)} = f(t, x, x', x'', \dots, x^{(n-1)}), \quad (1.1.10)$$

where f is a real-valued function of $n + 1$ variables. In addition, we will always find only real-valued solutions unless we remind the reader otherwise.

If $x = \varphi(t)$, $t \in I$ satisfies the equation (1.1.9), that is,

$$F(t, \varphi(t), \varphi'(t), \varphi''(t), \dots, \varphi^{(n)}(t)) = 0, t \in I,$$

then $x = \varphi(t)$ is called a **solution** of equation (1.1.9) which is defined on the interval I .

A solution of equation (1.1.9) is called a **general solution** provided that the solution

$$x = \varphi(t, c_1, c_2, \dots, c_n) \quad (1.1.11)$$

contains n independent arbitrary constants $c_1, c_2, \dots, c_n$.

Here, independence about the constants $c_1, c_2, \dots, c_n$ means that the determinant

$$\frac{D[\varphi, \varphi', \dots, \varphi^{(n-1)}]}{D[c_1, c_2, \dots, c_n]} \stackrel{\text{def}}{=} \begin{vmatrix} \frac{\partial \varphi}{\partial c_1} & \frac{\partial \varphi}{\partial c_2} & \dots & \frac{\partial \varphi}{\partial c_n} \\ \frac{\partial \varphi'}{\partial c_1} & \frac{\partial \varphi'}{\partial c_2} & \dots & \frac{\partial \varphi'}{\partial c_n} \\ \vdots & \vdots & & \vdots \\ \frac{\partial \varphi^{(n-1)}}{\partial c_1} & \frac{\partial \varphi^{(n-1)}}{\partial c_2} & \dots & \frac{\partial \varphi^{(n-1)}}{\partial c_n} \end{vmatrix} \neq 0,$$

where

$$\begin{cases} \varphi = \varphi(t, c_1, c_2, \dots, c_n), \\ \varphi' = \varphi'(t, c_1, c_2, \dots, c_n), \\ \vdots \\ \varphi^{(n-1)} = \varphi^{(n-1)}(t, c_1, c_2, \dots, c_n). \end{cases}$$

Any one solution of the equation is called a **particular solution** of the differential equation.

It is easy to check that $y = ce^{-x^2}$ is a general solution of equation (1.1.7) and $x(t) = c_1 \sin 3t + c_2 \cos 3t$ is a general solution of equation (1.1.8). While $x_1(t) = 4 \sin 3t$ and

$x_2(t) = \cos 3t$ are the particular solutions of $x''(t) + 9x(t) = 0$.

In application we are usually interested in a solution of the differential equation (1.1.10) which satisfies some additional requirements named initial or boundary conditions. By initial conditions for equation (1.1.10), we mean n conditions of the form

$$x(t_0) = x_0, x'(t_0) = x_1, \dots, x^{(n-1)}(t_0) = x_{n-1}, \quad (1.1.12)$$

where $x_0, x_1, \dots, x_{n-1}$ and t_0 are given constants.

A problem consisting of the differential equation (1.1.10) together with the initial conditions (1.1.12) is called an **initial value problem** or **Cauchy problem**. If the subsidiary conditions are given at more than one value of the independent variable, then the conditions are called **boundary conditions** and the problem including the differential equation together with boundary conditions is called a **boundary value problem**.

Example 1.1.6 The problem

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases} \quad (1.1.13)$$

is an initial value problem. While the problem

$$\begin{cases} x''(t) + \lambda x(t) = 0 \\ x(0) = x(\pi) = 0 \end{cases} \quad (1.1.14)$$

is a boundary value problem.

If F is linear about $x, x', \dots, x^{(n)}$, then the n th-order differential equation (1.1.9) is said to be linear. Otherwise, we call it a nonlinear differential equation.

Obviously, the differential equations

$$\begin{aligned} xy' + y &= x^3 \\ x^2 y'' - 3y' + 3y &= e^x \\ x^{(n)} + p_1(t)x^{(n-1)} + \dots + p_n(t)x &= f(t) \end{aligned}$$

are linear differential equations. While the differential equations

$$y'' + yy' = e^x$$

and

$$\frac{d^2\theta}{dt^2} + a^2 \sin \theta = 0$$

are nonlinear differential equations.

If the unknown function that appears in a differential equation depends on only one independent variable, then the differential equation is called an **ordinary differential equation**. Indeed, the differential equations in above mentioned examples are all ordinary differential equations. If two or more independent variables and hence also partial derivatives are present, then the equation is called a **partial differential equation**.

For example,

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2x + 2y$$

is a partial differential equation of first order for an unknown function $u(x, y)$. An

important example of a second-order partial differential equation is the potential equation

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0,$$

where $u = u(x, y, z)$.

In this textbook, we will be concerned only with ordinary differential equations. The emphasis will be on differential equations in the real domain where the independent variable is a real variable and the unknown function is a real function.

The expression integral of a differential equation is another term used for a solution, and the terms **solution curve** and **integral curve** are used to give the geometric interpretation of a solution as a curve.

To sum up, differential equations play an important role not only in natural science and technology, particularly in physics, for the reason that many physical laws take the form of a differential equation, but also in other scientific domains where mathematical models and theories are used.

Exercise 1.1

In Problems 1 through 5, verify by substitution that each given function is a solution of the given differential equation.

1. $y' = 4x^3; y = x^4 + 2$.
2. $y' + 4y = 0; y = 5e^{-4x}$.
3. $y'' + 16y = 0; y = 4\sin 4x + \cos 4x$.
4. $x^2 y'' + xy' - y = \ln x; y = \frac{1}{x} - \ln x$.
5. $y'' + 4y' + 3y = 0; y = c_1 e^{-x} + c_2 e^{-3x}$.

In Problems 6 through 10, first verify that $y(x)$ satisfies the given differential equation. Then determine a value of the constant c so that $y(x)$ satisfies the given initial condition.

6. $y' + 3y = 0; y(x) = c e^{-3x}, y(0) = 2$.
7. $y' - y = 2; y(x) = c e^x - 2, y(0) = 1$.
8. $y' = 3x^2(y^2 + 1); y = \tan(x^3 + c), y(0) = 1$.
9. $e^y y' = 2; y(x) = \ln(2x + c), y(0) = 2$.
10. $y' + 4x^3 y = 0; y(x) = c e^{-x^4}, y(0) = 6$.

Determine the order and linearity in Problems 11 through 15.

11. $x''' - 5t(x')^2 = e^t + 1$.
12. $y' + p(x)y = q(x)y^n, (n \neq 0, 1)$.
13. $x' + 2tx = t^2 + 2$.
14. $xy'' - 2y' + 2xy = \cos x$.
15. $e^{y'} + y = x$.

In Problems 16 through 20, a function $y = f(x)$ is described by some geometric

property of its graph. Write a differential equation of the form $\frac{dy}{dx} = g(x, y)$ having the function f as its solution.

16. The slope of the graph of f at the point (x, y) is the difference of x and y .

17. The line tangent to the graph of f at the point (x, y) intersects the x -axis at the point $(x, 0)$.

18. The graph of f is normal to every curve of the form $y = x^2 + c$ (c is a constant) where they meet.

19. The slope of the graph of f at the point (x, y) is proportional to the square root of y .

20. The line tangent to the graph of f at the point (x, y) passes through the point (y, x) .

21. Find a differential equation such that the function $y = cx + 3x^3$ satisfies this differential equation, where c is a constant.

22. Show that $x = \frac{1}{t^4 - 1}$ is a solution of the differential equation $x' + 4t^3x^2 = 0$ in the interval $(-1, 1)$, but not in any larger interval containing $(-1, 1)$.

In Problems 23 through 26, determine at least one solution of the given differential equation by making an intelligent guess.

23. $y''' = x$.

24. $y' = 3y$.

25. $y' + y = 2e^x$.

26. $2xy' + y = 4x^2$.

1.2 Separable Equations

The first-order differential equation

$$\frac{dy}{dx} = f(x, y)$$

is called a **separable equation** provided that the function $f(x, y)$ can be written as a product of a function of x and a function of y , that is,

$$\frac{dy}{dx} = g(x)h(y), \tag{1.2.1}$$

where $g(x)$ and $h(y)$ are continuous functions.

The solving method is as follows.

If there exists y_0 such that $h(y_0) = 0$, then $y = y_0$ is a solution of equation (1.2.1).

If $h(y) \neq 0$, then the variables x and y can be separated by writing the following form

$$\frac{dy}{h(y)} = g(x)dx. \tag{1.2.2}$$

It is easy to solve this equation simply by integrating both sides with respect to x

$$\int \frac{1}{h(y(x))} \frac{dy}{dx} dx = \int g(x) dx + c$$

that is,

$$\int \frac{1}{h(y)} dy = \int g(x) dx + c. \quad (1.2.3)$$

This is the implicit general solution of the differential equation (1.2.1).

Remark 1.2.1 The symbol $\int g(x) dx$ is different in ordinary differential equation and mathematical analysis. Here, $\int g(x) dx$ means some antiderivative, while $\int g(x) dx$ indicates all antiderivatives in mathematical analysis.

Example 1.2.1 Solve the differential equation

$$\frac{dy}{dx} = 4x^3 y. \quad (1.2.4)$$

Solution This is a separable equation. When $y \neq 0$, we separate variables, integrate both sides and get

$$\begin{aligned} \int \frac{1}{y} dx &= \int 4x^3 dx, \\ \ln |y| &= x^4 + c_1, \end{aligned}$$

where c_1 is an arbitrary constant of integration. Applying the exponential function to both sides of last equation yields

$$y = c e^{x^4},$$

where $c = \pm e^{c_1}$. Note that $y \equiv 0$ is also a solution.

Example 1.2.2 Solve the differential equation

$$\frac{dy}{dx} = 3x^2 (1 - y^2)^{\frac{1}{2}}. \quad (1.2.5)$$

Solution It is obvious that $y = \pm 1$ are two particular solutions. When $y \neq \pm 1$, separation of variables gives

$$(1 - y^2)^{-\frac{1}{2}} dy = 3x^2 dx.$$

We integrate both sides and get the implicit general solution

$$\arcsin y = x^3 + c,$$

where c is an arbitrary constant of integration. This solution can be written as explicit form

$$y = \sin(x^3 + c).$$

Example 1.2.3 Solve the initial value problem

$$\frac{dy}{dx} = \frac{4 - 2x}{3y^2 - 5}, \quad y(1) = 3. \quad (1.2.6)$$

Solution Separation of variables gives

$$\begin{aligned} \int (3y^2 - 5) dy &= \int (4 - 2x) dx, \\ y^3 - 5y &= 4x - x^2 + c. \end{aligned} \quad (1.2.7)$$

This is an implicit general solution of equation (1.2.6). It is not readily solved for y as an explicit function of x .

We substitute the initial condition in equation (1.2.7) and get $c = 9$. Thus the desired particular solution $y(x)$ is defined implicitly by the equation

$$y^3 - 5y = 4x - x^2 + 9.$$

Example 1.2.4 Solve the differential equation

$$x^2(y^3 - 1)dx + y^2(x^3 - 1)dy = 0. \quad (1.2.8)$$

Solution This is a separable equation. Obviously, $x = 1$ and $y = 1$ are two particular solutions of this equation. When $x \neq 1$ and $y \neq 1$, we separate the variables and get

$$\int \frac{x^2}{x^3 - 1} dx + \int \frac{y^2}{y^3 - 1} dy = 0,$$

$$\ln |x^3 - 1| + \ln |y^3 - 1| = \ln |c|. (c \neq 0)$$

Hence

$$(x^3 - 1)(y^3 - 1) = c$$

is an implicit general solution of equation (1.2.8).

The differential equation

$$\frac{dx}{dt} = kx \quad (1.2.9)$$

serves as a mathematical model for a wide range of natural phenomena, that is, any involving a quantity whose time rate of change is proportional to its current size.

Besides Example 1.1.3, here are some other examples.

Example 1.2.5 Phenomenon of radioactivity was discovered by the physicist Rutherford and his colleagues. They proved that the radioactivity of a material is directly proportional to the current number of atoms of the material. Let $N(t)$ be the current number of atoms at time t .

Since $\frac{dN}{dt}$, the number of atoms that disintegrate per unit time, is proportional to N , hence we have

$$\frac{dN}{dt} = -\lambda N, \quad (1.2.10)$$

where $\lambda > 0$ is known as the decay constant of the substance, in addition, $N(t_0) = N_0$ is the initial condition.

Example 1.2.6 In many cases the amount $Q(t)$ of a certain drug in the bloodstream, measured by the excess over the natural level of the drug, will decrease at a rate proportional to the current excess amount. Thus we have

$$\frac{dQ}{dt} = -\mu Q, \quad (1.2.11)$$

where $\mu > 0$ is called the elimination constant of the drug.

In fact, the differential equations in Example 1.1.3, Example 1.2.5 and Example 1.2.6 all serve as the same mathematical model (1.2.9).

It is easy to get the general solution of the differential equation (1.2.9) with $x(t) > 0$ and k a constant (either positive or negative) by separating variables and integrating both sides

$$\int \frac{1}{x} dx = \int k dt,$$

$$\ln x = kt + c_1,$$

hence

$$x = x(t) = e^{c_1} e^{kt} = c e^{kt}.$$

The particular solution of the equation (1.2.9) with initial condition $x(0) = x_0$ is

$$x(t) = x_0 e^{kt}. \quad (1.2.12)$$

The differential equation $\frac{dx}{dt} = kx$ is often named the exponential or natural growth equation because the natural exponential function appears in its solution (1.2.12). Figure 1.2.1 shows a typical graph of $x(t)$ in the case $k > 0$; while the case $k < 0$ is illustrated in Figure 1.2.2.

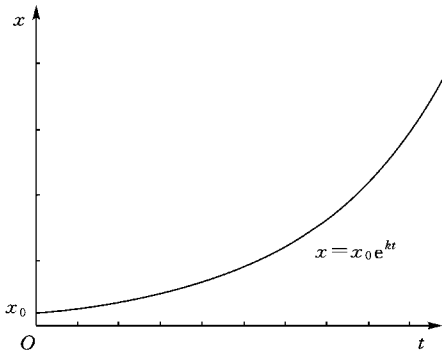


Figure 1.2.1 $k > 0$, Natural growth

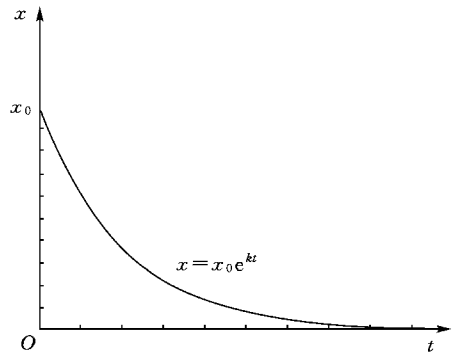


Figure 1.2.2 $k < 0$, Natural decay

Exercise 1.2

Find general solutions (implicit if necessary, explicit if convenient) of the differential equations in Problems 1 through 10.

1. $\frac{dy}{dx} + 4xy = 0.$
2. $\frac{dy}{dx} + 3x^2 y^2 = 0.$
3. $\frac{dy}{dx} = e^y \sin x.$
4. $y^3 \frac{dy}{dx} = (y^4 + 1) \sin x.$
5. $x^2 y' = 1 - 2x^2 + y^2 - 2x^2 y^2.$
6. $x \sin y + (x^2 + 1) \cos y \cdot y' = 0.$
7. $(x^2 + 1) \cdot (\tan y) y' = 2x.$
8. $y' = 1 + x + y + xy.$
9. $\frac{dy}{dx} = y^2 \cos x.$

$$10. \frac{dy}{dx} = \frac{x + x^{\frac{3}{2}}}{1 + y^{\frac{5}{2}}}.$$

Find the particular solutions of the initial value problems in Problems 11 through 15.

$$11. \frac{dy}{dx} = 2ye^x, \quad y(0) = e.$$

$$12. \frac{dy}{dx} = 4x^2y - y, \quad y(1) = 3.$$

$$13. \frac{dy}{dx} = e^{2x-y}, \quad y(0) = 0.$$

$$14. \frac{dy}{dx} = xy^2 + 2x^2y^2, \quad y(1) = 1.$$

$$15. (y^2 + xy^2)dx - (x^2yx^2)dy = 0, \quad y(1) = 2.$$

16. Show that the transformation $y = xu$ changes the differential equation

$$y^n f(x) + g\left(\frac{y}{x}\right)(y - xy') = 0$$

into a separable equation.

17. When sugar is dissolved in water, the amount A that remains undissolved after t minutes satisfies the differential equation

$$\frac{dA}{dt} = -kA \quad (k > 0).$$

If 25% of the sugar dissolves after 1 minute, how long does it take for half of the sugar to dissolve?

18. Solve the differential equation

$$(y - xy^2)dx - (x + x^2y)dy = 0$$

by making the substitution $y = ux^n$ for an appropriate n .

1.3 Homogeneous Equations

A **homogeneous** first-order differential equation is one that can be written in the form

$$\frac{dy}{dx} = g\left(\frac{y}{x}\right). \quad (1.3.1)$$

The following two types of differential equation are also homogeneous equations.

Type 1

$$\frac{dy}{dx} = \frac{P(x, y)}{Q(x, y)} \quad (1.3.2)$$

where $P(x, y)$ and $Q(x, y)$ are homogeneous functions of same degree, that is, for $t > 0$,

$$P(tx, ty) = t^m P(x, y), \quad Q(tx, ty) = t^m Q(x, y).$$

In fact, by choosing $t = x$, the differential equation (1.3.2) can be changed into

$$\frac{dy}{dx} = \frac{x^m P\left(1, \frac{y}{x}\right)}{x^m Q\left(1, \frac{y}{x}\right)} = \frac{P\left(1, \frac{y}{x}\right)}{Q\left(1, \frac{y}{x}\right)}.$$

Type 2

$$\frac{dy}{dx} = h(x, y) \quad (1.3.3)$$

where $h(x, y)$ satisfies

$$h(tx, ty) = h(x, y) \text{ for } t > 0.$$

Indeed, the differential equation (1.3.3) can be written in the form

$$\frac{dy}{dx} = h\left(1, \frac{y}{x}\right).$$

If we make the substitutions

$$u = \frac{y}{x}, y = xu, \frac{dy}{dx} = u + x \frac{du}{dx} \quad (1.3.4)$$

then the equation (1.3.1) is transformed into the separable equation

$$x \frac{du}{dx} = g(u) - u.$$

Example 1.3.1 Solve the differential equation

$$\frac{dy}{dx} = \frac{xy - y^2}{x^2}.$$

Solution This is a homogeneous equation. Making substitution $y = xu$, we have

$$u + x \frac{du}{dx} = u - u^2,$$

that is

$$x \frac{du}{dx} = -u^2. \quad (1.3.5)$$

Obviously, $u = 0$ is a solution of the equation (1.3.5), thus, $y = 0$ is a solution of the original differential equation. When $u \neq 0$, separating variables and integrating both sides, we get

$$\begin{aligned} -\frac{du}{u^2} &= \frac{dx}{x}, \\ \frac{1}{u} &= \ln |x| + c. \end{aligned}$$

Hence,

$$y = \frac{x}{\ln |x| + c}$$

is a general solution of the original differential equation.

Example 1.3.2 Solve the initial value problem

$$t \frac{dx}{dt} = x + \sqrt{t^2 - x^2}, x(t_0) = 0,$$

where $t_0 > 0$.

Solution Dividing both sides by t , we find

$$\frac{dx}{dt} = \frac{x}{t} + \sqrt{1 - \left(\frac{x}{t}\right)^2}.$$

Substitution in $u = \frac{x}{t}$ gives

$$u + t \frac{du}{dt} = u + \sqrt{1 - u^2}.$$

Integrating both sides of the last equation, we get

$$\int \frac{1}{\sqrt{1 - u^2}} du = \int \frac{1}{t} dt,$$

$$\arcsin u = \ln t + c.$$

Noting that $t_0 > 0$, we need not write $\ln |t|$ because $t > 0$ near t_0 . Together with the initial condition $u(t_0) = \frac{x(t_0)}{t_0} = 0$, we get $c = -\ln t_0$.

Therefore,

$$x(t) = t \sin\left(\ln \frac{t}{t_0}\right)$$

is the desired particular solution.

Example 1.3.3 Solve the differential equation

$$\frac{dy}{dx} = \frac{y}{x} + \frac{\sqrt{x^2 + y^2}}{x}, x > 0.$$

Solution We rewrite the differential equation in the form

$$\frac{dy}{dx} = \frac{y}{x} + \sqrt{1 + \left(\frac{y}{x}\right)^2}.$$

Substitution in $u = \frac{y}{x}$ yields

$$\frac{du}{\sqrt{1 + u^2}} = \frac{dx}{x}.$$

Integrating both sides, we have

$$\ln(u + \sqrt{1 + u^2}) = \ln x + c_1,$$

$$u + \sqrt{1 + u^2} = c_2 x.$$

Substituting $u = \frac{y}{x}$ back in the last equation, we get the general solution of the original differential equation

$$y = \frac{c}{2} x^2 - \frac{1}{2c},$$

where $c = c_2$.

Next, we consider a specific type of differential equation in the form

$$\frac{dy}{dx} = f\left(\frac{a_1 x + b_1 y + c_1}{a_2 x + b_2 y + c_2}\right) \quad (1.3.6)$$

where $a_i, b_i, c_i, i = 1, 2$ are constants.

Indeed, we can transform it into a homogeneous equation.

Case 1 When $c_1 = c_2 = 0$, this is a homogeneous equation.

Case 2 When $c_1^2 + c_2^2 \neq 0$ and the determinant

$$\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0,$$

we consider the system of linear equations

$$\begin{cases} a_1x + b_1y + c_1 = 0 \\ a_2x + b_2y + c_2 = 0 \end{cases}$$

and obtain a unique solution $x = \alpha, y = \beta$. By introducing the new variables

$$x = X + \alpha, y = Y + \beta,$$

we change the differential equation (1.3.6) into

$$\frac{dY}{dX} = f\left(\frac{a_1X + b_1Y}{a_2X + b_2Y}\right).$$

Case 3 When $c_1^2 + c_2^2 \neq 0$ and $\Delta = 0$, there exists a real number λ such that either $a_1 = \lambda a_2, b_1 = \lambda b_2$ or $a_2 = \lambda a_1, b_2 = \lambda b_1$. If the latter holds, then we have

$$\frac{dy}{dx} = f\left(\frac{(a_1x + b_1y) + c_1}{\lambda(a_1x + b_1y) + c_2}\right).$$

Making substitution $u = a_1x + b_1y$, we transform it into a separable equation

$$\frac{du}{dx} = a_1 + b_1f\left(\frac{u + c_1}{\lambda u + c_2}\right).$$

Case 4 If the differential equation (1.3.6) has the following particular form

$$\frac{dy}{dx} = f(a_1x + b_1y + c_1),$$

then we denote $u = a_1x + b_1y + c_1$ and change it directly into a separable equation

$$\frac{du}{dx} = a_1 + b_1f(u).$$

Example 1.3.4 Solve the differential equation

$$\frac{dy}{dx} = \frac{-x + y + 4}{x + y - 8}.$$

Solution We solve the system of linear equations

$$\begin{cases} -x + y + 4 = 0 \\ x + y - 8 = 0 \end{cases}$$

and get the solution $x = 6, y = 2$. The substitutions

$$x = X + 6, y = Y + 2$$

transform it into

$$\frac{dY}{dX} = \frac{-X + Y}{X + Y}.$$

It is easy to obtain the general solution of last equation

$$X^2 + Y^2 = ce^{-2\arctan \frac{Y}{X}}.$$

Thus the implicit general solution of the original differential equation is

$$(x - 6)^2 + (y - 2)^2 = ce^{-2\arctan \frac{y-2}{x-6}}.$$

Example 1.3.5 Solve the differential equation

$$\frac{dy}{dx} = \cos^2(x - y).$$

Solution The substitution $u = x - y$ leads to

$$\frac{du}{dx} = \sin^2 u.$$

When $\sin u \neq 0$, we separate variables, integrate both sides and obtain the implicit general solution

$$-\cot(x - y) = x + c.$$

When $\sin u = \sin(x - y) = 0$, we easily get a family of particular solutions

$$y = x + k\pi,$$

where k is any integer.

Exercise 1.3

Find general solutions of the differential equations in Problems 1 through 15.

1. $(x + y)y' = (x - y).$

2. $x^2 y' = xy + x^2 e^{\frac{y}{x}}.$

3. $2x^2 y' = xy + y^2.$

4. $y' = \frac{2xy}{x^2 - y^2}.$

5. $xyy' = y^2 + x\sqrt{4x^2 + y^2}.$

6. $2xyy' = x^2 + 3y^2.$

7. $y' = (x + y + 1)^3.$

8. $y' = x^2 + 2xy + y^2.$

9. $xy' - y = (x + y)\ln \frac{x + y}{x}.$

10. $(2x - 4y + 6)dx + (x + y - 3)dy = 0.$

11. $(2x + y + 1)dx + (4x + 2y + 3)dy = 0.$

12. $(x + 3y)y' = x + y + 5.$

13. $(x^2 + 2y^2 - 1)x dx - (x^2 + y^2 + 2)y dy = 0.$

14. $y' = \left(\frac{x + y - 1}{x + 2} \right)^2.$

15. $y' = (x + y + 1)^2 - 4.$

16. Show that the substitution $u = \ln y$ transforms the differential equation

$$\frac{dy}{dx} + p(x)y = q(x)(y \ln y)$$

into the linear equation

$$\frac{du}{dx} + p(x)u = q(x)u(x).$$

17. Show that if the differential equation

$$P(x, y)dx + Q(x, y)dy = 0$$

is homogeneous and P and Q possess continuous partial derivatives in some domain D , then

$$\frac{x \frac{\partial P}{\partial x} + y \frac{\partial P}{\partial y}}{x \frac{\partial Q}{\partial x} + y \frac{\partial Q}{\partial y}} = \frac{P}{Q}.$$

18. Show that the substitution $y = x^n u$ in the differential equation

$$y' = x^{n-1} f\left(\frac{y}{x^n}\right)$$

leads to a separable equation. Hence, solve the differential equation

$$x^3 y' = 2y(x^2 - y).$$

1.4 First-Order Linear Differential Equations

A first-order linear differential equation is an equation of the form

$$\frac{dy}{dx} + p(x)y = q(x) \quad (1.4.1)$$

where the two given functions $p(x)$ and $q(x)$ are continuous on an interval I . When $q(x) \equiv 0$, the equation (1.4.1) is called **homogeneous**, otherwise, **nonhomogeneous**.

The differential operator

$$Ly = y' + p(x)y \quad (1.4.2)$$

can be used to write the differential equation (1.4.1) in the form

$$Ly = y' + p(x)y = 0 \quad (\text{homogeneous}) \quad (1.4.3)$$

and

$$Ly = y' + p(x)y = q(x) \quad (\text{nonhomogeneous}) \quad (1.4.4)$$

We first consider the equation (1.4.3). Obviously, this equation is a separable equation. Note that $y = 0$ is a solution. When $y \neq 0$, separation of variables yields

$$\frac{dy}{y} = -p(x)dx.$$

We easily obtain the general solution of the equation (1.4.3)

$$y = ce^{-\int p(x)dx}. \quad (1.4.5)$$

Next, we solve the nonhomogeneous equation (1.4.4). Solutions to the nonhomogeneous equation can be derived with the help of the **method of variation of parameters**. In this method, the constant c in the general solution (1.4.5) of the homogeneous equation is replaced by a function $c(x)$. The proper choice of $c(x)$ gives a solution of the nonhomogeneous equation. Indeed, the substitution

$$y(x) = c(x)e^{-\int p(x)dx} \quad (1.4.6)$$

leads to

$$\begin{aligned} Ly = y' + p(x)y &= [c'(x) - p(x)c(x) + p(x)c(x)]e^{-\int p(x)dx} \\ &= c'(x)e^{-\int p(x)dx}. \end{aligned}$$

Hence $Ly = q(x)$ holds if and only if

$$c'(x) = q(x)e^{\int p(x)dx},$$

equivalently,

$$c(x) = \int q(x)e^{\int p(x)dx} dx + c.$$

Thus the general solution of the nonhomogeneous equation (1.4.4) is

$$y = ce^{-\int p(x)dx} + e^{-\int p(x)dx} \int q(x)e^{\int p(x)dx} dx. \quad (1.4.7)$$

In order to solve the initial value problem

$$\begin{cases} \frac{dy}{dx} + p(x)y = q(x), \\ y(x_0) = y_0, \end{cases} \quad (1.4.8)$$

we use the definite integral in equation (1.4.6), that is,

$$y = c(x)e^{-\int_{x_0}^x p(t)dt}. \quad (1.4.9)$$

Substitution back in the nonhomogeneous equation (1.4.4) gives

$$c'(x) = q(x)e^{\int_{x_0}^x p(t)dt}.$$

Making definite integral from x_0 to x , we get

$$c(x) = \int_{x_0}^x q(s)e^{\int_{x_0}^s p(t)dt} ds + c.$$

So the general solution of the equation (1.4.4) is

$$y(x) = e^{-\int_{x_0}^x p(t)dt} [c + \int_{x_0}^x q(s)e^{\int_{x_0}^s p(t)dt} ds].$$

The initial condition $y(x_0) = y_0$ gives $c = y_0$. Thus the solution of the initial value problem (1.4.8) is

$$\begin{aligned} y(x) &= e^{-\int_{x_0}^x p(t)dt} [y_0 + \int_{x_0}^x q(s)e^{\int_{x_0}^s p(t)dt} ds] \\ &= y_0 e^{-\int_{x_0}^x p(t)dt} + \int_{x_0}^x q(s)e^{-\int_s^x p(t)dt} ds. \end{aligned} \quad (1.4.10)$$

Example 1.4.1 Solve the differential equation

$$\frac{dx}{dt} + x = e^t.$$

Solution It is easy to obtain the general solution of the corresponding homogeneous equation

$$x = ce^{-t}.$$

Let $x(t) = c(t)e^{-t}$ be the general solution of the nonhomogeneous equation. Substitution leads to

$$c'(t) = e^{2t},$$

$$c(t) = \frac{1}{2}e^{2t} + c.$$

So the general solution of the nonhomogeneous equation is given by

$$x(t) = ce^{-t} + \frac{1}{2}e^t.$$

Example 1.4.2 Suppose that $f(x)$ is a continuous and bounded function for $x \geq 0$. Prove all solutions of the differential equation

$$\frac{dy}{dx} + y = f(x)$$

are bounded for $x \geq 0$.

Solution Let $y = y(x)$ be any one solution satisfying the initial condition $y(x_0) = y_0$, $x_0 \in [0, +\infty)$. By using the formula (1.4.10), we obtain

$$y(x) = y_0 e^{-(x-x_0)} + \int_{x_0}^x f(t) e^{t-x} dt.$$

In fact, it suffices to prove that $y(x)$ is bounded for $x \geq x_0$. Suppose that

$$|f(x)| \leq M, x \in [0, +\infty),$$

then

$$\begin{aligned} |y(x)| &\leq |y_0| e^{-(x-x_0)} + \int_{x_0}^x |f(t)| e^{t-x} dt \\ &\leq |y_0| + M e^{-x} \int_{x_0}^x e^t dt \\ &= |y_0| + M [1 - e^{(x_0-x)}] \\ &\leq |y_0| + M. \end{aligned}$$

Next we consider a specific type of the differential equation which can be transformed into a linear differential equation. The differential equation of the form

$$\frac{dy}{dx} + p(x)y = q(x)y^n \tag{1.4.11}$$

is called **Bernoulli's Equation** (named after Jacob Bernoulli, 1654—1705), where n is a real number and $n \neq 0, 1$, $p(x), q(x)$ are two given continuous functions in the interval I and $q(x) \not\equiv 0$.

When $n > 0$, $y = 0$ is a solution of Bernoulli's Equation.

When $y \neq 0$, division by y^n gives

$$y^{-n} \frac{dy}{dx} + p(x)y^{1-n} = q(x),$$

that is

$$\frac{dy^{1-n}}{dx} + (1-n)p(x)y^{1-n} = (1-n)q(x).$$

This is a first-order linear differential equation with the unknown function $z = y^{1-n}$.

Example 1.4.3 Solve the differential equation

$$xy' - 3y = x^5 y^{\frac{1}{3}}.$$

Solution This is a Bernoulli's Equation with $p(x) = -\frac{3}{x}$, $q(x) = x^4$ and $n = \frac{1}{3}$.

Substitution $z = y^{\frac{2}{3}}$, $z' = \frac{2}{3}y^{-\frac{1}{3}}y'$ leads to

$$\frac{dz}{dx} - \frac{2}{x}z = \frac{2}{3}x^4.$$

It is easy to get the general solution of the last linear differential equation is

$$z = cx^2 + \frac{2}{9}x^5.$$

So the general solution of the original problem is given implicitly by

$$y^{\frac{2}{3}} = cx^2 + \frac{2}{9}x^5.$$

Exercise 1.4

Find general solutions of the differential equations in Problems 1 through 10. If an initial condition is given, find the corresponding particular solution.

1. $y' + 2y = 1, y(0) = 0.$

2. $y' + 2y = xe^{-2x}.$

3. $y' - 2xy = e^{x^2}.$

4. $y' + \frac{3}{x}y = 2x^4, y(2) = 1.$

5. $y' - (\cot x)y = 2x \sin x.$

6. $y' + y + 2x + x^2 + x^3 = 0.$

7. $y' + y \tan x = \sec x.$

8. $2x(y' - y) = e^x.$

9. $y' + xy = 2x, y(0) = 1.$

10. $xy' + 5y = 6x^2, y(1) = 4.$

Find general solutions of the differential equations in Problems 11 through 15.

11. $\frac{dy}{dx} + y = xy^5.$

12. $y' + xy + xy^3 = 0.$

13. $y' + 2xy = x^3y^3.$

14. $y' + y = y^2e^{-x}.$

15. $xy' + 6y = 3xy^{4/3}.$

Solve the differential equations in Problems 16 through 20 by regarding y as the independent variable rather than x .

16. $(2x + ye^y) \frac{dy}{dx} = 1.$

17. $(1 - 2xy^2) \frac{dy}{dx} = y^3.$

18. $(y + 2xy) \frac{dy}{dx} = 1 + y^2.$

19. $y'(x^3 - xy) = 1.$

20. $\frac{dx}{dy} + xy = x^4.$

21. If $y_1(x)$ and $y_2(x)$ are solutions of $y' + p(x)y = q_i(x)$, $i = 1, 2$, respectively, then $c_1 y_1(x) + c_2 y_2(x)$ is a solution of the differential equation

$$y' + p(x)y = c_1 q_1(x) + c_2 q_2(x)$$

where c_1 and c_2 are constants.

22. Show that the following properties hold.

(1) If $\varphi_1(x)$ and $\varphi_2(x)$ are solutions of the homogeneous linear equation (1.4.3), then a linear combination $c_1 \varphi_1(x) + c_2 \varphi_2(x)$ is also a solution of equation (1.4.3), where c_1 and c_2 are constants.

(2) Suppose that $\varphi_c(x)$ and $\varphi_p(x)$ are solutions of the differential equations (1.4.3) and (1.4.4) respectively. Then $\varphi_c(x) + \varphi_p(x)$ is a solution of equation (1.4.4).

(3) Let $\varphi_1(x)$ and $\varphi_2(x)$ be solutions of the differential equation (1.4.4). Then $\varphi_1(x) - \varphi_2(x)$ is a solution of equation (1.4.3).

(4) Let $\varphi(x)$ be any one solution of the differential equation (1.4.3). Then either $\varphi(x) \equiv 0$ or $\varphi(x) \neq 0$ at each point of I .

(5) The initial value problem (1.4.8) has a unique solution.

23. Let $g(x)$ be continuous in $[0, \infty)$ and $\lim_{x \rightarrow \infty} g(x) = \beta$. For the differential equation

$$y' + by = g(x),$$

show the following:

(1) If $b > 0$, every solution tends to $\frac{\beta}{b}$ as $x \rightarrow \infty$.

(2) If $b < 0$, there exists a unique solution which approaches $\frac{\beta}{b}$ as $x \rightarrow \infty$.

24. Let $y(x)$ be continuously differentiable in $[0, \infty)$ and

$$\lim_{x \rightarrow +\infty} [y'(x) + 2y(x)] = 0.$$

Show that

$$\lim_{x \rightarrow +\infty} y(x) = 0.$$

25. Consider the differential equation

$$\frac{dy}{dx} + p(x)y = q(x)$$

where $p(x)$ and $q(x)$ are continuous and ω -periodic. Show the following:

(1) If $q(x) \equiv 0$, then any nonzero solution of this differential equation is ω -periodic if and only if $\bar{p} = \frac{1}{\omega} \int_0^\omega p(x) dx = 0$.

(2) If $q(x) \neq 0$, then this differential equation exists a unique ω -periodic solution if

and only if $\bar{p} \neq 0$. Find this solution.

1.5 Exact Differential Equations

A differential equation of the symmetric form

$$M(x, y)dx + N(x, y)dy = 0 \quad (1.5.1)$$

is called an **exact equation** if there exists a function $F(x, y)$ such that

$$dF(x, y) = M(x, y)dx + N(x, y)dy,$$

that is

$$\frac{\partial F}{\partial x} = M(x, y), \frac{\partial F}{\partial y} = N(x, y). \quad (1.5.2)$$

If the differential equation is an exact equation, then we rewrite the equation

$$dF(x, y) = 0.$$

Thus $F(x, y) \equiv c$ is the implicit general solution of the equation (1.5.1).

For example, the differential equation

$$x dx + y dy = 0$$

is an exact equation. Because we easily find a function of two variables x and y

$$F(x, y) = \frac{1}{2}(x^2 + y^2)$$

such that

$$dF(x, y) = x dx + y dy.$$

Example 1.5.1 Solve the differential equation

$$3x^2y^2 dx + 2x^3y dy = 0.$$

Solution It is easy to notice that the left-hand side of this equation is the total differential of the function x^3y^2 , that is,

$$d(x^3y^2) = 3x^2y^2 dx + 2x^3y dy.$$

So the implicit general solution is given by

$$x^3y^2 = c.$$

Natural questions are these:

1. How can we determine whether the differential equation of the form (1.5.1) is exact?

2. If it is exact, how can we find the function $F(x, y)$ such that $\frac{\partial F}{\partial x} = M$ and $\frac{\partial F}{\partial y} = N$?

3. If it is not exact, can we find a function $\mu(x, y)$ such that the equivalent differential equation

$$\mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = 0$$

is exact?

In this section, we answer the first two questions, we will consider the third question in next section.

Theorem 1.5.1 (Criterion for Exactness)

Suppose that the functions $M(x, y)$ and $N(x, y)$ are continuous and have continuous first-order partial derivatives in the open rectangle:

$$R: a < x < b, c < y < d.$$

Then the differential equation (1.5.1) is exact in R if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad (1.5.3)$$

at each point of R . Furthermore, when the equation (1.5.3) holds, we can obtain that the implicit general solution of the equation (1.5.1) is

$$\int_{x_0}^x M(x, y) dx + \int_{y_0}^y N(x_0, y) dy = c, \quad (1.5.4)$$

or

$$\int_{x_0}^x M(x, y_0) dx + \int_{y_0}^y N(x, y) dy = c, \quad (1.5.5)$$

where (x_0, y_0) is any one point in R .

Proof (1) **Necessity.** Suppose that the equation (1.5.1) is exact, then there exists a function $F(x, y)$ such that

$$\frac{\partial F}{\partial x} = M(x, y), \quad \frac{\partial F}{\partial y} = N(x, y). \quad (1.5.6)$$

Taking the partial derivatives of the last two equations with respect to y and x respectively, we have

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial M}{\partial y}, \quad \frac{\partial^2 F}{\partial y \partial x} = \frac{\partial N}{\partial x}. \quad (1.5.7)$$

Since $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ are continuous, that is, $\frac{\partial^2 F}{\partial x \partial y}$ and $\frac{\partial^2 F}{\partial y \partial x}$ are continuous, so

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x}.$$

Hence

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

(2) **Sufficiency.** If $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then we construct a function $F(x, y)$ such that (1.5.6)

holds. By $\frac{\partial F}{\partial x} = M$, we choose

$$F(x, y) = \int_{x_0}^x M(x, y) dx + \varphi(y). \quad (1.5.8)$$

Together with (1.5.3), we get

$$\begin{aligned} \frac{\partial F}{\partial y} &= \frac{\partial}{\partial y} \int_{x_0}^x M(x, y) dx + \varphi'(y) \\ &= \int_{x_0}^x \frac{\partial M}{\partial y} dx + \varphi'(y) \end{aligned}$$

$$\begin{aligned} &= \int_{x_0}^x \frac{\partial N}{\partial x} dx + \varphi'(y) \\ &= N(x, y) - N(x_0, y) + \varphi'(y). \end{aligned}$$

In order to make $F(x, y)$ satisfy $\frac{\partial F}{\partial y} = N$, we have

$$\varphi'(y) = N(x_0, y).$$

We choose

$$\varphi(y) = \int_{y_0}^y N(x_0, y) dy.$$

Thus

$$F(x, y) = \int_{x_0}^x M(x, y) dx + \int_{y_0}^y N(x_0, y) dy \quad (1.5.9)$$

is the desired antiderivative. Similarly, we can obtain another function

$$\bar{F}(x, y) = \int_{x_0}^x M(x, y_0) dx + \int_{y_0}^y N(x, y) dy$$

which satisfies (1.5.6). Hence we get the implicit general solution (1.5.4) or (1.5.5).

Example 1.5.2 Solve the differential equation

$$(3x^2 \sin y + 6x^2 y^2) dx + (x^3 \cos y + 4x^3 y + 2) dy = 0. \quad (1.5.10)$$

Solution Here $M(x, y) = 3x^2 \sin y + 6x^2 y^2$, $N(x, y) = x^3 \cos y + 4x^3 y + 2$. Since

$$\frac{\partial M}{\partial y} = 3x^2 \cos y + 12x^2 y = \frac{\partial N}{\partial x},$$

so the differential equation (1.5.10) is exact. That is, there exists a function $F(x, y)$ such that

$$(1) \quad \frac{\partial F}{\partial x} = 3x^2 \sin y + 6x^2 y^2,$$

$$(2) \quad \frac{\partial F}{\partial y} = x^3 \cos y + 4x^3 y + 2.$$

We will apply three different methods to solve the antiderivative $F(x, y)$.

Method 1 Integrating both sides of the equation (1) yields

$$F(x, y) = x^3 \sin y + 2x^3 y^2 + \varphi(y), \quad (1.5.11)$$

where $\varphi(y)$ is an undetermined function. Then by using the equation (2), we get

$$x^3 \cos y + 4x^3 y + \varphi'(y) = x^3 \cos y + 4x^3 y + 2,$$

that is,

$$\varphi'(y) = 2.$$

Thus

$$\varphi(y) = 2y + c_1.$$

Therefore, the general solution of (1.5.10) is

$$x^3 \sin y + 2x^3 y^2 + 2y = c.$$

(We have absorbed the constant c_1 into the constant c).

Similarly, we can also start from the equation $\frac{\partial F}{\partial y} = N$ to obtain $F(x, y)$.

Method 2 Choose $(x_0, y_0) = (0, 0)$. By using the equation (1.5.4), we get the general solution

$$\begin{aligned} F(x, y) &= \int_0^x M(x, y) dx + \int_0^y N(0, y) dy \\ &= \int_0^x (3x^2 \sin y + 6x^2 y^2) dx + \int_0^y 2 dy \\ &= x^3 \sin y + 2x^3 y^2 + 2y = c. \end{aligned}$$

Similarly, we can obtain the general solution by using the equation (1.5.5).

Method 3 We rewrite the differential equation (1.5.10) in the form

$$(3x^2 \sin y \, dx + x^3 \cos y \, dy) + (6x^2 y^2 dx + 4x^3 y \, dy) + 2dy = 0.$$

It is easy to obtain

$$d(x^3 \sin y) + d(2x^3 y^2) + d(2y) = 0,$$

that is

$$d(x^3 \sin y + 2x^3 y^2 + 2y) = 0.$$

So the general solution of the equation (1.5.10) is given by

$$x^3 \sin y + 2x^3 y^2 + 2y = c.$$

Exercise 1.5

In Problems 1 through 10, verify that the given differential equation is exact, then solve it.

1. $(x^2 + 3y)dx + (3x + e^y)dy = 0$.
2. $(x - y)dx + (5y - x)dy = 0$.
3. $(x^3 + 2y^3)dx + (6xy^2 + \cos y)dy = 0$.
4. $\left(x^2 + \frac{y}{x}\right)dx + (2y + \ln x)dy = 0$.
5. $(3x^2 y^3 + y^4)dx + (3x^3 y^2 + y^4 + 4xy^3)dy = 0$.
6. $(e^x + 2xy)dx + (x^2 - 3y^2)dy = 0$.
7. $(1 + y^2 \sin 2x)dx + (e^y - y \cos 2x)dy = 0$.
8. $(x^2 + \arctan y)dx + \left(\frac{x + y}{1 + y^2}\right)dy = 0$.
9. $(2e^x \sin y + \tan y)dx + (2e^x \cos y + x \sec^2 y)dy = 0$.
10. $(\sin x + 2 \ln y)dx + \left(\frac{2x}{y} + 2 \cos y\right)dy = 0$.

Solve the following initial value problems.

11. $(4x^3 y + 8xy^2)dx + (x^4 + 8x^2 y + y^2)dy = 0, y(2) = 1$.
12. $(2x - y \cos x)dx + (y - \sin x)dy = 0, y\left(\frac{\pi}{2}\right) = 1$.
13. $(2xy + x^2)dx + (x^2 + 2y)dy = 0, y(0) = 1$.
14. $(ye^{xy} + 4y^3)dx + (xe^{xy} + 12xy^2 - 2y)dy = 0, y(0) = 2$.

In Problems 15 through 18, determine the constant λ so that each equation is exact

and solve the resulting differential equations.

$$15. (t^3 + 3tx)dt + (\lambda t^2 + 4x)dx = 0.$$

$$16. (x^{-2} + 2y^{-2})dx + (\lambda x + 2)y^{-3}dy = 0.$$

$$17. (x^2 + \lambda xy^3)dx + (2x^2y^2 + 2y)dy = 0.$$

$$18. (e^x + \lambda y^2 \sin x)dx + (y^3 - y \cos x)dy = 0.$$

1.6 Integrating Factor Method

In this section, we will answer the third question of last section. That is, if a differential equation

$$M(x, y)dx + N(x, y)dy = 0 \quad (1.6.1)$$

is not exact, can we make it be exact?

For example, the equation $2ydx + xdy = 0$ is not exact, but if we multiply it by the function x , then $2xydx + x^2dy = 0$ is an exact equation. Indeed, $d(x^2y) = 2xydx + x^2dy$, so $x^2y = c$ is the general solution of this equation.

Definition 1.6.1 For the differential equation (1.6.1), a nonzero function $\mu(x, y)$ is called an **integrating factor** if the equivalent differential equation

$$\mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = 0 \quad (1.6.2)$$

is exact.

Example 1.6.1 A separable equation of the form

$$P_0(x)Q_1(y)dx + P_1(x)Q_0(y)dy = 0. \quad (1.6.3)$$

If $P_1(x)Q_1(y) \neq 0$, then we multiply it by $\mu(x, y) = [P_1(x)Q_1(y)]^{-1}$. The equivalent differential equation

$$\frac{P_0(x)}{P_1(x)}dx + \frac{Q_0(y)}{Q_1(y)}dy = 0$$

is an exact equation. So $\mu(x, y) = [P_1(x)Q_1(y)]^{-1}$ is an integrating factor for the given equation (1.6.3).

Suppose that the functions $M(x, y)$, $N(x, y)$ and $\mu(x, y)$ are continuously differentiable. By Theorem 1.5.1, the differential equation (1.6.2) is exact, that is, $\mu(x, y)$ is an integrating factor if and only if

$$\begin{aligned} \frac{\partial(\mu M)}{\partial y} &= \frac{\partial(\mu N)}{\partial x}, \\ N \frac{\partial \mu}{\partial x} - M \frac{\partial \mu}{\partial y} &= \mu \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right). \end{aligned} \quad (1.6.4)$$

This is a first-order linear partial differential equation with unknown function $\mu(x, y)$. A solution of (1.6.4) gives an integrating factor of (1.6.1), but finding a solution of the partial differential equation (1.6.4) is by no means an easier task. In the following, we will give some particular cases.

Theorem 1.6.1 An integrating factor $\mu(x, y)$ is a function of x alone, that is,

$\mu = \mu(x)$ if and only if

$$\frac{1}{N(x, y)} \left(\frac{\partial M(x, y)}{\partial y} - \frac{\partial N(x, y)}{\partial x} \right)$$

is a function of x alone. And $\mu(x) = e^{\int g(x) dx}$ is an integrating factor of (1.6.1), where

$$g(x) \triangleq \frac{1}{N(x, y)} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right).$$

Proof $\mu(x, y)$ only depends on x , by (1.6.4), we have

$$N \frac{d\mu}{dx} = \mu \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right).$$

That is

$$\frac{1}{\mu(x)} \frac{d\mu(x)}{dx} = \frac{1}{N(x, y)} \left(\frac{\partial M(x, y)}{\partial y} - \frac{\partial N(x, y)}{\partial x} \right). \quad (1.6.5)$$

This shows that the expression in the right side of (1.6.5) is a function of x alone.

We rewrite (1.6.5) in the form

$$\frac{d\mu}{\mu} = g(x) dx.$$

Thus

$$\mu = e^{\int g(x) dx}.$$

Similarly, we have the following result.

Theorem 1.6.2 An integrating factor $\mu(x, y)$ is a function of y alone if and only if the expression

$$\frac{1}{M(x, y)} \left(\frac{\partial N(x, y)}{\partial x} - \frac{\partial M(x, y)}{\partial y} \right) \triangleq h(y)$$

is a function of y alone. And $\mu(y) = e^{\int h(y) dy}$ is an integrating factor of the differential equation (1.6.1).

One may also look for an integrating factor of the form $\mu = \mu(\varphi)$, where φ is a known function of x and y .

Theorem 1.6.3 Suppose that the functions $M(x, y), N(x, y)$ and $\varphi(x, y)$ are continuously differentiable in the open rectangle

$$R: a < x < b, c < y < d.$$

Then, there exists an integrating factor in the form $\mu = \mu[\varphi(x, y)]$ of the differential equation (1.6.1) if and only if the expression

$$\frac{\frac{\partial M(x, y)}{\partial y} - \frac{\partial N(x, y)}{\partial x}}{N(x, y) \frac{\partial \varphi(x, y)}{\partial x} - M(x, y) \frac{\partial \varphi(x, y)}{\partial y}} \quad (1.6.6)$$

is a function of φ alone. In addition, let the function (1.6.6) be a function of $\varphi(x, y)$ alone and denote (1.6.6) by $g = g[\varphi(x, y)]$, then the function

$$\mu = e^{G(\varphi(x, y))} \quad (1.6.7)$$

is an integrating factor of the differential equation (1.6.1), where $G(t) = \int g(t) dt$.

Proof If $\mu = \mu(\varphi)$ is an integrating factor, by (1.6.4), we have

$$N \frac{d\mu}{d\varphi} \cdot \frac{\partial \varphi}{\partial x} - M \frac{d\mu}{d\varphi} \cdot \frac{\partial \varphi}{\partial y} = \mu \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right),$$

that is

$$\frac{\mu'}{\mu} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N \frac{\partial \varphi}{\partial x} - M \frac{\partial \varphi}{\partial y}}. \tag{1.6.8}$$

The left side of (1.6.8) is a function of φ alone, so the right side is also a function of φ alone. Conversely, if the expression (1.6.6) is a function of φ alone, then we rewrite (1.6.8) in the form

$$\frac{\mu'}{\mu} = g(\varphi).$$

It is easy to obtain

$$\mu = e^{G(\varphi(x,y))}.$$

Some special cases of φ and the corresponding $g(\varphi)$ are given in the following Table 1.6.1:

Table 1.6.1

Cases (φ)	Conditions ($g(\varphi)$)	Integrating Factors ($e^{G(\varphi)}$)
x	$\frac{M_y - N_x}{N} \equiv g(x)$	$e^{\int g(x) dx}$
y	$\frac{N_x - M_y}{N} \equiv g(y)$	$e^{\int g(y) dy}$
$x - y$	$\frac{M_y - N_x}{N + M} \equiv g(x - y)$	$e^{\int g(t) dt} \Big _{t=x-y}$
$x + y$	$\frac{M_y - N_x}{N - M} \equiv g(x + y)$	$e^{\int g(t) dt} \Big _{t=x+y}$
xy	$\frac{M_y - N_x}{yN - xM} \equiv g(xy)$	$e^{\int g(t) dt} \Big _{t=xy}$
$\frac{x}{y}$	$\frac{(M_y - N_x)y^2}{yN + xM} \equiv g\left(\frac{x}{y}\right)$	$e^{\int g(t) dt} \Big _{t=\frac{x}{y}}$
$x^a y^\beta$	$\frac{M_y - N_x}{\alpha x^{-1} N - \beta y^{-1} M} \cdot \frac{1}{x^a y^\beta} \equiv g(x^a y^\beta)$	$e^{\int g(t) dt} \Big _{t=x^a y^\beta}$
$x^2 + y^2$	$\frac{M_y - N_x}{2(xN - yM)} \equiv g(x^2 + y^2)$	$e^{\int g(t) dt} \Big _{t=x^2 + y^2}$
$\varphi(x, y)$	$\frac{M_y - N_x}{N\varphi_x - M\varphi_y} \equiv g(\varphi(x, y))$	$e^{\int g(t) dt} \Big _{t=\varphi(x, y)}$

Example 1.6.2 Solve the differential equation

$$(x^2y + y + 1)dx + x(1 + x^2)dy = 0.$$

Solution Here $M(x, y) = x^2y + y + 1$, $N(x, y) = x(1 + x^2)$. Computation leads to

$$\frac{M_y - N_x}{N} = \frac{1}{x(1 + x^2)}(x^2 + 1 - 3x^2 - 1) = \frac{-2x}{1 + x^2}$$

which is a function of x alone. By Theorem 1.6.1, we find

$$\mu(x) = e^{-\int \frac{2x}{1+x^2} dx} = (1 + x^2)^{-1}.$$

Thus, the differential equation

$$\left(y + \frac{1}{1 + x^2}\right)dx + xdy = 0$$

is exact. The general solution is given by

$$xy + \arctan x = c.$$

Example 1.6.3 Solve the differential equation

$$(xy + y^2)dx + (xy + y + 2)dy = 0.$$

Solution Here $M(x, y) = xy + y^2$, $N(x, y) = xy + y + 2$. Direct computation leads to

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{y - x - 2y}{xy + y^2} = -\frac{1}{y}.$$

By Theorem 1.6.2, we obtain an integrating factor

$$\mu = e^{\int \frac{1}{y} dy} = \frac{1}{y}.$$

Multiplying both sides by $\frac{1}{y}$ gives

$$(x + y)dx + \left(x + 1 + \frac{2}{y}\right)dy = 0.$$

Thus the implicit general solution is given by

$$\frac{1}{2}x^2 + xy + y + 2 \ln|y| = c.$$

Example 1.6.4 Solve the differential equation

$$(xy^3 + 2x^2y^2 - y^2)dx + (x^2y^2 + 2x^3y - 2x^2)dy = 0.$$

Solution Here $M(x, y) = xy^3 + 2x^2y^2 - y^2$, $N(x, y) = x^2y^2 + 2x^3y - 2x^2$. We have

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{yN - xM} = 1 - \frac{2}{xy}$$

which is a function of xy . Hence, from the above table, we get

$$\mu = (xy)^{-2} e^{xy}.$$

The resulting exact differential equation

$$(yx^{-1} + 2 - x^{-2})e^{xy}dx + (1 + 2xy^{-1} - 2y^{-2})e^{xy}dy = 0$$

has a general solution

$$e^{xy}(x^{-1} + 2y^{-1}) = c.$$

Example 1.6.5 Suppose that $M(x, y)dx + N(x, y)dy = 0$ is a homogeneous equation, that is, $M(x, y)$ and $N(x, y)$ are homogeneous functions of the same degree, say, n . Prove the function

$$\mu = [xM(x, y) + yN(x, y)]^{-1} \quad (1.6.9)$$

is an integrating factor of the differential equation, provided

$$xM(x, y) + yN(x, y) \neq 0.$$

Solution The homogeneous equation can be written as

$$M\left(x, x \frac{y}{x}\right) dx + N\left(x, x \frac{y}{x}\right) dy = 0. \quad (1.6.10)$$

Substitution $y(x) = xu(x)$ in (1.6.10) gives

$$x^n [M(1, u) + uN(1, u)] dx + x^{n+1} N(1, u) du = 0. \quad (1.6.11)$$

The equation (1.6.11) is a separable equation and admits the integrating factor

$$\mu = \frac{1}{x^{n+1} [M(1, u) + uN(1, u)]} = \frac{1}{xM(x, y) + yN(x, y)}$$

provided $xM + yN \neq 0$.

Exercise 1.6

1. Solve the following differential equations.

(1) $(3x^2y + 2xy + y^3)dx + (x^2 + y^2)dy = 0.$

(2) $ydx + (4xy - e^{-4y})dy = 0.$

(3) $ydx - (2x^2 + 2y^2 + x)dy = 0.$

(4) $xy^3dx + (x^2y^2 - 4)dy = 0.$

(5) $y(1 + e^xy)dx - xdy = 0.$

(6) $(x^2 + 2y^2 + 4x)dx + 2xydy = 0.$

(7) $(x^4 + y^4)dx - 2xy^3dy = 0.$

(8) $\frac{dy}{dx} = \frac{xy - y + y^2}{x + 2y}.$

(9) $x \frac{dy}{dx} + y \ln y - x^2y = 0.$

(10) $\frac{dy}{dx} = \frac{e^{2x} - e^y - y}{1 + e^y}.$

2. Solve the following differential equations for which the type of integrating factor has been indicated.

(1) $y' = \frac{y^2 - x}{xy} [\mu(x)].$

(2) $2ydx + (y^2 - 2x)dy = 0 [\mu(y)].$

(3) $y + x(1 - 3x^2y^2) \frac{dy}{dx} = 0 [\mu(xy)].$

(4) $y' = -\frac{3xy + y^2}{3xy + x^2} [\mu(x + y)].$

$$(5) (x + x^4 + 2x^2y^2 + y^4)dx + ydy = 0 \quad [\mu(x^2 + y^2)].$$

$$(6) (y^2 + 2x^2y)dx + (xy + x^3)dy = 0 \quad [\mu(x^m y^n)].$$

3. Find an integrating factor of the following equations.

(1) First-order linear differential equation

$$y' + p(x)y = q(x).$$

(2) Bernoulli's equation

$$y' + p(x)y = q(x)y^n \quad (n \neq 0, 1).$$

4. The differential equation $(e^x \sec y - \tan y)dx + dy = 0$ has an integrating factor of the form $e^{-ax} \cos y$ for some constant a . Find a , and solve this equation.

5. Suppose that $f(u), g(u)$ are continuously differentiable and $\varphi(x, y) = [f(xy) - g(xy)]xy \neq 0$. Prove that

$$\mu(x, y) = \frac{1}{\varphi(x, y)}$$

is an integrating factor of the equation

$$f(xy)ydx + g(xy)x dy = 0.$$

6. Verify that $\mu(x, y) = (x^2 + y^2)^{-1}$ is an integrating factor of a differential equation of the form

$$[y + xf(x^2 + y^2)]dx + [yf(x^2 + y^2) - x]dy = 0.$$

Hence, solve the differential equation

$$[y + x(x^2 + y^2)^2]dx + [y(x^2 + y^2)^2 - x]dy = 0.$$

7. Suppose that $\mu = \mu(x, y)$ is an integrating factor of the differential equation of

$$M(x, y)dx + N(x, y)dy = 0$$

such that

$$d\Phi(x, y) = \mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy.$$

Show that $\mu(x, y)g[\Phi(x, y)]$ is also an integrating factor of this equation, where $g(t)$ is any one differentiable non-zero function.

8. Suppose that

$$M(x, y)dx + N(x, y)dy = 0$$

is exact and has an integrating factor $\mu(x, y)$ ($\mu(x, y) \neq \text{constant}$), then prove $\mu(x, y) = c$ is a solution of this equation.

9. If $\mu_1(x, y)$ and $\mu_2(x, y)$ are two integrating factors of

$$M(x, y)dx + N(x, y)dy = 0$$

such that their ratio is not a constant, then $\mu_1(x, y) = c\mu_2(x, y)$ is a solution of this equation.

10. Show the following:

(1) $u(x, y) = c$ is the general solution of the differential equation

$$M(x, y)dx + N(x, y)dy = 0$$

if and only if

$$M(x, y) \frac{\partial u}{\partial y} = N(x, y) \frac{\partial u}{\partial x}.$$

(2) If $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$, $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$, then the equation $M(x, y)dx + N(x, y)dy = 0$ has an integrating factor

$$\mu(x, y) = (M^2 + N^2)^{-1}.$$

1.7 Flexibility in Problem Solving

In this section, we introduce some ideas or methods which can transform a typical differential equation into a solvable equation.

1.7.1 Applying Substitution Methods

In previous sections, we have used substitution method to solve some specific types of differential equation. For example, if $y' = f(x, y)$ is a homogeneous equation, then we make substitution $u = \frac{y}{x}$, $y = xu$ and change it into a separable equation. When solving Bernoulli's equation $y' + p(x)y = q(x)y^n$, substitution $z = y^{1-n}$ can transform it into a linear equation. In this section, we will introduce some more proper substitution to solve the differential equation.

$$\text{I. } \frac{dy}{dx} = f(x \pm y). \quad (1.7.1)$$

Substitution $z = x \pm y$ leads to a separable equation

$$\frac{dz}{dx} = f(z) + 1.$$

Example 1.7.1 Solve the differential equation

$$\frac{dy}{dx} = \sin^2(x - y).$$

Solution Substitution $z = x - y$ gives

$$\frac{dz}{dx} = 1 - \frac{dy}{dx} = 1 - \sin^2 z = \cos^2 z$$

which is separable. The general solution is given implicitly by

$$\tan(x - y) = x + c.$$

$$\text{II. } \frac{dy}{dx} = -\frac{y}{x} + f(xy) \quad (1.7.2)$$

Making substitution $z = xy$, we have a separable equation

$$\frac{dz}{dx} = xf(z).$$

Example 1.7.2 Solve the differential equation

$$\frac{dy}{dx} = -\frac{y}{x} - 4x^2y^2 - 2.$$

Solution The substitution $z = xy$ can transform this equation into the separable

equation

$$\frac{dz}{dx} = -x(2 + 4z^2).$$

The general solution is given by

$$y = \frac{x}{\sqrt{2}} \tan(-\sqrt{2}x^2 + c).$$

$$\text{III. } \frac{dy}{dx} = xg\left(ax + b \frac{y}{x}\right) + \frac{y}{x}, \text{ } a, b \text{ are constants.} \quad (1.7.3)$$

By substitution $u = \frac{y}{x}$, we get

$$\frac{du}{dx} = g(ax + bu)$$

which is the first type of the differential equation.

Example 1.7.3 Solve the differential equation

$$\frac{dy}{dx} = \frac{4(x^2 + y)^2 + y}{x}.$$

Solution We rewrite the differential equation in the form

$$\frac{dy}{dx} = 4x \left(x + \frac{y}{x}\right)^2 + \frac{y}{x}.$$

Substitution $z = \frac{y}{x}$ gives

$$\frac{dz}{dx} = 4(x + z)^2.$$

Making another substitution $u = x + z$, we have

$$\frac{du}{dx} = 1 + 4u^2.$$

The general solution of the original differential equation is

$$y = \frac{1}{2}x \tan(2x + c) - x^2.$$

$$\text{IV. } \frac{dy}{dx} = p(x) + q(x)e^{ay}, \text{ } a \neq 0 \text{ is constant.} \quad (1.7.4)$$

Multiplying both sides by e^{ay} , we have

$$e^{ay} \frac{dy}{dx} = p(x)e^{ay} + q(x)e^{2ay}.$$

Substitution $u = e^{ay}$ yields

$$\frac{du}{dx} = ap(x)u + aq(x)u^2$$

which is a Bernoulli's equation with unknown function u .

Example 1.7.4 Solve the differential equation

$$\frac{dy}{dx} = xe^{2y} - \frac{1}{x}.$$

Solution Multiplying it by e^{2y} and making substitution $u = e^{2y}$ yield

$$\frac{du}{dx} = 2xu^2 - \frac{2}{x}u,$$

which is a Bernoulli's equation. The implicit general solution is given by

$$(e^{2y})^{-1} = cx^2 - 2x^2 \ln x.$$

$$V. \frac{dy}{dx} = kx \sqrt{ax^4 + bx^2 + cy}, \text{ where } a > 0, b, c, k \text{ are constants.} \quad (1.7.5)$$

First substitution $v = x^2$ gives

$$2 \frac{dy}{dv} = k \sqrt{av^2 + bv + cy}.$$

Making another substitution $u = \sqrt{av^2 + bv + cy}$, we have

$$\frac{du}{dv} = \frac{\frac{ck}{2} + 2av + b}{2u}$$

which is a differential equation that can be transformed into the homogeneous equation.

V. Riccati's Equation

The equation of the form

$$\frac{dy}{dx} = p(x)y^2 + q(x)y + r(x) \quad (1.7.6)$$

is called **Riccati's equation**, where the functions $p(x), q(x), r(x)$ are continuous in the interval I , $p(x) \neq 0$.

Generally, Riccati's equation can not be solved by elementary integration method. However, if one solution is known, then the remaining solutions can be explicitly calculated.

Theorem 1.7.1 If $y = \varphi(x)$ is a particular solution of Riccati's equation (1.7.6), then we can solve it.

Proof The substitution $y = u + \varphi(x)$ gives

$$\frac{du}{dx} + \frac{d\varphi}{dx} = p(x)[u^2 + 2\varphi u + \varphi^2] + q(x)(u + \varphi) + r(x), \quad (1.7.7)$$

where u is an unknown function. Since $y = \varphi(x)$ is a solution, that is,

$$\frac{d\varphi}{dx} = p(x)\varphi^2 + q(x)\varphi + r(x).$$

So the equation (1.7.7) is changed into

$$\frac{du}{dx} = [2p(x)\varphi(x) + q(x)]u + p(x)u^2,$$

which is a Bernoulli's equation. The proof is completed.

Theorem 1.7.2 Consider Riccati's equation

$$\frac{dy}{dx} + ay^2 = bx^m, \quad (1.7.8)$$

where $a \neq 0, b, m$ are constants. If $x \neq 0, y \neq 0$, then Riccati's equation (1.7.8) can be

transformed into a separable equation by using the appropriate substitution provided

$$m = 0, -2, \frac{-4k}{2k+1}, \frac{-4k}{2k-1} \quad (k = 1, 2, \dots). \quad (1.7.9)$$

Remark 1.7.1 Theorem 1.7.2 was proved by Daniel Bernoulli (1700—1784) in 1725. This shows that Riccati's equation can be solved by elementary integral method and the condition (1.7.9) is sufficient. In fact, Liouville proved that the condition (1.7.9) is also necessary in 1841. Liouville's work plays an important role in development of differential equation. The result reminds us the nonlinear differential equation in the simplest form $\frac{dy}{dx} = x^2 + y^2$ can not be solved by elementary integral method. This makes people find new ways to study differential equation.

Example 1.7.5 Solve the differential equation

$$y' = -y^2 - \frac{1}{4x^2}.$$

Solution This is Riccati's equation. It is easy to find a particular solution

$$\varphi(x) = \frac{1}{2x}.$$

Substitution $y = u(x) + \frac{1}{2x}$ gives

$$\frac{du}{dx} = -\frac{u}{x} - u^2,$$

which is a Bernoulli's equation. The general solution of the original differential equation is

$$y = \frac{1}{2x} + \frac{1}{cx + x \ln x}.$$

1.7.2 Interchanging the Roles of x and y

In fact, the solutions of $y' = f(x, y)$ and $\frac{dx}{dy} = \frac{1}{f(x, y)}$ determine the same curves in a domain in xOy plane provided the function is defined, continuous and nonzero. For this, if $y = \varphi(x)$ is a solution of $y' = f(x, y)$ in the interval I and $\varphi'(x) = f(x, \varphi(x)) \neq 0$, then $\varphi(x)$ is a monotonic function in I and hence has an inverse function $x = \varphi^{-1}(y)$. This function x is such that $\frac{dx}{dy} = \frac{1}{\varphi'(x)} = \frac{1}{f(x, \varphi(x))}$ in I .

Example 1.7.6 Solve the differential equation

$$\frac{dy}{dx} = \frac{1}{x + x^2 y}.$$

Solution Regarding x as unknown function and y as independent variable, one has

$$\frac{dx}{dy} = x + yx^2,$$

which is a Bernoulli's equation with $n = 2$. The general solution is given by

$$x^{-1} = ce^{-y} - y + 1.$$

Example 1.7.7 Solve the differential equation

$$\frac{dy}{dx} = \frac{1}{e^y - x}.$$

Solution The differential equation can be written as

$$\frac{dx}{dy} + x = e^y.$$

This can be solved to obtain

$$x = ce^{-y} + \frac{1}{2}e^y.$$

1.7.3 Changing the Form of Differential Equation

Sometimes, it is convenient to solve the differential equation by changing the form of it.

Example 1.7.8 Solve the differential equation

$$\frac{dy}{dx} = \frac{2x^2 - y^2}{1 + 2xy}.$$

Solution Changing the equation in the symmetric or differential form yields

$$(2x^2 - y^2)dx - (1 + 2xy)dy = 0,$$

which is exact. The implicit general solution is

$$\frac{2}{3}x^3 - xy^2 - y = c.$$

Example 1.7.9 Solve the differential equation

$$(\ln x + xy^2)dx + 2x^2ydy = 0.$$

Solution We change the differential equation in the form

$$\frac{dy}{dx} = -\frac{\ln x + xy^2}{2x^2y},$$

that is

$$\frac{dy^2}{dx} + \frac{1}{x}y^2 = -\frac{\ln x}{x^2}.$$

The implicit general solution is given by

$$y^2 = \frac{1}{x} \left(c - \frac{1}{2} \ln^2 x \right).$$

1.7.4 Introducing Polar Coordinate

Sometimes, the introduction of polar coordinate makes the differential equation be solved easily.

Example 1.7.10 Solve the differential equation

$$\frac{dy}{dx} = \frac{ax + by}{bx - ay}.$$

Solution We introduce the polar coordinates $x = r \cos \theta$, and $y = r \sin \theta$, so the total

differentials are

$$dx = \cos \theta dr - r \sin \theta d\theta, \quad dy = \sin \theta dr + r \cos \theta d\theta.$$

Thus the original differential equation can be transformed into

$$\frac{dr}{d\theta} = \frac{b}{a} r,$$

which is separable. The general solution of this equivalent equation is

$$r = c e^{\frac{b}{a}\theta},$$

that is

$$\sqrt{x^2 + y^2} = c e^{\frac{b}{a} \arctan \frac{y}{x}}.$$

Exercise 1.7

Solve the following differential equations.

1. $\frac{dy}{dx} = x^2 + 4xy + 4y^2.$

2. $\frac{dy}{dx} = \cos^2(x - y).$

3. $\frac{dy}{dx} = 2x e^{x+y} - 1.$

4. $\frac{dy}{dx} = 4x \tan(x + y) - 1.$

5. $\frac{dy}{dx} = -\frac{y}{x} - x^2 y^2 - 4.$

6. $xy' + y = \sin^2(xy).$

7. $y' = x \left(x + \frac{y}{x} \right)^2 + \frac{y}{x}.$

8. $xy' - x^2 e^{\frac{y}{x}} - y = 0.$

9. $y' = \frac{y}{2x} + \frac{1}{2y} \tan \frac{y^2}{x}.$

10. $y' = 6x \sqrt{x^4 + y}.$

11. $y' = \frac{xy^2 + y}{x^2 y - x}.$

12. $y' = 3x e^{xy} - \frac{y}{x}.$

13. $\frac{dy}{dx} = 2\sqrt{y} - 2x.$

14. $\frac{dy}{dx} = e^{-2y} \cos x - 2.$

15. $\frac{dy}{dx} = \frac{1}{2x + x^2 y}.$

16. $\frac{dy}{dx} = \frac{x^4 y}{x^5 - y^3}.$

$$17. \frac{dy}{dx} = \frac{y}{x + y^2 \sin y}.$$

$$18. \frac{dy}{dx} = \frac{y}{e^{2x} - y}.$$

$$19. \frac{dy}{dx} = -\frac{3x^2 y^2 + 1}{2x^3 y + e^y}.$$

$$20. \frac{dy}{dx} = \frac{2x^2 - y^2}{1 + 2xy}.$$

$$21. \frac{dy}{dx} = \frac{x^3 - 2y}{2x + y^2}.$$

$$22. dx - (xy + x^2 y^2) dy = 0.$$

$$23. (x^2 + y^2) dx + (4xy + y^2 - x^2) dy = 0.$$

$$24. \frac{dy}{dx} = \frac{x + 2e^y + 3}{y - 2xe^y}.$$

$$25. x(x^2 + 4y^2) dx + y(3x^2 + y^2) dy = 0.$$

$$26. \frac{dy}{dx} = \frac{2x + 3y}{3x - 2y}.$$

$$27. \frac{dy}{dx} = \frac{x(ax^2 + by^2)}{y(bx^2 - ay^2)}.$$

$$28. x^2 y' = x^2 y^2 + xy + 1. \quad (\text{Hint: } y = -\frac{1}{x} \text{ is a solution})$$

29. Show that second-order differential equation

$$y'' + p(x)y' + q(x)y = 0$$

can be transformed into the Riccati's equation.

30. Show that the Riccati's equation

$$y' + p(x)y + q(x)y^2 = h(x)$$

with $p \in C^0(I)$, $q \in C^1(I)$, $q(x) \neq 0$ in I , can be transformed into a second-order linear differential equation

$$u'' + u' \left(p - \frac{q'}{q} \right) - hqu = 0$$

using the transformation

$$u(x) = \exp \left[\int q(x)y(x) dx \right].$$

1.8 Implicit First-Order Differential Equations

In this section, we consider the implicit differential equation

$$F(x, y, y') = 0. \quad (1.8.1)$$

Throughout this section, we assume that the function $F(x, y, p)$ is continuous in a region D of three-dimensional space.

Generally, it is hard to solve the differential equation (1.8.1), but we can try to get

the general solution of it by the following steps.

Step 1 The equation (1.8.1) may be viewed as a surface in three-dimensional space (x, y, y') . Suppose that

$$x = f(s, t), y = g(s, t), p = y' = h(s, t) \quad (1.8.2)$$

are parametric representations of (1.8.1) with s, t as the parameters.

Step 2 Substituting these equations

$$dx = f_s ds + f_t dt,$$

$$dy = g_s ds + g_t dt,$$

$$p = y' = h(s, t)$$

in the differential expression $dy = y' dx$, we get the following differential equation

$$(g_s - hf_s)ds + (g_t - hf_t)dt = 0. \quad (1.8.3)$$

Step 3 Try to find the general solution of (1.8.3). If the general solution of it is

$$s = u(t, c) \text{ or } t = v(s, c),$$

then substitution back in the first two equations of (1.8.2) gives the general solution of (1.8.1) in parametric form

$$x = f(u(t, c), t), y = g(u(t, c), t)$$

or

$$x = f(s, v(s, c)), y = g(s, v(s, c)).$$

However, finding the proper parametric representation of (1.8.1) and solving the equation (1.8.3) are by no means easier tasks. In the following sections, we will discuss some specific cases of implicit differential equations which can be solved by using this procedure.

Case I If we solve y' from (1.8.1) in the form

$$y' = h(x, y), \quad (1.8.4)$$

then we choose $f(s, t) = s$ and $g(s, t) = t$. This type of equation has been discussed in the previous sections.

Case II From (1.8.1), if we can obtain

$$y = g(x, y'), \quad (1.8.5)$$

then we choose $f(s, t) = s$ and $h(s, t) = t$.

Substitution $y' = p$ in (1.8.5) yields

$$y = g(x, p). \quad (1.8.6)$$

Differentiating both sides of (1.8.6) with respect to x , one has

$$p = g_x(x, p) + g_p(x, p) \frac{dp}{dx} \quad (1.8.7)$$

which is the differential equation with two variables x and p .

If $p = \omega(x, c)$ is the solution of (1.8.7), then substitution $p = \omega(x, c)$ in (1.8.6) gives a general solution

$$y = g(x, \omega(x, c)).$$

If we solve the implicit general solution of (1.8.7)

$$G(x, p, c) = 0,$$

then the general solution of (1.8.5) in parametric form is given by

$$\begin{cases} G(x, p, c) = 0, \\ y = g(x, p). \end{cases}$$

Example 1.8.1 Solve the differential equation

$$y = (y')^2 - xy' + \frac{x^2}{2}.$$

Solution We rewrite the differential equation in the form

$$y = p^2 - xp + \frac{x^2}{2}, (p = y'). \quad (1.8.8)$$

Differentiating both sides with respect to x gives

$$\left(\frac{dp}{dx} - 1\right)(2p - x) = 0.$$

If $2p - x = 0$, that is, $p = \frac{x}{2}$, then substituting $p = \frac{x}{2}$ in (1.8.8) yields a particular solution

$$y = \frac{x^2}{4}.$$

If $\frac{dp}{dx} - 1 = 0$, then integrating both sides gives $p = x + c$. Substituting $p = x + c$ in (1.8.8), the general solution of the original differential equation is

$$y = \frac{x^2}{2} + cx + c^2.$$

Example 1.8.2 Solve Clairaut's equation

$$y = xp + f(p), \left(p = \frac{dy}{dx}\right) \quad (1.8.9)$$

where $f'' \neq 0$.

Solution Differentiating both sides of (1.8.9) with respect to x , we have

$$p = p + x \frac{dp}{dx} + f'(p) \frac{dp}{dx},$$

that is

$$[x + f'(p)] \frac{dp}{dx} = 0.$$

If $\frac{dp}{dx} = 0$, then $p = c$. Thus the general solution of Clairaut's equation is

$$y = cx + f(c) \quad (1.8.10)$$

where c is an arbitrary constant.

If $x + f'(p) = 0$, we obtain a particular solution

$$\begin{cases} x = -f'(p) \\ y = -pf'(p) + f(p) \end{cases} \quad (1.8.11)$$

where p is a parameter.

By $f''(p) \neq 0$, we get the inverse function $p = w(x)$ from $x = -f'(p)$. The particular

solution can be written as

$$y = xw(x) + f(w(x)) \quad (1.8.12)$$

whose derivative $\frac{dy}{dx} = w(x)$.

We easily obtain the tangent line of the particular solution (1.8.12) at the point (x_0, y_0)

$$y = c_0 x + f(c_0)$$

where $c_0 = w(x_0)$. This shows that the particular solution (1.8.12) is tangent to some one solution of (1.8.10) at any point (x_0, y_0) . In addition, $w'(x) = -\frac{1}{f''(w(x))} \neq 0$, so $w(x)$ is not a constant. Thus the particular solution (1.8.12) is not included in the general solution (1.8.10). The “extra solution” which can not be obtained by assigning particular values of the constants, is called a **singular solution** of Clairaut’s equation.

Case III From (1.8.1), if we get

$$x = f(y, y'), (y' = p) \quad (1.8.13)$$

then we take $g(s, t) = s$, $h(s, t) = t$.

Applying similar method in Case II, we have

$$\frac{1}{p} = f_y(y, p) + f_p(y, p) \frac{dp}{dy} \quad (1.8.14)$$

by differentiating both sides of (1.8.13) with respect to y .

If we solve the general solution of (1.8.14)

$$p = w(y, c),$$

then substitution $p = w(y, c)$ in (1.8.13) gives the general solution of the original differential equation

$$x = f(y, w(y, c)).$$

If we get the implicit general solution of (1.8.14)

$$F(y, p, c) = 0,$$

then the general solution in parametric form is given by

$$\begin{cases} x = f(y, p), \\ F(y, p, c) = 0. \end{cases}$$

Example 1.8.3 Solve the differential equation

$$4xyy' = (y')^3 + 8y^2.$$

Solution We easily obtain

$$x = \frac{(y')^2}{4y} + \frac{2y}{y'} = \frac{p^2}{4y} + \frac{2y}{p}, (p = y'). \quad (1.8.15)$$

Differentiating both sides with respect to y yields

$$\left(\frac{p^3 - 4y^2}{2yp^2} \right) \left(\frac{dp}{dy} - \frac{p}{y} \right) = 0.$$

If $p^3 - 4y^2 = 0$, that is, $p = (4y^2)^{\frac{1}{3}}$, then substituting this back in (1.8.15), we get the

particular solution

$$y = \frac{4}{27}x^3.$$

If $\frac{dp}{dy} = \frac{p}{2y}$, then the solution is $p = cy^{\frac{1}{2}}$. So the general solution of the original differential equation is

$$x = \frac{c^2}{4} + \frac{2y^{\frac{1}{2}}}{c}.$$

Example 1.8.4 Solve the differential equation

$$y = x\left(y' + \frac{1}{y}\right) + (y')^4.$$

Solution Consider pairs $(x(p), y(p))$ of continuously differentiable functions with the property

$$\frac{dy}{dp} = p \frac{dx}{dp} \tag{1.8.16}$$

where $p = \frac{dy}{dx}$.

Differentiation of $y(p) = x(p)\left(p + \frac{1}{p}\right) + p^4$ with respect to p leads to

$$\frac{dy}{dp} = \frac{dx}{dp}\left(p + \frac{1}{p}\right) + x(p)\left(1 - \frac{1}{p^2}\right) + 4p^3.$$

Together with (1.8.16), one has

$$\frac{dx}{dp} = x(p)\left(\frac{1}{p} - p\right) - 4p^4 \tag{1.8.17}$$

which is a linear differential equation with unknown function $x(p)$.

The general solution of (1.8.17) is given by

$$x(p) = cpe^{-\frac{p^2}{2}} + 8p - 4p^3.$$

Thus the general solution of the original differential equation reads

$$\begin{cases} x(p) = cpe^{-\frac{p^2}{2}} + 8p - 4p^3 \\ y(p) = x(p)\left(p + \frac{1}{p}\right) + p^4 \end{cases}$$

where p is a parameter.

Case IV Consider

$$F(x, y') = 0, \quad y \text{ is missing.}$$

Here, we choose $x = f(s)$, $y = t$, $y' = h(s)$. Since $dy = y' dx$, we have

$$dy = h(s)f'(s)ds, \quad y = \int h(s)f'(s)ds + c.$$

So the general solution of the differential equation in parametric form reads

$$\begin{cases} x = f(s) \\ y = \int h(s)f'(s)ds + c \end{cases}$$

where c is a parameter.

Case V Similarly, we consider

$$F(y, y') = 0, \quad x \text{ is missing.}$$

Here we choose $x = s, y = g(t), y' = h(t)$. The expression $dy = y' dx$ gives

$$g'(t) dt = h(t) dx,$$

$$x = \int \frac{g'(t)}{h(t)} dt + c.$$

So the general solution of this equation is given by

$$\begin{cases} x = \int \frac{g'(t)}{h(t)} dt + c, \\ y = g(t). \end{cases}$$

Example 1.8.5 Solve the differential equation

$$x^3 + (y')^3 - 3xy' = 0.$$

Solution Let $y' = tx$, we have

$$x^3 + t^3 x^3 - 3tx^2 = 0$$

which leads to

$$x = \frac{3t}{1+t^3}.$$

So the differential equation in parametric form is

$$\begin{cases} x = \frac{3t}{1+t^3}, \\ y' = \frac{3t^2}{1+t^3}. \end{cases}$$

Substitution in $dy = y' dx$ yields

$$\begin{aligned} dy &= \frac{9t^2(1-2t^3)}{(1+t^3)^3} dt \\ y &= \int \frac{9t^2(1-2t^3)}{(1+t^3)^3} dt \\ &= \frac{3(1+4t^3)}{2(1+t^3)^2} + c. \end{aligned}$$

Thus the general solution of the original equation reads

$$\begin{cases} x = \frac{3t}{1+t^3}, \\ y = \frac{3(1+4t^3)}{2(1+t^3)^2} + c. \end{cases}$$

Example 1.8.6 Solve the differential equation

$$2y = \sqrt{1 + (y')^2}.$$

Solution Suppose that $y' = \text{sh } t$, then $y = \frac{1}{2} \text{ch } t$. Thus

$$dx = \frac{1}{y} dy = \frac{1}{\operatorname{sh} t} \frac{\operatorname{sh} t}{2} dt = \frac{1}{2} dt,$$

$$x = \frac{1}{2}t + c.$$

The general solution in parametric form reads

$$\begin{cases} x = \frac{1}{2}t + c \\ y = \frac{1}{2}\operatorname{ch} t \end{cases}$$

that is

$$y = \frac{1}{2}\operatorname{ch}(2x - 2c).$$

Exercise 1.8

Find the general solutions of the following equations.

1. $(y')^2 - 2y^2 = 0.$

2. $3x^2 - (y')^2 = 1.$

3. $y^2 + 2(y')^2 = 3.$

4. $2y - x^2 + (y')^2 = 0.$

5. $y - (y')^3 = 0.$

6. $y = (y' - 2)e^{y'}.$

7. $y^2(y' - 2) = (4 - y')^2.$

8. $x^3 + (y')^3 - xy' = 0.$

9. $(y')^2 - 2xy' + y = 0.$

10. $y = xy' + x^2(y')^4.$

11. Determine all the solutions for the following Clairaut's differential equations in explicit form.

(1) $y = xy' + (y')^2.$

(2) $y = 2xy' - \sqrt{y' - 1}.$

12. Prove that all solutions of the differential equation

$$y = xy' + ay' + b \quad (a, b \text{ are constants})$$

are lines through a fixed point.

13. Solve Lagrange's equation $y = xf(y') + g(y')$, where $f(p)$ and $g(p)$ are continuously differentiable.

14. Determine the solutions of the following differential equations in parametric form

(1) $y = 2xy' + y' - (y')^2.$

(2) $x = yy' + \frac{1}{y'}.$

1.9 High-Order Differential Equations

In this section, we introduce several kinds of high-order differential equation that can be solved by elementary integral method. The basic thought is to reduce the order of the differential equation.

$$I \quad F(x, y^{(k)}, y^{(k+1)}, \dots, y^{(n)}) = 0 \quad (1.9.1)$$

Since the lowest order of this differential equation is k , so we assume that $u = y^{(k)}$. Then substitution leads to

$$F(x, u, u', \dots, u^{(n-k)}) = 0. \quad (1.9.2)$$

If the general solution of (1.9.2) is given by

$$u = f(x, c_1, c_2, \dots, c_{n-k}),$$

then integrating the last equation k times yields the general solution of the original differential equation (1.9.1).

Example 1.9.1 Solve the differential equation

$$xy^{(6)} - y^{(5)} = 0.$$

Solution Making substitution $u = y^{(5)}$, we have

$$\frac{du}{dx} - \frac{u}{x} = 0,$$

whose general solution is $u = cx$, that is, $y^{(5)} = cx$. By integrating it five times, we get the general solution of the original differential equation

$$y(x) = c_1 x^6 + c_2 x^4 + c_3 x^3 + c_4 x^2 + c_5 x + c_6.$$

Example 1.9.2 Solve the differential equation.

$$y'' = a \sqrt{1 + (y')^2}, \quad a \text{ is constant.} \quad (1.9.3)$$

Solution The equation (1.9.3) is called the **catenary equation**. Substitution $z = y'$ transforms the equation (1.9.3) into

$$\frac{dz}{dx} = a \sqrt{1 + z^2}$$

which is a separable equation. Separating variables and integrating both sides give

$$\ln(z + \sqrt{1 + z^2}) = a(x - c_1)$$

or

$$z + \sqrt{1 + z^2} = e^{a(x - c_1)}.$$

Furthermore

$$y' = z = \operatorname{sh} a(x - c_1). \quad (1.9.4)$$

Integrating (1.9.4) gives the general solution of (1.9.3)

$$y = \frac{1}{a} \operatorname{ch} a(x - c_1) + c_2.$$

If the boundary value conditions are given by

$$y(x_0) = A \quad \text{and} \quad y(x_1) = B,$$

then we can determine the integral constants c_1, c_2 and obtain a particular solution.

$$\text{II} \quad F(y, y', \dots, y^n) = 0. \quad (1.9.5)$$

In this differential equation, the independent variable x is missing. Let $u = \frac{dy}{dx}$, then one has

$$\begin{cases} \frac{d^2 y}{dx^2} = \frac{du}{dx} = \frac{du}{dy} \cdot \frac{dy}{dx} = u \frac{du}{dy}, \\ \frac{d^3 y}{dx^3} = \frac{d}{dx} \left(u \frac{du}{dy} \right) = u^2 \frac{d^2 u}{dy^2} + u \left(\frac{du}{dy} \right)^2, \\ \vdots \\ \frac{d^n y}{dx^n} = \varphi \left(u, \frac{du}{dy}, \dots, \frac{d^{n-1} u}{dy^{n-1}} \right). \end{cases} \quad (1.9.6)$$

Now the substitution (1.9.6) leads to the $n-1$ th-order differential equation

$$F_0 \left(y, u, \frac{du}{dy}, \dots, \frac{d^{n-1} u}{dy^{n-1}} \right) = 0 \quad (1.9.7)$$

where y is independent variable, u is unknown function.

Example 1.9.3 Solve the differential equation

$$4yy'' = 1.$$

Solution Substitutions $u = y', y'' = u \frac{du}{dy}$ lead to

$$4yu \frac{du}{dy} = 1, \quad (1.9.8)$$

which is a separable equation. The general solution of (1.9.8) is

$$y = c_1 e^{2u^2} = c_1 e^{2(y')^2}, \quad c_1 \neq 0.$$

Applying the corresponding method, we obtain the general solution of the differential equation in parametric form

$$\begin{cases} x = 4c_1 \int e^{p^2} dp + c_2, \\ y = c_1 e^{2p^2}. \end{cases}$$

Example 1.9.4 Solve the differential equation

$$y'' + 4y = 0.$$

Solution Substitutions $y' = u, y'' = u \frac{du}{dy}$ give

$$u \frac{du}{dy} + 4y = 0,$$

that is

$$u du + 4y dy = 0.$$

Making integration, one has

$$u^2 + 4y^2 = a^2 \quad (a \text{ is constant}).$$

Furthermore

$$\frac{dy}{\sqrt{a^2 - 4y^2}} = \pm dx.$$

Integrating both sides of last equation yields

$$\arcsin \frac{2y}{a} = b \pm 2x, \quad b \text{ is constant.}$$

That is

$$y = \frac{a}{2} \sin(b \pm 2x)$$

or

$$y = c_1 \sin 2x + c_2 \cos 2x, \quad c_1 \text{ and } c_2 \text{ are constants.}$$

$$\text{III} \quad F(x, y, y', y'') = 0 \quad (1.9.9)$$

Here F satisfies

$$F(x, ty, ty', ty'') = t^m F(x, y, y', y''),$$

that is, F is a homogeneous function about variables y, y', y'' of the degree m .

Making substitution $y' = uy, y'' = (u' + u^2)y$, we have

$$F(x, y, y', y'') = y^m F(x, 1, u, u' + u^2).$$

Thus the substitution $y' = uy$ transforms (1.9.9) into

$$F(x, 1, u, u' + u^2) = 0.$$

Example 1.9.5 Solve the differential equation

$$xyy'' - x(y')^2 - yy' = 0.$$

Solution The left side is a homogeneous function about variables y, y', y'' of the degree two. Substitutions $y' = uy, y'' = (u' + u^2)y$ in the original differential equation yield

$$x \frac{du}{dx} - u = 0$$

which is a separable equation. The general solution is

$$u = 2c_1 x.$$

Together with $y' = uy$, we obtain the general solution

$$y = c_2 e^{c_1 x^2}.$$

$$\text{IV} \quad \text{High-Order Exact Equation } F(x, y, y', \dots, y^{(n)}) = 0 \quad (1.9.10)$$

The differential equation (1.9.10) is called a high-order exact equation provided the left side is the derivative of some function with respect to x , that is,

$$\frac{d}{dx} G(x, y, y', \dots, y^{(n-1)}) = F(x, y, y', \dots, y^{(n)}).$$

Thus the differential equation (1.9.10) is transformed into

$$\frac{d}{dx} G(x, y, y', \dots, y^{(n-1)}) = 0.$$

Hence

$$G(x, y, y', \dots, y^{(n-1)}) = c.$$

Then we try to solve the last differential equation.

Example 1.9.6 Solve the differential equation

$$yy' + x(y')^2 + xyy'' = 0.$$

Solution The differential equation is rewritten as

$$d(xyy') = 0,$$

so

$$xyy' = c_1,$$

that is

$$y dy = \frac{c_1}{x} dx.$$

Integrating both sides gives the general solution

$$\frac{1}{2}y^2 = c_1 \ln|x| + c_2.$$

Example 1.9.7 Solve the differential equation

$$2y(y')^2 + y^2y'' = 0.$$

Solution The left side is the derivative of the function y^2y' with respect to x , that is

$$\frac{d(y^2y')}{dx} = 2y(y')^2 + y^2y'' = 0,$$

so

$$y^2y' = c_1.$$

This is a separable differential equation. The general solution is

$$\frac{1}{3}y^3 = c_1x + c_2.$$

Example 1.9.8 Solve the differential equation

$$yy'' + (y')^2 + 3 = 0.$$

Solution We rewrite the differential equation as

$$\frac{d}{dx}(yy' + 3x) = 0.$$

Thus

$$yy' + 3x = c_1.$$

The general solution of the original equation is

$$\frac{1}{2}y^2 = -\frac{3}{2}x^2 + c_1x + c_2.$$

Exercise 1.9

Solve the following differential equations.

1. $(xy''' - y'')^2 = (y''')^2 + 1.$

2. $(y'')^2 = (1 + (y')^2)^3.$

3. $yy'' - (y')^2 = 0.$

4. $yy'' - (y')^2 - y^2y' = 0.$

5. $yy'' - (y')^2 - 4xy^2 = 0.$

6. $y'' - (y')^2 = e^{-y}.$

$$7. y''' + \ln y''' = x + y''.$$

$$8. 2x(yy'' + (y')^2) + 3yy' = 4x^3. \text{ (Hint: } u = y^2 \text{)}$$

$$9. 2x^2yy'' + (xy' - y)^2 = 0.$$

$$10. xy'' + (x^2 - 1)(y' - 1) = 0.$$

$$11. x^3y'' + 2xyy' - x^2(y')^2 - y^2 = 0.$$

$$12. x^2yy'' - x^2(y')^2 - 4xyy' + 8y^2 = 0.$$

Chapter 2 Linear Systems of Differential Equations

In the preceding chapter, we have discussed elementary integral method for solving an ordinary differential equation which involves only one dependent variable. Many applications, however, require the use of two or more dependent variables, each a function of a single independent variable. Such a problem leads naturally to a system of simultaneous ordinary differential equations.

In this chapter, we are concerned to a special kind of the system of differential equations, named after linear system of differential equations. Although, this type of system can not be solved by using elementary integral method, but we can conclude the simple and clear structure of the general solution in theory. This important fact is not only a cornerstone of the theory about linear system of differential equation, but also widely applied in the study of nonlinear differential equation.

2.1 First-Order System of Differential Equations

We will restrict our attention to systems in which the number of equations is the same as the number of dependent variables (unknown functions). We will denote usually the independent variable by t and the dependent variables (the unknown functions of t) by $x_1, x_2, \dots, x_n$ or by $x, y, z, \dots$.

The first-order system of differential equations involving n unknown functions $x_1, x_2, \dots, x_n$ has the standard form

$$\left\{ \begin{array}{l} \frac{dx_1}{dt} = f_1(t, x_1, x_2, \dots, x_n), \\ \frac{dx_2}{dt} = f_2(t, x_1, x_2, \dots, x_n), \\ \vdots \\ \frac{dx_n}{dt} = f_n(t, x_1, x_2, \dots, x_n), \end{array} \right. \quad (2.1.1)$$

where $f_1, f_2, \dots, f_n$ are continuous functions in the region D of $n+1$ dimensional space.

A solution of this system (2.1.1) is a group of functions $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ of t that satisfy all equations identically over some interval of values of t , that is,

$$\frac{d\varphi_i(t)}{dt} = f_i(t, \varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)) \quad (i = 1, 2, \dots, n).$$

The solution of the form that contains n arbitrary constants

$$\begin{cases} x_1 = \varphi_1(t, c_1, c_2, \dots, c_n), \\ x_2 = \varphi_2(t, c_1, c_2, \dots, c_n), \\ \vdots \\ x_n = \varphi_n(t, c_1, c_2, \dots, c_n) \end{cases} \quad (2.1.2)$$

is called the general solution of the system (2.1.1). If the general solution satisfies the system of equations

$$\begin{cases} \Phi_1(t; \varphi_1(t), \varphi_2(t), \dots, \varphi_n(t); c_1, c_2, \dots, c_n) = 0, \\ \Phi_2(t; \varphi_1(t), \varphi_2(t), \dots, \varphi_n(t); c_1, c_2, \dots, c_n) = 0, \\ \vdots \\ \Phi_n(t; \varphi_1(t), \varphi_2(t), \dots, \varphi_n(t); c_1, c_2, \dots, c_n) = 0, \end{cases} \quad (2.1.3)$$

then the system (2.1.3) is called the implicit general solution (or general integral) of the system (2.1.1).

If we solve the initial value problem, that is, the system (2.1.1) together with the initial condition

$$x_1(t_0) = x_{10}, x_2(t_0) = x_{20}, \dots, x_n(t_0) = x_{n0}, \quad (2.1.4)$$

then we only need to substitute the condition (2.1.4) back in the general solution and further obtain the values c_i , $i = 1, 2, \dots, n$. So, we can solve the initial value problem.

Example 2.1.1 An n th-order differential equation is written as

$$\frac{d^n x}{dt^n} = F(t, x, x', \dots, x^{(n-1)}). \quad (2.1.5)$$

Making substitution $x_1 = x, x_2 = x', \dots, x_n = x^{(n-1)}$, we have the following equivalent system of differential equations

$$\begin{cases} \frac{dx_1}{dt} = x_2, \\ \frac{dx_2}{dt} = x_3, \\ \vdots \\ \frac{dx_{n-1}}{dt} = x_n, \\ \frac{dx_n}{dt} = F(t, x_1, x_2, \dots, x_n). \end{cases} \quad (2.1.6)$$

The “equivalence” means that if $x = \varphi(t)$ is a solution of (2.1.5), then the family of functions $x_1 = \varphi(t), x_2 = \varphi'(t), \dots, x_n = \varphi^{(n-1)}(t)$ is a solution of the system (2.1.6). Conversely, the group of functions $x_1 = \varphi_1(t), x_2 = \varphi_2(t), \dots, x_n = \varphi_n(t)$ is a solution of (2.1.6), then the first function $x = \varphi_1(t)$ is a solution of (2.1.5).

Example 2.1.2 Consider a particle of mass m that moves in space under the influence of a force field $\mathbf{F} = (F_1, F_2, F_3)$ that depends on time t , the position $(x(t), y(t), z(t))$ of the particle and its velocity $(x'(t), y'(t), z'(t))$. Applying Newton's law $m\mathbf{a} = \mathbf{F}$

componentwise, we have the system

$$\begin{cases} mx'' = F_1(t, x, y, z, x', y', z'), \\ my'' = F_2(t, x, y, z, x', y', z'), \\ mz'' = F_3(t, x, y, z, x', y', z'), \end{cases} \quad (2.1.7)$$

of three second-order equations with independent variable t and dependent variables x, y, z , where F_1, F_2, F_3 are the components of the vector-valued function $\mathbf{F}$.

Substitutions $x_1 = x, x_2 = x', x_3 = y, x_4 = y', x_5 = z$ and $x_6 = z'$ in (2.1.7) give the following first-order system of differential equations

$$\begin{cases} \frac{dx_1}{dt} = x_2, \\ \frac{dx_2}{dt} = \frac{1}{m}F_1(t, x_1, x_2, x_3, x_4, x_5, x_6), \\ \frac{dx_3}{dt} = x_4, \\ \frac{dx_4}{dt} = \frac{1}{m}F_2(t, x_1, x_2, x_3, x_4, x_5, x_6), \\ \frac{dx_5}{dt} = x_6, \\ \frac{dx_6}{dt} = \frac{1}{m}F_3(t, x_1, x_2, x_3, x_4, x_5, x_6). \end{cases} \quad (2.1.8)$$

It is of both practical and theoretical importance that any such higher-order system can be transformed into an equivalent system of first-order system of differential equations.

In the following, we introduce the notation and terminology of matrices and vectors. The use of matrix notation will be the simplification of computations with systems of differential equation, especially those computations that would be burdensome in scalar notation.

Let the column vectors be

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}, \quad \mathbf{f}(t, \mathbf{x}) = \begin{bmatrix} f_1(t, x_1, x_2, \dots, x_n) \\ f_2(t, x_1, x_2, \dots, x_n) \\ \vdots \\ f_n(t, x_1, x_2, \dots, x_n) \end{bmatrix},$$

and define

$$\frac{d\mathbf{x}(t)}{dt} = \begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \\ \vdots \\ \frac{dx_n}{dt} \end{bmatrix}, \quad \int_{t_0}^t \mathbf{f}(t) dt = \begin{bmatrix} \int_{t_0}^t f_1(t) dt \\ \int_{t_0}^t f_2(t) dt \\ \vdots \\ \int_{t_0}^t f_n(t) dt \end{bmatrix},$$

and then the system (2.1.1) reads in vector form

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}(t, \mathbf{x}) \quad (2.1.9)$$

and the initial value problem is written as

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{f}(t, \mathbf{x}), \\ \mathbf{x}(t_0) = \mathbf{x}_0, \end{cases} \quad (2.1.10)$$

where

$$\mathbf{x}_0 = \begin{bmatrix} x_{10} \\ x_{20} \\ \vdots \\ x_{n0} \end{bmatrix}.$$

2.2 First-Order Linear System of Differential Equations

The system of the form

$$\begin{cases} \frac{dx_1}{dt} = a_{11}(t)x_1 + a_{12}(t)x_2 + \cdots + a_{1n}(t)x_n + f_1(t), \\ \frac{dx_2}{dt} = a_{21}(t)x_1 + a_{22}(t)x_2 + \cdots + a_{2n}(t)x_n + f_2(t), \\ \vdots \\ \frac{dx_n}{dt} = a_{n1}(t)x_1 + a_{n2}(t)x_2 + \cdots + a_{nn}(t)x_n + f_n(t) \end{cases} \quad (2.2.1)$$

is called the first-order linear system of differential equations.

The system (2.2.1) can be written in matrix notation

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t), \quad (2.2.2)$$

where

$$\mathbf{A}(t) = \begin{bmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & & \vdots \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{bmatrix}, \quad \mathbf{f}(t) = \begin{bmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{bmatrix} \quad (2.2.3)$$

are a matrix and a column vector respectively.

If $\mathbf{f}(t) \equiv 0$, the system is called **homogeneous**, otherwise, it is called **nonhomogeneous**.

Example 2.2.1 Consider n th-order linear differential equation

$$x^{(n)} + p_1(t)x^{(n-1)} + \cdots + p_{n-1}(t)x' + p_n(t)x = f(t). \quad (2.2.4)$$

Substitutions $x_1 = x(t)$, $x_2 = x'(t)$, $\cdots$, $x_{n-1} = x^{(n-2)}(t)$, $x_n = x^{(n-1)}(t)$ lead to

$$\left\{ \begin{array}{l} \frac{dx_1}{dt} = x_2, \\ \frac{dx_2}{dt} = x_3, \\ \vdots \\ \frac{dx_{n-1}}{dt} = x_n, \\ \frac{dx_n}{dt} = -p_n(t)x_1 - p_{n-1}(t)x_2 - \cdots - p_2(t)x_{n-1} - p_1(t)x_n + f(t) \end{array} \right. \quad (2.2.5)$$

or

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t) \quad (2.2.6)$$

where

$$\mathbf{A}(t) = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ -p_n & -p_{n-1} & -p_{n-2} & \cdots & -p_2 & -p_1 \end{bmatrix}, \quad \mathbf{f}(t) = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ f(t) \end{bmatrix}.$$

2.3 Existence and Uniqueness for First-Order System

In this section, we discuss existence and uniqueness of solutions of first-order system of differential equation (2.1.1) and linear system of differential equation (2.2.1), which lays a foundation for the theory of linear differential equation.

We denote the matrix $\mathbf{A}(t)$ and n -dimensional column vector $\mathbf{f}(t)$ as shown in (2.2.3). In this notation, statements of the type “ $\mathbf{A}(t), \mathbf{f}(t)$ are continuous, differentiable, $\cdots$ ” mean that each component of $\mathbf{A}(t), \mathbf{f}(t)$ is continuous, differentiable, $\cdots$. Derivatives and integrals of a matrix-valued function $\mathbf{A}(t)$ are also defined componentwise.

$$\mathbf{A}'(t) = [a'_{ij}(t)]_{n \times n} \text{ and } \int_a^b \mathbf{A}(t) dt = \left[\int_a^b a_{ij}(t) dt \right]_{n \times n}.$$

If

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix},$$

then we call

$$\|\mathbf{x}\| = \sum_{i=1}^n |x_i|, \quad \|\mathbf{A}\| = \sum_{i=1}^n \sum_{j=1}^n |a_{ij}|$$

the norms of $\mathbf{x}$ and $\mathbf{A}$ respectively. Further, we have the following properties.

1. $\|\mathbf{x}\| \geq 0$ and $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

2. $\|\mathbf{A}\mathbf{x}\| \leq \|\mathbf{A}\| \cdot \|\mathbf{x}\|$.
3. $\|\mathbf{A}\mathbf{B}\| \leq \|\mathbf{A}\| \cdot \|\mathbf{B}\|$, where $\mathbf{B}$ is also a $n \times n$ matrix, especially, for any natural number m , we have $\|\mathbf{A}^m\| \leq \|\mathbf{A}\|^m$.
4. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$, where $\mathbf{y}$ is also a n dimensional vector.
5. If $\mathbf{x}(t)$ is a n dimensional vector-valued function and continuous in the interval $[a, b]$, then

$$\left\| \int_a^b \mathbf{x}(t) dt \right\| \leq \int_a^b \|\mathbf{x}(t)\| dt.$$

Consider the initial value problem

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t), \\ \mathbf{x}(t_0) = \mathbf{x}_0, \end{cases} \quad (2.3.1)$$

where $\mathbf{A}(t)$ is $n \times n$ matrix-valued function, $\mathbf{f}(t)$ is a vector-valued function. We have the following conclusion.

Theorem 2.3.1 Suppose that $\mathbf{A}(t)$ and $\mathbf{f}(t)$ are continuous in the interval I . Then, for any $t_0 \in I$ and any one n -dimensional constant vector $\mathbf{x}_0$, the initial value problem (2.3.1) has one and only one solution in the whole interval I .

Proof We only give the simple procedure.

Step 1 The initial value problem (2.3.1) is equivalent to the integral equation

$$\mathbf{x}(t) = \mathbf{x}_0 + \int_{t_0}^t [\mathbf{A}(s)\mathbf{x}(s) + \mathbf{f}(s)] ds. \quad (2.3.2)$$

“Equivalence” means that if $\mathbf{x} = \boldsymbol{\varphi}(t)$ is a solution of (2.3.1), then integrating both sides of the following equation

$$\boldsymbol{\varphi}'(t) = \mathbf{A}(t)\boldsymbol{\varphi}(t) + \mathbf{f}(t)$$

from t_0 to t yields the integral equation. Conversely, if $\mathbf{x} = \boldsymbol{\varphi}(t)$ satisfies (2.3.2), that is

$$\boldsymbol{\varphi}(t) = \mathbf{x}_0 + \int_{t_0}^t [\mathbf{A}(s)\boldsymbol{\varphi}(s) + \mathbf{f}(s)] ds,$$

then we easily obtain that $\boldsymbol{\varphi}(t)$ is a solution of (2.3.1).

Step 2 Construct successive approximations or Picard iterates.

$$\begin{aligned} \boldsymbol{\varphi}_0(t) &= \mathbf{x}_0, \\ \boldsymbol{\varphi}_1(t) &= \mathbf{x}_0 + \int_{t_0}^t [\mathbf{A}(s)\boldsymbol{\varphi}_0(s) + \mathbf{f}(s)] ds, \\ &\vdots \\ \boldsymbol{\varphi}_k(t) &= \mathbf{x}_0 + \int_{t_0}^t [\mathbf{A}(s)\boldsymbol{\varphi}_{k-1}(s) + \mathbf{f}(s)] ds, k = 1, 2, \dots. \end{aligned} \quad (2.3.3)$$

Applying mathematical induction, we easily prove that Picard iterates $\boldsymbol{\varphi}_k(t), k = 1, 2, \dots$ are continuous in the interval I .

Step 3 Picard iterates (2.3.3) converges uniformly on any finite closed subinterval I_1 of I .

Assume that I_1 is any finite closed subinterval of I and $t_0 \in I_1$. It suffices to prove the

infinite series

$$\sum_{k=1}^{\infty} (\boldsymbol{\varphi}_k(t) - \boldsymbol{\varphi}_{k-1}(t)) \quad (2.3.4)$$

converges uniformly on I_1 . Let M be common upper bound of $\|\mathbf{A}(t)\|$ and $\|\mathbf{A}(t)\mathbf{x}_0 + \mathbf{f}(t)\|$ on I_1 . We have

$$\|\boldsymbol{\varphi}_1(t) - \boldsymbol{\varphi}_0(t)\| = \left\| \int_{t_0}^t [\mathbf{A}(s)\mathbf{x}_0(s) + \mathbf{f}(s)] ds \right\| \leq M |t - t_0|.$$

By mathematical induction, for each natural number k , we get

$$\|\boldsymbol{\varphi}_k(t) - \boldsymbol{\varphi}_{k-1}(t)\| \leq \frac{M^k |t - t_0|^k}{k!}, t \in I_1$$

which shows that Picard iterates $\boldsymbol{\varphi}_k(t), k = 1, 2, \dots$ converges uniformly on I_1 by Weierstrass Criterion.

Step 4 Integral equation (2.3.2) has at least one solution.

Assume that

$$\boldsymbol{\varphi}_k(t) \rightrightarrows \boldsymbol{\varphi}(t), k \rightarrow \infty, t \in I_1.$$

Taking limit in (2.3.3) as $k \rightarrow \infty$ gives

$$\boldsymbol{\varphi}(t) = \mathbf{x}_0 + \int_{t_0}^t [\mathbf{A}(s)\boldsymbol{\varphi}(s) + \mathbf{f}(s)] ds, t \in I_1.$$

That is, $\boldsymbol{\varphi}(t)$ is a solution of the integral equation (2.3.2) on the interval I_1 . Since I_1 is any one subinterval of I , so $\mathbf{x} = \boldsymbol{\varphi}(t)$ is a continuous solution of (2.3.2) in I .

Step 5 Integral equation (2.3.2) has only one solution. It suffices to prove that if $\mathbf{x} = \boldsymbol{\psi}(t)$ is also a continuous solution of (2.3.2) in the interval $I_0 \subset I$, then

$$\boldsymbol{\psi}(t) \equiv \boldsymbol{\varphi}(t), t \in I_0. \quad (2.3.5)$$

Let $I_2 \subset I_0$ be any one finite closed subinterval and $t_0 \in I_2$. By (2.3.2), we have

$$\begin{aligned} \|\boldsymbol{\psi}(t) - \boldsymbol{\varphi}(t)\| &= \left\| \int_{t_0}^t \mathbf{A}(s)[\boldsymbol{\psi}(s) - \boldsymbol{\varphi}(s)] ds \right\| \\ &\leq N \left| \int_{t_0}^t \|\boldsymbol{\psi}(s) - \boldsymbol{\varphi}(s)\| ds \right|, t \in I_2 \end{aligned} \quad (2.3.6)$$

where N is a common upper bound of $\|\mathbf{A}(t)\|$ and $\|\boldsymbol{\psi}(t) - \boldsymbol{\varphi}(t)\|$ on the interval I_2 . Thus

$$\|\boldsymbol{\psi}(t) - \boldsymbol{\varphi}(t)\| \leq N^2 |t - t_0|, t \in I_2.$$

Substitution this back in the right side of (2.3.6) yields

$$\|\boldsymbol{\psi}(t) - \boldsymbol{\varphi}(t)\| \leq \frac{N^3 |t - t_0|^2}{2!}, t \in I_2.$$

Applying mathematical induction gives

$$\|\boldsymbol{\psi}(t) - \boldsymbol{\varphi}(t)\| \leq \frac{N^{k+1}}{k!} |t - t_0|^k, t \in I_2 \quad (2.3.7)$$

for any natural number k .

The inequality (2.3.7) shows that (2.3.5) holds on I_2 . Since $I_2 \subset I_0$ is any one subinterval, so

$$\boldsymbol{\psi}(t) \equiv \boldsymbol{\varphi}(t), t \in I_0.$$

Remark 2.3.1 Any solution of linear system (2.2.2) can be extended to entire interval I .

Remark 2.3.2 Picard iterates provides a solving method for approximate solution of the initial value problem (2.3.1).

Applying similar method yields the existence and uniqueness of the initial value problem

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{f}(t, \mathbf{x}), \\ \mathbf{x}(t_0) = \mathbf{x}_0, \end{cases}$$

where $\mathbf{f}(t, \mathbf{x})$ is a proper function in the region R of $n + 1$ dimensional space. The detail will be discussed in Chapter 4. Throughout this chapter, we always assume that $\mathbf{A}(t)$ and $\mathbf{f}(t)$ are continuous on I .

Exercise 2.3

1. If $\mathbf{x} = \boldsymbol{\varphi}(t)$ is a solution of $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}$ and there exists some $t_0 \in I$ such that

$$\boldsymbol{\varphi}(t_0) = \mathbf{0},$$

then show that

$$\boldsymbol{\varphi}(t) \equiv \mathbf{0}, t \in I.$$

2. Show that the initial value problem

$$x''(t) = f(t, x), x(t_0) = x_0, x'(t_0) = x_1,$$

where $f(t, x)$ is continuous in a domain D containing the point (t_0, x_0) , is equivalent to the integral equation

$$x(t) = x_0 + (t - t_0)x_1 + \int_{t_0}^t (t - s)f[s, x(s)]ds.$$

3. Suppose that $y = \varphi(x)$ is a solution of the initial value problem

$$y' = yg(x, y), y(0) = 1$$

on the interval $[0, 1]$, where $g(x, y)$ is a bounded and continuous function in the (x, y) plane. Show that there exists a constant c such that

$$|\varphi(x)| \leq e^{cx}, \text{ for all } x \in [0, 1].$$

4. Let $\varphi(x), p(x)$ and $q(x)$ be nonnegative continuous functions on the interval $[x_0 - a, x_0 + a]$ and

$$\varphi(x) \leq p(x) + \left| \int_{x_0}^x q(t)\varphi(t)dt \right| \quad \text{for } |x - x_0| \leq a.$$

Then the following inequality holds

$$\varphi(x) \leq p(x) + \left| \int_{x_0}^x p(t)q(t)\exp\left[\int_t^x q(s)ds\right]dt \right| \quad \text{for } |x - x_0| \leq a.$$

5. Show that the initial value problem

$$\begin{cases} y' + p(x)y = q(x) \\ y(x_0) = y_0 \end{cases}$$

possesses a unique solution for all real x , where $p(x), q(x)$ are continuous in R .

6. Suppose that $\mathbf{x} = \boldsymbol{\varphi}(t)$ is a solution of

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t), t \in R,$$

where $\mathbf{A}(t) = \mathbf{A}(t + \omega)$ and $\mathbf{f}(t) = \mathbf{f}(t + \omega)$, $\omega > 0$. Then $\mathbf{x} = \boldsymbol{\varphi}(t)$ is a periodic function with period ω if and only if $\boldsymbol{\varphi}(0) = \boldsymbol{\varphi}(\omega)$.

7. Let $\boldsymbol{\varphi}(t, t_0, \mathbf{x}_0)$ be a solution of $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}$ in the interval I . Show that

$$\boldsymbol{\varphi}(t, t_0, \alpha\mathbf{x}_1 + \beta\mathbf{x}_2) = \alpha\boldsymbol{\varphi}(t, t_0, \mathbf{x}_1) + \beta\boldsymbol{\varphi}(t, t_0, \mathbf{x}_2),$$

where α, β are arbitrary constants, $t \in I$.

2.4 General Theory of Linear Homogeneous Systems

In this section, we discuss linear homogeneous system of differential equation

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} \quad (2.4.1)$$

and study the structure of the set of all solutions.

Theorem 2.4.1 (Principle of Superposition)

If

$$\boldsymbol{\varphi}_1(t) = \begin{bmatrix} \varphi_{11}(t) \\ \varphi_{21}(t) \\ \vdots \\ \varphi_{n1}(t) \end{bmatrix}, \boldsymbol{\varphi}_2(t) = \begin{bmatrix} \varphi_{12}(t) \\ \varphi_{22}(t) \\ \vdots \\ \varphi_{n2}(t) \end{bmatrix}, \dots, \boldsymbol{\varphi}_m(t) = \begin{bmatrix} \varphi_{1m}(t) \\ \varphi_{2m}(t) \\ \vdots \\ \varphi_{nm}(t) \end{bmatrix} \quad (2.4.2)$$

are m solutions of (2.4.1), then any linear combination of m solutions $\mathbf{x} = c_1\boldsymbol{\varphi}_1(t) + \dots + c_m\boldsymbol{\varphi}_m(t)$ is also a solution of (2.4.1), where $c_i (i = 1, 2, \dots, m)$ are arbitrary constants.

Proof Since $\boldsymbol{\varphi}_i(t) (i = 1, 2, \dots, m)$ are solutions of (2.4.1), that is,

$$\frac{d\boldsymbol{\varphi}_i(t)}{dt} = \mathbf{A}(t)\boldsymbol{\varphi}_i(t),$$

thus

$$\begin{aligned} & \frac{d}{dt}(c_1\boldsymbol{\varphi}_1(t) + c_2\boldsymbol{\varphi}_2(t) + \dots + c_m\boldsymbol{\varphi}_m(t)) \\ &= c_1 \frac{d\boldsymbol{\varphi}_1}{dt} + c_2 \frac{d\boldsymbol{\varphi}_2}{dt} + \dots + c_m \frac{d\boldsymbol{\varphi}_m}{dt} \\ &= c_1\mathbf{A}(t)\boldsymbol{\varphi}_1(t) + c_2\mathbf{A}(t)\boldsymbol{\varphi}_2(t) + \dots + c_m\mathbf{A}(t)\boldsymbol{\varphi}_m(t) \\ &= \mathbf{A}(t)[c_1\boldsymbol{\varphi}_1(t) + c_2\boldsymbol{\varphi}_2(t) + \dots + c_m\boldsymbol{\varphi}_m(t)]. \end{aligned}$$

The proof is completed.

If $\boldsymbol{\varphi}_i(t)$, $(i = 1, 2, \dots, n)$ are n solutions of (2.4.1), then whether the linear combination

$$\mathbf{x}(t) = c_1\boldsymbol{\varphi}_1(t) + c_2\boldsymbol{\varphi}_2(t) + \dots + c_n\boldsymbol{\varphi}_n(t)$$

which contains n arbitrary constants is a general solution of (2.4.1) ?

The question involves the concept of linear independence and dependence of vector-

valued functions.

Definition 2.4.1 The vector-valued functions $\boldsymbol{\varphi}_1(t), \dots, \boldsymbol{\varphi}_m(t)$ defined in an interval I are said to be **linearly dependent** provided there exist constants $c_1, \dots, c_m$ not all zero such that

$$c_1 \boldsymbol{\varphi}_1(t) + c_2 \boldsymbol{\varphi}_2(t) + \dots + c_m \boldsymbol{\varphi}_m(t) = 0 \quad \text{for all } t \in I. \quad (2.4.3)$$

Otherwise, we call these functions **linearly independent**, that is, if the relation

$$c_1 \boldsymbol{\varphi}_1(t) + c_2 \boldsymbol{\varphi}_2(t) + \dots + c_m \boldsymbol{\varphi}_m(t) = 0 \quad \text{for all } t \in I$$

implies that $c_1 = c_2 = \dots = c_m = 0$, then $\boldsymbol{\varphi}_1(t), \dots, \boldsymbol{\varphi}_m(t)$ are said to be linearly independent in I .

The concept of linear independence allows us to distinguish when the given functions are “essentially” different.

Remark 2.4.1 If two functions are linearly dependent in I , then one of these functions is a constant multiple of the other function, while if two functions are linearly independent, then it is impossible to express either function as a constant times the other.

Remark 2.4.2 The functions $\boldsymbol{\varphi}_1(t), \boldsymbol{\varphi}_2(t), \dots, \boldsymbol{\varphi}_m(t)$ are linearly dependent if there exists some one function $\boldsymbol{\varphi}_i(t)$ such that $\boldsymbol{\varphi}_i(t) \equiv 0$.

Example 2.4.1 Show that the functions $1, x, \dots, x^{m-1}$ are linearly independent in the real line.

Solution In fact, the expression $c_1 + c_2 x + \dots + c_m x^{m-1} = 0$ implies that $c_1 = c_2 = \dots = c_m = 0$. If any $c_k \neq 0$, then the equation $c_1 + c_2 x + \dots + c_m x^{m-1} = 0$ could hold at most $m - 1$ values of x , whereas it must hold for all x in the real line.

Example 2.4.2 Show that the functions

$$\boldsymbol{\varphi}_1(t) = \begin{bmatrix} e^{2t} \\ 2e^{2t} \end{bmatrix}, \boldsymbol{\varphi}_2(t) = \begin{bmatrix} 3e^{-t} \\ e^{-t} \end{bmatrix}$$

are linearly independent in every interval I .

Solution Indeed,

$$c_1 \boldsymbol{\varphi}_1(t) + c_2 \boldsymbol{\varphi}_2(t) = c_1 \begin{bmatrix} e^{2t} \\ 2e^{2t} \end{bmatrix} + c_2 \begin{bmatrix} 3e^{-t} \\ e^{-t} \end{bmatrix} = 0$$

implies that

$$\begin{cases} c_1 e^{2t} + 3c_2 e^{-t} = 0, \\ 2c_1 e^{2t} + c_2 e^{-t} = 0, \end{cases}$$

which is possible only for $c_1 = c_2 = 0$.

Theorem 2.4.2 Linear homogeneous system (2.4.1) has n linearly independent solutions

$$\boldsymbol{\varphi}_1(t), \boldsymbol{\varphi}_2(t), \dots, \boldsymbol{\varphi}_n(t) \quad (2.4.4)$$

and the general solution of (2.4.1) is given by

$$\mathbf{x}(t) = c_1 \boldsymbol{\varphi}_1(t) + c_2 \boldsymbol{\varphi}_2(t) + \dots + c_n \boldsymbol{\varphi}_n(t) \quad (2.4.5)$$

where $c_1, c_2, \dots, c_n$ are arbitrary constants.

Proof Let X be the set of all solutions of (2.4.1). Theorem 2.4.1 implies that X is a

linear space. We can prove that the dimension of the linear space X is n . Indeed, for fixed $t_0 \in I$, Theorem 2.3.1 implies that there exists a unique element $\boldsymbol{\varphi}(t)$ in X such that $\boldsymbol{\varphi}(t_0) = \mathbf{x}_0$ for any n dimensional constant vector $\mathbf{x}_0 \in R^n$. That is, we can define a mapping

$$L: \mathbf{x}_0 \mapsto \boldsymbol{\varphi}(t); R^n \rightarrow X.$$

Applying the existence and uniqueness of solutions, it is not difficult to obtain that L is a bijective mapping. In addition, from principle of superposition and uniqueness of solution, we show that the mapping L is linear, that is,

$$L(c_1 \mathbf{x}_1^0 + c_2 \mathbf{x}_2^0) = c_1 L(\mathbf{x}_1^0) + c_2 L(\mathbf{x}_2^0).$$

Thus the mapping L defines an isomorphism between R^n and the space of solutions X , namely, $X \cong R^n$. Hence the dimension of X is n and we can obtain n linearly independent solutions written in the form (2.4.4). So the general solution is given by

$$\mathbf{x}(t) = c_1 \boldsymbol{\varphi}_1(t) + c_2 \boldsymbol{\varphi}_2(t) + \cdots + c_n \boldsymbol{\varphi}_n(t).$$

Every such set of n linearly independent solutions is called a fundamental system of solutions. If $\boldsymbol{\varphi}_1(t), \boldsymbol{\varphi}_2(t), \cdots, \boldsymbol{\varphi}_n(t)$ is a fundamental system, then every solution $\mathbf{x}(t)$ can be written in a unique way as a linear combination

$$\mathbf{x} = c_1 \boldsymbol{\varphi}_1 + c_2 \boldsymbol{\varphi}_2 + \cdots + c_n \boldsymbol{\varphi}_n.$$

For the given n vector-valued functions

$$\boldsymbol{\varphi}_1(t) = \begin{bmatrix} \varphi_{11}(t) \\ \varphi_{21}(t) \\ \vdots \\ \varphi_{n1}(t) \end{bmatrix}, \boldsymbol{\varphi}_2(t) = \begin{bmatrix} \varphi_{12}(t) \\ \varphi_{22}(t) \\ \vdots \\ \varphi_{n2}(t) \end{bmatrix}, \cdots, \boldsymbol{\varphi}_n(t) = \begin{bmatrix} \varphi_{1n}(t) \\ \varphi_{2n}(t) \\ \vdots \\ \varphi_{nn}(t) \end{bmatrix},$$

the determinant $W(t)$, defined by

$$W(t) = \begin{vmatrix} \varphi_{11}(t) & \varphi_{12}(t) & \cdots & \varphi_{1n}(t) \\ \varphi_{21}(t) & \varphi_{22}(t) & \cdots & \varphi_{2n}(t) \\ \vdots & \vdots & & \vdots \\ \varphi_{n1}(t) & \varphi_{n2}(t) & \cdots & \varphi_{nn}(t) \end{vmatrix} \quad (2.4.6)$$

is called the **Wronskian determinant** or Wronskian of these functions. This determinant is closely related to the question of whether $\boldsymbol{\varphi}_1(t), \boldsymbol{\varphi}_2(t), \cdots, \boldsymbol{\varphi}_n(t)$ are linearly independent.

Theorem 2.4.3 (Liouville's Formula) Let $\boldsymbol{\varphi}_1(t), \boldsymbol{\varphi}_2(t), \cdots, \boldsymbol{\varphi}_n(t)$ be the solutions of the linear system (2.4.1) in I and $t_0 \in I$. Then for all $t \in I$,

$$W(t) = W(t_0) \exp\left[\int_{t_0}^t \text{tr} \mathbf{A}(s) ds\right], \quad (2.4.7)$$

where

$$\text{tr} \mathbf{A}(t) = a_{11}(t) + a_{22}(t) + \cdots + a_{nn}(t) = \sum_{j=1}^n a_{jj}(t)$$

is the trace of $\mathbf{A}(t)$.

Proof From the fundamental properties of the determinant, we have

$$\begin{aligned}
 \frac{dW}{dt} &= \sum_{i=1}^n \begin{vmatrix} \varphi_{11} & \varphi_{12} & \cdots & \varphi_{1n} \\ \vdots & \vdots & & \vdots \\ \frac{d\varphi_{i1}}{dt} & \frac{d\varphi_{i2}}{dt} & \cdots & \frac{d\varphi_{in}}{dt} \\ \vdots & \vdots & & \vdots \\ \varphi_{n1} & \varphi_{n2} & \cdots & \varphi_{nn} \end{vmatrix} \\
 &= \sum_{i=1}^n \begin{vmatrix} \varphi_{11} & \varphi_{12} & \cdots & \varphi_{1n} \\ \vdots & \vdots & & \vdots \\ \sum_{j=1}^n a_{ij}\varphi_{j1} & \sum_{j=1}^n a_{ij}\varphi_{j2} & \cdots & \sum_{j=1}^n a_{ij}\varphi_{jn} \\ \vdots & \vdots & & \vdots \\ \varphi_{n1} & \varphi_{n2} & \cdots & \varphi_{nn} \end{vmatrix} \\
 &= a_{11}(t)W(t) + a_{22}(t)W(t) + \cdots + a_{nn}(t)W(t) \\
 &= \text{tr} \mathbf{A}(t)W(t), \tag{2.4.8}
 \end{aligned}$$

which is a linear differential equation with unknown function $W(t)$. Solving the equation (2.4.8) yields Liouville's formula (2.4.7).

Remark 2.4.3 The Wronskian determinant is either identically zero or never zero in I .

The nonvanishing of the Wronskian determinant is a necessary and sufficient condition for the functions $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ to be a fundamental system of solutions.

We will prove the following results.

Theorem 2.4.4 The solutions $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ of linear system of differential equations (2.4.1) are linearly independent if and only if

$$W(t) \neq 0, t \in I. \tag{2.4.9}$$

Proof By Liouville's Formula, the condition $W(t) \neq 0$ is equivalent to $W(t_0) \neq 0$ which is again equivalent to the fact that the initial value vectors

$$\varphi_1(t_0), \varphi_2(t_0), \dots, \varphi_n(t_0) \tag{2.4.10}$$

are linearly independent in R^n . Together with the proof of Theorem 2.4.2, we have

$$\begin{aligned}
 &L[c_1\varphi_1(t_0) + c_2\varphi_2(t_0) + \cdots + c_n\varphi_n(t_0)] \\
 &= c_1\varphi_1(t) + c_2\varphi_2(t) + \cdots + c_n\varphi_n(t).
 \end{aligned}$$

Thus, from $L(0) = 0$, we obtain that the initial value vectors (2.4.10) are linearly independent if and only if the solutions (2.4.4) are linearly independent in linear space X .

Corollary 2.4.1 The solutions $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ of linear system (2.4.1) are linearly dependent if and only if the corresponding Wronskian determinant

$$W(t) \equiv 0, t \in I.$$

Example 2.4.3 It is easily verified that

$$\varphi_1(t) = \begin{bmatrix} 2e^t \\ 2e^t \\ e^t \end{bmatrix}, \varphi_2(t) = \begin{bmatrix} 2e^{3t} \\ 0 \\ -e^{3t} \end{bmatrix} \quad \text{and} \quad \varphi_3(t) = \begin{bmatrix} 2e^{5t} \\ -2e^{5t} \\ e^{5t} \end{bmatrix}$$

are solutions of the equation

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} 3 & -2 & 0 \\ -1 & 3 & -2 \\ 0 & -1 & 3 \end{bmatrix} \mathbf{x}.$$

The Wronskian of these solutions is

$$W(t) = \begin{vmatrix} 2e^t & 2e^{3t} & 2e^{5t} \\ 2e^t & 0 & -2e^{5t} \\ e^t & -e^{3t} & e^{5t} \end{vmatrix} = e^{9t} \begin{vmatrix} 2 & 2 & 2 \\ 2 & 0 & -2 \\ 1 & -1 & 1 \end{vmatrix} = -16e^{9t},$$

which is never zero. Hence Theorem 2.4.4 implies that these solutions $\boldsymbol{\varphi}_1(t), \boldsymbol{\varphi}_2(t)$ and $\boldsymbol{\varphi}_3(t)$ are linearly independent on the real line.

A system of n solutions $\boldsymbol{\varphi}_1(t), \boldsymbol{\varphi}_2(t), \dots, \boldsymbol{\varphi}_n(t)$ can be assembled into an $n \times n$ **solution matrix**

$$\boldsymbol{\Phi}(t) = [\boldsymbol{\varphi}_1(t) \quad \boldsymbol{\varphi}_2(t) \quad \cdots \quad \boldsymbol{\varphi}_n(t)].$$

In this notation, the n differential equations

$$\boldsymbol{\varphi}'_i(t) = \mathbf{A}(t)\boldsymbol{\varphi}_i(t) \quad (i = 1, 2, \dots, n)$$

can be written as a matrix differential equation

$$\frac{d\boldsymbol{\Phi}(t)}{dt} = \mathbf{A}(t)\boldsymbol{\Phi}(t). \quad (2.4.11)$$

The equation (2.4.11) shows that the solution matrix of (2.4.1) is the matrix-value solution of (2.4.1).

If these solutions are linearly independent, then the corresponding solution matrix is called the **fundamental matrix** of (2.4.1). Indeed, if $\boldsymbol{\Phi}(t)$ is a fundamental matrix, then each solution of (2.4.1) can be written in the form

$$\mathbf{x}(t) = \boldsymbol{\Phi}(t)\mathbf{c}, \quad (2.4.12)$$

where $\mathbf{c}$ is a n dimensional constant column vector. This question is identical to

$$\mathbf{x}(t) = c_1\boldsymbol{\varphi}_1(t) + c_2\boldsymbol{\varphi}_2(t) + \cdots + c_n\boldsymbol{\varphi}_n(t).$$

For this matrix $\boldsymbol{\Phi}(t)$, we have the following result.

Theorem 2.4.5 If $\boldsymbol{\Phi}(t)$ is a fundamental matrix of the differential system (2.4.1), then for any constant nonsingular $n \times n$ matrix $\mathbf{P}$, $\boldsymbol{\Phi}(t)\mathbf{P}$ is also a fundamental matrix of (2.4.1), and every fundamental matrix of (2.4.1) is of the form $\boldsymbol{\Phi}(t)\mathbf{P}$ for some constant nonsingular $n \times n$ matrix $\mathbf{P}$.

Proof Obviously, $[\boldsymbol{\Phi}(t)\mathbf{P}]' = \mathbf{A}(t)[\boldsymbol{\Phi}(t)\mathbf{P}]$, namely, $\boldsymbol{\Phi}(t)\mathbf{P}$ is a solution of the matrix differential system

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}(t)\mathbf{x}(t).$$

Further, it follows that $\det(\boldsymbol{\Phi}(t)\mathbf{P}) \neq 0$ from $\det \boldsymbol{\Phi}(t) \neq 0$ and $\det \mathbf{P} \neq 0$. Hence $\boldsymbol{\Phi}(t)\mathbf{P}$ is also a fundamental matrix solution of (2.4.1). Conversely, suppose that $\boldsymbol{\Phi}_1(t)$ and $\boldsymbol{\Phi}_2(t)$ are two fundamental matrix solutions of (2.4.1). If $\boldsymbol{\Phi}_2^{-1}(t)\boldsymbol{\Phi}_1(t) = \mathbf{P}(t)$, that is, $\boldsymbol{\Phi}_1(t) = \boldsymbol{\Phi}_2(t)\mathbf{P}(t)$, then we have

$$\Phi_1'(t) = \Phi_2'(t)P(t) + \Phi_2(t)P'(t)$$

which is the same as

$$\begin{aligned} A(t)\Phi_1(t) &= A(t)\Phi_2(t)P(t) + \Phi_2(t)P'(t) \\ &= A(t)\Phi_1(t) + \Phi_2(t)P'(t), \end{aligned}$$

namely,

$$\Phi_2(t)P'(t) \quad \text{or} \quad P'(t) = 0.$$

Thus $P(t)$ is a constant matrix. Further, since $\Phi_1(t)$ and $\Phi_2(t)$ are nonsingular, the constant matrix is also nonsingular.

Exercise 2.4

Find the general solution of the following systems.

$$1. \begin{cases} \frac{dx}{dt} = \frac{x}{t}, t \neq 0, \\ \frac{dy}{dt} = \frac{y}{t}. \end{cases}$$

$$2. \begin{cases} \frac{dx}{dt} = x + y, \\ \frac{dy}{dt} = y. \end{cases}$$

$$3. \begin{cases} \frac{dr}{dt} = -r(r^2 - 1), \\ \frac{d\theta}{dt} = 2. \end{cases}$$

$$4. \begin{cases} \frac{dx}{dt} = p(t)x + q(t)y, \\ \frac{dy}{dt} = q(t)x + p(t)y, \end{cases}$$

where $p(t), q(t)$ are continuous on the interval $[a, b]$.

$$5. \begin{cases} \frac{dx}{dt} = \frac{2t}{1+t^2}x, \frac{dy}{dt} = -\frac{1}{t}y + x + t (t > 0), \\ x(1) = 0, y(1) = \frac{4}{3}. \end{cases}$$

6. Show that the linear differential system

$$\frac{dy}{dx} = A(x)y + f(x)$$

remains linearly after the change of the independent variable $x = p(t)$.

7. Show that the vector-valued functions

$$\phi_1(t) = \begin{bmatrix} t \\ 1 \end{bmatrix} \quad \text{and} \quad \phi_2(t) = \begin{bmatrix} t^2 \\ t \end{bmatrix}$$

are linearly independent in any interval I , and $W(\phi_1(t), \phi_2(t)) \equiv 0$ in I . What does the fact show?

8. If the following systems

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}_1(t)\mathbf{x} \quad \text{and} \quad \frac{d\mathbf{x}}{dt} = \mathbf{A}_2(t)\mathbf{x}$$

have the same fundamental matrix $\Phi(t)$, then show that

$$\mathbf{A}_1(t) \equiv \mathbf{A}_2(t),$$

where $\mathbf{A}_1(t), \mathbf{A}_2(t)$ are continuous $n \times n$ matrices.

9. The fundamental matrix $\Phi(t)$ of (2.4.1) is called a **transition matrix** provided $\Phi(t)$ satisfies the initial condition $\Phi(t_0) = \mathbf{E}$ (identity matrix), where $t_0 \in I$. Obviously, for any one fundamental matrix $\Phi(t)$ of (2.4.1), we get

$$\Psi(t, t_0) = \Phi(t)\Phi^{-1}(t_0) \quad (2.4.13)$$

is a transition matrix. Show the following :

(1) The transition matrix $\Psi(t, t_0)$ is independent of the choice of the fundamental matrix.

(2) $\Psi(t, t_0)$ with respect to t is a unique solution of matrix equation

$$\frac{d\mathbf{Y}}{dt} = \mathbf{A}(t)\mathbf{Y}$$

which satisfies the initial condition $\mathbf{Y}(t_0) = \mathbf{E}$.

(3) For any $t, t_0 \in I$, $\Psi(t, t_0)$ is nonsingular and

$$\Psi^{-1}(t, t_0) = \Psi(t_0, t).$$

(4) For any $t, t_0, t_1 \in I$,

$$\Psi(t, t_0) = \Psi(t, t_1)\Psi(t_1, t_0).$$

(5) The solution of initial value problem

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} \\ \mathbf{x}(t_0) = \mathbf{x}_0 \end{cases}$$

can be written as

$$\mathbf{x}(t, t_0, \mathbf{x}_0) = \Psi(t, t_0)\mathbf{x}_0.$$

10. Suppose that $\Phi(t)$ is a fundamental matrix of $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}$. Show that $\Phi(t)$ determines the matrix $\mathbf{A}(t)$ uniquely by the relation $\mathbf{A}(t) = \Phi'(t)\Phi^{-1}(t)$.

11. The system

$$\frac{d\mathbf{x}}{dt} = -\mathbf{A}^T(t)\mathbf{x}(t) \quad (2.4.14)$$

is called the adjoint system to the differential system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t),$$

where $\mathbf{A}^T(t)$ is a transpose of $\mathbf{A}(t)$. This relationship is symmetric in the sense that the system (2.4.1) is the adjoint system to (2.4.14) and vice versa. Show the following:

(1) Let $\Phi(t)$ be a fundamental matrix of (2.4.1). Then $[\Phi^{-1}(t)]^T$ is a fundamental matrix of its adjoint system (2.4.14).

(2) If $\Phi(t)$ is a fundamental matrix of the differential system (2.4.1), then $\Psi(t)$ is a fundamental matrix of its adjoint system (2.4.14) if and only if

$$\Psi^T(t)\Phi(t) = C, \quad (2.4.15)$$

where C is a constant nonsingular $(n \times n)$ matrix.

12. The differential system (2.4.1) is said to be self-adjoint when $A(t) = -A^T(t)$ for all t in I . Show the following:

(1) If $\Phi(t)$ is a fundamental matrix of the self-adjoint system (2.4.1), then

$$\Phi^T(t)\Phi(t) = C,$$

where C is a constant nonsingular $n \times n$ matrix.

(2) Let $\Phi(t, t_0)$ be the transition matrix of the system (2.4.1), and $\Psi(t, t_0)$ be the transition matrix of the adjoint system (2.4.14). If the differential system (2.4.1) is self-adjoint, then

$$\Phi(t, t_0) = \Psi^T(t_0, t).$$

2.5 General Theory of Nonhomogeneous Linear System

In this section, we consider the following nonhomogeneous linear system

$$\frac{dx}{dt} = A(t)x + f(t) \quad (2.5.1)$$

and show that the method of variation of parameters used in previous chapter to find the solutions of nonhomogeneous first-order differential equations is equally applicable for the nonhomogeneous system (2.5.1).

First, we study the structure of the general solution of (2.5.1) and give the following conclusions.

Theorem 2.5.1 Let $\bar{\varphi}(t)$ be a fixed solution of the nonhomogeneous system (2.5.1). If $\varphi(t)$ is an arbitrary solution of the corresponding homogeneous system (2.4.1), then

$$x(t) = \varphi(t) + \bar{\varphi}(t)$$

is a solution of the nonhomogeneous system (2.5.1).

Proof Indeed, we have

$$\frac{d\varphi(t)}{dt} = A(t)\varphi(t), \quad \frac{d\bar{\varphi}(t)}{dt} = A(t)\bar{\varphi}(t) + f(t).$$

Thus

$$\begin{aligned} \frac{d}{dt}[\varphi(t) + \bar{\varphi}(t)] &= A(t)\varphi(t) + A(t)\bar{\varphi}(t) + f(t) \\ &= A(t)[\varphi(t) + \bar{\varphi}(t)] + f(t) \end{aligned}$$

which shows the desired conclusion.

Theorem 2.5.2 The difference of any two solutions $\varphi_1(t)$ and $\varphi_2(t)$ of the nonhomogeneous system (2.5.1) is a solution of the corresponding homogeneous system (2.4.1).

Proof Let $\boldsymbol{\varphi}(t) = \boldsymbol{\varphi}_1(t) - \boldsymbol{\varphi}_2(t)$. Since

$$\frac{d\boldsymbol{\varphi}_1(t)}{dt} = \mathbf{A}(t)\boldsymbol{\varphi}_1(t) + \mathbf{f}(t), \quad \frac{d\boldsymbol{\varphi}_2(t)}{dt} = \mathbf{A}(t)\boldsymbol{\varphi}_2(t) + \mathbf{f}(t),$$

we obtain

$$\begin{aligned} \frac{d}{dt} [\boldsymbol{\varphi}_1(t) - \boldsymbol{\varphi}_2(t)] &= \frac{d\boldsymbol{\varphi}_1(t)}{dt} - \frac{d\boldsymbol{\varphi}_2(t)}{dt} \\ &= \mathbf{A}(t)\boldsymbol{\varphi}_1(t) + \mathbf{f}(t) - \mathbf{A}(t)\boldsymbol{\varphi}_2(t) - \mathbf{f}(t) \\ &= \mathbf{A}(t)[\boldsymbol{\varphi}_1(t) - \boldsymbol{\varphi}_2(t)]. \end{aligned}$$

The proof is completed.

Theorem 2.5.3 Let $\boldsymbol{\varphi}_p$ be a particular solution of the nonhomogeneous system (2.5.1). Let $\boldsymbol{\varphi}_1, \boldsymbol{\varphi}_2, \dots, \boldsymbol{\varphi}_n$ be a fundamental system of solutions of the associated homogeneous system (2.4.1). If $\mathbf{x}(t)$ is any solution whatsoever of the system (2.5.1), then there exist constants $c_1, c_2, \dots, c_n$ such that

$$\mathbf{x}(t) = c_1\boldsymbol{\varphi}_1(t) + c_2\boldsymbol{\varphi}_2(t) + \dots + c_n\boldsymbol{\varphi}_n(t) + \boldsymbol{\varphi}_p(t). \quad (2.5.2)$$

Proof Theorem 2.5.1 shows that the expression (2.5.2) is a solution of the nonhomogeneous system (2.5.1). In addition, by Theorem 2.5.2, $\mathbf{x}(t) - \boldsymbol{\varphi}_p(t)$ is a solution of the associated homogeneous system (2.4.1). So there exist constants $c_1, c_2, \dots, c_n$ such that

$$\mathbf{x}(t) - \boldsymbol{\varphi}_p(t) = c_1\boldsymbol{\varphi}_1 + c_2\boldsymbol{\varphi}_2 + \dots + c_n\boldsymbol{\varphi}_n$$

which is the desired conclusion.

Theorem 2.5.3 gives the relationship between solutions of the nonhomogeneous differential system (2.5.1) and solutions of the associated homogeneous differential system (2.4.1). Thus finding a general solution of a nonhomogeneous linear system involves two separate steps:

Step 1 Finding the general solution

$$\boldsymbol{\varphi}_c(t) = c_1\boldsymbol{\varphi}_1 + c_2\boldsymbol{\varphi}_2 + \dots + c_n\boldsymbol{\varphi}_n$$

of the associated homogeneous system.

Step 2 Finding a single particular solution $\boldsymbol{\varphi}_p(t)$ of the nonhomogeneous system.

The sum $\mathbf{x}(t) = \boldsymbol{\varphi}_c(t) + \boldsymbol{\varphi}_p(t)$ will be a general solution of the nonhomogeneous system.

Next, our task is to find just one solution of the nonhomogeneous equation. We make use of a procedure that originated with Lagrange, namely, the method of variation of parameters.

If $\boldsymbol{\Phi}(t)$ is a fundamental matrix to the homogeneous differential system (2.4.1), then every solution of the homogeneous system can be written in the form $\boldsymbol{\Phi}(t)\mathbf{c}$, where $\mathbf{c}$ runs through all constant vectors. In the method of variation of parameters, the constants $(c_1, c_2, \dots, c_n)$ are replaced by functions of t ,

$$\mathbf{y}(t) = \boldsymbol{\Phi}(t)\mathbf{c}(t). \quad (2.5.3)$$

The function $\mathbf{c}(t)$ is to be determined such that $\mathbf{y}(t)$ is a solution of the nonhomogeneous

differential system (2.5.1). Substituting $\mathbf{y}(t)$ into (2.5.1) yields

$$\begin{aligned}\frac{d\mathbf{y}}{dt} &= \frac{d}{dt}(\Phi(t))\mathbf{c}(t) + \Phi(t) \frac{d\mathbf{c}(t)}{dt} \\ &= \mathbf{A}(t)\Phi(t)\mathbf{c}(t) + \Phi(t) \frac{d\mathbf{c}(t)}{dt} \\ &= \mathbf{A}(t)\Phi(t)\mathbf{c}(t) + \mathbf{f}(t)\end{aligned}$$

which leads to

$$\Phi(t)\mathbf{c}'(t) = \mathbf{f}(t). \quad (2.5.4)$$

Since $\Phi(t)$ is a fundamental matrix, therefore, the inverse matrix $\Phi^{-1}(t)$ exists and is continuous in I .

Multiplying the equation (2.5.4) on the left by $\Phi^{-1}(t)$ and integrating both sides yield

$$\mathbf{c}(t) = \mathbf{c}(t_0) + \int_{t_0}^t \Phi^{-1}(s)\mathbf{f}(s)ds. \quad (2.5.5)$$

Since we need only a single particular solution, we choose $\mathbf{c}(t_0) = \mathbf{0}$ and further obtain one solution of (2.5.1)

$$\mathbf{y}(t) = \Phi(t) \int_{t_0}^t \Phi^{-1}(s)\mathbf{f}(s)ds. \quad (2.5.6)$$

Theorem 2.5.4 Let $\Phi(t)$ be a fundamental matrix to the homogeneous system (2.4.1). Then the general solution of the nonhomogeneous system (2.5.1) is given by

$$\mathbf{x}(t) = \Phi(t)\mathbf{c} + \Phi(t) \int_{t_0}^t \Phi^{-1}(s)\mathbf{f}(s)ds, \quad (2.5.7)$$

where $\mathbf{c}$ is arbitrary constant column vector.

Further, the initial value problem

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \end{cases}$$

has the uniquely determined solution

$$\mathbf{x}(t) = \Phi(t)\Phi^{-1}(t_0)\mathbf{x}_0 + \Phi(t) \int_{t_0}^t \Phi^{-1}(s)\mathbf{f}(s)ds. \quad (2.5.8)$$

Hence, we have an explicit representation of the general solution of (2.5.1) in terms of any fundamental matrix of the associated homogeneous system (2.4.1).

Example 2.5.1 Solve the differential system

$$\begin{cases} \frac{dx}{dt} = y - 5\cos t, \\ \frac{dy}{dt} = 2x + y. \end{cases}$$

Solution Here $\mathbf{A}(t) = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$, $\mathbf{f}(t) = \begin{bmatrix} -5\cos t \\ 0 \end{bmatrix}$. A fundamental matrix of the associated homogeneous system is given by

$$\Phi(t) = \begin{bmatrix} e^{-t} & e^{2t} \\ -e^{-t} & 2e^{2t} \end{bmatrix}.$$

Suppose that the particular solution of this system has the form

$$\begin{bmatrix} \bar{x}(t) \\ \bar{y}(t) \end{bmatrix} = c_1(t) \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix} + c_2(t) \begin{bmatrix} e^{2t} \\ 2e^{2t} \end{bmatrix}.$$

Substitution back in the original differential system gives

$$\begin{cases} c_1'(t)e^{-t} + c_2'(t)e^{2t} = -5\cos t \\ -c_1'(t)e^{-t} + 2c_2'(t)e^{2t} = 0 \end{cases}$$

which leads to

$$\begin{cases} c_1'(t) = -\frac{10}{3}e^t \cos t, \\ c_2'(t) = -\frac{5}{3}e^{-2t} \cos t. \end{cases}$$

Thus we obtain

$$\begin{cases} c_1(t) = -\frac{5}{3}e^t (\cos t + \sin t), \\ c_2(t) = \frac{1}{3}e^{-2t} (2\cos t - \sin t). \end{cases}$$

The general solution is given by

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} e^{-t} & e^{2t} \\ -e^{-t} & 2e^{2t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \begin{bmatrix} -\cos t - 2\sin t \\ 3\cos t + \sin t \end{bmatrix}.$$

Exercise 2.5

1. Find a solution of the nonhomogeneous differential system

$$\begin{cases} \frac{dx_1}{dt} = \frac{1}{t}x_1 - x_2 + t, \\ \frac{dx_2}{dt} = \frac{1}{t^2}x_1 + \frac{2}{t}x_2 - t^2. \end{cases}$$

The known fundamental matrix is given by $\Phi(t) = \begin{bmatrix} t^2 & -t^2 \ln t \\ -t & t + t \ln t \end{bmatrix}$.

2. Determine the general solution of the system

$$\begin{cases} \frac{dx}{dt} = (3t-1)x - (1-t)y + te^{t^2}, \\ \frac{dy}{dt} = -(t+2)x + (t-2)y - e^{t^2}. \end{cases}$$

Hint. The homogeneous system has a solution of the form

$$[x(t), y(t)] = [\varphi(t), -\varphi(t)].$$

3. Let $\Phi(t, t_0)$ be a transition matrix of the associated homogeneous system of the system (2.4.1). Show that the initial value problem

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \end{cases}$$

has a unique solution of the form

$$\mathbf{x}(t) = \Phi(t, t_0)\mathbf{x}_0 + \int_{t_0}^t \Phi(t, s)\mathbf{f}(s) \, ds.$$

(A fundamental matrix $\Phi(t)$ is called a transition matrix provided $\Phi(t_0) = \mathbf{E}$, where $\mathbf{E}$ is identity matrix.)

4. Show that the nonhomogeneous differential system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{f}(t) \quad (\mathbf{f}(t) \not\equiv 0)$$

has $n + 1$ linearly independent solutions.

5. Let $\phi_1(t), \phi_2(t), \dots, \phi_{n+1}(t)$ be $n + 1$ linearly independent solutions of the nonhomogeneous system (2.5.1) in the interval I . Give the expression of general solution of (2.5.1).

6. If $\Phi(t)$ is a fundamental matrix of $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}$, and n -dimensional vector-valued function $\mathbf{f}(t, \mathbf{x})$ is continuous in the domain $R: a < t < b, |\mathbf{x}| < \infty$, then the initial value problem

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t, \mathbf{x}) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \end{cases}$$

is equivalent to the integral equation

$$\mathbf{x}(t) = \Phi(t)\Phi^{-1}(t_0)\mathbf{x}_0 + \int_{t_0}^t \Phi(t)\Phi^{-1}(s)\mathbf{f}(s, \mathbf{x}(s)) \, ds,$$

where $t_0 \in (a, b)$.

2.6 Linear Differential System with Constant Coefficients

In this section, we consider the linear differential system with constant coefficients

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} + \mathbf{f}(t), \tag{2.6.1}$$

where $\mathbf{A}$ is an $(n \times n)$ constant matrix, $\mathbf{f}(t)$ is a given continuous vector-valued function in the interval I . The critical point in solving (2.6.1) is to find a fundamental matrix $\Phi(t)$ of the associated homogeneous system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}. \tag{2.6.2}$$

2.6.1 Exponential Matrix $e^{\mathbf{A}}$

If $\mathbf{A}$ is an $(n \times n)$ matrix, then the **exponential matrix** $e^{\mathbf{A}}$ is the $n \times n$ matrix defined by the series

$$e^{\mathbf{A}} = \mathbf{E} + \mathbf{A} + \frac{\mathbf{A}^2}{2!} + \dots + \frac{\mathbf{A}^n}{n!} + \dots \tag{2.6.3}$$

which converges absolutely for all $\mathbf{A}$, where $\mathbf{E}$ is identity matrix.

Especially, if $n = 1$, then $\mathbf{A} = \mathbf{a}$, and $e^{\mathbf{A}} = e^{\mathbf{a}}$ is the exponential function. Indeed, the exponential matrix $e^{\mathbf{A}}$ satisfies most of the exponential relations that are familiar in the case of scalar exponents. We have the following properties.

Proposition 2.6.1

(1) If $\mathbf{AB} = \mathbf{BA}$, then $e^{\mathbf{A+B}} = e^{\mathbf{A}} \cdot e^{\mathbf{B}}$.

(2) If $\mathbf{P}$ is a nonsingular square matrix, then

$$e^{\mathbf{PAP}^{-1}} = \mathbf{P}e^{\mathbf{A}}\mathbf{P}^{-1}.$$

(3) The matrix $e^{\mathbf{A}}$ is nonsingular for every $n \times n$ matrix $\mathbf{A}$ and $(e^{\mathbf{A}})^{-1} = e^{-\mathbf{A}}$.

Proof (1) Because of the absolute convergence of the series for $e^{\mathbf{A}}$ and $e^{\mathbf{B}}$, these series can be multiplied termwise, and we have

$$\begin{aligned} e^{\mathbf{A+B}} &= \sum_{n=0}^{\infty} \frac{(\mathbf{A+B})^n}{n!} = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{\mathbf{A}^k \mathbf{B}^{n-k}}{k!(n-k)!} \\ &= \sum_{p=0}^{\infty} \frac{\mathbf{A}^p}{p!} \cdot \sum_{q=0}^{\infty} \frac{\mathbf{B}^q}{q!} = e^{\mathbf{A}} \cdot e^{\mathbf{B}}. \end{aligned}$$

(2) By induction, we can show that

$$(\mathbf{PAP}^{-1})^k = \mathbf{PA}^k\mathbf{P}^{-1}, k = 0, 1, 2, \dots.$$

Hence

$$\sum_{k=0}^n \frac{1}{k!} (\mathbf{PAP}^{-1})^k = \mathbf{P} \left(\sum_{k=0}^n \frac{1}{k!} \mathbf{A}^k \right) \mathbf{P}^{-1}.$$

The conclusion (2) follows from this relation by taking limits as $n \rightarrow \infty$.

(3) Because of $\mathbf{A}(-\mathbf{A}) = (-\mathbf{A})\mathbf{A}$, one obtains

$$e^{\mathbf{A}} e^{-\mathbf{A}} = e^{\mathbf{A-A}} = e^{\mathbf{0}} = \mathbf{E},$$

namely, $e^{\mathbf{A}}$ is nonsingular and $(e^{\mathbf{A}})^{-1} = e^{-\mathbf{A}}$.

Example 2.6.1 Let $\mathbf{A} = \text{diag}(\lambda_1, \dots, \lambda_n)$ be a diagonal matrix. Solve exponential matrix $e^{\mathbf{A}}$.

Solution It can be shown by induction that

$$[\text{diag}(\lambda_1, \dots, \lambda_n)]^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k).$$

Thus

$$\begin{aligned} e^{\mathbf{A}} &= \mathbf{E} + \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix} + \frac{1}{2!} \begin{pmatrix} \lambda_1^2 & & \\ & \lambda_2^2 & \\ & & \ddots \\ & & & \lambda_n^2 \end{pmatrix} + \dots \\ &= \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_n}). \end{aligned}$$

Especially, if $\mathbf{A} = \lambda \mathbf{E}$, then $e^{\mathbf{A}} = e^{\lambda} \mathbf{E}$.

Example 2.6.2 If

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

then solve e^A .

Solution One easily obtains

$$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

So $A^n = 0$ for $n \geq 3$. It therefore follows that

$$\begin{aligned} e^A &= E + A + \frac{1}{2!}A^2 \\ &= \begin{bmatrix} 1 & 1 & \frac{1}{2!} \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

Remark 2.6.1 If $A^n = 0$ for some positive integer n , then the exponential series in (2.6.3) terminates after a finite number of terms. Such a matrix with vanishing power is said to be nilpotent.

Example 2.6.3 If

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix},$$

then solve e^A .

Solution $A = E + B$, where B is the nilpotent matrix of Example 2.6.2. Therefore, one obtains

$$e^A = e^E \cdot e^B = e \begin{bmatrix} 1 & 1 & \frac{1}{2!} \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

2.6.2 Fundamental Matrix of the Homogeneous System

The substitution of At for A in (2.6.3) yields

$$e^{At} = E + At + \frac{A^2}{2!}t^2 + \cdots + \frac{A^n}{n!}t^n + \cdots \quad (2.6.4)$$

where At is obtained by multiplying each element by t .

Similarly, we have:

Proposition 2.6.2

$$(1) (e^{At})^{-1} = e^{-At}.$$

$$(2) Ae^{At} = e^{At}A.$$

$$(3) \text{ If } AB = BA, \text{ then}$$

$$e^{At} \cdot e^{Bt} = e^{(A+B)t}.$$

Especially, $e^{At} \cdot e^{-At_0} = e^{A(t-t_0)}$. Now, we can obtain a fundamental matrix of (2.6.2) and further a general solution of (2.6.1) by the exponential matrix.

Theorem 2.6.1 The exponential matrix

$$\Phi(t) = e^{At}$$

is a fundamental matrix of the homogeneous system of (2.6.2) which satisfies $\Phi(0) = E$.

Proof It happens that term by term differentiation of series in (2.6.4) is valid, with the result

$$\begin{aligned} \frac{d}{dt}(e^{At}) &= A + A^2 t + A^3 \frac{t^2}{2!} + \dots \\ &= A(E + At + \frac{A^2}{2!} t^2 + \dots) \\ &= A e^{At} \end{aligned}$$

which shows that $\Phi(t) = e^{At}$ is a solution matrix of (2.6.2). Meanwhile, $\Phi(0) = E$, so $\det[\Phi(0)] = 1$. Hence $\Phi(t)$ is a fundamental matrix of (2.6.2).

Theorem 2.6.2 The general solution of the nonhomogeneous system (2.6.1) is given by

$$x = e^{tA} c + \int_{t_0}^t e^{(t-s)A} f(s) ds, \quad (2.6.5)$$

where c is an arbitrary constant column vector.

The initial value problem

$$\begin{cases} \frac{dx}{dt} = Ax + f(t) \\ x(t_0) = x_0 \end{cases}$$

has a uniquely determined solution

$$x(t) = e^{(t-t_0)A} x_0 + \int_{t_0}^t e^{(t-s)A} f(s) ds. \quad (2.6.6)$$

2.6.3 Structure of the Fundamental Matrix e^{tA}

Example 2.6.4 Solve the exponential matrix e^{tA} .

(1) $A_1 = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$

$$(2) A_2 = \begin{bmatrix} \lambda & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{bmatrix}$$

Solution (1) We easily obtain

$$e^{A_1 t} = \begin{bmatrix} e^{\lambda_1 t} & & & \\ & e^{\lambda_2 t} & & \\ & & \ddots & \\ & & & e^{\lambda_n t} \end{bmatrix}.$$

$$(2) \mathbf{A}_2 = \lambda \mathbf{E} + \mathbf{B}, \text{ where } \mathbf{B} = \begin{bmatrix} 0 & 1 & & & \\ & \ddots & \ddots & & \\ & & \ddots & \ddots & \\ & & & \ddots & 1 \\ & & & & 0 \end{bmatrix}. \text{ Since } (\lambda \mathbf{E}t)(\mathbf{B}t) = (\mathbf{B}t)(\lambda \mathbf{E}t), \text{ we have}$$

$$\mathbf{e}^{\mathbf{A}_2 t} = \mathbf{e}^{\lambda \mathbf{E}t} \mathbf{e}^{\mathbf{B}t}.$$

Notice that

$$\mathbf{B}^2 = \begin{bmatrix} 0 & 0 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & \ddots & 0 \\ & & & & 0 \end{bmatrix}, \dots, \mathbf{B}^{n-1} = \begin{bmatrix} 0 & \cdots & \cdots & 0 & 1 \\ & 0 & \cdots & \cdots & 0 \\ & & \ddots & \ddots & \\ & & & \ddots & \vdots \\ & & & & 0 \end{bmatrix}, \mathbf{B}^n = \mathbf{0}.$$

Thus

$$\begin{aligned} \mathbf{e}^{\mathbf{A}t} &= \mathbf{e}^{\lambda t} \left(\mathbf{E} + \mathbf{B}t + \mathbf{B}^2 \frac{t^2}{2!} + \cdots + \mathbf{B}^{n-1} \frac{t^{n-1}}{(n-1)!} \right) \\ &= \mathbf{e}^{\lambda t} \begin{bmatrix} 1 & t & \frac{t^2}{2!} & \cdots & \frac{t^{n-1}}{(n-1)!} \\ & \ddots & \ddots & \ddots & \vdots \\ & & \ddots & \ddots & \frac{t^2}{2!} \\ & & & \ddots & t \\ & & & & 1 \end{bmatrix}. \end{aligned}$$

In order to handle the general case, we make use of a result from linear algebra. It says that for every real square matrix $\mathbf{A}$, there exists a nonsingular matrix $\mathbf{P}$ such that $\mathbf{J} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ has the so-called Jordan normal form

$$\mathbf{J} = \begin{bmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \\ & & & J_m \end{bmatrix}$$

where Jordan block $\mathbf{J}_i$ is a square matrix of the form

$$\mathbf{J}_i = \begin{bmatrix} \lambda_i & 1 & & & \\ & \lambda_i & 1 & & \\ & & \ddots & \ddots & \\ & & & \ddots & 1 \\ & & & & \lambda_i \end{bmatrix}, \quad i = 1, 2, \dots, m$$

with n_i rows and columns, outside of Jordan blocks, $\mathbf{J}$ consists entirely of zeros. Here $n_1 + n_2 + \cdots + n_m = n$ and the characteristic polynomial

$$P_n(\lambda) = \det(\lambda \mathbf{E} - \mathbf{A}) = (\lambda - \lambda_1)^{n_1} (\lambda - \lambda_2)^{n_2} \cdots (\lambda - \lambda_m)^{n_m}.$$

Notice that the main diagonal of $\mathbf{J}$ consists of eigenvalues of $\mathbf{A}$ (that is, the zero points of $P_n(\lambda)$) and that each block is made up of one and the same eigenvalue. However, the same eigenvalue can appear in more than one block. The Jordan block $\mathbf{J}_i$ with n_i rows and diagonal element λ_i is rewritten as

$$\begin{aligned}\mathbf{J}_i &= \lambda_i \mathbf{E} + \mathbf{B}_i \\ &= \begin{pmatrix} \lambda_i & & & \\ & \lambda_i & & \\ & & \ddots & \\ & & & \lambda_i \end{pmatrix} + \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix}.\end{aligned}$$

From Example 2.6.4, one has

$$e^{t\mathbf{J}_i} = e^{\lambda_i t} \begin{bmatrix} 1 & t & \cdots & \frac{t^{n_i-1}}{(n_i-1)!} \\ & 1 & t & \ddots \\ & & \ddots & \ddots \\ & & & t \\ & & & & 1 \end{bmatrix}. \quad (2.6.7)$$

Further

$$e^{t\mathbf{J}} = \begin{bmatrix} e^{t\mathbf{J}_1} & & \\ & e^{t\mathbf{J}_2} & \\ & & \ddots \\ & & & e^{t\mathbf{J}_m} \end{bmatrix}.$$

Together with the properties of exponential matrix, we have

$$e^{t\mathbf{A}} = e^{t\mathbf{P}\mathbf{J}\mathbf{P}^{-1}} = \mathbf{P} e^{t\mathbf{J}} \mathbf{P}^{-1}. \quad (2.6.8)$$

That is

$$e^{t\mathbf{A}} \mathbf{P} = \mathbf{P} e^{t\mathbf{J}}. \quad (2.6.9)$$

Theorem 2.6.3 $\Phi(t) = \mathbf{P} e^{t\mathbf{J}}$ is a fundamental matrix of the homogeneous system with constant coefficients (2.6.2).

Proof From (2.6.9), one has

$$\Phi(t) = \mathbf{P} e^{t\mathbf{J}} = e^{t\mathbf{A}} \mathbf{P}.$$

It suffices to prove that $e^{t\mathbf{A}} \mathbf{P}$ is a fundamental matrix. Indeed, $\frac{d}{dt}(e^{t\mathbf{A}} \mathbf{P}) = \mathbf{A} e^{t\mathbf{A}} \mathbf{P}$ and $\Phi(0) = \mathbf{P}$ is nonsingular, thus $\Phi(t) = e^{t\mathbf{A}} \mathbf{P}$ is a fundamental matrix of (2.6.2).

2.6.4 The Method of Undetermined Coefficients

Since $\Phi(t) = \mathbf{P} e^{t\mathbf{J}}$ is a fundamental matrix of (2.6.2), together with the form of $\mathbf{P} e^{t\mathbf{J}}$, one obtains that $\mathbf{x} = \boldsymbol{\varphi}(t)$ is a solution of the system (2.6.2) of the form

$$\mathbf{x}(t) = \mathbf{P}(t) e^{\lambda t}$$

with

$$\mathbf{P}(t) = [P_1^m(t), P_2^m(t), \dots, P_n^m(t)]^T,$$

where $P_i^m(t)$ ($i = 1, 2, \dots, n$) is a polynomial of degree $\leq m$. Consequently, we have

Theorem 2.6.4 If the characteristic polynomial $P_n(\lambda)$ is written as

$$P_n(\lambda) = (\lambda - \lambda_1)^{n_1} (\lambda - \lambda_2)^{n_2} \cdots (\lambda - \lambda_m)^{n_m},$$

then for every n_i -fold eigenvalue λ_i of A , $i = 1, 2, \dots, m$, the homogeneous system (2.6.2) has n_i linearly independent solutions of the form

$$\boldsymbol{\varphi}_1(t) = \mathbf{P}_1(t)e^{\lambda_1 t}, \dots, \boldsymbol{\varphi}_{n_i}(t) = \mathbf{P}_{n_i}(t)e^{\lambda_i t}, \quad (2.6.10)$$

in which every component of

$$\mathbf{P}_k(t) = (P_1^k(t), P_2^k(t), \dots, P_n^k(t))^T \quad (k = 1, 2, \dots, n_i)$$

is a polynomial of degree $< n_i$. Further, if carried out for every eigenvalue, this construction leads to n solutions, which form a fundamental system of (2.6.2).

Proof A fundamental matrix of (2.6.2) can be written as

$$\boldsymbol{\Phi}(t) = \mathbf{P} \text{diag}(e^{tJ_1}, \dots, e^{tJ_m}),$$

which shows the first desired conclusion. Since for every eigenvalue λ_i , we have obtained n_i linearly independent solutions of the form (2.6.10). Together with the fact that $n_1 + n_2 + \dots + n_m = n$, it suffices to prove that n solutions formed by all eigenvalues λ_i are linearly independent. The proof of this is left to the reader.

Next, we will discuss the following two cases.

Case 1 A has only single eigenvalue.

Theorem 2.6.5 Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the distinct eigenvalues of the coefficient matrix A and $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n$ be the corresponding eigenvectors. Then the matrix-valued function

$$\boldsymbol{\Phi}(t) = [e^{\lambda_1 t} \mathbf{r}_1, e^{\lambda_2 t} \mathbf{r}_2, \dots, e^{\lambda_n t} \mathbf{r}_n] \quad (2.6.11)$$

is a fundamental matrix of (2.6.2).

Proof Since $\mathbf{r}_i$ is an eigenvector of A corresponding to the eigenvalue λ_i , we have

$$(e^{\lambda_i t} \mathbf{r}_i)' = \lambda_i e^{\lambda_i t} \mathbf{r}_i = A e^{\lambda_i t} \mathbf{r}_i$$

which shows that every column $e^{\lambda_i t} \mathbf{r}_i$ of $\boldsymbol{\Phi}(t)$ is a solution of (2.6.2). To show that $\boldsymbol{\Phi}(t)$ is a fundamental matrix, we notice that

$$W(0) = \det[\boldsymbol{\Phi}(0)] = \det[\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n] \neq 0,$$

since $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n$ are linearly independent. Thus, it follows the desired result.

Remark 2.6.2 Obviously, from Theorem 2.6.5, it follows that

$$e^{At} = \boldsymbol{\Phi}(t) \boldsymbol{\Phi}^{-1}(0). \quad (2.6.12)$$

Remark 2.6.3 We are interested in real-valued solutions. Here one runs into the difficulty that a real matrix may have complex eigenvalues, which lead to complex-valued solution $\mathbf{x}_1(t) = \mathbf{u}(t) + i\mathbf{v}(t)$. It is obvious that for real A both the real part $\mathbf{u}(t)$ and imaginary part $\mathbf{v}(t)$ of a complex-valued solution are real-valued solutions to (2.6.2). Thus one obtains two real-valued solutions from a complex eigenvalue.

Example 2.6.5 Find a general solution of the system

$$\begin{cases} \frac{dx_1}{dt} = 4x_1 - 3x_2, \\ \frac{dx_2}{dt} = 3x_1 + 4x_2. \end{cases}$$

Solution Here the coefficient matrix $\mathbf{A} = \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}$. Its characteristic equation

$$|\lambda \mathbf{E} - \mathbf{A}| = \begin{vmatrix} \lambda - 4 & 3 \\ -3 & \lambda - 4 \end{vmatrix} = (\lambda - 4)^2 + 9 = 0.$$

Hence $\mathbf{A}$ has the complex conjugate eigenvalues $\lambda = 4 + 3i$ and $\bar{\lambda} = 4 - 3i$. Substituting $\lambda = 4 + 3i$ in the eigenvector equation $(\lambda \mathbf{E} - \mathbf{A})\mathbf{r} = 0$, we get the corresponding eigenvector is $\mathbf{r} = [1, -i]^T$. The complex-valued solution of

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$$

is

$$\mathbf{x}(t) = e^{(4+3i)t} \mathbf{r} = e^{4t} \begin{bmatrix} \cos 3t + i \sin 3t \\ -i \cos 3t + \sin 3t \end{bmatrix}.$$

The real part and imaginary part of $\mathbf{x}(t)$ are the real-valued solutions

$$\mathbf{x}_1(t) = e^{4t} \begin{bmatrix} \cos 3t \\ \sin 3t \end{bmatrix} \quad \text{and} \quad \mathbf{x}_2(t) = e^{4t} \begin{bmatrix} \sin 3t \\ -\cos 3t \end{bmatrix}.$$

A real-valued general solution of $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ is given by

$$\mathbf{x}(t) = e^{4t} \begin{bmatrix} \cos 3t & \sin 3t \\ \sin 3t & -\cos 3t \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

Example 2.6.6 Find a general solution of the system

$$\begin{cases} \frac{dx}{dt} = 3x - y + z, \\ \frac{dy}{dt} = -x + 5y - z, \\ \frac{dz}{dt} = x - y + 3z. \end{cases}$$

Solution The coefficient matrix

$$\mathbf{A} = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

has the characteristic equation

$$\det(\lambda \mathbf{E} - \mathbf{A}) = \lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$$

and hence has the eigenvalues

$$\lambda_1 = 2, \lambda_2 = 3, \lambda_3 = 6.$$

Substituting $\lambda_1 = 2$ in the eigenvector equation $(\lambda \mathbf{E} - \mathbf{A})\mathbf{r} = 0$ gives the corresponding eigenvector

$$\mathbf{r}_1 = [1, 0, -1]^T.$$

Thus the associated solution with $\lambda_1 = 2$ is

$$\boldsymbol{\phi}_1(t) = [e^{2t}, 0, -e^{2t}]^T.$$

Similarly, we obtain the corresponding particular solutions with $\lambda_2 = 3, \lambda_3 = 6$

$$\boldsymbol{\varphi}_2(t) = [e^{3t}, e^{3t}, e^{3t}]^T$$

and

$$\boldsymbol{\varphi}_3(t) = [e^{6t}, -2e^{6t}, e^{6t}]^T.$$

The general solution of the system is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = c_1 \begin{bmatrix} e^{2t} \\ 0 \\ -e^{2t} \end{bmatrix} + c_2 \begin{bmatrix} e^{3t} \\ e^{3t} \\ e^{3t} \end{bmatrix} + c_3 \begin{bmatrix} e^{6t} \\ -2e^{6t} \\ e^{6t} \end{bmatrix}.$$

Example 2.6.7 Solve the initial value problem

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} 3 & -2 & 0 \\ -1 & 3 & -2 \\ 0 & -1 & 3 \end{bmatrix} \mathbf{x}, \mathbf{x}(0) = \begin{bmatrix} 0 \\ 2 \\ 6 \end{bmatrix}.$$

Solution Applying similar method gives the general solution in scalar form

$$\begin{cases} x_1(t) = 2c_1 e^t + 2c_2 e^{3t} + 2c_3 e^{5t}, \\ x_2(t) = 2c_1 e^t - 2c_3 e^{5t}, \\ x_3(t) = c_1 e^t - c_2 e^{3t} + c_3 e^{5t}. \end{cases}$$

Substituting these initial values in the three preceding scalar equations yields algebraic linear system

$$\begin{cases} 2c_1 + 2c_2 + 2c_3 = 0, \\ 2c_1 - 2c_3 = 2, \\ c_1 - c_2 + c_3 = 6. \end{cases}$$

We solve for $c_1 = 2, c_2 = -3$ and $c_3 = 1$. Thus the particular solution is given by

$$\mathbf{x}(t) = 2\mathbf{x}_1 - 3\mathbf{x}_2 + \mathbf{x}_3 = \begin{bmatrix} 4e^t - 6e^{3t} + 2e^{5t} \\ 4e^t - 2e^{5t} \\ 2e^t + 3e^{3t} + e^{5t} \end{bmatrix}.$$

Example 2.6.8 Find e^{At} if

$$\mathbf{A} = \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix}.$$

Solution We easily find that the system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$$

has a fundamental matrix

$$\boldsymbol{\Phi}(t) = \begin{bmatrix} e^{-2t} & 2e^{5t} \\ -3e^{-2t} & e^{5t} \end{bmatrix} \quad \text{with} \quad \boldsymbol{\Phi}^{-1}(0) = \frac{1}{7} \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}.$$

The equation $e^{At} = \boldsymbol{\Phi}(t)\boldsymbol{\Phi}^{-1}(0)$ gives

$$\begin{aligned} e^{At} &= \frac{1}{7} \begin{bmatrix} e^{-2t} & 2e^{5t} \\ -3e^{-2t} & e^{5t} \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix} \\ &= \frac{1}{7} \begin{bmatrix} e^{-2t} + 6e^{5t} & -2e^{-2t} + 2e^{5t} \\ -3e^{-2t} + 3e^{5t} & 6e^{-2t} + e^{5t} \end{bmatrix}. \end{aligned}$$

Case 2 $\mathbf{A}$ has repeated eigenvalues

If the characteristic polynomial

$$P_n(\lambda) = (\lambda - \lambda_1)^{n_1} (\lambda - \lambda_2)^{n_2} \cdots (\lambda - \lambda_m)^{n_m},$$

then by Theorem 2.6.4, for every n_i -fold eigenvalue λ_i , the homogeneous system (2.6.2) has n_i linearly independent solutions of the form (2.6.10), that is

$$\mathbf{x}(t) = e^{\lambda_i t} \left(\mathbf{r}_0 + t\mathbf{r}_1 + \frac{t^2}{2!}\mathbf{r}_2 + \cdots + \frac{t^{n_i-1}}{(n_i-1)!}\mathbf{r}_{n_i-1} \right), \quad (2.6.13)$$

where $\mathbf{r}_j$ ($j = 0, 1, \dots, n_i - 1$) are n -dimensional constant column vectors.

Theorem 2.6.6 If λ_i is n_i -fold eigenvalue of $\mathbf{A}$, then the homogeneous system (2.6.2) has the nonzero solution of the form (2.6.13) if and only if $\mathbf{r}_0$ is a nonzero solution of this system

$$(\mathbf{A} - \lambda_i \mathbf{E})^{n_i} \mathbf{r} = 0 \quad (2.6.14)$$

and $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_{n_i-1}$ will be determined by

$$\begin{cases} \mathbf{r}_1 = (\mathbf{A} - \lambda_i \mathbf{E}) \mathbf{r}_0, \\ \mathbf{r}_2 = (\mathbf{A} - \lambda_i \mathbf{E}) \mathbf{r}_1, \\ \vdots \\ \mathbf{r}_{n_i-1} = (\mathbf{A} - \lambda_i \mathbf{E}) \mathbf{r}_{n_i-2}. \end{cases} \quad (2.6.15)$$

Proof Substituting (2.6.13) in the homogeneous system gives

$$\begin{aligned} & \lambda_i e^{\lambda_i t} \left(\mathbf{r}_0 + t\mathbf{r}_1 + \frac{t^2}{2!}\mathbf{r}_2 + \cdots + \frac{t^{n_i-1}}{(n_i-1)!}\mathbf{r}_{n_i-1} \right) \\ & + e^{\lambda_i t} \left(\mathbf{r}_1 + t\mathbf{r}_2 + \cdots + \frac{t^{n_i-2}}{(n_i-2)!}\mathbf{r}_{n_i-1} \right) \\ & = \mathbf{A} e^{\lambda_i t} \left(\mathbf{r}_0 + t\mathbf{r}_1 + \cdots + \frac{t^{n_i-1}}{(n_i-1)!}\mathbf{r}_{n_i-1} \right), \end{aligned}$$

that is

$$(\mathbf{A} - \lambda_i \mathbf{E}) \left(\mathbf{r}_0 + t\mathbf{r}_1 + \cdots + \frac{t^{n_i-1}}{(n_i-1)!}\mathbf{r}_{n_i-1} \right) = \mathbf{r}_1 + t\mathbf{r}_2 + \cdots + \frac{t^{n_i-2}}{(n_i-2)!}\mathbf{r}_{n_i-1}.$$

Equating the coefficients of like terms yields

$$\begin{cases} (\mathbf{A} - \lambda_i \mathbf{E}) \mathbf{r}_0 = \mathbf{r}_1, \\ (\mathbf{A} - \lambda_i \mathbf{E}) \mathbf{r}_1 = \mathbf{r}_2, \\ \vdots \\ (\mathbf{A} - \lambda_i \mathbf{E}) \mathbf{r}_{n_i-2} = \mathbf{r}_{n_i-1}, \\ (\mathbf{A} - \lambda_i \mathbf{E}) \mathbf{r}_{n_i-1} = 0, \end{cases}$$

namely

$$\begin{cases} (\mathbf{A} - \lambda_i \mathbf{E}) \mathbf{r}_0 = \mathbf{r}_1, \\ (\mathbf{A} - \lambda_i \mathbf{E})^2 \mathbf{r}_0 = \mathbf{r}_2, \\ \vdots \\ (\mathbf{A} - \lambda_i \mathbf{E})^{n_i-1} \mathbf{r}_0 = \mathbf{r}_{n_i-1}, \\ (\mathbf{A} - \lambda_i \mathbf{E})^{n_i} \mathbf{r}_0 = 0. \end{cases}$$

The reasoning process is reversible.

In order to obtain a fundamental system of (2.6.2), we have the following procedure.

Step 1 Solve all distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_r$ and the corresponding multiplicity $n_1, n_2, \dots, n_r$.

Step 2 For every λ_i , we first find n_i linearly independent $\mathbf{r}_0$ from $(\mathbf{A} - \lambda_i \mathbf{E})^{n_i} \mathbf{r}_0 = 0$, and obtain $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_{n_i-1}$. Secondly, substituting these values in (2.6.13) yields n_i linearly independent solutions with λ_i . Finally, if carried out for every eigenvalue, the construction leads to n solutions, which form a fundamental system.

Example 2.6.9 Find a general solution of the following system

$$\begin{cases} \frac{dx}{dt} = 4x - y, \\ \frac{dy}{dt} = 3x + y - z, \\ \frac{dz}{dt} = x + z. \end{cases}$$

Solution The characteristic polynomial is

$$P_n(\lambda) = \det(\mathbf{A} - \lambda \mathbf{E}) = \begin{vmatrix} 4 - \lambda & -1 & 0 \\ 3 & 1 - \lambda & -1 \\ 1 & 0 & 1 - \lambda \end{vmatrix} = -(\lambda - 2)^3$$

which shows that $\mathbf{A}$ has the repeated eigenvalue $\lambda = 2$ of multiplicity 3. So the original system has three linearly independent solutions of the form

$$\mathbf{x} = \left(\mathbf{r}_0 + \mathbf{r}_1 t + \mathbf{r}_2 \frac{t^2}{2!} \right) e^{2t},$$

where $\mathbf{r}_k$ ($k = 0, 1, 2$) satisfies

$$\begin{cases} (\mathbf{A} - 2\mathbf{E})^3 \mathbf{r}_0 = 0, \\ (\mathbf{A} - 2\mathbf{E}) \mathbf{r}_0 = \mathbf{r}_1, \\ (\mathbf{A} - 2\mathbf{E}) \mathbf{r}_1 = \mathbf{r}_2. \end{cases} \quad (2.6.16)$$

We easily obtain

$$(\mathbf{A} - 2\mathbf{E})^3 \mathbf{r}_0 = 0$$

has three linearly independent solutions

$$\mathbf{r}_0 : \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

From (2.6.16), we get the corresponding $\mathbf{r}_1$ and $\mathbf{r}_2$ respectively:

$$\mathbf{r}_1 : \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix}, \mathbf{r}_2 : \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

Thus a fundamental system is given by

$$\boldsymbol{\varphi}_1(t) = e^{2t} \left[\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \frac{t^2}{2} \right] = e^{2t} \begin{bmatrix} 1 + 2t + \frac{t^2}{2} \\ 3t + t^2 \\ t + \frac{t^2}{2} \end{bmatrix},$$

$$\boldsymbol{\varphi}_2(t) = e^{2t} \left[\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} t + \begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix} \frac{t^2}{2} \right] = e^{2t} \begin{bmatrix} -t - \frac{t^2}{2} \\ 1 - t - t^2 \\ -\frac{t^2}{2} \end{bmatrix},$$

$$\boldsymbol{\varphi}_3(t) = e^{2t} \left[\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \frac{t^2}{2} \right] = e^{2t} \begin{bmatrix} \frac{t^2}{2} \\ -t + t^2 \\ 1 - t + \frac{t^2}{2} \end{bmatrix}.$$

Hence a general solution is

$$\mathbf{x}(t) = c_1 \boldsymbol{\varphi}_1(t) + c_2 \boldsymbol{\varphi}_2(t) + c_3 \boldsymbol{\varphi}_3(t).$$

Exercise 2.6

Find general solutions of the homogeneous systems in Problems 1 through 14.

$$1. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} -2 & 1 \\ -1 & -4 \end{bmatrix} \mathbf{x}.$$

$$2. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix} \mathbf{x}.$$

$$3. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 2 & -3 \\ 1 & -2 \end{bmatrix} \mathbf{x}.$$

$$4. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 7 & 1 \\ -4 & 3 \end{bmatrix} \mathbf{x}.$$

$$5. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \mathbf{x}.$$

$$6. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 2 & 0 & 0 \\ -7 & 9 & 7 \\ 0 & 0 & 2 \end{bmatrix} \mathbf{x}.$$

$$7. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \mathbf{x}.$$

$$8. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} -3 & 0 & -4 \\ -1 & -1 & -1 \\ 1 & 0 & 1 \end{bmatrix} \mathbf{x}.$$

$$9. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 0 & 0 & 1 \\ -5 & -1 & -5 \\ 4 & 1 & -2 \end{bmatrix} \mathbf{x}.$$

$$10. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 3 & 1 \\ -2 & -4 & -1 \end{bmatrix} \mathbf{x}.$$

$$11. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 0 & 1 & -5 \\ -5 & 3 & 0 \\ 1 & 0 & -3 \end{bmatrix} \mathbf{x}.$$

$$12. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} \mathbf{x}.$$

$$13. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \mathbf{x}.$$

$$14. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \mathbf{x}.$$

Find general solutions of the nonhomogeneous systems in Problems 15 through 20.

$$15. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 0 & -n^2 \\ -n^2 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} \cos nt \\ \sin nt \end{bmatrix}.$$

$$16. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 2e^t \end{bmatrix}.$$

$$17. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} t^2 \\ 2t \\ t \end{bmatrix}.$$

$$18. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 3 \\ -2 \end{bmatrix}.$$

$$19. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 5 \\ -2t \end{bmatrix}.$$

$$20. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2e^t \\ -3e^t \end{bmatrix}.$$

Apply the method of undetermined coefficients to solve the initial value problems 21 through 25.

$$21. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 3 & 4 \\ 3 & 2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ t^2 \end{bmatrix}, \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$$22. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 4 & 1 \\ 6 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} e^t \\ -e^t \end{bmatrix}, \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

$$23. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

$$24. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} \sin t \\ -2\cos t \end{bmatrix}, \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$$25. \frac{d\mathbf{x}}{dt} = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 3t \\ 2 \end{bmatrix}, \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

$$26. \text{ If } \mathbf{A} = \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix}, \text{ show that}$$

$$e^{\mathbf{A}t} = e^{\alpha t} \begin{bmatrix} \cos \beta t & \sin \beta t \\ -\sin \beta t & \cos \beta t \end{bmatrix}.$$

$$27. \text{ If } \mathbf{A}^2 = \alpha \mathbf{A}, \text{ show that } e^{\mathbf{A}t} = \mathbf{E} + [(e^{\alpha t} - 1)/\alpha] \mathbf{A}.$$

28. Show the following:

(1) Every solution of the differential system (2.6.2) tends to zero as $t \rightarrow +\infty$, if and only if the real parts of the eigenvalues of the matrix $\mathbf{A}$ are negative.

(2) Every solution of the differential system (2.6.2) is bounded in the interval $[0, \infty)$ if and only if the real parts of the multiple eigenvalues of the matrix $\mathbf{A}$ are negative, and the real parts of the simple eigenvalues of the matrix $\mathbf{A}$ are nonpositive.

29. If the homogeneous system (2.6.2) has a solution of the form $x = \mathbf{P}(t)e^{\lambda t}$, where λ is a constant, $\mathbf{P}(t)$ is an n -dimensional vector valued function in which every component is a polynomial of degree $\leq m$, then show that

$$e^{\lambda t} \mathbf{P}(t), e^{\lambda t} \frac{d\mathbf{P}}{dt}, \dots, e^{\lambda t} \frac{d^m \mathbf{P}(t)}{dt^m}$$

are $(m+1)$ linearly independent solutions of (2.6.2).

30. If $\Phi(t)$ is a fundamental matrix of the system (2.6.2) with $\Phi(0) = \mathbf{E}$, then show that for any $\tau \in R$

$$\Phi(t)\Phi^{-1}(\tau) = \Phi(t - \tau).$$

31. Show that the vector valued function

$$\mathbf{x}(t) = \exp \left[\int_{t_0}^t \mathbf{A}(s) ds \right] \mathbf{x}_0$$

is not a solution of the differential system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x},$$

unless $\mathbf{A}(t)$ and $\int_{t_0}^t \mathbf{A}(s) ds$ commute for all t , where $\mathbf{x}(t_0) = \mathbf{x}_0$.

2.7 Periodic Linear Systems

Periodicity of solutions of differential systems is an interesting and critical aspect of qualitative study. In this section, we will deal with linear systems with periodic coefficients.

A function $x(t)$ is called periodic of period $\omega > 0$ if

$$x(t + \omega) = x(t), t \in R, \quad (2.7.1)$$

which means that the graph of $x(t)$ repeats itself in successive intervals of length ω . For convenience, we assume that ω is the smallest number for which (2.7.1) holds. If every component $\varphi_i(t), i = 1, 2, \dots, n$ of vector valued function $\boldsymbol{\varphi}(t)$ and every element $a_{ij}(t), 1 \leq i, j \leq n$ of $\mathbf{A}(t)$ are periodic of period ω , then $\boldsymbol{\varphi}(t)$ and $\mathbf{A}(t)$ are said to be periodic of period ω . For example, $\sin 2t, \cos 2t$ are periodic of period π . The matrix valued function

$$\mathbf{A}(t) = \begin{bmatrix} \sin 2t & \cos t \\ \cos t & \cos 2t \end{bmatrix}$$

is periodic of period 2π .

2.7.1 Homogeneous Systems with Periodic Coefficients

We consider the following systems with a continuous, ω -periodic real coefficient matrix

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} \text{ with } \mathbf{A}(t + \omega) = \mathbf{A}(t). \quad (2.7.2)$$

A main result states that the system with periodic coefficient (2.7.2) can be reduced to a system with constant coefficient.

Firstly, we provide a necessary and sufficient condition for the homogeneous system (2.7.2) to have a periodic solution of period ω .

Theorem 2.7.1 Let $\boldsymbol{\Phi}(t)$ be a fundamental matrix of the system (2.7.2). Then the periodic linear system (2.7.2) has a nontrivial periodic solution $\mathbf{x}(t)$ of period ω if and only if

$$\det[\boldsymbol{\Phi}(0) - \boldsymbol{\Phi}(\omega)] = 0.$$

Proof The general solution of the differential system (2.7.2) is

$$\mathbf{x}(t) = \boldsymbol{\Phi}(t)\mathbf{c},$$

where $\mathbf{c}$ is an arbitrary constant vector.

$\mathbf{x}(t)$ is periodic of period ω if and only if

$$\boldsymbol{\Phi}(0)\mathbf{c} = \boldsymbol{\Phi}(\omega)\mathbf{c}.$$

That is, the system

$$[\boldsymbol{\Phi}(0) - \boldsymbol{\Phi}(\omega)]\mathbf{c} = 0$$

has a nontrivial solution vector $\mathbf{c}$. This shows that the system has a nontrivial solution if and only if

$$\det[\boldsymbol{\Phi}(0) - \boldsymbol{\Phi}(\omega)] = 0.$$

Corollary 2.7.1 The homogeneous system with constant coefficient

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \quad (2.7.3)$$

has a nontrivial periodic solution of period ω if and only if the matrix $(\mathbf{E} - e^{\mathbf{A}\omega})$ is singular.

Theorem 2.7.2 (Floquet's Theorem) Let $\boldsymbol{\Phi}(t)$ be a fundamental matrix of (2.7.2)

with $\Phi(0) = E$. Then the following conclusions hold:

- (1) The matrix $\Psi(t) = \Phi(t + \omega)$ is also a fundamental matrix of (2.7.2).
- (2) $\Phi(t)$ has a Floquet representation

$$\Phi(t) = P(t)e^{Rt} \quad (2.7.4)$$

where $P(t)$ is ω -periodic nonsingular matrix and R satisfies the equation $C = \Phi(\omega) = e^{\omega R}$.

Proof Since $\Phi(t)$ is a fundamental matrix of (2.7.2), it follows that

$$\Psi'(t) = \Phi'(t + \omega) = A(t + \omega)\Phi(t + \omega) = A(t)\Psi(t),$$

that is, $\Psi(t)$ is a solution matrix of the differential system (2.7.2). Further, we have $\det(\Psi(t)) \neq 0$ for all t , since $\det[\Phi(t + \omega)] \neq 0$ for all t . Thus $\Psi(t)$ is also a fundamental matrix of (2.7.2). This completes the proof of part (1).

Next, we shall prove part (2). Since $\Phi(t)$ and $\Phi(t + \omega)$ are both fundamental matrices of (2.7.2), there exists a nonsingular constant matrix C such that

$$\Phi(t + \omega) = \Phi(t)C \quad (2.7.5)$$

which shows that $C = \Phi(\omega)$ because of $\Phi(0) = E$. From the given result in linear algebra, there exists a constant matrix R such that $C = e^{\omega R}$. The equation (2.7.5) reads

$$\Phi(t + \omega) = \Phi(t)e^{\omega R} \quad (2.7.6)$$

Let $P(t)$ be a matrix defined by the relation $P(t) = \Phi(t)e^{-Rt}$. Together with (2.7.6), we have

$$\begin{aligned} P(t + \omega) &= \Phi(t + \omega)e^{-R(t + \omega)} = \Phi(t)e^{\omega R}e^{-\omega R}e^{-Rt} \\ &= \Phi(t)e^{-Rt} = P(t). \end{aligned}$$

Hence, $P(t)$ is periodic of period ω . Since $\Phi(t)$ and e^{-Rt} are nonsingular, we know $\det(P(t)) \neq 0$ in R .

By Floquet's Theorem, we can use a transformation which reduces the differential system (2.7.2) to a differential system of the type (2.7.3). Precisely, we have the following result.

Theorem 2.7.3 Let $P(t)$ and R be the matrices obtained in Theorem 2.7.2. Then the substitution $x(t) = P(t)y(t)$ reduces the differential system (2.7.2) to the system

$$\frac{dy}{dt} = Ry. \quad (2.7.7)$$

Proof Together with $\frac{d\Phi(t)}{dt} = A(t)\Phi(t)$, the expression (2.7.4) gives

$$(P(t)e^{Rt})' = A(t)P(t)e^{Rt}$$

which yields

$$P'(t) + P(t)R - A(t)P(t) = 0. \quad (2.7.8)$$

Substituting $x(t) = P(t)y$ in the differential system (2.7.2), we obtain

$$P'(t)y + P(t)y' = A(t)P(t)y,$$

that is,

$$P(t)y' + [P'(t) - A(t)P(t)]y = 0. \quad (2.7.9)$$

Combining the relations (2.7.8) and (2.7.9), we obtain

$$\frac{dy}{dt} = Ry.$$

2.7.2 Nonhomogeneous System with Periodic Coefficients

We consider the system

$$\frac{dx}{dt} = A(t)x + f(t) \quad (2.7.10)$$

where the matrix $A(t)$ and the function $f(t)$ are continuous and ω -periodic in R .

Theorem 2.7.4 The differential system (2.7.10) has a periodic solution $x(t)$ of period ω if and only if $x(0) = x(\omega)$.

Proof If $x(t)$ is a periodic solution of period ω , namely, $x(t) = x(t + \omega)$, then it is necessary that $x(0) = x(\omega)$. To show sufficiency, let $x(t)$ be a solution of (2.7.10) which satisfies $x(0) = x(\omega)$. We denote $y(t)$ by $y(t) = x(t + \omega)$, then it follows that

$$\begin{aligned} y'(t) &= x'(t + \omega) = A(t + \omega)x(t + \omega) + f(t + \omega) \\ &= A(t)y(t) + f(t) \end{aligned}$$

which shows that $y(t)$ is a solution of (2.7.10). But, since $y(0) = x(\omega) = x(0)$, the uniqueness of the initial value problem implies that

$$x(t) = y(t) = x(t + \omega).$$

This shows that $x(t)$ is ω -periodic.

Corollary 2.7.2 Let the conditions of Theorem 2.7.4 be satisfied. Then the system (2.7.10) has a unique periodic solution of period ω if and only if the associated homogeneous system (2.7.2) does not have a periodic solution of period ω other than the trivial solution.

Proof If $\Phi(t)$ is a fundamental matrix of the system (2.7.2), then the general solution of (2.7.10) is given by

$$x(t) = \Phi(t)c + \int_0^t \Phi(t)\Phi^{-1}(s)f(s)ds,$$

where c is an arbitrary constant. Thus $x(t)$ is periodic of period ω if and only if

$$\Phi(0)c = \Phi(\omega)c + \int_0^\omega \Phi(\omega)\Phi^{-1}(s)f(s)ds$$

that is, the linear system

$$[\Phi(0) - \Phi(\omega)]c = \int_0^\omega \Phi(\omega)\Phi^{-1}(s)f(s)ds$$

has a unique solution vector c . However, this system has a unique solution if and only if $\det(\Phi(0) - \Phi(\omega)) \neq 0$.

From Theorem 2.7.1, we obtain the desired conclusion.

The following theorem clarifies the relationship of (2.7.10) with the system

$$\frac{dy}{dt} = Ry + g(t), g(t) = P^{-1}(t)f(t). \quad (2.7.11)$$

Theorem 2.7.5 The solutions $x(t)$ of the system (2.7.10) and $y(t)$ of the system

(2.7.11) with the same initial condition $\mathbf{x}(0) = \mathbf{y}(0) = \boldsymbol{\mu}$ are coupled by relation

$$\mathbf{x}(t) = \mathbf{P}(t)\mathbf{y}(t).$$

Proof By **Floquet's Theorem**, the fundamental matrix $\boldsymbol{\Phi}(t)$ is given by

$$\boldsymbol{\Phi}(t) = \mathbf{P}(t)e^{Rt}$$

which follows that

$$\boldsymbol{\Phi}'(t) = (\mathbf{P}'(t) + \mathbf{P}(t)\mathbf{R})e^{Rt} = \mathbf{A}(t)\boldsymbol{\Phi} = \mathbf{A}(t)\mathbf{P}(t)e^{Rt}.$$

Thus

$$\mathbf{P}'(t) + \mathbf{P}(t)\mathbf{R} = \mathbf{A}(t)\mathbf{P}(t). \quad (2.7.12)$$

Combining $\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t)$ and $\mathbf{x} = \mathbf{P}(t)\mathbf{y}$, namely $\mathbf{x}' = \mathbf{P}'(t)\mathbf{y} + \mathbf{P}(t)\mathbf{y}'$, we have

$$\mathbf{P}'(t)\mathbf{y} + \mathbf{P}(t)\mathbf{y}' = \mathbf{A}(t)\mathbf{P}(t)\mathbf{y} + \mathbf{f}(t).$$

From (2.7.12),

$$\mathbf{P}'(t)\mathbf{y} + \mathbf{P}(t)\mathbf{y}' = \mathbf{P}'(t)\mathbf{y} + \mathbf{P}(t)\mathbf{R}\mathbf{y} + \mathbf{f}(t).$$

Using multiplication on the left by $\mathbf{P}^{-1}(t)$, one obtains (2.7.11). Similarly, we can prove the reverse direction.

Exercise 2.7

1. Consider the system (2.7.2). The following hold:

(1) If $\mathbf{x}(t)$ is a solution of (2.7.2), then so is $\mathbf{y}(t) = \mathbf{x}(t + \omega)$.

(2) If $\mathbf{x}(t)$ is a solution of (2.7.2) and $\mathbf{x}(\omega) = \lambda\mathbf{x}(0)$ ($\lambda \in R$), then it follows that $\mathbf{x}(t + \omega) = \lambda\mathbf{x}(t)$ and more generally $\mathbf{x}(t + k\omega) = \lambda^k\mathbf{x}(t)$ for all t (k an integer).

2. In Theorem 2.7.2 (Floquet's Theorem), $\mathbf{C} = \boldsymbol{\Phi}(\omega)$ is called the transition matrix. Its eigenvalues λ_i are called Floquet multipliers. Together with the part (2) in problem 1, show the following results:

(1) There exist nontrivial solutions to (2.7.2) satisfying $\mathbf{x}(t + \omega) = \lambda\mathbf{x}(t)$ if and only if λ is an eigenvalue of $\mathbf{C} = \boldsymbol{\Phi}(\omega)$.

(2) Every such solution has the form

$$\mathbf{x} = \boldsymbol{\Phi}(t)\mathbf{r},$$

where $\mathbf{r}$ is an eigenvector of $\mathbf{C}$ corresponding to λ .

(3) If the matrix $\mathbf{C}$ is diagonalizable, then one obtains a fundamental system of solution in this manner.

(4) There exist nontrivial ω -periodic solutions if and only if $\lambda = 1$ is an eigenvalue of $\mathbf{C}$.

(5) There exist periodic solutions with minimal period $k\omega > 0$ ($k \in \mathbb{N}$) if and only if $\mathbf{C}$ has an eigenvalue λ satisfying $\lambda^k = 1$ and $\lambda^j \neq 1$ for $1 \leq j < k$.

3. Consider the equation

$$x' = ax + \sin t.$$

Discuss the cases (1) $a = 0$, (2) $a > 0$, (3) $a < 0$, separately for the existence of a unique periodic solution.

4. Verify that in the differential equation

$$x' = (\cos^2 t)x$$

even though the function $\cos^2 t$ is periodic of period π , the solution is not periodic.

5. Show that the differential equation

$$x'' + x = 0$$

admits nontrivial periodic solutions of period 2π .

6. Consider the linear equation

$$\frac{dx}{dt} + p(t)x = q(t), \quad (2.7.13)$$

where $p(t)$ and $q(t)$ are both continuous, ω -periodic functions. Show the following:

(1) If $q(t) \equiv 0$, then each nontrivial solution of (2.7.13) is periodic of period ω if and only if the mean value of the function $p(t)$

$$\bar{p} = \frac{1}{\omega} \int_0^\omega p(t) dt = 0.$$

(2) If $q(t) \neq 0$, then the equation (2.7.13) has a unique periodic solution of period ω if and only if $\bar{p} \neq 0$. Find this solution.

Chapter 3 High-Order Linear Differential Equations

In this chapter, we now turn to linear equations of high order $n \geq 2$. This type of differential equation plays an important role in the natural sciences and technology.

3.1 Introduction

Recall that an n th-order differential equation in the unknown function $x(t)$ is one of the form

$$F(t, x, x', \dots, x^{(n)}) = 0.$$

This differential equation is said to be linear provided that F is linear in the dependent variable x and its derivatives $x', \dots, x^{(n)}$.

Thus a linear n th-order differential equation takes the form

$$a_0(t)x^{(n)} + a_1(t)x^{(n-1)} + \dots + a_{n-1}(t)x' + a_n(t)x = g(t). \quad (3.1.1)$$

We will always assume that the coefficient functions $a_i(t), i = 0, 1, 2, \dots, n$ and $g(t)$ are continuous on the interval I . If $a_0(t) \neq 0$ at each point of I , we can divide each term in the equation (3.1.1) by $a_0(t)$ to obtain an equation of the form

$$x^{(n)} + p_1(t)x^{(n-1)} + \dots + p_{n-1}(t)x' + p_n(t)x = f(t). \quad (3.1.2)$$

If $f(t) \neq 0$, then we call the equation (3.1.2) a **nonhomogeneous linear equation**. Its associated homogeneous linear equation is

$$x^{(n)} + p_1(t)x^{(n-1)} + \dots + p_{n-1}(t)x' + p_n(t)x = 0. \quad (3.1.3)$$

In the case, the differential equation in (3.1.2) models a physical system, the nonhomogeneous term $f(t)$ usually corresponds to some external influence on the system.

Linear differential equations frequently appear as mathematical models of mechanical systems and electrical circuits.

Example 3.1.1 (Spring Vibration)

Assume that a mass m is attached both to a spring that exerts on it a force f_s and to a dashpot, namely shock absorber, that exerts a force f_d on the mass. If the restoring force f_s of the spring is proportional to the position x of the mass from equilibrium (positive to the right, negative to the left), and the force f_d is proportional to the velocity $v = \frac{dx}{dt}$ of the mass, then

$$f_s = -kx \quad \text{and} \quad f_d = -\mu \frac{dx}{dt} \quad (k, \mu > 0).$$

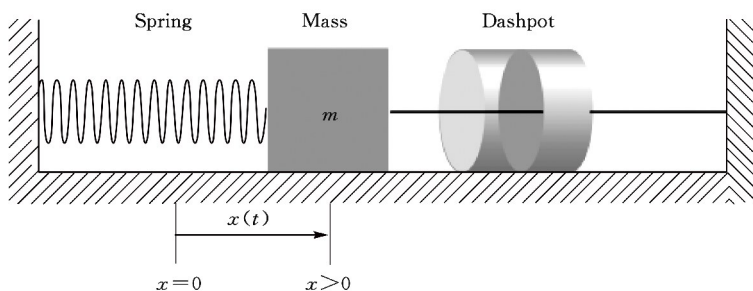


Figure 3.1.1

The minus signs show that f_s is negative when x is positive, f_d is negative when $\frac{dx}{dt}$ is positive. Using Newton's law, we obtain

$$mx'' = f_s + f_d,$$

that is

$$m \frac{d^2 x}{dt^2} + \mu \frac{dx}{dt} + kx = 0, \quad (3.1.4)$$

which is a differential equation with the position function $x(t)$ of the mass as unknown function. If, besides f_s and f_d , the mass m is acted on by an external force $f(t)$, then the resulting equation is

$$m \frac{d^2 x}{dt^2} + \mu \frac{dx}{dt} + kx = f(t). \quad (3.1.5)$$

The homogeneous linear equation (3.1.4) governs the free vibrations of the mass. However, the nonhomogeneous linear equation governs the forced vibrations of the mass under the influence of the external force $f(t)$.

Example 3.1.2 (Electrical Circuits)

We discuss the RLC circuit that is a fundamental building block in more complicated electrical circuits. It consists of a resistor with a resistance of R , an inductor with an inductance of L , a capacitor with a capacitance of C and a battery or a generator that supplies a voltage of E .

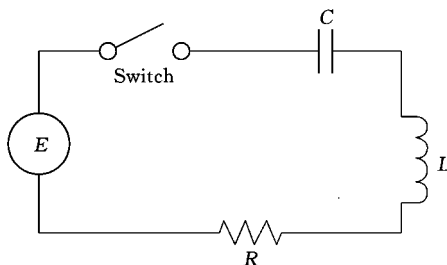


Figure 3.1.2

If the switch in the circuit is closed, then this results in a current of $I(t)$ in the circuit

and a charge of $q(t)$ on the capacitor at time t . The relation between $q(t)$ and $I(t)$ is

$$\frac{dq}{dt} = I(t) \quad (3.1.6)$$

with the aid of Kirchhoff's second law, that is, the algebraic sum of the voltage drops across the elements in a simple loop of an electrical circuit is equal to the applied voltage, the current and charge in the simple RLC circuit satisfy the basic circuit equation

$$L \frac{dI}{dt} + RI + \frac{q}{C} = E(t). \quad (3.1.7)$$

Because of $\frac{dq}{dt} = I$, the equation (3.1.7) reads

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E(t), \quad (3.1.8)$$

which is a second-order linear differential equation with unknown function $q(t)$.

In most practical problems, the primary interest is the current $I(t)$ rather than the charge $q(t)$. So differentiating both sides of (3.1.8) and substituting (3.1.6) give

$$L \frac{d^2I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = E'(t). \quad (3.1.9)$$

We can learn all we need to know about these constants and unknown functions by studying the differential equations (3.1.6) through (3.1.9).

It is of deep impression that the equations (3.1.8) and (3.1.9) have the same form as the equation (3.1.5) of a mass-spring-dashpot system with external force $f(t)$. Consequently, most of the results obtained from mechanical systems can be applied immediately to electrical circuits.

For convenience, we introduce the following operator

$$L[x] = x^{(n)} + p_1(t)x^{(n-1)} + \cdots + p_{n-1}(t)x' + p_n(t)x, \quad (3.1.10)$$

which is called linear differential operator. Applying (3.1.10), we rewrite respectively the equations (3.1.2) and (3.1.3) of the form

$$L[x] = f(t)$$

and

$$L[x] = 0.$$

Further, we easily obtain

$$L[kx] = kL[x], k \text{ is a constant.} \quad (3.1.11)$$

and

$$L[x_1 + x_2] = L[x_1] + L[x_2]. \quad (3.1.12)$$

Exercise 3.1

1. Show that the equations (3.1.11) and (3.1.12) hold.
2. Verify that $x_1(t) = \cos 2t$ and $x_2(t) = \sin 2t$ are two solutions of

$$x'' + 4x = 0.$$

Show that

$$x(t) = c_1 \cos 2t + c_2 \sin 2t$$

is a solution of this equation, where c_1 and c_2 are constants.

First verify that x_1 and x_2 are solutions of the differential equation. Then find a particular solution of the form $x = c_1 x_1 + c_2 x_2$ that satisfies the given initial conditions in Problems 3 through 6.

$$3. x'' - x = 0; x_1(t) = e^t, x_2(t) = e^{-t}; x(0) = 0, x'(0) = 5.$$

$$4. x'' + 25x = 0; x_1(t) = \cos 5t, x_2(t) = \sin 5t; x(0) = 10, x'(0) = -10.$$

$$5. x'' - 3x' = 0; x_1 = 1, x_2 = e^{3t}; x(0) = 4, x'(0) = -2.$$

$$6. x'' - 10x' + 25x = 0; x_1(t) = e^{5t}, x_2(t) = t e^{5t}; x(0) = 3, x'(0) = 13.$$

3.2 Existence and Uniqueness for High-Order Equations

In the first section, we introduce the second order linear differential equation

$$mx'' + \mu x' + kx = f(t)$$

as a mathematical model of the motion of the mass. Physical considerations show the motion of the mass should be determined by its initial position and initial velocity. Hence, the equations (3.1.2) and (3.1.3) ought to have a unique solution satisfying these initial conditions. We will give the following existence and uniqueness theorems.

Theorem 3.2.1 Consider the following initial value problem

$$\begin{cases} x^{(n)} + p_1(t)x^{(n-1)} + \cdots + p_{n-1}(t)x' + p_n(t)x = f(t), \\ x(0) = x_0, x'(0) = x_0^1, \dots, x^{(n-1)}(0) = x_0^{n-1}, \end{cases} \quad (3.2.1)$$

where $p_i(t) i = 1, 2, \dots, n$ and $f(t)$ are continuous on some open interval I containing the point zero, and $x_0, x_0^1, \dots, x_0^{n-1}$ are n given numbers. Then the initial value problem (3.2.1) has a unique solution on the entire interval I .

Proof Substitutions

$$x_1(t) = x, x_2(t) = x', \dots, x_n(t) = x^{(n-1)}$$

in the nonhomogeneous linear equation (3.1.2) yield a corresponding equivalent system of the form

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{f}(t),$$

where

$$\mathbf{A}(t) = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ -p_n & -p_{n-1} & -p_{n-2} & \cdots & -p_2 & -p_1 \end{bmatrix}, \mathbf{f}(t) = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ f(t) \end{bmatrix}.$$

By Theorem 2.3.1, we obtain the desired result.

Remark 3.2.1 When $n = 1$, the initial value problem reads

$$\begin{cases} x'(t) + p(t)x = f(t), \\ x(0) = x_0. \end{cases} \quad (3.2.2)$$

Then the initial value problem (3.2.2) has one and only one solution provided $p(t)$ and $f(t)$ are continuous on the open interval I .

For the general case, we also obtain similar result.

Theorem 3.2.2 Consider the initial value problem

$$\begin{cases} x^{(n)} = f(t, x, x', \dots, x^{(n-1)}), \\ x(t_0) = x_0, x'(t_0) = x_0^1, \dots, x^{(n-1)}(t_0) = x_0^{n-1}. \end{cases} \quad (3.2.3)$$

Let the function $f(t, x, x', \dots, x^{(n-1)})$ be continuous in some neighborhood U containing the point $P_0(t_0, x_0, x_0^1, \dots, x_0^{n-1})$. If $f(t, x, x', \dots, x^{(n-1)})$ satisfies a Lipschitz condition with respect to $x, x', \dots, x^{(n-1)}$, that is, there exists a constant $N > 0$, such that in the neighborhood U , one has

$$\begin{aligned} & |f(t, x_1, x_1', \dots, x_1^{(n-1)}) - f(t, x_2, x_2', \dots, x_2^{(n-1)})| \\ & \leq N(|x_1 - x_2| + |x_1' - x_2'| + \dots + |x_1^{(n-1)} - x_2^{(n-1)}|), \end{aligned}$$

then the initial value problem (3.2.3) has a unique solution on some neighborhood containing t_0 .

The proof will be given in Chapter 4. Throughout this chapter, we will always assume that $p_i(t), i = 1, 2, \dots, n$ and $f(t)$ are continuous functions.

Exercise 3.2

1. Let $x = \varphi(t)$ be a solution of (3.1.3) satisfying $\varphi(t_0) = \varphi'(t_0) = \dots = \varphi^{(n-1)}(t_0) = 0$, $t_0 \in I$. Then show that $\varphi(t) \equiv 0$.

2. Verify that

$$x_1(t) = t^2 \quad \text{and} \quad x_2(t) = t^3$$

are two solutions of $t^2 x'' - 4tx' + 6x = 0$ and that both satisfy the initial conditions $x(0) = x'(0) = 0$. Why does this not contradict the uniqueness part of Theorem 3.2.1.

3.3 General Solutions of High-Order Homogeneous Linear Equations

In this section, we consider the homogeneous linear differential equation

$$x^{(n)}(t) + p_1(t)x^{(n-1)}(t) + \dots + p_{n-1}(t)x'(t) + p_n(t)x(t) = 0 \quad (3.3.1)$$

and study the structure of a general solution.

Substituting

$$x_1(t) = x, x_2(t) = x', \dots, x_n(t) = x^{(n-1)}$$

in (3.3.1) gives the corresponding equivalent system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t) \quad (3.3.2)$$

where

$$A(t) = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ -p_n & -p_{n-1} & -p_{n-2} & \cdots & -p_2 & -p_1 \end{bmatrix} \quad \text{and} \quad \mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}.$$

The word “equivalence” means that if $x = \varphi(t)$, $t \in I$ is a solution of (3.3.1), then $\mathbf{x}(t) = [\varphi(t), \varphi'(t), \dots, \varphi^{(n-1)}(t)]^T$ is a solution of (3.3.2). Conversely, if $\mathbf{x}(t) = [\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)]^T$ is a solution of (3.3.2), then $x = \varphi_1(t)$ is a solution of (3.3.1).

From the given results of (3.3.2) in Chapter 2, we have the following conclusions.

Theorem 3.3.1 (Principle of Superposition) Let $x_1, x_2, \dots, x_n$ be n solutions of (3.3.1) on the interval I . Then any linear combination

$$x = c_1 x_1 + c_2 x_2 + \cdots + c_n x_n \quad (3.3.3)$$

is also a solution of the equation (3.3.1) on I , where $c_1, c_2, \dots, c_n$ are arbitrary constants.

Definition 3.3.1 The n functions $x_1, x_2, \dots, x_n$ are said to be linearly dependent on the interval I provided that there exist constants $c_1, c_2, \dots, c_n$ not all zero such that

$$c_1 x_1 + c_2 x_2 + \cdots + c_n x_n = 0$$

on I . Otherwise, we say that these functions are linearly independent.

Remark 3.3.1 The functions $x_1, x_2, \dots, x_n$ are linearly dependent if and only if at least one of them is a linear combination of the others.

Remark 3.3.2 The n functions $x_1, x_2, \dots, x_n$ are linearly independent on the interval I provided that the identity

$$c_1 x_1 + c_2 x_2 + \cdots + c_n x_n = 0$$

holds on I only in the trivial case

$$c_1 = c_2 = \cdots = c_n = 0.$$

That is, no one of them is a linear combination of the others.

Remark 3.3.3 The two functions x_1 and x_2 are linearly dependent if one is a constant multiple of the other. Two functions defined on I are said to be linearly independent provided that neither is a constant multiple of the other.

Theorem 3.3.2 The n solutions $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ of (3.3.1) are linearly dependent (independent) on the interval I if and only if the n solutions

$$\begin{bmatrix} \varphi_1(t) \\ \varphi_1'(t) \\ \vdots \\ \varphi_1^{(n-1)}(t) \end{bmatrix}, \begin{bmatrix} \varphi_2(t) \\ \varphi_2'(t) \\ \vdots \\ \varphi_2^{(n-1)}(t) \end{bmatrix}, \dots, \begin{bmatrix} \varphi_n(t) \\ \varphi_n'(t) \\ \vdots \\ \varphi_n^{(n-1)}(t) \end{bmatrix}$$

of the equivalent system (3.3.2) are linearly dependent (independent) on I .

Theorem 3.3.3 The n th-order homogeneous linear equation (3.3.1) has n linearly independent solutions $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ on the interval I . Then a general solution is given by

$$x(t) = c_1 \varphi_1(t) + c_2 \varphi_2(t) + \cdots + c_n \varphi_n(t).$$

Definition 3.3.2 Suppose that the n functions $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ are each $(n-1)$ times differentiable. Then we call the $n \times n$ determinant

$$W(t) = \begin{vmatrix} \varphi_1 & \varphi_2 & \cdots & \varphi_n \\ \varphi'_1 & \varphi'_2 & \cdots & \varphi'_n \\ \vdots & \vdots & & \vdots \\ \varphi_1^{(n-1)} & \varphi_2^{(n-1)} & \cdots & \varphi_n^{(n-1)} \end{vmatrix}$$

the Wronskian of n given functions $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$.

Theorem 3.3.4 Let $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ be n solutions of the homogeneous linear equation (3.3.1). Then the following hold:

(1) $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ are linearly dependent if and only if

$$W(t) \equiv 0 \quad \text{on } I.$$

(2) $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ are linearly independent if and only if $W(t) \neq 0$ at each point of I .

Theorem 3.3.5 (Liouville's Formula) Suppose that $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ are n linearly independent solutions of homogeneous linear equation (3.3.1). Then the Wronskian is given by

$$W(t) = W(t_0) \exp \left\{ - \int_{t_0}^t p_1(s) ds \right\}, \quad t \in I,$$

where $t_0 \in I$.

By Liouville's Formula, there are just two possibilities: either $W(t) = 0$ everywhere on I or $W(t) \neq 0$ everywhere on I .

Example 3.3.1 Prove directly that the functions

$$x_1(t) \equiv 1, x_2(t) = t, x_3(t) = t^2$$

are linearly independent on the whole real line.

Solution Assume that

$$c_1 + c_2 t + c_3 t^2 = 0.$$

Differentiating this equation twice gives

$$c_2 + 2c_3 t = 0 \quad \text{and} \quad 2c_3 = 0.$$

Combining the above three equations yields

$$\begin{cases} c_1 + c_2 t + c_3 t^2 = 0, \\ c_2 + 2c_3 t = 0, \\ 2c_3 = 0. \end{cases}$$

We easily obtain $c_1 = c_2 = c_3 = 0$.

Example 3.3.2 Show that the n functions $e^{\lambda_1 t}, e^{\lambda_2 t}, \dots, e^{\lambda_n t}$ ($\lambda_i \neq \lambda_j$ if $i \neq j$) are linearly independent on the whole real line.

Proof The Wronskian is given by

$$W(t) = \begin{vmatrix} e^{\lambda_1 t} & e^{\lambda_2 t} & \cdots & e^{\lambda_n t} \\ \lambda_1 e^{\lambda_1 t} & \lambda_2 e^{\lambda_2 t} & \cdots & \lambda_n e^{\lambda_n t} \\ \vdots & \vdots & & \vdots \\ \lambda_1^{n-1} e^{\lambda_1 t} & \lambda_2^{n-1} e^{\lambda_2 t} & \cdots & \lambda_n^{n-1} e^{\lambda_n t} \end{vmatrix}$$

$$\begin{aligned}
 &= e^{(\lambda_1 + \dots + \lambda_n)t} \begin{vmatrix} 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \cdots & \lambda_n \\ \vdots & \vdots & & \vdots \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \cdots & \lambda_n^{n-1} \end{vmatrix} \\
 &= e^{(\lambda_1 + \dots + \lambda_n)t} \prod_{1 \leq j < i \leq n} (\lambda_i - \lambda_j) \neq 0.
 \end{aligned}$$

So we obtain the desired conclusion.

Example 3.3.3 Show first that the three solutions

$$x_1(t) = t, x_2(t) = t \ln t \quad \text{and} \quad x_3(t) = t^2$$

of the third-order linear equation

$$t^3 x''' - t^2 x'' + 2tx' - 2x = 0$$

are linearly independent on the open interval $t > 0$. Then find a particular solution of this equation satisfying the initial conditions

$$x(1) = 3, x'(1) = 2, x''(1) = 1.$$

Solution Notice that $t > 0$. Dividing each term by t^3 gives

$$x''' - \frac{1}{t}x'' + \frac{2}{t^2}x' - \frac{2}{t^3}x = 0,$$

which is a homogeneous linear equation of the standard form. We obtain

$$W(t) = \begin{vmatrix} t & t \ln t & t^2 \\ 1 & 1 + \ln t & 2t \\ 0 & \frac{1}{t} & 2 \end{vmatrix} = t.$$

Since $W(t) \neq 0$ for $t > 0$, so $x_1(t), x_2(t)$ and $x_3(t)$ are linearly independent on the interval $t > 0$. Thus a general solution of this equation is

$$x(t) = c_1 t + c_2 t \ln t + c_3 t^2.$$

Substituting the initial conditions yields the simultaneous equations

$$\begin{cases} c_1 + c_3 = 3, \\ c_1 + c_2 + 2c_3 = 2, \\ c_2 + 2c_3 = 1. \end{cases}$$

We solve to find $c_1 = 1, c_2 = -3$ and $c_3 = 2$. Thus the particular solution is

$$x(t) = t - 3t \ln t + 2t^2.$$

Example 3.3.4 Suppose that $\varphi_1(t)$ is a nontrivial solution of second-order linear equation

$$x'' + p(t)x' + q(t)x = 0.$$

Find another solution $\varphi_2(t)$ such that $\varphi_1(t)$ and $\varphi_2(t)$ are linearly independent.

Solution Let $\varphi_2(t)$ be another solution which is different with the given solution $\varphi_1(t)$. By Liouville's Formula, we have

$$W'(t) = \begin{vmatrix} \varphi_1 & \varphi_2 \\ \varphi_1' & \varphi_2' \end{vmatrix} = c e^{-\int p(t) dt},$$

that is

$$\varphi_1 \varphi_2' - \varphi_1' \varphi_2 = c e^{-\int p(t) dt}.$$

Multiplying both sides by $\frac{1}{\varphi_1^2}$ gives

$$\frac{d}{dt} \left(\frac{\varphi_2}{\varphi_1} \right) = \frac{c}{\varphi_1^2} e^{-\int p(t) dt}.$$

Integrating both sides of the last equation yields

$$\varphi_2 = \varphi_1 \left[\int \frac{c e^{-\int p(t) dt}}{\varphi_1^2} dt + c_0 \right].$$

Thus, we get another particular solution

$$\varphi_2 = \varphi_1 \int \frac{1}{\varphi_1^2} e^{-\int p(t) dt} dt, (c = 1, c_0 = 0).$$

Hence a general solution of this equation is given by

$$x(t) = c_1 \varphi_1(t) + c_2 \varphi_1(t) \int \frac{1}{\varphi_1^2} e^{-\int p(t) dt} dt.$$

Exercise 3.3

Determine whether the pairs of functions in Problems 1 through 4 are linear independent or linearly dependent on the real line.

1. $t^3, t^2 \mid t \mid$.
2. $\sin^2 t, 1 - \cos 2t$.
3. $e^t \sin t, e^t \cos t$.
4. $t e^t, t^2 e^t$.

In Problems 5 through 8, use the Wronskian to prove that the given functions are linearly independent on the indicated interval.

5. e^t, e^{2t}, e^{3t} ; the real line.
6. $e^x, x^{-2}, x^{-2} \ln x$; $x > 0$.
7. $e^x, \sin x, \cos x$; the real line.
8. $x, \cos(\ln x), \sin(\ln x)$; $x > 0$.

In Problems 9 through 12, a third-order homogeneous linear equation and three linearly independent solutions are given. Find a particular solution satisfying the given initial conditions.

9. $x''' - 3x'' + 3x' - x = 0$; $x(0) = 2, x'(0) = 0, x''(0) = 0$; $x_1 = e^t, x_2 = t e^t, x_3 = t^2 e^t$.
10. $t^3 x''' - 3t^2 x'' + 6t x' - 6x = 0$; $x(1) = 6, x'(1) = 14, x''(1) = 22$; $x_1 = t, x_2 = t^2, x_3 = t^3$.
11. $x''' + 9x' = 0$; $x(0) = 3, x'(0) = -1, x''(0) = 2$; $x_1 = 1, x_2 = \cos 3t, x_3 = \sin 3t$.
12. $x''' + 2x'' - x' - 2x = 0$; $x(0) = 1, x'(0) = 2, x''(0) = 0$; $x_1 = e^t, x_2 = e^{-t}, x_3 = e^{-2t}$.

In Problems 13 through 20, a differential equation and one solution $x_1(t)$ are given. Use the method as in Example 3.3.4 to find a second linearly independent solution $x_2(t)$.

$$13. t^2 x'' + tx' - 9x = 0 \quad (t > 0); x_1(t) = t^3.$$

$$14. t^2 x'' - t(t+2)x' + (t+2)x = 0 \quad (t > 0); x_1(t) = t.$$

$$15. (1-t^2)x'' + 2tx' - 2x = 0 \quad (-1 < t < 1); x_1(t) = t.$$

$$16. (t-1)x'' - tx' + x = 0; x_1(t) = t.$$

$$17. (1 - \ln t)x'' + \frac{1}{t}x' - \frac{1}{t^2}x = 0; x_1(t) = \ln t.$$

$$18. (t^2 - t)x'' + (3t - 1)x' + x = 0 \quad (x \neq 0, 1); x_1(t) = \frac{1}{t-1}.$$

$$19. tx'' - x' - 4t^3x = 0 \quad (x \neq 0); x_1(t) = e^{t^2}.$$

$$20. y'' + \lambda(xy' + y) = 0; y_1(x) = \exp\left(-\frac{\lambda x^2}{2}\right).$$

$$21. \text{ Let } x_1 = t \text{ and } x_2 = t^2 \text{ be two particular solutions of the linear equation} \\ t^3 x''' - 3t^2 x'' + 6tx' - 6x = 0.$$

Find a general solution of this equation.

$$22. \text{ Let } \varphi_1(t) \text{ be a nontrivial solution of homogeneous } n\text{th-order linear equation}$$

$$x^{(n)} + p_1(t)x^{(n-1)} + \cdots + p_{n-1}(t)x' + p_n(t)x = 0.$$

Then use the substitution $x = \varphi_1(t)y$ to transform this equation into $(n-1)$ th-order homogeneous linear equation.

23. Verify that $x_1 = t$ and $x_2 = t^2$ are linearly independent solutions on the entire real line of the equation

$$t^2 x'' - 2tx' + 2x = 0,$$

but that $W(t)$ vanishes at $t = 0$. Why do these observations not contradict part (2) of Theorem 3.3.4.

$$24. \text{ Prove that any two solutions } \varphi_1(t) \text{ and } \varphi_2(t) \text{ of the differential equation}$$

$$x'' + p(t)x' + q(t)x = 0$$

satisfying the same initial condition $x(0) = 0$ are linearly dependent, where $p(t)$ and $q(t)$ are continuous on the interval $0 \leq t \leq 1$.

3.4 Nonhomogeneous High-Order Linear Differential Equation

We now consider the nonhomogeneous n th-order linear differential equation

$$Lx = x^{(n)} + p_1(t)x^{(n-1)} + \cdots + p_{n-1}(t)x' + p_n(t)x = f(t) \quad (3.4.1)$$

with associated homogeneous equation

$$Lx = x^{(n)} + p_1(t)x^{(n-1)} + \cdots + p_{n-1}(t)x' + p_n(t)x = 0, \quad (3.4.2)$$

where L is an operator defined in the first section of this chapter.

By the equations (3.1.11) and (3.1.12), we have

$$L(c_1 x_1 + c_2 x_2) = c_1 L(x_1) + c_2 L(x_2), c_1, c_2 \text{ are constants,}$$

which shows that the operator L is linear.

Theorem 3.4.1 Suppose that φ_p is a particular solution of the nonhomogeneous

equation (3.4.1) on an open interval I . Let $\varphi_1, \varphi_2, \dots, \varphi_n$ be linearly independent solutions of the associated homogeneous equation (3.4.2). If $x(t)$ is any solution whatsoever of the equation (3.4.1) on I , then there exist n constants $c_1, c_2, \dots, c_n$ such that

$$x(t) = c_1\varphi_1(t) + c_2\varphi_2(t) + \dots + c_n\varphi_n(t) + \varphi_p(t).$$

Proof The conditions imply that

$$L(x - \varphi_p) = Lx - L\varphi_p = f - f = 0.$$

This shows that $\varphi_c = x - \varphi_p$ is a solution of the associated homogeneous equation (3.4.2). It follows from Theorem 3.3.3 that

$$\varphi_c = x - \varphi_p = c_1\varphi_1(t) + c_2\varphi_2(t) + \dots + c_n\varphi_n(t), \quad (3.4.3)$$

where $c_1, c_2, \dots, c_n$ are n numbers. Thus we rewrite (3.4.3) as

$$x = \varphi_c + \varphi_p.$$

Remark 3.4.1 We call φ_c a complementary function of the nonhomogeneous equation. Theorem 3.4.1 shows that a general solution of the nonhomogeneous equation (3.4.1) is a sum of its complementary solution φ_c and a single particular solution φ_p of the equation (3.4.1).

Next we will find a particular solution of the nonhomogeneous equation (3.4.1) by means of the method of variation of parameters.

Let

$$\varphi_p(t) = c_1(t)\varphi_1(t) + c_2(t)\varphi_2(t) + \dots + c_n(t)\varphi_n(t),$$

where $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ are n linearly independent solutions of (3.4.2) and $c_1(t), c_2(t), \dots, c_n(t)$ are functions that are to be determined. We refer to the result of the corresponding equivalent system with (3.4.1)

$$\frac{dx}{dt} = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{f}(t), \quad (3.4.4)$$

where

$$\mathbf{A}(t) = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -p_n & -p_{n-1} & -p_{n-2} & \dots & -p_2 & -p_1 \end{bmatrix}, \mathbf{f}(t) = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ f(t) \end{bmatrix}.$$

There in (2.5.6) it was shown that

$$\varphi_p(t) = \Phi(t) \int_{t_0}^t \Phi^{-1}(s) \mathbf{f}(s) ds \quad (3.4.5)$$

is a solution of the nonhomogeneous differential system (3.4.4), where

$$\varphi_p(t) = \begin{bmatrix} \varphi_p(t) \\ \varphi'_p(t) \\ \vdots \\ \varphi_p^{(n-1)}(t) \end{bmatrix}, \Phi(t) = \begin{bmatrix} \varphi_1 & \varphi_2 & \dots & \varphi_n \\ \varphi'_1 & \varphi'_2 & \dots & \varphi'_n \\ \vdots & \vdots & & \vdots \\ \varphi_1^{(n-1)} & \varphi_2^{(n-1)} & \dots & \varphi_n^{(n-1)} \end{bmatrix}.$$

Indeed, we only find the first component of the vector $\varphi_p(t)$ which is a desired particular

solution of (3.4.1). Due to the special form of $f(s)$, we easily get the expression $\Phi^{-1}(t)f(t) = g(t)$, that is,

$$\Phi(t)g(t) = f(t), \quad (3.4.6)$$

which shows that $g(t)$ is a solution of the linear system of equations (3.4.6). Applying Cramer's rule, we obtain the components of $g(t)$ in the form

$$g_i(t) = \frac{V_i}{W} \quad (3.4.7)$$

with $W = \det(\Phi(t))$ and

$$V_i = \det \begin{pmatrix} \varphi_1 & \varphi_2 & \cdots & \varphi_{i-1} & 0 & \varphi_{i+1} & \cdots & \varphi_n \\ \varphi'_1 & \varphi'_2 & \cdots & \varphi'_{i-1} & 0 & \varphi'_{i+1} & \cdots & \varphi'_n \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ \varphi_1^{(n-1)} & \varphi_2^{(n-1)} & \cdots & \varphi_{i-1}^{(n-1)} & f(t) & \varphi_{i+1}^{(n-1)} & \cdots & \varphi_n^{(n-1)} \end{pmatrix}.$$

Expanding V_i in terms of cofactors of the i th column gives

$$V_i(t) = (-1)^{n+i} f(t) W_i(t) \quad (3.4.8)$$

where

$$W_i(t) = \begin{vmatrix} \varphi_1 & \cdots & \varphi_{i-1} & \varphi_{i+1} & \cdots & \varphi_n \\ \varphi'_1 & \cdots & \varphi'_{i-1} & \varphi'_{i+1} & \cdots & \varphi'_n \\ \vdots & & \vdots & \vdots & & \vdots \\ \varphi_1^{(n-2)} & \cdots & \varphi_{i-1}^{(n-2)} & \varphi_{i+1}^{(n-2)} & \cdots & \varphi_n^{(n-2)} \end{vmatrix}$$

is the Wronskian determinant of order $(n-1)$ of the functions

$$\varphi_1, \dots, \varphi_{i-1}, \varphi_{i+1}, \dots, \varphi_n.$$

Substituting (3.4.7) and (3.4.8) back in (3.4.5) and taking the first component of $\Phi_p(t)$ yield

$$\varphi_p(t) = \sum_{i=1}^n \varphi_i(t) (-1)^{n+i} \int_{t_0}^t \frac{f(s)}{W(s)} W_i(s) ds. \quad (3.4.9)$$

Theorem 3.4.2 If $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$ are linearly independent solutions of (3.4.2), then $\varphi_p(t)$ in the form (3.4.9) is a particular solution of the nonhomogeneous equation (3.4.1).

Remark 3.4.2 Especially, when $n=2$, a particular solution of the nonhomogeneous equation

$$x'' + p_1(t)x' + p_2(t)x = f(t)$$

is given by

$$\begin{aligned} \varphi_p(t) &= -\varphi_1(t) \int_{t_0}^t \frac{f(s)\varphi_2(s)}{W(s)} ds + \varphi_2(t) \int_{t_0}^t \frac{f(s)\varphi_1(s)}{W(s)} ds \\ &= \int_{t_0}^t \frac{\varphi_2(t)\varphi_1(s) - \varphi_1(t)\varphi_2(s)}{W(s)} f(s) ds, \end{aligned} \quad (3.4.10)$$

where $\varphi_1(t), \varphi_2(t)$ is a fundamental system of the associated homogeneous equation

$$x'' + p_1(t)x' + p_2(t)x = 0.$$

Example 3.4.1 Suppose that $\sin t$ and $\cos t$ are two given solutions of

$$x'' + x = 0.$$

Solve a general solution of the following equation

$$x'' + x = \frac{1}{\cos t}. \quad (3.4.11)$$

Solution Since $\sin t$ and $\cos t$ are linearly independent, a general solution of $x'' + x = 0$ reads

$$x = c_1 \cos t + c_2 \sin t.$$

Applying the method of variation of parameters, we find a particular solution of the nonhomogeneous equation. Let

$$\varphi_p(t) = c_1(t) \cos t + c_2(t) \sin t$$

be a solution of (3.4.11). Then we know that $c'_1(t)$ and $c'_2(t)$ satisfy the following linear system

$$\begin{cases} c'_1(t) \cos t + c'_2(t) \sin t = 0, \\ -c'_1(t) \sin t + c'_2(t) \cos t = \frac{1}{\cos t}. \end{cases}$$

Solving this system yields

$$c'_1(t) = -\tan t \quad \text{and} \quad c'_2(t) = 1.$$

Further we take

$$c_1(t) = \ln |\cos t|, c_2(t) = t.$$

So a general solution of (3.4.11) reads

$$x(t) = c_1 \cos t + c_2 \sin t + \cos t \ln |\cos t| + t \sin t.$$

Example 3.4.2 Let $\varphi(t)$ be a nontrivial solution of the homogeneous equation

$$x'' + p_1(t)x' + p_2(t)x = 0.$$

Then find a general solution of

$$x'' + p_1(t)x' + p_2(t)x = f(t).$$

Solution Example 3.3.4 shows that we can solve another nontrivial solution $\psi(t)$ of the associated homogeneous equation and that $\varphi(t)$, $\psi(t)$ are linearly independent. By Remark 3.4.2, a particular solution $\varphi_p(t)$ of the nonhomogeneous equation is given by the form (3.4.10). Thus a general solution of the nonhomogeneous equation reads

$$x(t) = c_1 \varphi(t) + c_2 \psi(t) + \varphi_p(t).$$

Exercise 3.4

Find a general solution of the following nonhomogeneous differential equations.

1. $x'' + 4x = \sin 2t.$

2. $x'' + 4x' + 3x = e^{-3t}.$

3. $x'' + x = \tan t.$

4. Suppose that $x_1(t)$ and $x_2(t)$ are two functions such that

$$L[x_1(t)] = f_1(t) \quad \text{and} \quad L[x_2(t)] = f_2(t).$$

Show that their sum $x = x_1(t) + x_2(t)$ satisfies the nonhomogeneous equation

$$L[x(t)] = f_1(t) + f_2(t).$$

5. Let the functions

$$3e^t + te^t, 7e^t + te^t, 5e^t + te^t + e^{-t^3}$$

be three solutions of the second-order linear equation

$$x''(t) + p(t)x' + q(t)x = f(t),$$

where $p(t)$, $q(t)$ and $f(t)$ are continuous on the real line. Find a general solution and a particular solution satisfying the initial condition $x(0) = 0, x'(0) = 1$ of this equation.

6. Verify that $x_1(t) = t$ and $x_2(t) = \frac{1}{t}$ are two solutions of

$$t^3 x'' + t^2 x' - tx = 0.$$

Use this information and the method of variation of parameters to find the general solution of

$$t^3 x'' + t^2 x' - tx = \frac{t}{1+t}.$$

7. Prove that for any solution $\varphi(t)$ of the differential equation

$$x'' + 3x' + 2x = f(t),$$

we have

$$\lim_{t \rightarrow +\infty} \varphi(t) = 0,$$

where $f(t)$ is continuous on the interval $[a, +\infty)$ and $\lim_{t \rightarrow +\infty} f(t) = 0$.

3.5 High-Order Homogeneous Linear Equations with Constant Coefficients.

The solution of a linear differential equation with variable coefficients usually requires numerical methods or infinite series methods. In this section, we shall show how to find explicitly a fundamental system of a given n th-order homogeneous linear equation with constant coefficients.

In general, such equation may be written in the form

$$Lx = x^{(n)} + a_1 x^{(n-1)} + \cdots + a_{n-1} x' + a_n x = 0, \quad (3.5.1)$$

where a_i ($i = 1, 2, \dots, n$) are all real constants.

Indeed, to find the solution of the equation (3.5.1), we first begin with the observation that the equation

$$x' + ax = 0 \quad (a \text{ is a constant})$$

for which all solutions are multiples of e^{-at} . Thus, for (3.5.1), some form of exponential function would be a reasonable choice. Notice that any derivative of $e^{\lambda t}$ is a constant multiple of $e^{\lambda t}$, that is,

$$\frac{d^k}{dt^k} (e^{\lambda t}) = \lambda^k e^{\lambda t}. \quad (3.5.2)$$

We substitute $x = e^{\lambda t}$ in (3.5.1) and obtain

$$\begin{aligned} L(e^{\lambda t}) &= \lambda^n e^{\lambda t} + a_1 \lambda^{n-1} e^{\lambda t} + \cdots + a_{n-1} \lambda e^{\lambda t} + a_n e^{\lambda t} \\ &= (\lambda^n + a_1 \lambda^{n-1} + \cdots + a_{n-1} \lambda + a_n) e^{\lambda t} \\ &= P(\lambda) e^{\lambda t} = 0 \end{aligned}$$

which gives

$$P(\lambda) = \lambda^n + a_1 \lambda^{n-1} + \cdots + a_{n-1} \lambda + a_n = 0. \quad (3.5.3)$$

We see that $x = e^{\lambda t}$ will be the solution of the equation (3.5.1) precisely if λ is a root of the algebraic equation (3.5.3).

The equation (3.5.3) is called the characteristic equation of the differential equation in (3.5.1). The problem is reduced to the root of this purely algebraic equation. For the roots of (3.5.3), we have the following three cases.

Case 1 Distinct Real Roots

The characteristic equation (3.5.3) has n distinct real roots $\lambda_1, \lambda_2, \dots, \lambda_n$. Then the functions

$$e^{\lambda_1 t}, e^{\lambda_2 t}, \dots, e^{\lambda_n t}$$

are all solutions of the differential equation (3.5.1) and these n solutions are linearly independent on the real line.

Theorem 3.5.1 Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be n real distinct roots of the characteristic equation (3.5.3). Then

$$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \cdots + c_n e^{\lambda_n t}$$

is a general solution of the equation (3.5.1).

Example 3.5.1 Find a general solution of the equation

$$x'' - 9x = 0.$$

Solution The characteristic equation is $\lambda^2 - 9 = 0$ whose roots are $\lambda_1 = 3, \lambda_2 = -3$. So by Theorem 3.5.1, we get a general solution is

$$x(t) = c_1 e^{3t} + c_2 e^{-3t}.$$

Case 2 Repeated Real Roots

If λ is a repeated real root of multiplicity k , then the part of a general solution of the differential equation in (3.5.1) corresponding to λ is of the form

$$(c_1 + c_2 t + \cdots + c_k t^{k-1}) e^{\lambda t}. \quad (3.5.4)$$

Indeed, it suffices to prove that

$$L(t^m e^{\lambda t}) \equiv 0, (m = 0, 1, \dots, k-1). \quad (3.5.5)$$

First, since λ is a k -fold root of the characteristic equation, we have

$$P(\lambda) = P'(\lambda) = \cdots = P^{(k-1)}(\lambda) = 0, P^{(k)}(\lambda) \neq 0. \quad (3.5.6)$$

Secondly, let $u = e^{\lambda t}$, then $\frac{\partial^m u}{\partial \lambda^m} = t^m e^{\lambda t}$, thus

$$\begin{aligned} L(t^m e^{\lambda t}) &= L\left(\frac{\partial^m u}{\partial \lambda^m}\right) = \frac{\partial^m}{\partial \lambda^m} L(u) \\ &= \frac{\partial^m}{\partial \lambda^m} L(e^{\lambda t}) \end{aligned}$$

$$= \frac{\partial^m}{\partial \lambda^m} (P(\lambda) e^{\lambda t}). \quad (3.5.7)$$

Finally, let $f(\lambda) = P(\lambda)$, $g(\lambda) = e^{\lambda t}$. Applying Leibniz's formula

$$(f \cdot g)^m = \sum_{\mu=0}^m C_m^\mu f^{(\mu)} g^{(m-\mu)},$$

we have

$$\begin{aligned} L(t^m e^{\lambda t}) &= \frac{\partial^m}{\partial \lambda^m} (P(\lambda) e^{\lambda t}) \\ &= \sum_{\mu=0}^m C_m^\mu P^{(\mu)}(\lambda) t^{m-\mu} e^{\lambda t}. \end{aligned} \quad (3.5.8)$$

Together with $P^{(\mu)}(\lambda) = 0, \mu = 0, 1, 2, \dots, k-1$, one obtains

$$L(t^m e^{\lambda t}) = 0, m = 0, 1, 2, \dots, k-1,$$

which shows that

$$e^{\lambda t}, t e^{\lambda t}, \dots, t^{k-1} e^{\lambda t}$$

are k solutions of the differential equation (3.5.1) corresponding to λ . Further, we easily prove that k solutions $e^{\lambda t}, t e^{\lambda t}, \dots, t^{k-1} e^{\lambda t}$ are linearly independent on the real line. Hence, a root of multiplicity k corresponds to k linearly independent solutions of (3.5.1).

Theorem 3.5.2 Suppose that

$$P(\lambda) = (\lambda - \lambda_1)^{k_1} (\lambda - \lambda_2)^{k_2} \cdots (\lambda - \lambda_m)^{k_m} = 0,$$

that is, $\lambda_i (i = 1, 2, \dots, m)$ is a repeated real root of multiplicity k_i , $k_1 + k_2 + \cdots + k_m = n$.

Then the following functions

$$\begin{aligned} &e^{\lambda_1 t}, t e^{\lambda_1 t}, \dots, t^{k_1-1} e^{\lambda_1 t}; \\ &e^{\lambda_2 t}, t e^{\lambda_2 t}, \dots, t^{k_2-1} e^{\lambda_2 t}; \\ &\dots \\ &e^{\lambda_m t}, t e^{\lambda_m t}, \dots, t^{k_m-1} e^{\lambda_m t}; \end{aligned}$$

are n linearly independent solutions of the equation (3.5.1).

Example 3.5.2 Solve the equation

$$x'' + 6x' + 9x = 0.$$

Solution The characteristic equation

$$\lambda^2 + 6\lambda + 9 = (\lambda + 3)^2 = 0$$

which shows that $\lambda = -3$ is a repeated real root of multiplicity 2. Then a general solution is

$$x(t) = c_1 e^{-3t} + c_2 t e^{-3t}.$$

Example 3.5.3 Solve the equation

$$9x^{(5)} - 6x^{(4)} + x^{(3)} = 0.$$

Solution The characteristic equation is

$$9\lambda^5 - 6\lambda^4 + \lambda^3 = \lambda^3(3\lambda - 1)^2 = 0.$$

It has the triple root $\lambda = 0$ and the double root $\lambda = \frac{1}{3}$. The part of a general solution corresponding to $\lambda = 0$ is of the form

$$c_1 e^{0t} + c_2 t e^{0t} + c_3 t^2 e^{0t} = c_1 + c_2 t + c_3 t^2,$$

while the double root $\lambda = \frac{1}{3}$ contributes

$$c_4 e^{\frac{t}{3}} + c_5 t e^{\frac{t}{3}} \quad (3.5.9)$$

to the solution. Thus a general solution of the original differential equation is

$$x(t) = c_1 + c_2 t + c_3 t^2 + c_4 e^{\frac{t}{3}} + c_5 t e^{\frac{t}{3}}.$$

Case 3 Complex Roots

Because the characteristic equation is real, any complex roots appear in complex conjugate pairs $a \pm bi$ where a, b are real and $i = \sqrt{-1}$. The complex roots lead to complex valued solutions of the differential equation. The particular complex-valued functions of interest here are of the form $x(t) = e^{\lambda t}$, where $\lambda = a \pm bi$. By Euler's formula, we obtain

$$e^{(a \pm bi)t} = e^{at} (\cos bt \pm i \sin bt). \quad (3.5.10)$$

In fact, the result $e^{it} = \cos t + i \sin t$ is known as Euler's formula. Since for the differential equation (3.5.1) the real part $e^{at} \cos bt$ and the imaginary part $e^{at} \sin bt$ both are solutions, we obtain two real-valued solutions of (3.5.1).

Theorem 3.5.3 Suppose that the characteristic equation (3.5.3) has an unrepeatd pair of complex conjugate roots $a \pm bi$ ($b \neq 0$). Then the corresponding part of a general solution of the equation (3.5.1) has the form

$$e^{at} (c_1 \cos bt + c_2 \sin bt).$$

Remark 3.5.1 Indeed, Theorem 3.5.2 holds for repeated complex roots. If the conjugate pair $a \pm bi$ has multiplicity k , then the corresponding part of the general solution has the form

$$(a_1 + a_2 t + \cdots + a_k t^{k-1}) e^{(a+bi)t} + (b_1 + b_2 t + \cdots + b_k t^{k-1}) e^{(a-bi)t}.$$

Thus $2k$ real-valued solutions can be obtained by splitting the k solutions corresponding to a complex root $a + bi$ into real and imaginary parts

$$t^m e^{at} \cos bt, t^m e^{at} \sin bt \quad (m = 0, 1, \dots, k-1). \quad (3.5.11)$$

Example 3.5.4 Find a general solution of

$$x'' - 4x' + 13x = 0.$$

Solution The characteristic equation is

$$\lambda^2 - 4\lambda + 13 = (\lambda - 2)^2 + 9 = 0$$

with the complex conjugate roots $2 \pm 3i$. Thus a general solution is given by

$$x(t) = e^{2t} (c_1 \cos 3t + c_2 \sin 3t).$$

Example 3.5.5 Find a general solution of

$$x^{(4)} + 8x''' + 42x'' + 104x' + 169x = 0.$$

Solution The characteristic equation

$$\lambda^4 + 8\lambda^3 + 42\lambda^2 + 104\lambda + 169 = [(\lambda + 2)^2 + 9]^2 = 0$$

has the conjugate pair $-2 \pm 3i$ of multiplicity $k = 2$. Thus a general solution is

$$x(t) = e^{-2t} (c_1 \cos 3t + c_2 \sin 3t) + t e^{-2t} (c_3 \cos 3t + c_4 \sin 3t).$$

Exercise 3.5

Find the general solutions of the differential equations in Problems 1 through 10.

1. $x'' - 4x' = 0$.
2. $x'' - 4x = 0$.
3. $x'' + 6x' + 9x = 0$.
4. $x^{(4)} + 3x^{(3)} = 0$.
5. $x^{(4)} - 8x'' + 16x = 0$.
6. $x^{(4)} - 16x = 0$.
7. $x^{(7)} - 2x^{(5)} + x^{(3)} = 0$.
8. $x''' - 5x'' + 8x' - 4x = 0$.
9. $x''' + x'' - x' - x = 0$.
10. $x'' + 8x' + 25x = 0$.

Solve the initial value problem given in Problems 11 through 15.

11. $x'' - 4x' + 3x = 0; x(0) = 7, x'(0) = 11$.
12. $x'' - 6x' + 25x = 0; x(0) = 3, x'(0) = 1$.
13. $x'' + x' = 0; x(0) = 2, x'(0) = 5$.
14. $x'' + 4x' + 4x = 0; x(2) = 4, x'(2) = 0$.
15. $2x''' - 3x'' - 2x' = 0; x(0) = 1, x'(0) = -1, x''(0) = 3$.

In Problems 16 through 18, find a linear homogeneous constant coefficient equation with the given general solution.

16. $x(t) = (c_1 + c_2 t + c_3 t^2 + c_4 t^3)e^{2t}$.
17. $x(t) = c_1 e^{-2t} + c_2 \cos t + c_3 \sin t$.
18. $x(t) = (c_1 + c_2 t + c_3 t^2) \cos 2t + (c_4 + c_5 t + c_6 t^2) \sin 2t$.
19. Prove Theorem 3.5.2.

20. Show that any solution $x(t)$ of the differential equation $x'' + ax' + bx = 0$, one has

$$\lim_{t \rightarrow +\infty} x(t) = 0, \text{ where } a > 0, b > 0.$$

21. Show that if the real parts of all solutions of (3.5.3) are negative, then

$$\lim_{t \rightarrow +\infty} x(t) = 0$$

for every solution of (3.5.1).

22. Show that the substitution $x = e^t$ ($x > 0$) can transform the second-order Euler's equation

$$ax^2 y'' + bxy' + cy = 0$$

to a constant coefficient homogeneous linear equation. Further, show similarly that this same substitution transforms the third-order Euler's equation

$$ax^3 y''' + bx^2 y'' + cxy' + dy = 0$$

into the constant coefficient equation

$$a \frac{d^3 y}{dt^3} + (b - 3a) \frac{d^2 y}{dt^2} + (c - b + 2a) \frac{dy}{dt} + dy = 0,$$

where a, b, c, d are constants.

Find the general solutions (for $x > 0$) of the Euler's equations in Problems 23 through 26.

23. $x^2 y'' + xy' + 9y = 0$.
24. $x^3 y''' + 6x^2 y'' + 4xy' = 0$.

$$25. x^2 y'' + 7xy' + 25y = 0.$$

$$26. x^3 y''' - 3x^2 y'' + xy' = 0.$$

3.6 High-Order Nonhomogeneous Linear Equations with Constant Coefficients

The general n th-order nonhomogeneous linear differential equation with constant coefficients has the form

$$L(x) = x^{(n)} + a_1 x^{(n-1)} + \cdots + a_{n-1} x' + a_n x = f(t). \quad (3.6.1)$$

From Theorem 3.4.1, a general solution of the equation (3.6.1) is given by

$$x(t) = \varphi_c + \varphi_p \quad (3.6.2)$$

where the complementary function $\varphi_c(t)$ is a general solution of the associated homogeneous equation

$$x^{(n)} + a_1 x^{(n-1)} + \cdots + a_{n-1} x' + a_n x = 0, \quad (3.6.3)$$

and $\varphi_p(t)$ is a particular solution of the equation (3.6.1). In this section, our main task is to find $\varphi_p(t)$.

In Section 3.4, we introduce the method of variation of parameters to find $\varphi_p(t)$. This section is devoted to a straightforward way of finding $\varphi_p(t)$, namely, the method of undetermined coefficients provided the given function $f(t)$ in the equation (3.6.1) is sufficiently simple.

Example 3.6.1 Find a particular solution of

$$x'' + 3x' + 4x = t + 1.$$

Solution Here $f(t) = t + 1$ is a polynomial of degree 1, so we assume that

$$\varphi_p(t) = At + B.$$

Together with $\varphi_p' = A$ and $\varphi_p'' = 0$, thus φ_p is a solution of the differential equation provided that

$$0 + 3A + 4(At + B) = t + 1.$$

Equating coefficients of like powers of t on both sides of the last equation yields

$$4A = 1, 3A + 4B = 1$$

which give $A = \frac{1}{4}$ and $B = \frac{1}{16}$. So the particular solution is

$$\varphi_p(t) = \frac{1}{4}t + \frac{1}{16}.$$

Example 3.6.2 Find a particular solution of

$$x'' - 9x = 2e^{2t}.$$

Solution Here $f(t) = 2e^{2t}$. Any derivative of e^{2t} is a constant multiple of e^{2t} , so we try

$$\varphi_p(t) = Ae^{2t}.$$

In connection with $\varphi_p''(t) = 4Ae^{2t}$, so $\varphi_p(t) = Ae^{2t}$ satisfies the given differential equation provided that

$$4Ae^{2t} - 9Ae^{2t} = 2e^{2t},$$

which shows that $-5A = 2$, that is $A = -\frac{2}{5}$. Thus the particular solution is

$$\varphi_p(t) = -\frac{2}{5}e^{2t}.$$

Example 3.6.3 Find a particular solution of

$$x'' - x' - 6x = 2\sin 2t + \cos 2t.$$

Solution Here $f(t) = 2\sin 2t + \cos 2t$. Since any derivative of a linear combination of $\sin 2t$ and $\cos 2t$ has the same form, we try

$$\varphi_p(t) = A\sin 2t + B\cos 2t.$$

Differentiating both sides twice yields

$$\varphi_p'(t) = 2A\cos 2t - 2B\sin 2t$$

$$\varphi_p''(t) = -4A\sin 2t - 4B\cos 2t.$$

Then substitution of φ_p and its derivatives into the given differential equation gives

$$\begin{aligned} & (-4A\sin 2t - 4B\cos 2t) - (2A\cos 2t - 2B\sin 2t) - (6A\sin 2t + 6B\cos 2t) \\ &= (-10A + 2B)\sin 2t + (-2A - 10B)\cos 2t \\ &= 2\sin 2t + \cos 2t. \end{aligned}$$

We equate coefficients of $\cos 2t$ and $\sin 2t$ on both sides and obtain the following equations

$$\begin{cases} -10A + 2B = 2, \\ -2A - 10B = 1, \end{cases}$$

with solution $A = -\frac{11}{52}, B = -\frac{3}{52}$. Thus a particular solution is

$$\varphi_p(t) = -\frac{11}{52}\sin 2t - \frac{3}{52}\cos 2t.$$

Example 3.6.4 Find a particular solution of

$$x'' - 9x = 3e^{3t}.$$

Solution If we try $\varphi_p(t) = Ae^{3t}$, then

$$\varphi_p'' - 9\varphi_p = 9Ae^{3t} - 9Ae^{3t} = 0 \neq 3e^{3t}.$$

Thus no matter how A is chosen, Ae^{3t} cannot satisfy the given equation. Indeed, Ae^{3t} is a solution of the associated homogeneous equation. It is reasonable to try

$$\varphi_p(t) = At e^{3t}.$$

Substitution this into the original differential equation gives

$$(6Ae^{3t} + 9At e^{3t}) - 9At e^{3t} = 3e^{3t}$$

which gives $A = \frac{1}{2}$. Thus a particular solution is

$$\varphi_p(t) = \frac{1}{2}t e^{3t}.$$

Remark 3.6.1 If the nonhomogeneous term $f(t)$ is a polynomial, or an exponential function $e^{\lambda t}$, or $\cos at$, $\sin at$, then we try to find a particular solution by the method of undetermined coefficients. The reason is that all the derivatives of these types of function

$f(t)$ have the same form as $f(t)$ itself. For example, the derivatives of a polynomial are themselves polynomials of lower degree, and any derivative of $e^{\lambda t}$ is a constant multiple of $e^{\lambda t}$.

Indeed, the method of undetermined coefficients applies whenever the nonhomogeneous term $f(t)$ in (3.6.1) is a linear combination of finite products of functions of the following three types:

- (1) A polynomial in t ;
- (2) An exponential function $e^{\lambda t}$;
- (3) $\cos at$ or $\sin at$.

For example,

$$\begin{aligned} f_1(t) &= P_m(t)e^{\lambda t}, \\ f_2(t) &= e^{at} [P_m^{(1)}(t) \cos \beta t + P_m^{(2)}(t) \sin \beta t], \\ f_3(t) &= t^2 e^{5t} + t e^t \cos t. \end{aligned}$$

These functions have the critical property that only finitely many linearly independent functions appear as terms in $f(t)$ and its derivatives of all orders.

Remark 3.6.2 Consider the equation

$$x'' + x = \tan t.$$

Here $f(t) = \tan t$ has infinitely many linearly independent derivatives

$$\sec^2 t, 2\sec^2 t \tan t, 4\sec^2 t \tan^2 t + 2\sec^4 t, \dots$$

Thus, we do not have a finite linear combination to use as a trial solution. That is, in this kind of situation, the method of undetermined coefficients cannot be used to find a particular solution.

Remark 3.6.3 In Example 3.6.4, we cannot find a particular solution by a trial function of the form $\varphi_p(t) = Ae^{3t}$. The reason is the fact that $f(t) = Ae^{3t}$ is a solution of the associated homogeneous equation. But in other examples, this phenomenon does not occur.

In the following, we shall discuss two rules. **Rule 1** tells what to do when we do not have this phenomenon and **Rule 2** tells what to do when we do have it.

Rule 1 Assume that no term appearing either in $f(t)$ or in any of its derivatives satisfies the corresponding homogeneous equation (3.6.3). Then take a linear combination of all linearly independent such terms and their derivatives as a trial solution for $\varphi_p(t)$. Then determine the coefficients by substitution of this trial solution into the nonhomogeneous equation (3.6.1).

Example 3.6.5 Find a particular solution of

$$x'' + 9x = 3t^2.$$

Solution A general solution of the associated homogeneous equation is

$$\varphi_c(t) = c_1 \cos 3t + c_2 \sin 3t.$$

It is easy to check that the function $f(t) = 3t^2$ and its derivatives do not appear in $\varphi_c(t)$. We try $\varphi_p(t) = a_1 + a_2 t + a_3 t^2$. Substitution in the original equation gives

$$\begin{aligned}
 & 2a_3 + 9(a_1 + a_2 t + a_3 t^2) \\
 &= 9a_3 t^2 + 9a_2 t + 9a_1 + 2a_3 \\
 &= 3t^2.
 \end{aligned}$$

Equating coefficients of like power of t yields

$$9a_3 = 3, 9a_2 = 0, 9a_1 + 2a_3 = 0$$

with solution $a_1 = -\frac{2}{27}, a_2 = 0, a_3 = \frac{1}{3}$. Thus a particular solution is

$$\varphi_p(t) = \frac{1}{3}t^2 - \frac{2}{27}.$$

Example 3.6.6 Find a particular solution of

$$x'' + 9x = \sin 2t + t e^t.$$

Solution The general solution of the associated homogeneous equation is

$$\varphi_c(t) = c_1 \cos 3t + c_2 \sin 3t.$$

Since $f_1(t) = \sin 2t$ and its derivatives, $f_2(t) = t e^t$ and its derivatives do not satisfy the homogeneous equation, we take

$$\varphi_p(t) = A \sin 2t + B \cos 2t + (C + Dt) e^t$$

as a trial solution.

Substituting $\varphi_p(t)$ in the original equation and equating coefficient of like terms give

$$A = \frac{1}{5}, B = 0, C = -\frac{1}{50}, D = \frac{1}{10}.$$

Thus a particular solution is given by

$$\varphi_p(t) = \frac{1}{5} \sin 2t + \left(\frac{1}{10}t - \frac{1}{50} \right) e^t.$$

Rule 2 Suppose that some term appearing in $f(t)$ or in any of its derivatives satisfies the associated homogeneous equation (3.6.3).

If $f(t) = P_m(t) e^{\alpha t}$, where $P_m(t)$ is a polynomial in t of degree m , and α is a repeated root of multiplicity k of the characteristic equation, then we take the function

$$\varphi_p(t) = t^k Q_m(t) e^{\alpha t} \quad (3.6.4)$$

as a trial solution of (3.6.1), where

$$Q_m(t) = b_0 t^m + b_1 t^{m-1} + \cdots + b_{m-1} t + b_m, \quad b_i (i = 0, 1, \cdots, m)$$

are undetermined coefficients.

If $f(t) = e^{\alpha t} [P_{n_1}(t) \cos \beta t + P_{n_2}(t) \sin \beta t]$, where $P_{n_1}(t)$ and $P_{n_2}(t)$ are polynomials in t of degree n_1 and n_2 respectively, $m = \max\{n_1, n_2\}$, $\alpha + i\beta$ is a repeated root of multiplicity k of the characteristic equation, then the trial solution takes the form

$$\varphi_p(t) = t^k e^{\alpha t} [Q_m^{(1)}(t) \cos \beta t + Q_m^{(2)}(t) \sin \beta t], \quad (3.6.5)$$

where

$$Q_m^{(1)}(t) = p_0 t^m + p_1 t^{m-1} + \cdots + p_{m-1} t + p_m$$

and

$$Q_m^{(2)}(t) = q_0 t^m + q_1 t^{m-1} + \cdots + q_{m-1} t + q_m,$$

$p_i, q_i (i = 0, 1, \dots, m)$ are $(2m + 2)$ undetermined coefficients. Then determine the coefficients in (3.6.4) or (3.6.5) by substituting the trial solution $\varphi_p(t)$ into the nonhomogeneous equation (3.6.1).

Example 3.6.7 Find a general solution of

$$x'' + x = 2 \sin t.$$

Solution A general solution of the associated homogeneous equation is

$$\varphi_c(t) = c_1 \sin t + c_2 \cos t.$$

Notice that the imaginary unit i is a single root of the characteristic equation. We take the function

$$\varphi_p(t) = t(A \cos t + B \sin t)$$

as a trial solution.

Substitution of $\varphi_p(t)$ in the original equation gives

$$A = -1, B = 0.$$

So a general solution is

$$x(t) = c_1 \cos t + c_2 \sin t - t \cos t.$$

Exercise 3.6

In Problems 1 through 10, find a particular solution x_p of the given equation.

1. $x'' + 16x = 2e^{3t}.$
2. $x'' - x' - 2x = t^2.$
3. $x'' + x' + x = \sin^2 t.$
4. $2x'' + 4x' + 7x = 2t^2.$
5. $x'' + 2x' + 5x = 2e^t \sin t.$
6. $x^{(4)} - 2x'' + x = te^t.$
7. $x'' + x = \sin t + 5 \cos t.$
8. $x'' - 2x' + 10x = t \sin 2t.$
9. $x''' - x = 2e^t + 1.$
10. $x^{(4)} - 5x'' + 4x = e^t + te^{2t}.$

In Problems 11 through 18, set up the appropriate form of a particular solution x_p , but do not determine the values of the coefficients.

11. $x'' - 2x' + 2x = e^t \sin t + t^2.$
12. $x^{(5)} - x''' = 2e^t + t^2 + 4.$
13. $x'' + 4x = t^2 \cos 2t + 3 \sin t + 4.$
14. $x''' - x'' - 12x' = t^2 - 2te^{-3t}.$
15. $x'' + 3x' + 2x = t^2 e^{-t} - te^{2t}.$
16. $x^{(4)} - 2x'' + x = \cos t + e^t \sin t + t^2.$
17. $x^{(4)} + 9x'' = (t^2 + 1) \sin 3t + \cos t.$
18. $x^{(4)} + 5x'' + 4x = 3 \sin t + \cos 2t.$

Solve the initial value problems in Problems 19 through 23.

19. $x'' + 9x = \sin 2t; x(0) = 1, x'(0) = 0.$
20. $x'' + 2x' + 2x = \sin 3t; x(0) = 2, x'(0) = 0.$
21. $x'' + x = \cos t; x(0) = 1, x'(0) = -1.$
22. $x''' + x'' = t + e^{-t}; x(0) = 1, x'(0) = 0, x''(0) = 1.$
23. $x'' + 4x = 2t; x(0) = 1, x'(0) = 2.$

Use trigonometric identities to find general solutions of the equations in Problems 24 through 27.

$$24. x'' + 4x = 2 \cos^3 t.$$

$$25. x'' + x' + x = \cos t \sin 2t.$$

$$26. x'' + 9x = \cos^4 t.$$

$$27. x'' + x = t \sin^3 t.$$

Use the method of variation of parameters to find a particular solution of the given differential equation in Problems 28 through 30.

$$28. x'' + 9x = 3 \sec 3t.$$

$$29. x'' + x = \csc^2 t.$$

$$30. x'' + x = \tan t.$$

3.7 Power Series Methods

In Section 3.5, solving a homogeneous linear differential equation with constant coefficients can be reduced to the algebraic problem of finding the solutions of its characteristic equation. However, we can not apply similar procedure for solving linear differential equations with variable coefficients. In this section, we introduce power series techniques which suffice for many differential equations that appear frequently in applications. The most important applications are to second-order homogeneous linear differential equations of the form

$$x'' + P_1(t)x' + P_2(t)x = 0. \quad (3.7.1)$$

If the functions $P_1(t)$ and $P_2(t)$ are analytic at $t = t_0$, that is, these functions have convergent power series expansions in powers of $(t - t_0)$ on some open interval containing $t = t_0$. Then we have the following the theorem (without proof).

Theorem 3.7.1 For any x_0, x'_0 , the linear equation (3.7.1) has an analytic solution $x = \varphi(t)$ on some open interval containing $t = t_0$ which satisfies the initial condition

$$x(t_0) = x_0, x'(t_0) = x'_0,$$

that is, $\varphi(t)$ can be written as

$$\varphi(t) = \sum_{k=0}^{\infty} C_k (t - t_0)^k, \quad (3.7.2)$$

provided that the coefficient functions $P_1(t)$ and $P_2(t)$ are analytic at $t = t_0$.

Example 3.7.1 Solve Legendre's equation

$$(1 - t^2)x'' - 2tx' + \alpha(\alpha + 1)x = 0 \quad (3.7.3)$$

on some open interval containing $t = 0$, where α is a constant.

Solution It is obvious that the coefficient functions are analytic at $t = 0$. Let

$$\varphi(t) = c_0 + c_1 t + c_2 t^2 + \cdots + c_n t^n + \cdots = \sum_{k=0}^{\infty} c_k t^k$$

be an analytic solution. Then substitutions of

$$\varphi(t) = \sum_{k=0}^{\infty} c_k t^k,$$

$$\varphi'(t) = \sum_{k=1}^{\infty} k c_k t^{k-1},$$

and

$$\varphi''(t) = \sum_{k=2}^{\infty} k(k-1) c_k t^{k-2}$$

in the equation (3.7.3) yield

$$\begin{aligned} & (1-t^2)[2 \cdot 1c_2 + 3 \cdot 2c_3 t + \cdots + k(k-1)c_k t^{k-2} + \cdots] \\ & - 2t(1 \cdot c_1 + 2 \cdot c_2 t + \cdots + k c_k t^{k-1} + \cdots) \\ & + \alpha(\alpha+1)(c_0 + c_1 t + \cdots + c_k t^k + \cdots) \equiv 0, \end{aligned}$$

which leads to the following relations

$$2 \cdot 1c_2 + \alpha(\alpha+1)c_0 = 0,$$

$$3 \cdot 2c_3 - 2 \cdot 1c_1 + \alpha(\alpha+1)c_1 = 0,$$

$$(k+2)(k+1)c_{k+2} - k(k-1)c_k - 2kc_k + \alpha(\alpha+1)c_k = 0, k \geq 2.$$

Thus we have the recurrence relation

$$c_{k+2} = \frac{(k-\alpha)(\alpha+k+1)}{(k+2)(k+1)} c_k, k \geq 0. \quad (3.7.4)$$

Further, one obtains

$$c_{2m} = (-1)^m A_m c_0,$$

$$c_{2m+1} = (-1)^m B_m c_1,$$

where

$$A_m = \frac{\alpha(\alpha-2)(\alpha-4)\cdots(\alpha-2m+2)(\alpha+1)(\alpha+3)\cdots(\alpha+2m+1)}{(2m)!} \quad (3.7.5)$$

and

$$\begin{aligned} B_m &= \frac{(\alpha-1)(\alpha-3)\cdots(\alpha-2m+1)(\alpha+2)(\alpha+4)\cdots(\alpha+2m)}{(2m+1)!} \\ & \quad (m=1,2,3,\dots). \end{aligned} \quad (3.7.6)$$

With this notation, we obtain two linearly independent power series solutions

$$x_1(t) = c_0 \sum_{m=0}^{\infty} (-1)^m A_m t^{2m} = \sum_{m=0}^{\infty} c_{2m} t^{2m}$$

and

$$x_2(t) = c_1 \sum_{m=0}^{\infty} (-1)^m B_m t^{2m+1} = \sum_{m=0}^{\infty} c_{2m+1} t^{2m+1}$$

of Legendre's equation of order α .

Remark 3.7.1 Let $\alpha = n$ be a nonnegative integer. Then either $x_1(t)$ or $x_2(t)$ is a polynomial of degree n which is denoted by $P_n(t)$. It is called the Legendre polynomial of degree n . Indeed, if $\alpha = n$ is even, then we see from (3.7.5) that $c_{2m} = 0$ when $2m > n$. In this case, $x_1(t)$ is a polynomial of degree n and $x_2(t)$ is an infinite series. If $\alpha = n$ is odd, we obtain from (3.7.6) that $c_{2m+1} = 0$ when $(2m+1) > n$. In this case, $x_1(t)$ is an infinite

series and $x_2(t)$ is a polynomial of degree n .

Suppose that $P_1(t), P_2(t)$ in (3.7.1) are not analytic at $t = t_0$, for example,

$$P_1(t) = \frac{a(t)}{t - t_0}, P_2(t) = \frac{b(t)}{t - t_0},$$

where $a(t), b(t)$ are both analytic functions on some open interval containing $t = t_0$. Then it is a reasonable conjecture that the differential equation (3.7.1) might have a solution of the form

$$x(t) = (t - t_0)^\alpha \sum_{k=0}^{\infty} c_k (t - t_0)^k, \quad (3.7.7)$$

where $\alpha, c_k, (k = 0, 1, 2, \dots)$ are all undetermined constants, $c_0 \neq 0$.

An infinite series of the form in (3.7.7) is called a Frobenius series, named for the German mathematician Georg Frobenius (1848—1917). Notice that a Frobenius series is generally not a power series. For instance, with $\alpha = -\frac{1}{3}$ the series in (3.7.7) takes the form

$$x(t) = c_0 (t - t_0)^{-\frac{1}{3}} + c_1 (t - t_0)^{\frac{2}{3}} + c_2 (t - t_0)^{\frac{5}{3}} + \dots,$$

which is not a series in integral powers of $(t - t_0)$. The coefficients in these series can be determined by substituting the series in the differential equation.

Example 3.7.2 Find a solution of Bessel's equation of order n

$$t^2 x'' + tx' + (t^2 - n^2)x = 0, \quad (3.7.8)$$

on some open interval containing $t = 0$, where $n \geq 0$.

Solution Dividing both sides by t^2 yields the differential equation in standard form

$$x'' + \frac{1}{t}x' + \frac{t^2 - n^2}{t^2}x = 0. \quad (3.7.9)$$

Here $P_1(t) = \frac{1}{t}$ and $P_2(t) = 1 - \frac{n^2}{t^2}$ are not analytic at $t = 0$. So we try to find a solution of the form

$$x = c_0 t^\alpha + c_1 t^{\alpha+1} + \dots + c_k t^{\alpha+k} + \dots, c_0 \neq 0. \quad (3.7.10)$$

Substitution of the series (3.7.10) in Bessel's equation (3.7.8) gives

$$t^2 \sum_{k=0}^{\infty} c_k (\alpha + k)(\alpha + k - 1)t^{\alpha+k-2} + t \sum_{k=0}^{\infty} c_k (\alpha + k)t^{\alpha+k-1} + (t^2 - n^2) \sum_{k=0}^{\infty} c_k t^{\alpha+k} \equiv 0,$$

that is

$$\sum_{k=0}^{\infty} [(\alpha + n + k)(\alpha - n + k)c_k + c_{k-2}]t^{\alpha+k} = 0,$$

where $c_{-1} = c_{-2} = 0$.

Equating coefficients of like powers of t yields

$$\begin{cases} c_0[\alpha^2 - n^2] = 0, \\ c_1[(\alpha + 1)^2 - n^2] = 0, \\ c_2[(\alpha + 2)^2 - n^2] + c_0 = 0, \\ c_3[(\alpha + 3)^2 - n^2] + c_1 = 0, \\ \vdots \\ c_k[(\alpha + k)^2 - n^2] + c_{k-2} = 0, \\ \vdots \end{cases} \quad (3.7.11)$$

Since $c_0 \neq 0$, the first equation in (3.7.11) shows that

$$(\alpha - n)(\alpha + n) = 0$$

with the roots $\alpha_1 = n$ and $\alpha_2 = -n$.

If $\alpha = n \geq 0$, from the second equation in (3.7.11), we have

$$c_1 = c_3 = c_5 = \cdots = c_{2k+1} = 0,$$

$$c_2 = -\frac{c_0}{2^2(n+1)},$$

$$c_4 = \frac{c_0}{2^4(n+1)(n+2)1 \cdot 2},$$

$$\vdots$$

$$c_{2k} = \frac{(-1)^k c_0}{2^{2k} \cdot k! (n+1)(n+2)\cdots(n+k)},$$

$$\vdots$$

Thus, when $\alpha = n \geq 0$, we obtain the solution

$$x_1(t) = c_0 \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k+n}}{2^{2k} \cdot k! (n+1)(n+2)\cdots(n+k)}. \quad (3.7.12)$$

If we choose $c_0 = \frac{1}{2^n \Gamma(n+1)}$, where the gamma function $\Gamma(x)$ is defined for $x > 0$ by

$$\Gamma(x) = \int_0^{\infty} e^{-t} t^{x-1} dt,$$

then the formula in (3.7.12) can be simplified in the form

$$x_1(t) = \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{t}{2}\right)^{2k+n}}{k! \Gamma(n+k+1)}, \quad (3.7.13)$$

by the fact that $\Gamma(x+1) = x\Gamma(x)$.

The series in (3.7.13) is called Bessel function of the first kind of order n and denoted by $J_n(t)$.

If $\alpha = -n < 0$ and n is not a positive integer, we apply similar procedure to obtain

$$c_{2k+1} = 0,$$

$$c_{2k} = \frac{(-1)^k c_0}{2^{2k} \cdot k! (-n+1)(-n+2)\cdots(-n+k)}.$$

If we take

$$c_0 = \frac{1}{2^{-n} \Gamma(-n+1)},$$

then one obtains the linearly independent second solution

$$J_{-n}(t) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(-n+k+1)} \left(\frac{t}{2}\right)^{2k-n}, \quad (3.7.14)$$

which is called Bessel function of the first kind of order $-n$. Since $J_n(t), J_{-n}(t)$ are both convergent on the interval $(0, +\infty)$, we know that $J_n(t)$ and $J_{-n}(t)$ are linearly independent solutions of Bessel's equation provided that n is not an integer.

Thus a general solution is

$$x(t) = a_1 J_n(t) + a_2 J_{-n}(t), \quad (3.7.15)$$

where a_1, a_2 are arbitrary constants.

If n is an integer, we can still solve c_{2k} ($k = n+1, n+2, \dots$) from (3.7.11), where c_{2n} is arbitrary. However, we easily verify that $J_{-n}(t)$ and $J_n(t)$ are linearly dependent. Applying other methods, we can find the linearly independent second solution by $Y_n(t)$ which is called the Bessel function of the second kind of integral order $n \geq 0$.

A general solution of Bessel's equation of integral order n is given by

$$x(t) = a_1 J_n(t) + a_2 Y_n(t), \quad (3.7.16)$$

where a_1, a_2 are arbitrary constants.

Here we shall not discuss these contents in detail.

Exercise 3.7

Find general solutions in powers of t of the differential equations in Problems 1 through 4. State the recurrence relation and the guaranteed radius of convergence in each case.

1. $x'' - tx = 0$.
2. $x'' + tx' + x = 0$.
3. $x' + 2x = 0$.
4. $x'' + x = 0$.
5. Show that the following Bessel function identities hold:

$$\frac{d}{dt}[t^n J_n(t)] = t^n J_{n-1}(t)$$

and

$$\frac{d}{dt}[t^{-n} J_n(t)] = -t^{-n} J_{n+1}(t),$$

where n is nonnegative integer, $J_n(t)$ is defined in the form (3.7.13).

6. The Hermite equation of order α is

$$x'' - 2tx' + 2\alpha x = 0.$$

Find the two power series solutions

$$x_1 = 1 + \sum_{m=1}^{\infty} (-1)^m \frac{2^m \alpha(\alpha-2)\cdots(\alpha-2m+2)}{(2m)!} t^{2m}$$

and

$$x_2 = t + \sum_{m=1}^{\infty} (-1)^m \frac{2^m (\alpha - 1)(\alpha - 3) \cdots (\alpha - 2m + 1)}{(2m + 1)!} t^{2m+1}.$$

Show that x_1 is a polynomial if α is an even integer, whereas x_2 is a polynomial if α is an odd integer.

3.8 Laplace Transform Methods

In this section, we introduce Laplace transform method which is a powerful tool for solving the initial value problems. The Laplace transformation $\mathcal{L}$ contains the operation of integration and obtains a new function $\mathcal{L}[f(t)] = F(s)$ of a new independent variable s . The basic thought of this method is that Laplace transform converts a differential equation in the unknown function $f(t)$ into an algebraic equation in $F(s)$, then find a solution of the algebraic equation and further obtain a solution of the initial value problem by inverse transforms.

3.8.1 Laplace Transforms and Inverse Transforms

Definition 3.8.1 If the given function $f(t)$ is defined for all $t \geq 0$, then the **Laplace transform** of f is the function F defined as follows

$$\mathcal{L}[f(t)] = F(s) = \int_0^{\infty} e^{-st} f(t) dt \quad (3.8.1)$$

for all values of s for which the improper integral converges, where $f(t)$ is called **original function**, and $F(s)$ is called **image function**.

Remark 3.8.1 The improper integral is defined as a limit of integrals over bounded intervals, namely,

$$\int_0^{\infty} h(t) dt = \lim_{b \rightarrow \infty} \int_0^b h(t) dt. \quad (3.8.2)$$

If the limit in (3.8.2) exists, then the improper integral converges; otherwise, it diverges.

Definition 3.8.2 Suppose that $F(s) = \mathcal{L}[f(t)]$. Then $f(t)$ is called the inverse Laplace transform of $F(s)$ and denoted by

$$f(t) = \mathcal{L}^{-1}[F(s)]. \quad (3.8.3)$$

Example 3.8.1 Find $\mathcal{L}[f(t)]$, where $f(t) \equiv 1$ for $t \geq 0$.

Solution

$$\begin{aligned} \mathcal{L}[1] &= \int_0^{\infty} e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_0^{\infty} = \lim_{b \rightarrow \infty} \left[-\frac{1}{s} e^{-bs} + \frac{1}{s} \right] \\ &= \frac{1}{s} \quad \text{for } s > 0. \end{aligned}$$

In this computation, we use the common abbreviation

$$[h(t)]_a^{\infty} = \lim_{b \rightarrow \infty} [h(t)]_a^b.$$

Example 3.8.2 With $f(t) = e^{at}$ for $t \geq 0$, find $\mathcal{L}[e^{at}]$.

Solution

$$\begin{aligned}
 F(s) &= \mathcal{L}[e^{at}] = \int_0^{\infty} e^{-st} e^{at} dt \\
 &= \int_0^{\infty} e^{-(s-a)t} dt = \left[-\frac{e^{-(s-a)t}}{s-a} \right]_0^{\infty} \\
 &= \frac{1}{s-a} \quad \text{for } s > a.
 \end{aligned}$$

If $s \leq a$, the improper integral diverges.

Example 3.8.3 If $f(t) = t^n$, for $t \geq 0$, where n is a positive integer, then find $\mathcal{L}[t^n]$.

Solution when $n = 1$, one has

$$\begin{aligned}
 \mathcal{L}[t] &= \int_0^{\infty} t e^{-st} dt = -\frac{1}{s} \int_0^{\infty} t d e^{-st} \\
 &= \left[-\frac{1}{s} t e^{-st} \right]_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} dt \\
 &= \frac{1}{s^2} \quad \text{for } s > 0.
 \end{aligned}$$

when $n > 1$, we obtain

$$\begin{aligned}
 \mathcal{L}[t^n] &= \int_0^{\infty} t^n e^{-st} dt = \left[-\frac{1}{s} t^n e^{-st} \right]_0^{\infty} + \frac{n}{s} \int_0^{\infty} e^{-st} t^{n-1} dt \\
 &= \frac{n}{s} \mathcal{L}[t^{n-1}] = \frac{n}{s} \cdot \frac{n-1}{s} \mathcal{L}[t^{n-2}] = \dots \\
 &= \frac{n!}{s^n} \mathcal{L}[1] = \frac{n!}{s^{n+1}} \quad \text{for } s > 0.
 \end{aligned}$$

Example 3.8.4 Find $L^{-1}\left[\frac{1}{s+4}\right]$ and $L^{-1}\left[\frac{1}{s^4}\right]$.

Solution From the results in Example 3.8.2 and Example 3.8.3, we have

$$\mathcal{L}[e^{-4t}] = \frac{1}{s+4} \quad \text{and} \quad \mathcal{L}\left[\frac{1}{6} t^3\right] = \frac{1}{s^4}.$$

Thus

$$\mathcal{L}^{-1}\left[\frac{1}{s+4}\right] = e^{-4t} \quad \text{and} \quad \mathcal{L}^{-1}\left[\frac{1}{s^4}\right] = \frac{1}{6} t^3.$$

Theorem 3.8.1 (Existence of Laplace Transforms)

Suppose that the function $f(t)$ is piecewise continuous for $t \geq 0$ and is of exponential order as $t \rightarrow \infty$, that is, there exist constants $M > 0, s_0 \geq 0$ such that

$$|f(t)| \leq M e^{s_0 t}, \quad \text{for } t \geq 0.$$

Then $F(s)$ exists for all $s > s_0$.

Proof When $s > s_0$, one has

$$\begin{aligned}
 \left| \int_0^{\infty} e^{-st} f(t) dt \right| &\leq \int_0^{\infty} e^{-st} |f(t)| dt \\
 &\leq M \int_0^{\infty} e^{-(s-s_0)t} dt
 \end{aligned}$$

$$= \left[-\frac{M}{s-s_0} e^{-(s-s_0)t} \right]_0^\infty$$

$$= \frac{M}{s-s_0},$$

which shows that

$$|F(s)| \leq \frac{M}{s-s_0}. \quad (3.8.4)$$

Corollary 3.8.1 Suppose that $f(t)$ satisfies the conditions of Theorem 3.8.1. Then

$$\lim_{s \rightarrow \infty} F(s) = 0. \quad (3.8.5)$$

The condition in (3.8.5) shows that the function $F_1(s) = \frac{s+1}{s+2}$ can not be the Laplace transform of any function because as $s \rightarrow +\infty$ its limit is one, not zero. More generally, a rational function, that is, a quotient of two polynomials, can be a Laplace transform only if the degree of its numerator is less than that of the denominator.

Next, we state (without proof) the uniqueness of inverse Laplace transforms.

Theorem 3.8.2 Suppose that the functions $f_1(t)$ and $f_2(t)$ satisfy the conditions of Theorem 3.8.1. Then their Laplace transforms $F_1(s)$ and $F_2(s)$ both exist. If $F_1(s) = F_2(s)$ for all $s > s_0$, then $f_1(t) = f_2(t)$ wherever on $[0, +\infty)$, $f_1(t)$ and $f_2(t)$ are continuous.

3.8.2 Properties of Laplace Transforms and Inverse Transforms

In the following, we shall give some basic properties of Laplace transforms and inverse transforms by which the Laplace transforms of more functions can be found without performing tedious integrations.

Property 3.8.1 (Linearity) Let c_1 and c_2 be constants, then

$$\mathcal{L}[c_1 f_1(t) + c_2 f_2(t)] = c_1 \mathcal{L}[f_1(t)] + c_2 \mathcal{L}[f_2(t)] \quad (3.8.6)$$

for all s such that the Laplace transforms of functions f_1 and f_2 both exist. Meanwhile, we also have

$$\mathcal{L}^{-1}[c_1 F_1(s) + c_2 F_2(s)] = c_1 \mathcal{L}^{-1}[F_1(s)] + c_2 \mathcal{L}^{-1}[F_2(s)] \quad (3.8.7)$$

provided that the inverse transforms of two functions $F_1(s)$ and $F_2(s)$ exist.

The proof of this property follows from the linearity of the operations of integration and of taking limits.

Example 3.8.5 Find $\mathcal{L}[\cosh \lambda t]$ and $\mathcal{L}[\sinh \lambda t]$, where $\cosh \lambda t = \frac{1}{2}(e^{\lambda t} + e^{-\lambda t})$,

$\sinh \lambda t = \frac{1}{2}(e^{\lambda t} - e^{-\lambda t})$ and $\lambda > 0$.

Solution

$$\mathcal{L}[\cosh \lambda t] = \frac{1}{2} \mathcal{L}[e^{\lambda t}] + \frac{1}{2} \mathcal{L}[e^{-\lambda t}]$$

$$\begin{aligned}
 &= \frac{1}{2} \left(\frac{1}{s - \lambda} + \frac{1}{s + \lambda} \right) \\
 &= \frac{s}{s^2 - \lambda^2} \quad \text{for } s > \lambda > 0.
 \end{aligned}$$

Similarly

$$\mathcal{L}[\sinh \lambda t] = \frac{\lambda}{s^2 - \lambda^2} \quad \text{for } s > \lambda > 0.$$

Example 3.8.6 Find $\mathcal{L}[\cos \beta t]$ and $\mathcal{L}[\sin \beta t]$.

Solution By Euler's formula, we obtain

$$\cos \beta t = \frac{1}{2} (e^{i\beta t} + e^{-i\beta t})$$

and

$$\sin \beta t = \frac{1}{2i} (e^{i\beta t} - e^{-i\beta t}).$$

Thus

$$\begin{aligned}
 \mathcal{L}[\cos \beta t] &= \frac{1}{2} \mathcal{L}[e^{i\beta t}] + \frac{1}{2} \mathcal{L}[e^{-i\beta t}] \\
 &= \frac{1}{2} \left(\frac{1}{s - i\beta} + \frac{1}{s + i\beta} \right) \\
 &= \frac{s}{s^2 + \beta^2} \quad \text{for } s > 0.
 \end{aligned}$$

Similarly

$$\mathcal{L}[\sin \beta t] = \frac{\beta}{s^2 + \beta^2} \quad \text{for } s > 0.$$

Example 3.8.7 Find $\mathcal{L}[3e^{2t} + 2 \sin^2 3t]$.

$$\begin{aligned}
 \text{Solution} \quad \mathcal{L}[3e^{2t} + 2 \sin^2 3t] &= \mathcal{L}[3e^{2t} + 1 - \cos 6t] \\
 &= \mathcal{L}[3e^{2t}] + \mathcal{L}[1] - \mathcal{L}[\cos 6t] \\
 &= \frac{3}{s - 2} + \frac{1}{s} - \frac{s}{s^2 + 36} \\
 &= \frac{3s^3 + 144s - 72}{s(s - 2)(s^2 + 36)} \quad \text{for } s > 2.
 \end{aligned}$$

Example 3.8.8 Find $\mathcal{L}^{-1} \left[\frac{2s - 3}{s^2 - s - 6} \right]$.

Solution We easily obtain the partial fractions decomposition

$$\frac{2s - 3}{s^2 - s - 6} = \frac{2s - 3}{(s - 3)(s + 2)} = \frac{\frac{3}{5}}{s - 3} + \frac{\frac{7}{5}}{s + 2}.$$

Since $\mathcal{L}^{-1} \left[\frac{1}{s - a} \right] = e^{at}$, it follows that

$$\mathcal{L}^{-1} \left[\frac{2s - 3}{s^2 - s - 6} \right] = \frac{3}{5} \mathcal{L}^{-1} \left[\frac{1}{s - 3} \right] + \frac{7}{5} \mathcal{L}^{-1} \left[\frac{1}{s + 2} \right]$$

$$= \frac{3}{5}e^{3t} + \frac{7}{5}e^{-2t}.$$

Property 3.8.2 (Differentiation of Transforms)

Suppose that $\mathcal{L}[f(t)] = F(s)$, then

$$\frac{d}{ds}F(s) = -\int_0^{\infty} t e^{-st} f(t) dt = -\mathcal{L}[tf(t)]. \quad (3.8.8)$$

Equivalently

$$f(t) = \mathcal{L}^{-1}[F(s)] = -\frac{1}{t} \mathcal{L}^{-1}[F'(s)].$$

Mathematical induction gives

$$\frac{d^n F(s)}{ds^n} = (-1)^n \mathcal{L}[t^n f(t)], n = 1, 2, \dots. \quad (3.8.9)$$

Example 3.8.9 Find $\mathcal{L}[t^2 \cos at]$.

Solution

$$\begin{aligned} \mathcal{L}[t^2 \cos at] &= (-1)^2 \frac{d^2}{ds^2} \mathcal{L}[\cos at] \\ &= \frac{d^2}{ds^2} \left(\frac{s}{s^2 + a^2} \right) = \frac{2s^3}{(s^2 + a^2)^3}. \end{aligned}$$

Example 3.8.10 Find $\mathcal{L}^{-1} \left[\arctan \frac{1}{s} \right]$.

Solution Since $\frac{d}{ds} \left(\arctan \frac{1}{s} \right) = \frac{-\frac{1}{s^2}}{1 + \left(\frac{1}{s} \right)^2} = -\frac{1}{s^2 + 1}$ is a rational function, we apply

the equation (3.8.8)

$$\begin{aligned} \mathcal{L}^{-1} \left[\arctan \frac{1}{s} \right] &= -\frac{1}{t} \mathcal{L}^{-1} \left[\frac{d}{ds} \left(\arctan \frac{1}{s} \right) \right] \\ &= -\frac{1}{t} \mathcal{L}^{-1} \left[\frac{-1}{s^2 + 1} \right] = \frac{\sin t}{t}. \end{aligned}$$

This example shows that when the derivative of the transform is easier to find than the transform itself, we usually apply the form of the differentiation property in (3.8.9) to find an inverse transform.

Property 3.8.3 (Transforms of Derivatives)

Suppose that $f(t), f'(t), f''(t), \dots, f^{(n-1)}(t), f^{(n)}(t)$ satisfy the hypotheses in Theorem 3.8.1. Then $\mathcal{L}[f^{(n)}(t)]$ exists when $s > s_0$ and

$$\begin{aligned} \mathcal{L}[f^{(n)}(t)] &= s^n \mathcal{L}[f(t)] - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0) \\ &= s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0). \end{aligned} \quad (3.8.10)$$

Proof When $n = 1$, beginning with the definition of $\mathcal{L}[f'(t)]$ and integrating by parts, we obtain

$$\mathcal{L}[f'(t)] = \int_0^{+\infty} e^{-st} f'(t) dt = [e^{-st} f(t)]_0^{\infty} + s \int_0^{+\infty} e^{-st} f(t) dt$$

$$= sF(s) - f(0). \quad (3.8.11)$$

Further

$$\begin{aligned} \mathcal{L}[f''(t)] &= s \mathcal{L}[f'(t)] - f'(0) \\ &= s[sF(s) - f(0)] - f'(0) \\ &= s^2 F(s) - sf(0) - f'(0). \end{aligned}$$

Repeated application of the equation (3.8.11) yields

$$\mathcal{L}[f^{(n)}(t)] = s^n F(s) - s^{n-1} f(0) - \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0).$$

Example 3.8.11 Show that

$$\mathcal{L}[te^{at}] = \frac{1}{(s - \alpha)^2}.$$

Solution Let $f(t) = te^{at}$, then $f(0) = 0$ and $f'(t) = e^{at} + at e^{at}$. Thus Property 3.8.3 gives

$$\mathcal{L}[e^{at} + at e^{at}] = \mathcal{L}[f'(t)] = s \mathcal{L}[f(t)] = s \mathcal{L}[te^{at}].$$

That is

$$\mathcal{L}[e^{at}] + \alpha \mathcal{L}[te^{at}] = s \mathcal{L}[te^{at}].$$

So

$$\mathcal{L}[te^{at}] = \frac{1}{s - \alpha} \mathcal{L}[e^{at}] = \frac{1}{(s - \alpha)^2}.$$

Property 3.8.4 (Transforms of Integrals) If $f(t)$ satisfies the conditions in Theorem 3.8.1, then

$$\mathcal{L}\left[\int_0^t f(\tau) d\tau\right] = \frac{1}{s} \mathcal{L}[f(t)] = \frac{F(s)}{s}. \quad (3.8.12)$$

Proof If $g(t) = \int_0^t f(\tau) d\tau$, then $g'(t) = f(t)$. By Property 3.8.3, we obtain

$$\mathcal{L}[f(t)] = \mathcal{L}[g'(t)] = s \mathcal{L}[g(t)] - g(0).$$

Together with $g(0) = 0$, division by s gives

$$\mathcal{L}\left[\int_0^t f(\tau) d\tau\right] = \frac{1}{s} \mathcal{L}[f(t)] = \frac{F(s)}{s}.$$

Example 3.8.12 Find the inverse Laplace transform of

$$G(s) = \frac{1}{s^3(s - \alpha)}.$$

Solution The equation in (3.8.12) is equivalent to

$$\mathcal{L}^{-1}\left[\frac{F(s)}{s}\right] = \int_0^t f(\tau) d\tau. \quad (3.8.13)$$

Taking $F(s) = \frac{1}{s - \alpha}$ in (3.8.13) gives

$$\begin{aligned} \mathcal{L}^{-1}\left[\frac{1}{s(s - \alpha)}\right] &= \int_0^t \mathcal{L}^{-1}\left(\frac{1}{s - \alpha}\right) d\tau \\ &= \int_0^t e^{a\tau} d\tau = \frac{1}{a}(e^{at} - 1). \end{aligned}$$

Repeating the technique yields

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{1}{s^2(s-\alpha)}\right] &= \int_0^t \mathcal{L}^{-1}\left[\frac{1}{s(s-\alpha)}\right] d\tau \\ &= \int_0^t \frac{1}{a}(e^{a\tau} - 1) d\tau \\ &= \left[\frac{1}{a}\left(\frac{1}{a}e^{a\tau} - \tau\right)\right]_0^t \\ &= \frac{1}{a^2}(e^{at} - at - 1).\end{aligned}$$

Finally, we obtain

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{1}{s^3(s-\alpha)}\right] &= \int_0^t L^{-1}\left[\frac{1}{s^2(s-\alpha)}\right] d\tau \\ &= \int_0^t \frac{1}{a^2}(e^{a\tau} - a\tau - 1) d\tau \\ &= \left[\frac{1}{a^2}\left(\frac{1}{a}e^{a\tau} - \frac{a}{2}\tau^2 - \tau\right)\right]_0^t \\ &= \frac{1}{a^3}\left(e^{at} - \frac{1}{2}(at)^2 - at - 1\right).\end{aligned}$$

The technique is more convenient than the method of partial fractions for finding an inverse transform of a fraction of the form $\frac{P(s)}{s^n Q(s)}$.

Property 3.8.5 (Translation on the s -Axis) Suppose that $F(s) = \mathcal{L}[f(t)]$ exists for $s > s_0$, then Laplace transform $\mathcal{L}[e^{at}f(t)]$ exists for $s > a + s_0$ and

$$\mathcal{L}[e^{at}f(t)] = F(s - a), \quad (3.8.14)$$

which is equivalent to

$$\mathcal{L}^{-1}[F(s - a)] = e^{at}f(t). \quad (3.8.15)$$

Proof Using the definition $F(s)$, we obtain

$$\begin{aligned}F(s - a) &= \int_0^\infty e^{-(s-a)t} f(t) dt = \int_0^\infty e^{-st} [e^{at} f(t)] dt \\ &= \mathcal{L}[e^{at} f(t)].\end{aligned}$$

This property shows that the translation $s \rightarrow s - a$ in the Laplace transform corresponds to multiplication of the original function $f(t)$ by e^{at} .

Example 3.8.13 Find $\mathcal{L}[e^{at}t^n]$, $\mathcal{L}[e^{at}\cos bt]$ and $\mathcal{L}[e^{at}\sin bt]$.

Solution Since

$$\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}, s > 0,$$

$$\mathcal{L}[\cos bt] = \frac{s}{s^2 + b^2}, s > 0,$$

$$\mathcal{L}[\sin bt] = \frac{b}{s^2 + b^2}, s > 0,$$

from Property 3.8.5, we obtain

$$\mathcal{L}[e^{at}t^n] = \frac{n!}{(s-a)^{n+1}}, s > a,$$

$$\mathcal{L}[e^{at} \cos bt] = \frac{s-a}{(s-a)^2 + b^2}, s > a,$$

$$\mathcal{L}[e^{at} \sin bt] = \frac{b}{(s-a)^2 + b^2}, s > a.$$

Property 3.8.6 (Integration of Transforms) If $f(t)$ satisfies the conditions in Theorem 3.8.1, then

$$\mathcal{L}\left[\frac{f(t)}{t}\right] = \int_s^\infty F(\tau) d\tau \quad \text{for } s > s_0. \quad (3.8.16)$$

Equivalently

$$f(t) = t \mathcal{L}^{-1}\left[\int_s^\infty F(\tau) d\tau\right]. \quad (3.8.17)$$

This property is often helpful in finding an inverse transform when the indefinite integral of the transform is much easier to perform than the transform itself.

Example 3.8.14 Find $\mathcal{L}\left[\frac{2\sinh t}{t}\right]$.

Solution

$$\begin{aligned} \mathcal{L}\left[\frac{2\sinh t}{t}\right] &= 2 \int_s^\infty \mathcal{L}[\sinh t] d\sigma \\ &= 2 \int_s^\infty \frac{1}{\sigma^2 - 1} d\sigma \\ &= \left[\ln \frac{\sigma - 1}{\sigma + 1} \right]_s^\infty = \ln \frac{s + 1}{s - 1}. \end{aligned}$$

Property 3.8.7 (Transforms of Periodic Functions) If $f(t)$ is piecewise continuous for $t \geq 0$ and periodic with period ω , then $F(s) = \mathcal{L}[f(t)]$ exists for $s > 0$ and takes the form

$$F(s) = \frac{1}{1 - e^{-\omega s}} \int_0^\omega e^{-st} f(t) dt.$$

Proof The definition of Laplace transform gives

$$F(s) = \int_0^\infty e^{-st} f(t) dt = \sum_{n=0}^\infty \int_{n\omega}^{(n+1)\omega} e^{-st} f(t) dt.$$

Using the substitution $t = \tau + n\omega$, we obtain

$$\begin{aligned} \int_{n\omega}^{(n+1)\omega} e^{-st} f(t) dt &= \int_0^\omega e^{-s(\tau + n\omega)} f(\tau + n\omega) d\tau \\ &= e^{-nas} \int_0^\omega e^{-s\tau} f(\tau) d\tau. \end{aligned}$$

Thus

$$\begin{aligned} F(s) &= \sum_{n=0}^\infty (e^{-nas} \int_0^\omega e^{-s\tau} f(\tau) d\tau) \\ &= (1 + e^{-\omega s} + e^{-2\omega s} + \cdots) \int_0^\omega e^{-s\tau} f(\tau) d\tau \end{aligned}$$

$$= \frac{1}{1 - e^{-\omega s}} \int_0^{\omega} e^{-s\tau} f(\tau) d\tau, \quad \text{for } s > 0.$$

Example 3.8.15 Find $\mathcal{L}[f(t)]$, where Figure 3.8.1 shows the graph of a triangular wave function $f(t)$ of period $\omega = 2a$.

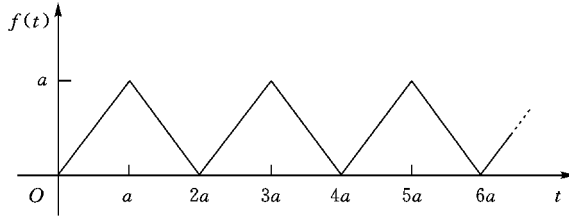


Figure 3.8.1

Solution From Figure 3.8.1, we have

$$f(t) = \begin{cases} t, & 0 \leq t < a, \\ 2a - t, & a \leq t < 2a. \end{cases}$$

By Property (3.8.7), one has

$$\mathcal{L}[f(t)] = \frac{1}{1 - e^{-2as}} \int_0^{2a} e^{-st} f(t) dt.$$

Direct computation shows that

$$\begin{aligned} \int_0^{2a} e^{-st} f(t) dt &= \int_0^a t e^{-st} dt + \int_a^{2a} (2a - t) e^{-st} dt \\ &= \frac{1}{s^2} (e^{-as} - 1)^2. \end{aligned}$$

Thus the Laplace transform is given by

$$F(s) = \mathcal{L}[f(t)] = \frac{1}{s^2} \left(\frac{1 - e^{-as}}{1 + e^{-as}} \right) = \frac{1}{s^2} \tanh \frac{as}{2}.$$

Definition 3.8.3 The convolution $f_1 * f_2$ of the piecewise continuous functions f_1 and f_2 is defined for $t \geq 0$ as follows

$$(f_1 * f_2)(t) = \int_0^t f_1(\tau) f_2(t - \tau) d\tau. \quad (3.8.18)$$

For convenience, we also write $f_1(t) * f_2(t)$.

Property 3.8.8 (Convolution Property)

If $f_1(t)$ and $f_2(t)$ satisfy the conditions in Theorem 3.8.1, then the Laplace transform of the convolution $f_1(t) * f_2(t)$ exists for $s > s_0$, and is given by

$$\mathcal{L}[f_1(t) * f_2(t)] = \mathcal{L}[f_1(t)] \cdot \mathcal{L}[f_2(t)]. \quad (3.8.19)$$

Equivalently

$$\mathcal{L}^{-1}[F_1(s) \cdot F_2(s)] = f_1(t) * f_2(t). \quad (3.8.20)$$

The proof is left to the reader. This property shows that we can solve the inverse transform of the product $F_1(s) \cdot F_2(s)$, provided that we can obtain the integral, that is,

$$\mathcal{L}^{-1}[F_1(s) \cdot F_2(s)] = \int_0^t f_1(\tau) f_2(t - \tau) d\tau.$$

Example 3.8.16 Find $\mathcal{L}^{-1}\left[\frac{2}{(s-1)(s^2+4)}\right]$ by two methods.

Solution Method 1 We easily obtain the partial fraction decomposition

$$\frac{2}{(s-1)(s^2+4)} = \frac{2}{5} \left(\frac{1}{s-1} - \frac{s}{s^2+4} - \frac{1}{s^2+4} \right).$$

By linearity of inverse transform, one has

$$\begin{aligned} \mathcal{L}^{-1}\left[\frac{2}{(s-1)(s^2+4)}\right] &= \frac{2}{5} \mathcal{L}^{-1}\left[\frac{1}{s-1}\right] - \frac{2}{5} \mathcal{L}^{-1}\left[\frac{s}{s^2+4}\right] - \frac{2}{5} \mathcal{L}^{-1}\left[\frac{1}{s^2+4}\right] \\ &= \frac{2}{5}e^t - \frac{1}{5}\sin 2t - \frac{2}{5}\cos 2t. \end{aligned}$$

Method 2 With $F_1(s) = \frac{1}{s-1}$ and $F_2(s) = \frac{2}{s^2+4}$, we have

$$f_1(t) = \mathcal{L}^{-1}[F_1(s)] = e^t, f_2(t) = \mathcal{L}^{-1}[F_2(s)] = \sin 2t.$$

The convolution property gives

$$\begin{aligned} \mathcal{L}^{-1}\left[\frac{2}{(s-1)(s^2+4)}\right] &= f_1(t) * f_2(t) = (\sin 2t) * e^t \\ &= \int_0^t e^{t-\tau} \sin 2\tau d\tau = e^t \int_0^t e^{-\tau} \sin 2\tau d\tau \\ &= e^t \left[\frac{1}{5} e^{-\tau} (-\sin 2\tau - 2\cos 2\tau) \right]_0^t \\ &= \frac{2}{5}e^t - \frac{1}{5}\sin 2t - \frac{2}{5}\cos 2t. \end{aligned}$$

3.8.3 Solving Initial Value Problems by Laplace Transforms

We now discuss the application of Laplace transforms. The method of Laplace transforms can be used to solve not only a linear differential equation with constant coefficients but also linear systems of differential equations satisfying the given initial conditions.

Example 3.8.17 Solve the initial value problem

$$x' - x = e^{2t}, x(0) = 0.$$

Solution Let $X(s) = \mathcal{L}[x(t)]$. Taking Laplace transform of both sides of the equation yields

$$sX(s) - X(s) = \frac{1}{s-2}.$$

Thus

$$X(s) = \frac{1}{(s-2)(s-1)}.$$

Taking inverse transform gives the particular solution

$$x(t) = \mathcal{L}^{-1}[X(s)] = \mathcal{L}^{-1}\left[\frac{1}{(s-2)(s-1)}\right]$$

$$\begin{aligned} &= \mathcal{L}^{-1} \left[\frac{1}{s-2} - \frac{1}{s-1} \right] \\ &= e^{2t} - e^t. \end{aligned}$$

Example 3.8.18 Solve the initial value problem

$$x'' - x = 4\sin t + 5\cos 2t; x(0) = -1, x'(0) = -2.$$

Solution Taking the Laplace transform of both sides of the differential equation yields

$$s^2 X(s) + s + 2 - X(s) = \frac{4}{s^2 + 1} + \frac{5s}{s^2 + 4}.$$

Furthermore

$$X(s) = \frac{1}{s^2 - 1} \left[\frac{4}{s^2 + 1} + \frac{5s}{s^2 + 4} - s - 2 \right].$$

By the method of partial fractions, we obtain

$$X(s) = -\frac{2}{s^2 + 1} - \frac{s}{s^2 + 4}.$$

Hence

$$\begin{aligned} x(t) &= \mathcal{L}^{-1}[X(s)] = \mathcal{L}^{-1} \left[-\frac{2}{s^2 + 1} - \frac{s}{s^2 + 4} \right] \\ &= -2\sin t - \cos 2t. \end{aligned}$$

Example 3.8.19 Solve the initial value problem

$$\begin{cases} \frac{dx_1}{dt} = x_1 - 3x_2, & x_1(0) = 0, \\ \frac{dx_2}{dt} = -2x_1 + 2x_2, & x_2(0) = 5. \end{cases}$$

Solution Let $X_1(s) = \mathcal{L}[x_1(t)]$, $X_2(s) = \mathcal{L}[x_2(t)]$. We have the transformed system

$$\begin{cases} sX_1(s) = X_1(s) - 3X_2(s), \\ sX_2(s) = -2X_1(s) + 2X_2(s) + 5. \end{cases}$$

Using Cramer's rule, we readily solve the system for

$$X_1(s) = \frac{3}{s+1} - \frac{3}{s-4}$$

and

$$X_2(s) = \frac{2}{s+1} + \frac{3}{s-4}.$$

Hence the solution of the initial value problem is

$$\begin{cases} x_1(t) = 3e^{-t} - 3e^{4t}, \\ x_2(t) = 2e^{-t} + 3e^{4t}. \end{cases}$$

Example 3.8.20 Solve the differential system

$$\begin{cases} x'' = -3x + y, \\ y'' = 2x - 2y + 40\sin 3t, \end{cases}$$

satisfying the initial conditions

$$x(0) = x'(0) = y(0) = y'(0) = 0.$$

Solution Let $X(s) = \mathcal{L}[x(t)]$ and $Y(s) = \mathcal{L}[y(t)]$. Combining the initial conditions, the transformed algebraic system is

$$\begin{cases} (s^2 + 3)X(s) - Y(s) = 0, \\ -2X(s) + (s^2 + 2)Y(s) = \frac{120}{s^2 + 9}. \end{cases}$$

By Cramer's rule, we obtain

$$\begin{cases} X(s) = \frac{120}{(s^2 + 1)(s^2 + 4)(s^2 + 9)}, \\ Y(s) = \frac{120(s^2 + 3)}{(s^2 + 1)(s^2 + 4)(s^2 + 9)}. \end{cases}$$

Using the method of partial fractions, we can write

$$X(s) = \frac{5}{s^2 + 1} - \frac{8}{s^2 + 4} + \frac{3}{s^2 + 9}$$

and

$$Y(s) = \frac{10}{s^2 + 1} + \frac{8}{s^2 + 4} - \frac{18}{s^2 + 9}.$$

Then taking the inverse Laplace transforms of the last two equations gives the particular solution

$$\begin{cases} x(t) = 5\sin t - 4\sin 2t + \sin 3t, \\ y(t) = 10\sin t + 4\sin 2t - 6\sin 3t. \end{cases}$$

Exercise 3.8

Find directly the Laplace transforms of the functions in Problems 1 through 10.

1. $f(t) = e^{4t+2}$.
2. $f(t) = 3t - 2t^4$.
3. $f(t) = \cos^2 4t$.
4. $f(t) = \sin 3t \cos 5t$.
5. $f(t) = (2+t)^3$.
6. $f(t) = t^2 \cos 3t$.
7. $f(t) = \sin 3t + 4\cos t$.
8. $f(t) = te^{at} \cos bt$.
9. $f(t) = te^{at} \sin bt$.
10. $f(t) = \sinh^2 3t$.

Find the inverse Laplace transforms of the functions in Problems 11 through 20.

11. $F(s) = \frac{3}{s-4}$.
12. $F(s) = \frac{5}{s^4}$.
13. $F(s) = \frac{4}{(s-2)^3}$.
14. $F(s) = \frac{2s+1}{s^2+4}$.
15. $F(s) = \frac{1+s}{4-s^2}$.
16. $F(s) = \frac{1}{s^2(s^2+1)}$.
17. $F(s) = \frac{1}{s^2(s^2-1)}$.
18. $F(s) = \frac{2s+1}{s(s^2+3)}$.
19. $F(s) = \frac{s^2-s}{s^4+5s^2+4}$.
20. $F(s) = \frac{s^2+3}{(s^2+2s+3)^2}$.

21. The gamma function $\Gamma(x)$ for $x > 0$ is defined by the formula

$$\Gamma(x) = \int_0^{\infty} e^{-t} t^{x-1} dt.$$

Show that the following hold:

(1) $\Gamma(x+1) = x\Gamma(x)$; especially, $\Gamma(n+1) = n!$, where n is a positive integer.

(2) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ is given. Find $\mathcal{L}\left[t^{\frac{1}{2}}\right]$ and $\mathcal{L}\left[t^{\frac{n}{2}}\right]$.

Use Laplace transforms to solve the initial value problems in Problems 22 through 30.

22. $x'' + 4x = 0; x(0) = 5, x'(0) = 1.$

23. $x'' - x' + 2x = 0; x(0) = 0, x'(0) = 2.$

24. $x'' + x = 2\cos 3t; x(0) = 1, x'(0) = 0.$

25. $x'' - 3x' + 2x = e^{3t}; x(0) = 1, x'(0) = 0.$

26. $x''' + 3x'' + 3x' + x = 1; x(0) = x'(0) = x''(0) = 0.$

27. $x'' + 3x' + 2x = t; x(0) = 0, x'(0) = 2.$

28. $x' = 2x + y, y' = 6x + 3y; x(0) = 1, y(0) = -2.$

29. $x' = x + 2y, y' = x + e^{-t}; x(0) = y(0) = 0.$

30. $x'' + 2x + 4y = 0, y'' + x + 2y = 0; x(0) = y(0) = 0, x'(0) = y'(0) = -1.$

Chapter 4 General Theory

In this chapter, we consider the following initial value problem

$$\begin{cases} \frac{dx}{dt} = f(t, x), \\ x(t_0) = x_0. \end{cases}$$

Before beginning with the general theory of the ordinary differential equations, we make a list of common questions facing an initial value problem.

Such questions are

- (1) Does the initial value problem has a solution if we cannot solve it in closed form?
- (2) If a solution exists, is it unique?
- (3) How long does a solution of the initial value problem last?
- (4) How much is a solution sensible with respect to the initial condition and parameters which appear in the differential equation?

In the following sections, we will give some satisfactory answers to these questions.

4.1 Peano's Existence Theorem

The first question is about the existence of solutions. It was first answered in the affirmative by the Italian mathematician Peano (1858—1932) provided that the function $f(t, x)$ is continuous.

Before studying the initial value problem

$$\begin{cases} x' = f(t, x), \\ x(t_0) = x_0, \end{cases} \quad (4.1.1)$$

we first introduce the concepts of **line element** and **direction field**. In this section, we always assume that the right-hand side $f(t, x)$ of the equation is continuous on the closed domain R :

$$|t - t_0| \leq a, \quad |x - x_0| \leq b$$

in the tx -plane.

If $x = \varphi(t)$ is a solution of the initial value problem (4.1.1), from geometric interpretation, then $\varphi(t)$ is an integral curve of the differential equation which passes through a point (t_0, x_0) . That is, the differential equation defines the slope of the curve at this point, $\varphi'(t_0) = f(t_0, x_0)$. This leads naturally to the concepts of line element and direction field.

Definition 4.1.1 Such a numerical triple of the form (t, x, p) is called a line element, where (t, x) gives a point in phase, the third component p gives the slope of a line through the point (t, x) with $p = f(t, x) = \tan \alpha$ (α is the angle of inclination of the line). All these line elements constitute a direction field for the equation

$$x' = f(t, x).$$

A solution $x(t)$ of a differential equation agrees with its direction field, that is, the slope at each point on the solution is equal to the slope of the line element at that point. Conversely, if a curve fits the direction field of a differential equation, then this curve will be an integral curve. The idea of sketching the direction field and then drawing curves which fit these line elements can be used to get approximate solution or numerical solution to a differential equation.

Theorem 4.1.1 (Peano's Existence Theorem) Let $f(t, x)$ be continuous on the rectangle domain $R : |t - t_0| \leq a, |x - x_0| \leq b$. Then the initial value problem (4.1.1) possesses at least one solution $x = \varphi(t)$ on the interval $I = [t_0 - h, t_0 + h]$, where $h = \min \left\{ a, \frac{b}{M} \right\}$, M is a bound for $|f(t, x)|$ on R .

We shall construct a sequence of approximate solutions $\{x_n(t)\}_{n=1}^{\infty}$ to the initial value problem in the interval $[t_0, t_0 + h]$, show that $\{x_n(t)\}_{n=1}^{\infty}$ contains at least a subsequence $\{x_{n_k}(t)\}_{k=1}^{\infty}$ which converges uniformly in $[t_0, t_0 + h]$ to a continuous function $x(t)$, and verify that the limit function $x(t)$ is a solution of the initial value problem in $[t_0, t_0 + h]$. A similar process will be performed to obtain a solution of the initial value problem in the interval $[t_0 - h, t_0]$.

For this, we partition the interval $[t_0, t_0 + h]$ into n equal subintervals, then the points of the partition are

$$t = t_0, t_1 = t_0 + \frac{h}{n}, t_2 = t_0 + \frac{2h}{n}, \dots, t_{n-1} = t_0 + \frac{n-1}{n}h, t_n = t_0 + h$$

and

$$\Delta t_i = \frac{h}{n}, i = 1, 2, \dots, n. \quad (4.1.2)$$

Next, we define a function $x_n(t)$ in the interval $[t_0, t_0 + h]$ by the recursive formula

$$x_n(t) = x_n(t_{i-1}) + f(t_{i-1}, x_n(t_{i-1}))(t - t_{i-1}), t_{i-1} \leq t \leq t_i, i = 1, 2, \dots, n \quad (4.1.3)$$

Clearly, this function $x_n(t)$ is continuous and has a piecewise continuous derivative

$$x'_n(t) = f[t_{i-1}, x_n(t_{i-1})], t_{i-1} < t < t_i, i = 1, 2, \dots, n$$

which cannot be defined only at the points $t_i, i = 1, 2, \dots, n - 1$.

This method of constructing a sequence of approximate solution $\{x_n(t)\}_{n=1}^{\infty}$ is known as Cauchy-Euler method.

In order to prove Theorem 4.1.1, we introduce several definitions and lemmas.

Definition 4.1.2 Let $\{f_n(t)\}_{n=1}^{\infty}$ be a sequence of functions defined in the interval I . If there exists a constant $K > 0$ such that

$$|f_n(t)| \leq K, t \in I, n=1,2,\dots$$

then this sequence of functions is said to be uniformly bounded in the interval I .

Definition 4.1.3 If for any given $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|f_n(t_1) - f_n(t_2)| \leq \epsilon, n=1,2,\dots$$

provided that $t_1, t_2 \in I, |t_1 - t_2| \leq \delta$, then the sequence of functions $\{f_n(t)\}_{n=1}^{\infty}$ is said to be equicontinuous in the interval I .

Next, for a given continuous function $f(t, x)$ in a domain R , we need the following definition.

Definition 4.1.4 If a function $\varphi(t)$ defined in the interval I satisfies the following conditions:

- (1) $\varphi(t)$ is continuous in I ;
- (2) the points $(t, \varphi(t)) \in R$ for all $t \in I$;
- (3) $\varphi(t)$ has a piecewise continuous derivative in I which may fail to be defined only for a finite number of points $t_i (i=1,2,\dots,m)$;
- (4) $|\varphi'(t) - f(t, \varphi(t))| \leq \epsilon$ for all $t \in I, t \neq t_i, i=1,2,\dots,m$, then the function $\varphi(t)$ is said to be an ϵ -approximate solution of the differential equation

$$\frac{dx}{dt} = f(t, x).$$

Lemma 4.1.1 Let $\{\varphi_n(t)\}$ be a sequence of continuous functions in the interval I which converges uniformly to $\varphi(t)$. Then $\varphi(t)$ is continuous in I .

Lemma 4.1.2 Let $\{\varphi_n(t)\}$ be a sequence converging uniformly to $\varphi(t)$ in $[\alpha, \beta]$ and $f(t, x)$ be a continuous function in the domain R such that for all n and $t \in [\alpha, \beta]$, the points $(t, \varphi_n(t))$ are in R . Then

$$\lim_{n \rightarrow \infty} \int_a^\beta f(t, \varphi_n(t)) dt = \int_a^\beta \lim_{n \rightarrow \infty} f(t, \varphi_n(t)) dt = \int_a^\beta f(t, \varphi(t)) dt.$$

Lemma 4.1.3 (Ascoli-Arzelà Lemma) Let a sequence $\{f_n(t)\}_{n=1}^{\infty}$ be uniformly bounded and equicontinuous in $[\alpha, \beta]$. Then the sequence contains at least a subsequence $\{f_{n_k}(t)\}_{k=1}^{\infty}$ converging uniformly in $[\alpha, \beta]$.

Next, we give the proof of Theorem 4.1.1.

Proof of Theorem 4.1.1

Step 1 Show that for any $\epsilon_m > 0$, the initial value problem (4.1.1) exists an ϵ_m -approximate solution $x_n(t)$, that is

$$|x'_n(t) - f(t, x_n(t))| \leq \epsilon_m, x_n(t_0) = x_0. \quad (4.1.4)$$

In fact, since $f(t, x)$ is continuous in the closed domain R , it is uniformly continuous. Thus, for a given $\epsilon_m > 0$, there exists a $\delta_m > 0$ such that

$$|f(t, x) - f(t_1, x_1)| \leq \epsilon_m \quad (4.1.5)$$

for all $(t, x), (t_1, x_1) \in R$ and $|t - t_1| \leq \delta_m, |x - x_1| \leq \delta_m$. For $\delta_m > 0$, there exists a natural number N , such that

$$\Delta t_i = \frac{h}{n} \leq \min\left\{\delta_m, \frac{\delta_m}{M}\right\}, n \geq N. \quad (4.1.6)$$

We construct $x_n(t)$ in (4.1.3). Obviously, $x_n(t)$ satisfies the conditions (1) and (3) in Definition 4.1.4.

Next, we prove $(t, x_n(t)) \in R$. It suffices to show that

$$|x_n(t_i) - x_0| \leq b \quad \text{for all } i = 1, 2, \dots, n \quad (4.1.7)$$

because in each subinterval $[t_{i-1}, t_i]$, $i = 1, 2, \dots, n$, the function $x_n(t)$ is a straight line. For this, if $i = 1$ and $t = t_1$ in (4.1.3), one has

$$|x_n(t_1) - x_0| = (t_1 - t_0) |f(t_0, x_0)| \leq Mh \leq b.$$

Now if the claim is true for $i = 1, 2, \dots, k-1 < n-1$, then from (4.1.3), we have

$$\begin{aligned} x_n(t_1) - x_0 &= (t_1 - t_0) f(t_0, x_0) \\ x_n(t_2) - x_n(t_1) &= (t_2 - t_1) f(t_1, x_n(t_1)) \\ &\vdots \\ x_n(t_k) - x_n(t_{k-1}) &= (t_k - t_{k-1}) f(t_{k-1}, x_n(t_{k-1})) \end{aligned}$$

and further

$$x_n(t_k) - x_0 = \sum_{q=1}^k (t_q - t_{q-1}) f(t_{q-1}, x_n(t_{q-1})),$$

which yields

$$|x_n(t_k) - x_0| \leq \sum_{q=1}^k (t_q - t_{q-1}) M = M(t_k - t_0) \leq Mh \leq b.$$

This shows that the inequality (4.1.7) holds.

Finally, if $t_{i-1} < t < t_i$, then from (4.1.3) and (4.1.6), we obtain

$$\begin{aligned} |x_n(t) - x_n(t_{i-1})| &\leq M |t - t_{i-1}| \leq M \frac{h}{n} \\ &\leq M \cdot \frac{\delta_m}{M} = \delta_m, \end{aligned}$$

and hence from (4.1.5), we have

$$|x'_n(t) - f(t, x_n(t))| = |f(t_{i-1}, x_n(t_{i-1})) - f(t, x_n(t))| \leq \epsilon_m$$

for all $t \in [t_0, t_0 + h]$, $t \neq t_i$, $i = 1, 2, \dots, n-1$.

This shows that $x_n(t)$ is an ϵ_m -approximate solution of the differential equation

$$\frac{dx}{dt} = f(t, x).$$

Step 2 We shall construct a sequence of approximate solutions $\{x_n(t)\}_{n=1}^{\infty}$ which contains a uniformly convergent subsequence in $[t_0, t_0 + h]$.

Let $\{\epsilon_n\}$ be a monotonically decreasing sequence of positive numbers such that $\epsilon_n \rightarrow 0$. For each $\epsilon_n > 0$, we shall construct an ϵ_n -approximate solution $x_n(t)$. Thus, we obtain a sequence of functions $\{x_n(t)\}_{n=1}^{\infty}$ in $[t_0, t_0 + h]$.

For any two points $t_1, t_2 \in [t_0, t_0 + h]$, we have

$$|x_n(t_1) - x_n(t_2)| \leq M |t_1 - t_2|$$

which shows that the sequence $\{x_n(t)\}$ is equicontinuous. In addition, for each t in $[t_0, t_0 + h]$, it is easy to prove that

$$|x_n(t)| \leq |x_0| + b,$$

and from this it follows that the sequence $\{x_n(t)\}$ is uniformly bounded. Hence, using Lemma 4.1.3, the sequence $\{x_n(t)\}_{n=1}^{\infty}$ contains a subsequence $\{x_{n_k}(t)\}_{k=1}^{\infty}$ converging uniformly in $[t_0, t_0 + h]$ to a continuous function $\varphi(t)$.

Step 3 Show that the limit function $\varphi(t)$ is a solution of the initial value problem.

We define

$$e_n(t) = \begin{cases} x'_{n_k}(t) - f(t, x_n(t)), & \text{at the points where } x'_{n_k}(t) \text{ exists,} \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, it follows that

$$x_n(t) = x_0 + \int_{t_0}^t [f(s, x_n(s)) + e_n(s)] ds \quad (4.1.8)$$

and $|e_n(t)| \leq \epsilon_n$.

Since $\{x_n(t)\}_{n=1}^{\infty}$ contains a uniformly convergent subsequence $\{x_{n_k}(t)\}_{k=1}^{\infty}$ and $|e_{n_k}(t)|$ uniformly converges to zero in $[t_0, t_0 + h]$ as $\epsilon_{n_k} \rightarrow 0$, we replace n by n_k in (4.1.8) and take the limit of both sides to yield

$$\varphi(t) = x_0 + \int_{t_0}^t f(s, \varphi(s)) ds. \quad (4.1.9)$$

This shows that $\varphi(t)$ is a solution of the integral equation

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds.$$

Differentiating both sides of (4.1.9) gives

$$\varphi'(t) = f(t, \varphi(t))$$

and $\varphi(t_0) = x_0$. So $x = \varphi(t)$ is a solution of the initial value problem. Similarly, we can obtain also a solution $x = \psi(t)$ of the initial value problem (4.1.1) in the interval $[t_0 - h, t_0]$.

We denote

$$\varphi_0(t) = \begin{cases} \varphi(t), & t \in [t_0, t_0 + h] \\ \psi(t), & t \in [t_0 - h, t_0] \end{cases}$$

which is a solution of (4.1.1) in the interval $[t_0 - h, t_0 + h]$. The proof is completed.

Remark 4.1.1 Any one subsequence $\{x_{n_k}(t)\}_{k=1}^{\infty}$ uniformly converging in I of the sequence $\{x_n(t)\}_{n=1}^{\infty}$ will tend to some one solution to the initial value problem. If the initial value problem exists a unique solution, then the sequence $\{x_n(t)\}_{n=1}^{\infty}$ uniformly converges to this unique solution.

Remark 4.1.2 The continuity of $f(t, x)$ can yield the existence of solution, but cannot guarantee the uniqueness of solution.

For example, let us consider the following initial value problem

$$\begin{cases} \frac{dx}{dt} = x^{\frac{2}{3}}, \\ x(0) = 0. \end{cases}$$

Here $f(t, x) = x^{\frac{2}{3}}$ defined on R^2 is continuous. Indeed, we have infinitely many solutions. Notice that $x \equiv 0$ is a solution. In addition, for every $\alpha, \beta \geq 0$, we also have the solution

$$x_{\alpha,\beta}(t) = \begin{cases} 0, & \text{if } -\beta \leq t \leq \alpha, \\ \left(\frac{t-\alpha}{3}\right)^3, & \text{if } t \geq \alpha, \\ \left(\frac{t+\beta}{3}\right)^3, & \text{if } t \leq -\beta. \end{cases}$$

Remark 4.1.3 Let us consider the following initial value problem

$$\begin{cases} \frac{dx}{dt} = f(x), \\ x(0) = 0. \end{cases}$$

where $f(x)$ is defined by

$$f(x) = \begin{cases} -2, & \text{if } x > 0, \\ 0, & \text{if } x = 0, \\ 2, & \text{if } x < 0. \end{cases}$$

Here $f(x)$ is discontinuous, but the initial value problem has a solution $x(t) \equiv 0$. This remark shows that the continuity of $f(t, x)$ is not a necessary condition for the existence of solution.

Exercise 4.1

1. Apply similar method to show that the initial value problem has at least one solution in the interval $[t_0 - h, t_0]$.
2. Let S be a set of functions $f(t)$ that satisfy a Lipschitz condition with a common Lipschitz constant L , that is

$$|f(t_1) - f(t_2)| \leq L |t_1 - t_2| \text{ for } t_1, t_2 \in I, f \in S.$$

Show that the set S is equicontinuous.

3. Picard's iterates satisfy the conditions of Ascoli-Arzelà Lemma, can we prove Peano's existence theorem by Picard's iterates? Give the reason.

4. Consider the initial value problem (4.1.1). Let $f(t, x)$ be continuous and bounded in the strip $S: t_0 \leq t \leq t_0 + a, |x| < \infty$. We divide the interval $[t_0, t_0 + a]$ into n equal subintervals, then $t_k = t_0 + k d_n$, where $d_n = \frac{a}{n}, k = 0, 1, 2, \dots, n$. We define a sequence of functions $\{x_n(t)\}$ by the scheme

$$x_n(t) = \begin{cases} x_0, & t_0 \leq t \leq t_0 + \frac{a}{n}, \\ x_0 + \int_{t_0}^{t-d_n} f(s, x_n(s)) ds, & t \in [t_1, t_0 + a], \end{cases}$$

which is called Tonelli's sequence. Show that the initial value problem has at least one solution by Tonelli's sequence and Ascoli-Arzelà Lemma.

4.2 Existence and Uniqueness of Solutions

In Section 4.1, we deal with the existence of a solution to the initial value problem

(4.1.1). The continuity of $f(t, x)$ can guarantee that the problem (4.1.1) has at least one solution. In this section, we shall discuss the existence and uniqueness of solutions to the problem (4.1.1) and give the following theorem.

Theorem 4.2.1 (Existence and Uniqueness Theorem)

Suppose that $f(t, x)$ satisfy the following conditions:

(1) $f(t, x)$ is continuous on the closed domain $R: |t - t_0| \leq a, |x - x_0| \leq b$.

(2) $f(t, x)$ satisfies a Lipschitz condition with respect to x in R , that is, there exists a constant $L > 0$ such that

$$|f(t, x_1) - f(t, x_2)| \leq L |x_1 - x_2| \quad (4.2.1)$$

for all $(t, x_1), (t, x_2)$ in the domain R , where L is called a Lipschitz constant.

Then the initial value problem (4.1.1) has one and only one solution $x = \varphi(t)$ on the interval $I = [t_0 - h, t_0 + h]$, where

$$h = \min \left\{ a, \frac{b}{M} \right\}, \quad (4.2.2)$$

M is a bound for $|f(t, x)|$ on the domain R .

Proof We shall divide this proof by the next steps.

Step 1 The function $x(t)$ satisfies the initial value problem (4.1.1) on the interval I if and only if it satisfies the integral equation

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds \quad (4.2.3)$$

for all t on I .

Any solution $x(t)$ of the differential equation $x' = f(t, x)$ converts it into an identity in t , namely, $x'(t) = f(t, x(t))$. Integrating this equality yields

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds.$$

Conversely, if $x(t)$ is one solution of the integral equation (4.2.3), then differentiating (4.2.3) gives the differential equation

$$x'(t) = f(t, x(t))$$

and we also obtain $x(t_0) = x_0$.

Step 2 Define iteratively a sequence $x_0(t), x_1(t), x_2(t), \dots$ of functions in t

$$x_{n+1}(t) = x_0 + \int_{t_0}^t f(s, x_n(s)) ds, t \in I \quad (4.2.4)$$

where $x_0(t) \equiv x_0, n = 0, 1, 2, \dots$.

These functions $x_n(t)$ are called Picard iterates after the French mathematician Picard (1856—1941). We can show that this sequence $x_n(t)$ defined in (4.2.4) is continuously differentiable and uniformly bounded on I .

Indeed, if $n = 1$, the expression (4.2.4) gives

$$x_1(t) = x_0 + \int_{t_0}^t f(s, x_0(s)) ds$$

Notice that $f[t, x_0(t)]$ is continuous on I . So $x_1(t)$ is continuously differentiable on I and

satisfies

$$|x_1(t) - x_0| \leq \left| \int_{t_0}^t |f(s, x_0(s))| ds \right| \leq M |t - t_0| \leq Mh \leq b.$$

Thus $f(t, x_1(t))$ is continuous on I . Further, when $n = 2$,

$$x_2(t) = x_0 + \int_{t_0}^t f(s, x_1(s)) ds$$

is continuously differentiable and

$$\begin{aligned} |x_2(t) - x_0| &\leq \left| \int_{t_0}^t |f(s, x_1(s))| ds \right| \\ &\leq M |t - t_0| \leq Mh \leq b, t \in I. \end{aligned}$$

By induction, we derive that Picard iterates $\{x_n(t)\}$ is continuously differentiable and satisfies

$$|x_n(t) - x_0| \leq M |t - t_0| \leq Mh \leq b, t \in I.$$

Step 3 We show that Picard iterates $\{x_n(t)\}$ converges uniformly on the interval I .

Obviously, the sequence $\{x_n(t)\}$ converges if and only if the infinite series

$$\sum_{n=1}^{\infty} [x_{n+1}(t) - x_n(t)] \quad (4.2.5)$$

converges. Using the Lipschitz condition (4.2.1), we have

$$|x_{n+1}(t) - x_n(t)| \leq \frac{M}{L} \frac{(L |t - t_0|)^{n+1}}{(n+1)!}. \quad (4.2.6)$$

In fact, if $n = 0$, one has

$$|x_1(t) - x_0(t)| \leq \left| \int_{t_0}^t |f(s, x_0(s))| ds \right| \leq \frac{M}{L} \frac{(L |t - t_0|)}{1!}.$$

Suppose that the inequality (4.2.6) holds with $n = k$, that is,

$$|x_{k+1}(t) - x_k(t)| \leq \frac{M}{L} \frac{(L |t - t_0|)^{k+1}}{(k+1)!}.$$

Then $n = k + 1$,

$$\begin{aligned} |x_{k+2}(t) - x_{k+1}(t)| &\leq \left| \int_{t_0}^t L |x_{k+1}(s) - x_k(s)| ds \right| \\ &\leq M \left| \int_{t_0}^t \frac{(L |s - t_0|)^{k+1}}{(k+1)!} ds \right| \\ &\leq \frac{M}{L} \frac{(L |t - t_0|)^{k+2}}{(k+2)!}. \end{aligned}$$

This shows that the inequality holds on the interval I . By Weierstrass's M-Test, the inequality (4.2.6) implies that the infinite series (4.2.5) converges uniformly on the interval I . So Picard iterates $\{x_n(t)\}$ converges uniformly on I .

Step 4 The initial value problem (4.1.1) has at least one solution.

We denote the limit of $\{x_n(t)\}$ by $\varphi(t)$, that is,

$$\varphi(t) = \lim_{n \rightarrow \infty} x_n(t).$$

By Lipschitz condition, we easily prove that the sequence of functions $f(t, x_n(t))$

converges uniformly to the limit function $f(t, \varphi(t))$. Then taking limit of both sides of the equality (4.2.4) gives

$$\varphi(t) = x_0 + \int_{t_0}^t f(s, \varphi(s)) ds,$$

which shows that $\varphi(t)$ is a solution of the integral equation (4.2.3). Thus $\varphi(t)$ is a solution of the initial value problem (4.1.1).

Step 5 The initial value problem (4.1.1) has only one solution on the interval I .

If $x = \varphi(t)$ and $x = \psi(t)$ are two solutions to the integral equation (4.2.3) on the interval I , then

$$|\varphi(t) - \psi(t)| = \left| \int_{t_0}^t |f(s, \varphi(s)) - f(s, \psi(s))| ds \right|.$$

Using Lipschitz condition, one has

$$|\varphi(t) - \psi(t)| \leq L \left| \int_{t_0}^t |\varphi(s) - \psi(s)| ds \right|. \quad (4.2.7)$$

Noticing that the continuity of the function $|\varphi(t) - \psi(t)|$, let K be a bound for $|\varphi(t) - \psi(t)|$ on I , then

$$|\varphi(t) - \psi(t)| \leq LK |t - t_0|. \quad (4.2.8)$$

Substitution (4.2.8) back into the right-hand side of the inequality in (4.2.7) and using induction give

$$|\varphi(t) - \psi(t)| \leq K \frac{(L |t - t_0|)^n}{n!} \leq \frac{K (Lh)^n}{n!} \text{ for any } n.$$

This implies that $\varphi(t) \equiv \psi(t)$ on I . So the initial value problem has a unique solution.

Remark 4.2.1 The choice of $h = \min\left\{a, \frac{b}{M}\right\}$ is natural. On the one hand, the requirement $h \leq a$ is necessary. On the other hand, if $x = x(t)$ is a solution of the initial value problem on $I = [t_0 - h, t_0 + h]$, then

$$|x'(t)| = |f(t, x(t))| \leq M$$

implies

$$|x(t) - x_0| \leq M |t - t_0|$$

which does not exceed b provided that $|t - t_0| \leq \frac{b}{M}$.

Remark 4.2.2 The continuity of the function $f(t, x)$ is sufficient for the existence of a solution to the initial value problem, but it does not imply uniqueness. For example, the function $f(t, x) = 6x^{\frac{2}{3}}$ is continuous in the entire tx -plane, however the initial value problem

$$\begin{cases} \frac{dx}{dt} = 6x^{\frac{2}{3}} \\ x(0) = 0 \end{cases}$$

has at least two solutions $x_1(t) \equiv 0$ and $x_2(t) = 8t^3$. The Lipschitz condition is a sufficient condition for the uniqueness of a solution to (4.1.1).

Remark 4.2.3 Suppose that $f(t, x)$ is differentiable with respect to x and $|f_x(t, x)| \leq L$. Then $f(t, x)$ satisfies the Lipschitz condition with respect to x . Indeed, by the mean value theorem, we have

$$|f(t, x_1) - f(t, x_2)| = |f_x(t, \sigma)| |x_1 - x_2| \leq L |x_1 - x_2|.$$

Especially, if the partial derivative $f_x(t, x)$ is continuous in R , then $f(t, x)$ satisfies the Lipschitz condition with respect to x . Conversely, if $f(t, x)$ satisfies the Lipschitz condition, then $f(t, x)$ is continuous with respect to x in R , but it is not necessarily differentiable with respect to x . For example, the function $f(t, x) = |x|$ is not differentiable in R^2 , but satisfies the Lipschitz condition with $L = 1$.

Remark 4.2.4 Let $x(t)$ be an n -dimensional vector-valued function. If $n = 1$, the sign $|\cdot|$ is regarded as the absolute value, otherwise, it is any norm on R^n with $n > 1$, not necessarily the Euclidean norm. Indeed, any two norms of a finite dimensional normed linear space are equivalent.

Remark 4.2.5 Notice that in Theorem 2.3.1, the initial value problem (2.3.1) of the form

$$\begin{cases} \frac{dx}{dt} = A(t)x + f(t) \\ x(t_0) = x_0 \end{cases}$$

has a unique solution on the entire interval I provided that $A(t)$ and $f(t)$ are continuous on I . This is a special case of Theorem 4.2.1. However, the initial value problem (4.1.1) exists a unique solution only on a neighborhood $[t_0 - h, t_0 + h]$ of t_0 , not necessarily on the entire interval $[t_0 - a, t_0 + a]$.

Remark 4.2.6 Beginning with an arbitrary continuous function $x_0(t)$ and calculating Picard iterates

$$x_{n+1}(t) = x_0 + \int_{t_0}^t f(s, x_n(s)) ds \quad (n = 0, 1, 2, \dots),$$

we obtain a sequence which converges uniformly on I to the solution $x(t)$ of the initial value problem. Using this iteration procedure, we can determine a numerical approximation to the solution. An estimate of the error of approximation is given by

$$|x(t) - x_n(t)| \leq \frac{M}{L} \frac{(L|t - t_0|)^{n+1}}{(n+1)!} \leq \frac{M}{L} \frac{(Lh)^{n+1}}{(n+1)!}. \quad (4.2.9)$$

Indeed, we can show easily that the inequality (4.2.9) holds by mathematical induction.

In numerical approximations, it is a good choice to start with a function $x_0(t)$ which is as close as possible to the solution. But if we know nothing about the solution, then $x_0(t) \equiv x_0$ is not a bad choice.

Example 4.2.1 Find the solution of the initial value problem

$$\begin{cases} \frac{dx}{dt} = 2tx \\ x(0) = 2 \end{cases}$$

by the method of successive approximations.

Solution We have

$$x_0(t) \equiv 2,$$

$$x_1(t) = 2 + \int_0^t 2sx_0(s)ds = 2 + 2t^2,$$

$$x_2(t) = 2 + \int_0^t 2sx_1(s)ds = 2 + \int_0^t 2s(2 + 2s^2)ds = 2 + 2t^2 + t^4,$$

$\vdots$

$$x_n(t) = 2 \left(1 + t^2 + \frac{(t^2)^2}{2!} + \cdots + \frac{(t^2)^n}{n!} \right) = 2 \sum_{k=0}^n \frac{(t^2)^k}{k!}.$$

Taking the limit of both sides of the last equation yields the particular solution

$$x(t) = 2e^{t^2}.$$

Example 4.2.2 (Gronwall's Inequality) Let $\varphi(t)$, $p(t)$ and $q(t)$ be nonnegative continuous functions in the interval $I = [t_0 - a, t_0 + a]$ and

$$\varphi(t) \leq p(t) + \left| \int_{t_0}^t q(s)\varphi(s)ds \right| \quad \text{for } |t - t_0| \leq a. \quad (4.2.10)$$

Then the following inequality holds:

$$\varphi(t) \leq p(t) + \left| \int_{t_0}^t p(s)q(s) \exp\left(\left| \int_s^t q(\tau)d\tau \right|\right) ds \right|, \quad |t - t_0| \leq a. \quad (4.2.11)$$

Proof We shall prove (4.2.11) for $t_0 \leq t \leq t_0 + a$ whereas the proof is similar for $t_0 - a \leq t \leq t_0$. We denote

$$u(t) = \int_{t_0}^t q(s)\varphi(s)ds.$$

So $u(t_0) = 0$ and $u'(t) = q(t)\varphi(t)$. The inequality (4.2.10) shows that

$$\varphi(t) \leq p(t) + u(t).$$

It follows that

$$u'(t) \leq p(t)q(t) + q(t)u(t).$$

Multiplying both sides by $\exp\left(-\int_{t_0}^t q(s)ds\right)$ gives

$$\left(u(t) \exp\left(-\int_{t_0}^t q(s)ds\right) \right)' \leq p(t)q(t) \exp\left(-\int_{t_0}^t q(s)ds\right).$$

Integrating the above inequality, we have

$$\begin{aligned} u(t) &\leq \exp\left(\int_{t_0}^t q(s)ds\right) \int_{t_0}^t p(s)q(s) \exp\left(-\int_{t_0}^s q(\tau)d\tau\right) ds \\ &= \int_{t_0}^t p(s)q(s) \exp\left(\int_s^t q(\tau)d\tau\right) ds. \end{aligned}$$

Thus, the inequality (4.2.11) follows from $\varphi(t) \leq p(t) + u(t)$. Especially, if the function $p(t) \equiv 0$ in (4.2.10), then we obtain $\varphi(t) \equiv 0$.

Example 4.2.3 Compute the first two iterates of the initial value problem

$$\begin{cases} x' = t^2 - x^2 - 1 \\ x(0) = 0 \end{cases}$$

with the initial approximation $x_0(t) \equiv 0$.

Solution Here $f(t, x) = t^2 - x^2 - 1$. We have

$$\begin{aligned}
 x_1(t) &= 0 + \int_0^t [s^2 - x_0^2(s) - 1] ds \\
 &= \int_0^t (s^2 - 1) ds \\
 &= \frac{1}{3} t^3 - t. \\
 x_2(t) &= \int_0^t [s^2 - x_1^2(s) - 1] ds \\
 &= \int_0^t \left[s^2 - \left(\frac{1}{3} s^3 - s \right)^2 - 1 \right] ds \\
 &= \int_0^t \left(-\frac{1}{9} s^6 + \frac{2}{3} s^4 - 1 \right) ds \\
 &= -\frac{1}{63} t^7 + \frac{2}{15} t^5 - t.
 \end{aligned}$$

Example 4.2.4 Show that the initial value problem

$$\begin{cases} \frac{dx}{dt} = t^2 + x^2 \\ x(t_0) = x_0 \end{cases}$$

exists a unique solution.

Proof Here $f(t, x) = t^2 + x^2$. Then the partial derivative with respect to x

$$\frac{\partial f}{\partial x} = 2x$$

is continuous on any closed rectangle domain R :

$$|t - t_0| \leq a, |x - x_0| \leq b.$$

Thus $f(t, x)$ satisfies the Lipschitz condition with respect to x on R . By Theorem 4.2.1, the initial value problem has a unique solution.

Example 4.2.5 Show that the initial value problem

$$\frac{dx}{dt} = f(t, x) = \begin{cases} 0, & x = 0, \\ x \ln |x|, & x \neq 0, \end{cases} \quad x(t_0) = x_0$$

has a unique solution, where (t_0, x_0) is any one point in tx -plane.

Solution We can verify that $f(t, x)$ satisfies the conditions in Theorem 4.2.1 provided that $x \neq 0$. Thus in such domain with $x \neq 0$, the initial value problem has exactly one solution. If $x_0 = 0$, we can obtain that $x \equiv 0$ is a solution. In addition, when $x \neq 0$, the differential equation reads

$$\frac{dx}{dt} = x \ln |x|$$

whose general solution is

$$x = \pm e^{e^t}.$$

$y = e^{e^t}$ is a general solution of the differential equation in the upper half plane, while $y = -e^{e^t}$ is a general solution in the lower half plane. These solutions are not possible to intersect with $x = 0$. Thus the differential equation with initial condition $x(t_0) = 0$ has a

unique solution $x(t) \equiv 0$.

But we can derive

$$|f(t, x) - f(t, 0)| = |x \ln |x|| = |\ln |x|| |x|.$$

Since $\lim_{x \rightarrow 0} |\ln |x|| = +\infty$, there does not exist a constant $L > 0$ such that

$$|f(t, x) - f(t, 0)| \leq L |x|.$$

Thus the function $f(t, x)$ does not satisfy the Lipschitz condition in any neighborhood containing $x = 0$.

The example indicates that the Lipschitz condition is not a necessary condition for the uniqueness of solutions.

Example 4.2.6 Find the existence interval of solution to the initial value problem

$$\begin{cases} \frac{dx}{dt} = t^2 - x^2, \\ x(-1) = 0, \end{cases}$$

on the rectangle domain $R: |t+1| \leq 1, |x| \leq 1$. Compute the first two approximate solutions $x_1(t), x_2(t)$ and give the error estimate.

Solution Here $f(t, x) = t^2 - x^2, a = 1, b = 1$. Let M be a bound for $|f(t, x)|$ on R .

We choose $M = 4$. So the existence interval is $I = \left[-\frac{5}{4}, -\frac{3}{4}\right]$, where $h = \min\left\{a, \frac{b}{M}\right\} = \frac{1}{4}$.

In addition, since $\frac{\partial f}{\partial x} = -2x$, we can take the Lipschitz constant

$$L = \max_{(t, x) \in R} \left| \frac{\partial f}{\partial x} \right| = 2.$$

The iteration relation gives

$$x_0(t) \equiv 0,$$

$$x_1(t) = 0 + \int_{-1}^t [s^2 - x_0^2(s)] ds = \int_{-1}^t s^2 ds = \frac{1}{3}t^3 + \frac{1}{3},$$

$$x_2(t) = 0 + \int_{-1}^t [s^2 - x_1^2(s)] ds = -\frac{1}{63}t^7 - \frac{1}{18}t^4 + \frac{1}{3}t^3 - \frac{1}{9}t + \frac{11}{42}.$$

By the inequality of error estimate in (4.2.9), we obtain the error estimate

$$|x_2(t) - x(t)| \leq \frac{M}{L} \frac{(Lh)^3}{3!} = \frac{1}{24} \approx 0.04.$$

Exercise 4.2

1. Prove the uniqueness of solution in Theorem 4.2.1 by Gronwall's inequality.
2. Show that the initial value problem

$$\begin{cases} x'' = f(t, x), \\ x(t_0) = x_0, x'(t_0) = x_1, \end{cases}$$

where $f(t, x)$ is continuous in a domain R containing the point (t_0, x_0) , is equivalent to the integral equation

$$x(t) = x_0 + (t - t_0)x_1 + \int_{t_0}^t (t - s)f(s, x(s))ds.$$

3. By computing appropriate Lipschitz constants, show that the following functions satisfy the Lipschitz condition in the given domains.

$$(1) x \sin y + y \cos x, R: |x| \leq a, |y| \leq b;$$

$$(2) t^2 e^{t+x}, R: |t| \leq a, |x| \leq b;$$

(3) $p(t)x + q(t)$, $R: |t| \leq 1, |x| < \infty$, where $p(t)$ and $q(t)$ are continuous in the interval $|t| \leq 1$.

4. Let $\varphi(t)$ be a nonnegative continuous function in the interval $|t - t_0| \leq b$, and $k \geq 0$ be a given constant, and

$$\varphi(t) \leq \left| \int_{t_0}^t k \varphi^a(t) dt \right|, 0 < a < 1.$$

Show that for all t in $|t - t_0| \leq b$,

$$\varphi(t) \leq [k(1 - \alpha)|t - t_0|]^{\frac{1}{1-a}}.$$

5. Compute the first three Picard iterates for the following initial value problems.

$$(1) \frac{dx}{dt} = t - x^2, x(0) = 0 \text{ with } x_0(t) \equiv 0;$$

$$(2) \frac{dx}{dt} = x^2 + t^2, x(0) = 1 \text{ with } x_0(t) \equiv 1;$$

$$(3) \frac{dx}{dt} = t^2 - x^2 - 1, x(0) = 0 \text{ with } x_0(t) \equiv t;$$

$$(4) \frac{dx}{dt} = t^2 + x^2, x(0) = 0 \text{ with } x_0(t) \equiv t.$$

6. Discuss the existence and uniqueness of solutions of the following initial value problems.

$$(1) x' = 1 + x^{\frac{2}{3}}, x(0) = 0;$$

$$(2) \frac{dx}{dt} = \frac{3}{2}x^{\frac{1}{3}}, x(0) = 0;$$

$$(3) \frac{dx}{dt} = \sqrt{|x|}, x(0) = 0;$$

$$(4) \frac{dx}{dt} = (t + x)t^2x^2, x(0) = 1.$$

7. Show that the initial value problem

$$\begin{cases} \frac{dx}{dt} = (t^2 - x^2) \sin x + x^2 \cos x, \\ x(0) = 0. \end{cases}$$

has a unique solution $x(t) \equiv 0$ in a closed domain R :

$$|t| \leq a, |x| \leq b.$$

8. Let $f(t, x)$ in (4.1.1) be continuous in $R^+: t_0 \leq t \leq t_0 + a, |x - x_0| \leq b$ and nonincreasing in x for each fixed t in $t_0 \leq t \leq t_0 + a$. Show that the initial value problem

(4.1.1) exists at most one solution in $t_0 \leq t \leq t_0 + a$.

9. Let $f(t, x)$ be continuous in R and for all $(t, x_1), (t, x_2) \in R$ it satisfies

$$|f(t, x_1) - f(t, x_2)| \leq \omega(|x_1 - x_2|),$$

where $\omega(s)$ is a continuous function and $\omega(0) = 0, \omega(s) > 0$ for $s > 0$ and the improper integral

$$\int_0^r \frac{ds}{\omega(s)} = \infty \quad (r > 0).$$

Then the initial value problem has at most one solution in $|t - t_0| \leq a$.

10. Let $f(t, x)$ be continuous and satisfy the generalized Lipschitz condition

$$|f(t, x_1) - f(t, x_2)| \leq L(t) |x_1 - x_2|$$

for all $(t, x_1), (t, x_2)$ in R , where $L(t)$ is such that the integral $\int_{t_0-a}^{t_0+a} L(s) ds$ exists. Show that the initial value problem has at most one solution in the interval

$$|t - t_0| \leq a.$$

11. Apply the iterative formula in (4.2.4) to compute the first three successive approximations to the solution of the initial value problem

$$\begin{cases} \frac{dx}{dt} = 2x - y, x(0) = 1, \\ \frac{dy}{dt} = 3x - 2y, y(0) = -1. \end{cases}$$

4.3 Extension Theorem

In this section, we shall answer the question that if a solution exists, then how long does it last. For example, let us consider the initial value problem

$$\frac{dx}{dt} = 1 + x^2, x(0) = 0. \quad (4.3.1)$$

Here $f(t, x) = 1 + x^2$ is continuous in the rectangle domain $R: |t| \leq a, |x| \leq b$. We can check easily that $f(t, x)$ satisfies the conditions of Theorem 4.2.1 in R . Indeed, we choose a bound $M = \max_{(t, x) \in R} |f(t, x)| = 1 + b^2$, the Lipschitz constant $L = 2b$ and the initial function $x_0(t) \equiv 0$. Thus, the initial value problem (4.3.1) has a unique solution in the interval

$|t| \leq h = \min\{a, \frac{b}{1+b^2}\}$. However, the optimum interval which Theorem 4.2.1 can give

is $|t| \leq \frac{1}{2}$ since

$$\frac{b}{1+b^2} \leq \frac{1}{2} \quad (\text{with equality for } b = 1).$$

While, separating variables gives the unique solution

$$x(t) = \tan t$$

which exists in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

This involves the extension of a solution. We first introduce several definitions.

Definition 4.3.1 The interval I is called a right maximal interval of existence for x provided that there does not exist an extension of $x(t)$ over the interval I_0 so that $x = x(t)$ remains a solution of the initial value problem (4.1.1), where I is a subset of I_0 and I, I_0 have different right endpoints. Similarly, we may define a left maximal interval of existence for x . A maximal interval of existence is an interval which is both a left and right maximal interval.

Remark 4.3.1 A maximal interval of existence is definitely an open interval (α, β) , where the left end point α and right endpoint β depend on the initial value $x(t_0) = x_0$.

Definition 4.3.2 The function $f(t, x)$ is said to satisfy a local Lipschitz condition with respect to x in a domain $D \subset R^2$ provided that for every point $(t_0, x_0) \in D$ there exists a neighborhood U and a Lipschitz constant $L = L(t_0, x_0)$ such that in $U \cap D$ the function f satisfies the Lipschitz condition

$$|f(t, x_1) - f(t, x_2)| \leq L |x_1 - x_2|. \quad (4.3.2)$$

Remark 4.3.2 If $f \in C(D)$ has a continuous derivative $\frac{\partial f}{\partial x}$ in an open domain D , then f satisfies a local Lipschitz condition in the set. Indeed, if U is a circular neighborhood of $(t_0, x_0) \in D$ with $\bar{U} \subset D$, then $\frac{\partial f}{\partial x}$ is bounded in U , say $\left| \frac{\partial f}{\partial x} \right| \leq L(t_0, x_0)$, and for $(t, x_1), (t, x_2) \in U$, then the mean value theorem gives

$$f(t, x_1) - f(t, x_2) = \frac{\partial f}{\partial x}(t, \bar{x})(x_1 - x_2) \quad \text{with } \bar{x} \in (x_1, x_2).$$

Thus, the inequality (4.3.2) holds.

Remark 4.3.3 A local Lipschitz condition is a weak requirement compared to the Lipschitz condition in Theorem 4.2.1 (sometimes called the global Lipschitz condition).

For instance, $f(t, x) = 2x^2$ satisfies

$$|f(t, x_1) - f(t, x_2)| = 2|x_1^2 - x_2^2| = 2|x_1 + x_2| |x_1 - x_2|.$$

Thus, f satisfies a local Lipschitz condition in R^2 , but does not satisfy a global Lipschitz condition in the entire tx -plane. Indeed, if D is a bounded and closed domain, then the local Lipschitz condition is equivalent to the global Lipschitz condition.

We shall consider the following initial value problem

$$\begin{cases} \frac{dx}{dt} = f(t, x), \\ x(t_0) = x_0, \end{cases} \quad (4.3.3)$$

where $f(t, x)$ is defined in an open domain $D \subseteq R^2$.

Our next objective is to derive some global statements on the unique extension of these local solutions.

Theorem 4.3.1 Let $f(t, x)$ be continuous and bounded in D with $M = \max_D |f(t, x)|$. If $f(t, x)$ satisfies the local Lipschitz condition with respect to x and the initial value

problem (4.3.3) has a unique solution $x(t)$ in a maximal interval of existence $I = (\alpha, \beta)$, where α and β are finite, then

(1) the limits

$$\lim_{t \rightarrow \alpha^+} x(t) = x(\alpha + 0) \quad \text{and} \quad \lim_{t \rightarrow \beta^-} x(t) = x(\beta - 0) \quad \text{exist};$$

(2) the points $[\alpha, x(\alpha + 0)]$ and $[\beta, x(\beta - 0)]$ are both on the boundary of D , say ∂D .

Proof (1) For $\alpha < t_1 < t_2 < \beta$, the equivalent integral equation

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds$$

yields that

$$|x(t_2) - x(t_1)| \leq \int_{t_1}^{t_2} |f(s, x(s))| ds \leq M(t_2 - t_1).$$

Thus, $x(t_2) - x(t_1) \rightarrow 0$ as $t_1, t_2 \rightarrow \alpha^+$. By the Cauchy criterion of convergence, $\lim_{t \rightarrow \alpha^+} x(t)$ exists. A similar argument holds for $\lim_{t \rightarrow \beta^-} x(t)$.

(2) If $[\beta, x(\beta - 0)]$ is not on ∂D , then it is an interior point of D . Thus the solution $x(t)$ can be extended to the right over the interval $(\alpha, \beta + \gamma)$ for some $\gamma > 0$. This contradicts with the maximal interval of existence (α, β) .

Theorem 4.3.2 (Extension Theorem)

Let $f(t, x) \in C(D)$ satisfy a local Lipschitz condition with respect to x in D , where $D \subset \mathbb{R}^2$ is open. Then the initial value problem (4.3.3) has a unique solution $\varphi(t)$ that cannot be extended, and that to the left and to the right comes arbitrarily close to the boundary of D .

Proof Let us define a set of the domains $D_n (n=1, 2, \dots)$ such that $(t_0, x_0) \in D_1 \subset D_2 \subset \dots \subset D_n \subset \dots \subset D$ and $\bar{D}_n \subset D_{n+1}, D_n \rightarrow D$ as $n \rightarrow +\infty$. Since $f(t, x)$ is bounded in the closed domain $\bar{D}_1$, by Theorem 4.3.1, the initial value problem exists a unique solution $\varphi(t)$ which can be extended to the right to some point $P_1 \in \partial \bar{D}_1$. Because $P_1 \in D_2$ and $f(t, x)$ is bounded in $\bar{D}_2$, applying Theorem 4.3.1, the solution can be extended to the right from the point P_1 to some one point $P_2 \in \partial \bar{D}_2$. Using similar procedure, we obtain a sequence of points $\{P_n\}, n=1, 2, \dots$. Since $D_n \rightarrow D$ as $n \rightarrow \infty$, then $\{P_n\}$ tends arbitrarily close to the boundary of D . A similar argument can be applied to the left.

Remark 4.3.4 If the maximal interval of existence of the solution $\varphi(t)$ is (α, β) , then the statement “ $\varphi(t)$ comes arbitrarily to the boundary of D to the right” is showed in the following three cases:

Case 1 $\beta = \infty$, the solution exists for all $t \geq t_0$;

Case 2 $\beta < \infty$ and $\lim_{t \rightarrow \beta^-} \sup |\varphi(t)| = \infty$, the solution is infinite;

Case 3 $\beta < \infty$ and $\lim_{t \rightarrow \beta^-} \inf \rho(t, \varphi(t)) = \infty$, where $\rho(t, x)$ denotes the distance from the point (t, x) to the boundary of D , the solution comes arbitrarily close to the boundary of D . To the left, we have also similar cases.

Remark 4.3.5 Let D be unbounded domain, for instance, $D = \mathbb{R}^2$. Then the sequence

of points $\{P_n\}$ is possible to tend to infinity, especially, the maximal interval of existence may be $(-\infty, \infty)$.

Example 4.3.1 Find the maximal interval of existence of the solution to the initial value problem

$$\begin{cases} \frac{dx}{dt} = \sqrt{1+x^2}, & |x| < +\infty, \\ x(0) = 0. \end{cases}$$

Solution Separating the variables yields the solution

$$\varphi(t) = \sinh t = \frac{e^t - e^{-t}}{2}, \quad |t| < +\infty.$$

The maximal interval of existence is $(-\infty, \infty)$. Here $f(t, x) = \sqrt{1+x^2}$ is continuous in R^2 . The result in this example corresponds to the first case.

Example 4.3.2 Find the maximal interval of existence of the solution to the initial value problem

$$\begin{cases} \frac{dx}{dt} = 2(1+x^2), & |x| < +\infty, \\ x(0) = 0. \end{cases}$$

Solution Here $f(t, x) = 2(1+x^2)$ is continuous in R^2 and satisfies the local Lipschitz condition with respect to x in R^2 . By Theorem 4.3.2, the initial value problem has a unique solution $\varphi(t)$ which can be extended to infinity.

Indeed, we separate the variables and obtain the solution

$$\varphi(t) = \tan 2t, \quad -\frac{\pi}{4} < t < \frac{\pi}{4}.$$

The maximal interval of existence is $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$. This agrees with the second case, that is,

$\beta = \frac{\pi}{4}$ is finite, and the solution $x = \tan 2t$ is infinite as t tends to $\frac{\pi}{4}$.

Example 4.3.3 Show that the maximal interval of existence of any one solution to the differential equation

$$\frac{dx}{dt} = t^2 + x^2, \quad (t, x) \in R^2$$

is finite.

Solution This is a Riccati's equation which can not be solved by the elementary integral method. Since $f(t, x)$ satisfies the conditions of Theorem 4.3.2, we obtain that this equation satisfying any initial condition has a unique solution which can be extended to infinity.

Let $x = \varphi(t)$ be a solution satisfying the initial condition $x(t_0) = x_0$ and the right maximal interval of existence be $[t_0, \beta)$. If $\beta < 0$, then $[t_0, \beta)$ is finite. If $\beta > 0$, we choose t_1 such that $0 < t_1 < \beta$ and obtain

$$\frac{dx(t)}{dt} \geq t_1^2 + x^2(t), t \in [t_1, \beta)$$

which reads

$$\frac{dx}{t_1^2 + x^2} \geq dt, t_1 \leq t < \beta.$$

Integrating both sides from t_1 to t gives

$$\frac{1}{t_1} \left[\arctan \frac{x(t)}{t_1} - \arctan \frac{x(t_1)}{t_1} \right] \geq t - t_1.$$

Thus

$$0 \leq t - t_1 \leq \frac{\pi}{t_1}, t \in [t_1, \beta).$$

Hence the right maximal interval of existence is finite. A corresponding argument holds to the left.

Example 4.3.4 If $f(t, x) \in C(R^2)$ is bounded and satisfies the local Lipschitz condition with respect to x , then the maximal interval of existence of solution to the initial value problem (4.3.3) is $(-\infty, +\infty)$. Show that this conclusion holds.

Proof By Theorem 4.3.2, any one solution of the initial value problem (4.3.3) comes arbitrarily close to the boundary of $D = R^2$. It suffices to prove that the solution is bounded in any one finite interval. Indeed, since

$$|x(t) - x_0| \leq \left| \int_{t_0}^t |f(s, x(s))| ds \right| \leq M |t - t_0|$$

where M is a bound for $|f(t, x)|$ in R^2 , we complete this proof.

Exercise 4.3

1. Show that the solution of the problem

$$\frac{dx}{dt} = -\frac{t}{x}, x(0) = 1$$

can not be extended beyond the interval $-1 < t < 1$.

2. Show that the solution of the problem

$$x' = x^2, x(0) = 2$$

is extendable only to the interval $-\infty < t < \frac{1}{2}$.

3. Show that the solution of the problem

$$x' = 2tx^2, x(0) = 1$$

exists only in the interval $|t| < 1$.

4. Find the maximal interval of existence of solutions of

$$x' + 3x^{\frac{4}{3}} \sin t = 0$$

which satisfies

$$(1) x\left(\frac{\pi}{2}\right) = 0; (2) x\left(\frac{\pi}{2}\right) = \frac{1}{8}; (3) x\left(\frac{\pi}{2}\right) = 8.$$

5. Let $f(t, x) \in C(R^2)$ satisfy the local Lipschitz condition with respect to x in R^2 and
- $$|f(t, x)| \leq N |x|, N > 0.$$

Then show that the maximal interval of existence of the solution of the initial value problem (4.3.3) is $(-\infty, +\infty)$.

6. Show that the solution of the problem

$$x' = 1 + x^2, x(0) = 1$$

can not be extended beyond the interval $-\frac{3\pi}{4} < t < \frac{\pi}{4}$.

7. Suppose that the maximal interval of existence of solution to the initial value problem

$$\begin{cases} \frac{dx}{dt} = (x^2 + x - 6)e^{(t+x)^2}, \\ x(t_0) = x_0 \end{cases}$$

is (α, β) , where (t_0, x_0) is any one point in tx -plane. Show that at least one of $\alpha = -\infty$ and $\beta = +\infty$ holds.

8. Find the maximal interval of existence of solution of the initial value problem

$$\begin{cases} \frac{dx}{dt} = x(x-1)e^{t+x}, (t, x) \in R^2, \\ x(t_0) = x_0, 0 < x_0 < 1. \end{cases}$$

4.4 Continuous Dependence on Initial Values and Parameters

The initial value problem (4.1.1) describes a model of physical problem in which some initial conditions and parameters are involved. For instance, linear single pendulum differential equation

$$\frac{d^2x}{dt^2} + a^2x = 0 \tag{4.4.1}$$

satisfies the initial conditions $x(t_0) = x_0$, $x'(t_0) = v_0$. These quantities x_0 , v_0 and a are obtained from measurements and, therefore, are not precisely known.

Based upon experience with the physical problems, we hope that the solutions are “insensitive” to small changes in these values. That is, if a small change is made in the initial values and parameters, then the solution changes only a little. Indeed, we obtain that the solution to the initial value problem is

$$x = x_0 \cos a(t - t_0) + \frac{v_0}{a} \sin a(t - t_0), \tag{4.4.2}$$

which is clearly a function of t, t_0, x_0, v_0 and the parameter a . Thus it is important to know how the solution changes when (t_0, x_0) and the parameter a are slightly altered. In this section, we shall investigate the problem of dependence of the solution on the initial values and parameters.

Definition 4.4.1 Suppose that the initial value problem (4.3.3) has a solution $x =$

$\varphi(t, t_0, x_0)$ on the interval $[a, b]$. If for any $\epsilon > 0$, there exists $\delta = \delta(\epsilon, t_0, x_0) > 0$ such that for any point (t_1, x_1) satisfying $|t_1 - t_0| < \delta$ and $|x_1 - x_0| < \delta$, the solution $x = \varphi(t, t_1, x_1)$ of the initial value problem

$$\begin{cases} \frac{dx}{dt} = f(t, x) \\ x(t_1) = x_1 \end{cases} \quad (4.4.3)$$

is defined on $[a, b]$ and

$$|\varphi(t, t_1, x_1) - \varphi(t, t_0, x_0)| < \epsilon, t \in (a, b).$$

Then the solution $\varphi(t, t_1, x_1)$ is said to be continuously dependent on the initial values t_1, x_1 at the point (t_0, x_0) .

Theorem 4.4.1 (Continuous Dependence on Initial Values) Let $f(t, x)$ be continuous in the domain D and satisfy the local Lipschitz condition with respect to x . If the interval $[a, b]$ is an interval of existence of the solution to the initial value problem (4.3.3), then the solution $\varphi(t, t_0, x_0)$ is continuous with the initial value in the interval $[a, b]$.

Proof For any given $\epsilon > 0$, we choose $0 < \delta_0 < \epsilon$ such that the closed domain

$$U: a \leq t \leq b, |x - \varphi(t, t_0, x_0)| \leq \delta_0$$

is contained in the domain D . By finite covering theorem, we can easily show that $f(t, x)$ satisfies the global Lipschitz condition on U with Lipschitz constant L .

Next, we shall choose δ such that

$$0 < \delta < \frac{\delta_0}{1+M} e^{-L(b-a)}, \quad (4.4.4)$$

where $M = \max_{(t,x) \in U} |f(t, x)|$ and the closed square domain $R: |t - t_0| \leq \delta, |x - x_0| \leq \delta$ is contained in U . By the existence and uniqueness theorem, for any point (t_1, x_1) in R , the initial value problem exists a unique solution $x = \varphi(t, t_1, x_1)$ which satisfies

$$\varphi(t, t_1, x_1) = x_1 + \int_{t_1}^t f(s, \varphi(s, t_1, x_1)) ds. \quad (4.4.5)$$

In addition

$$\varphi(t, t_0, x_0) = x_0 + \int_{t_0}^t f(s, \varphi(s, t_0, x_0)) ds.$$

Thus the difference

$$\begin{aligned} & |\varphi(t, t_1, x_1) - \varphi(t, t_0, x_0)| \\ & \leq |x_1 - x_0| + \left| \int_{t_0}^t f(s, \varphi(s, t_0, x_0)) ds - \int_{t_1}^t f(s, \varphi(s, t_1, x_1)) ds \right| \\ & \leq |x_1 - x_0| + \left| \int_{t_0}^t |f(s, \varphi(s, t_0, x_0)) - f(s, \varphi(s, t_1, x_1))| ds \right| \\ & \quad + \left| \int_{t_1}^{t_0} |f(s, \varphi(s, t_1, x_1))| ds \right| \\ & \leq (1+M)\delta + L \left| \int_{t_0}^t |\varphi(s, t_1, x_1) - \varphi(s, t_0, x_0)| ds \right|. \end{aligned}$$

Using Gronwall's inequality, we have

$$\begin{aligned}
 |\varphi(s, t_1, x_1) - \varphi(s, t_0, x_0)| &\leq (1 + M)\delta e^{L|t-t_0|} \\
 &\leq (1 + M)\delta e^{L|b-a|} \\
 &\leq \delta_0 < \varepsilon.
 \end{aligned} \tag{4.4.6}$$

Finally, we show that $\varphi(t, t_1, x_1)$ can be well defined in the interval $[a, b]$. In fact, the solution $x = \varphi(t, t_1, x_1)$ can not cross the curves $x = \varphi(t, t_0, x_0) + \varepsilon$ and $x = \varphi(t, t_0, x_0) - \varepsilon$. But the solution $x = \varphi(t, t_1, x_1)$ can be extended to the boundary of D by extension theorem. Thus, the solution $x = \varphi(t, t_1, x_1)$ is well defined in $[a, b]$. The proof is completed.

Theorem 4.4.2 (Continuous Dependence on the Right Side Function)

Let $f(t, x)$ satisfy the conditions of Theorem 4.4.1. Then the solution $x = \varphi(t)$ of the initial value problem (4.3.3) continuously depends on the right side function $f(t, x)$ in the interval $[a, b]$.

Proof Consider the following initial value problems

$$\begin{cases} \frac{dx}{dt} = f(t, x), \\ x(t_0) = x_0, \end{cases} \quad \text{and} \quad \begin{cases} \frac{dx}{dt} = f(t, x) + g(t, x), \\ x(t_0) = x_0 \end{cases}$$

where $f, g \in C(D)$ and both satisfy the local Lipschitz condition with respect to x with the common Lipschitz constant L . g is called the perturbed term and $|g| < \delta \ll 1, (t, x) \in D$. Suppose that $\varphi(t)$ and $\bar{\varphi}(t)$ are solutions of the initial value problems respectively in the common interval $[a, b]$. Then $\varphi(t)$ and $\bar{\varphi}(t)$ satisfy the corresponding equivalent integral equations respectively, that is,

$$\varphi(t) = x_0 + \int_{t_0}^t f(s, \varphi(s)) ds$$

and

$$\bar{\varphi}(t) = x_0 + \int_{t_0}^t [f(s, \bar{\varphi}(s)) + g(s, \bar{\varphi}(s))] ds.$$

Thus

$$\begin{aligned}
 |\bar{\varphi}(t) - \varphi(t)| &= \left| \int_{t_0}^t [f(s, \bar{\varphi}(s)) - f(s, \varphi(s))] ds + \int_{t_0}^t g(s, \bar{\varphi}(s)) ds \right| \\
 &\leq L \left| \int_{t_0}^t |\bar{\varphi}(s) - \varphi(s)| ds \right| + \delta |t - t_0| \\
 &\leq L \left| \int_{t_0}^t |\bar{\varphi}(s) - \varphi(s)| ds \right| + \delta(b - a).
 \end{aligned}$$

By Gronwall's inequality, we obtain

$$|\bar{\varphi}(t) - \varphi(t)| \leq \delta(b - a)e^{L|t-t_0|} \leq \delta(b - a)e^{L|b-a|}.$$

This shows that

$$|\bar{\varphi}(t) - \varphi(t)| < \varepsilon$$

provided that

$$\delta < \frac{\varepsilon}{b - a} e^{-L(b-a)} \quad \text{and} \quad |g| < \delta.$$

Theorem 4.4.3 (Continuous Dependence on the Parameters)

We investigate the initial value problem

$$\begin{cases} \frac{dx}{dt} = f(t, x, \mu), \\ x(t_0) = x_0, \end{cases} \quad (4.4.7)$$

where f is continuous in the domain $D \subset R \times R^n \times R^m$, and satisfies the local Lipschitz condition with respect to x , that is, for any $(t, x_1, \mu), (t, x_2, \mu) \in D$,

$$|f(t, x_1, \mu) - f(t, x_2, \mu)| \leq L |x_1 - x_2|, L > 0.$$

If the initial value problem (4.4.7) for the given μ_0 has a unique solution in the closed interval $[a, b]$, then the initial value problem (4.4.7) still exists a unique solution in the interval $[a, b]$ which is continuous with the parameter μ provided that

$$|\mu - \mu_0| \ll 1.$$

Proof We rewrite the differential equation in the initial value problem (4.4.7)

$$\begin{aligned} \frac{dx}{dt} &= f(t, x, \mu_0) + [f(t, x, \mu) - f(t, x, \mu_0)] \\ &= f(t, x, \mu_0) + g(t, x, \mu), \end{aligned}$$

where

$$g(t, x, \mu) = f(t, x, \mu) - f(t, x, \mu_0).$$

Since f is continuous with respect to μ , we obtain $|g| \ll 1$ if $|\mu - \mu_0| \ll 1$. Applying Theorem 4.4.2, we show that the solution continuously depends on the parameter μ .

Exercise 4.4

1. Give an example which shows that the solution of the initial value problem continuously depends on the initial value in the finite closed interval but not in infinite interval.

2. Let $x(t, \lambda)$ denote the solution of the initial value problem

$$x'(t) + p(t)x = q(t), x(t_0) = \lambda$$

in the interval $t_0 \leq t < \infty$. Show that for each fixed $t > t_0$, for each positive number ϵ , there exists a positive number δ such that

$$|x(t, \lambda + \Delta\lambda) - x(t, \lambda)| \leq \epsilon \quad \text{whenever} \quad |\Delta\lambda| \leq \delta,$$

that is, the solution $x(t, \lambda)$ is continuous with respect to the parameter λ .

Chapter 5 Qualitative Theory

In previous chapters we have used many methods to solve explicit solutions of differential equations. But when a given differential equation is difficult or impossible to solve explicitly, it often is possible to study qualitative information about general properties of its solutions. In this chapter, we will investigate the qualitative theory of ordinary differential equations.

5.1 Stability

We resume the problems investigated in Section 4.4. In contrast to the case treated there, we now study solutions defined on infinite intervals. The situation is much different when the finite interval is replaced by the infinite intervals. In this setting, continuous dependence on initial conditions is a more complicated matter than in Section 4.4.

For example, let us consider the initial value problem

$$\begin{cases} x'(t) = ax, \\ x(0) = x_0. \end{cases}$$

The unique solution of this problem is

$$x(t, 0, x_0) = x_0 e^{at}.$$

It follows that

$$\begin{aligned} |\Delta x| &= |x(t, 0, x_0 + \Delta x_0) - x(t, 0, x_0)| \\ &= |\Delta x_0| e^{at} \end{aligned}$$

for all $t \geq 0$. Thus, if $a \leq 0$, then $|\Delta x| = |\Delta x_0| e^{at} \leq \epsilon$ for all $t \geq 0$ provided $|\Delta x_0| \leq \epsilon$. But if $a > 0$, then $|\Delta x|$ becomes large for large t even for small values of $|\Delta x_0|$.

In this section, we consider the following initial value problem

$$\begin{cases} \mathbf{x}' = \mathbf{f}(t, \mathbf{x}), \\ \mathbf{x}(t_0) = \mathbf{x}_0, \end{cases} \quad (5.1.1)$$

where t is a real variable, $\mathbf{x} \in R^n$, whose unique solution is denoted by $\mathbf{x}(t, t_0, \mathbf{x}_0)$. A solution $\mathbf{x}(t, t_0, \mathbf{x}_0)$ of the initial value problem (5.1.1) existing in the interval $[t_0, \infty)$ is said to be stable if small changes in $\mathbf{x}_0$ bring only small changes in the solutions of (5.1.1) for all $t \geq t_0$. Otherwise, we say that the solution $\mathbf{x}(t, t_0, \mathbf{x}_0)$ is unstable.

In the following, we shall show a few definitions which classify various kinds of behavior of solutions.

Definition 5.1.1 A solution $\mathbf{x}(t) = \mathbf{x}(t, t_0, \mathbf{x}_0)$ of the initial value problem (5.1.1) is

said to be **stable**, if for each $\epsilon > 0$ there exists a $\delta = \delta(\epsilon, t_0) > 0$ such that $\|\Delta \mathbf{x}_0\| < \delta$ implies that

$$\|\mathbf{x}(t, t_0, \mathbf{x}_0 + \Delta \mathbf{x}_0) - \mathbf{x}(t, t_0, \mathbf{x}_0)\| < \epsilon \quad \text{for all } t \geq t_0.$$

Definition 5.1.2 A solution $\mathbf{x}(t) = \mathbf{x}(t, t_0, \mathbf{x}_0)$ of the initial value problem (5.1.1) is called **unstable** if it is not stable.

Definition 5.1.3 A solution $\mathbf{x}(t) = \mathbf{x}(t, t_0, \mathbf{x}_0)$ of the initial value problem (5.1.1) is called asymptotically stable if it is stable and there exists a $\delta_0 > 0$ such that $\|\Delta \mathbf{x}_0\| < \delta_0$ implies that

$$\|\mathbf{x}(t, t_0, \mathbf{x}_0 + \Delta \mathbf{x}_0) - \mathbf{x}(t, t_0, \mathbf{x}_0)\| \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

Definition 5.1.4 A solution $\mathbf{x}(t) = \mathbf{x}(t, t_0, \mathbf{x}_0)$ of the initial value problem (5.1.1) is called uniformly stable, if for each $\epsilon > 0$ there exists a $\delta = \delta(\epsilon) > 0$ such that $\|\Delta \mathbf{x}_0\| < \delta$ implies that

$$\|\mathbf{x}(t, t_0, \mathbf{x}_0 + \Delta \mathbf{x}_0) - \mathbf{x}(t, t_0, \mathbf{x}_0)\| < \epsilon \quad \text{for all } t \geq t_0.$$

Definition 5.1.5 A solution $\mathbf{x}(t) = \mathbf{x}(t, t_0, \mathbf{x}_0)$ of the initial value problem (5.1.1) is said to be exponentially stable if for each $\epsilon > 0$ there exists a $\delta = \delta(\epsilon) > 0$ such that $\|\Delta \mathbf{x}_0\| < \delta$ implies that

$$\|\mathbf{x}(t, t_0, \mathbf{x}_0 + \Delta \mathbf{x}_0) - \mathbf{x}(t, t_0, \mathbf{x}_0)\| < \epsilon e^{-\alpha(t-t_0)},$$

where α is a positive constant.

Remark 5.1.1 The norm $\|\cdot\|$ in these definitions is an arbitrary norm in R^n . It is easy to prove that the definitions are independent of the choice of norm.

Remark 5.1.2 The above definitions were introduced by A.M. Lyapunov in 1892 and hence a stable solution is called Lyapunov stable or stable in the sense of Lyapunov.

Example 5.1.1 Show that the trivial solution of the following system

$$\begin{cases} \frac{dx_1}{dt} = -x_2, \\ \frac{dx_2}{dt} = x_1, \end{cases}$$

is uniformly stable, but not asymptotically stable.

Solution We easily solve the system with the initial condition $(x_1(t_0), x_2(t_0)) = (x_1^0, x_2^0)$ for the explicit solution

$$\begin{cases} x_1(t) = x_1^0 \cos(t - t_0) - x_2^0 \sin(t - t_0), \\ x_2(t) = x_1^0 \sin(t - t_0) + x_2^0 \cos(t - t_0). \end{cases}$$

The direct computation yields

$$\begin{aligned} \|(x_1(t), x_2(t))\| &= \sqrt{x_1^2(t) + x_2^2(t)} \\ &= \sqrt{(x_1^0)^2 + (x_2^0)^2} = \|(x_1^0, x_2^0)\|. \end{aligned}$$

For each $\epsilon > 0$, we choose $\delta = \epsilon$ such that $\|(x_1^0, x_2^0)\| < \delta$ implies that

$$\|(x_1(t), x_2(t))\| < \epsilon \quad \text{for all } t \geq t_0.$$

This shows that the trivial solution is uniformly stable. However, it is easy to get

$\lim_{t \rightarrow +\infty} \|(x_1(t), x_2(t))\| \neq 0$. Hence, the trivial solution is not asymptotically stable.

Example 5.1.2 Show that the trivial solution of the differential equation

$$\frac{dx}{dt} = -\frac{x}{t+1}, t \geq 0$$

is uniformly stable and asymptotically stable.

Solution We readily solve (by separation of variables) for the explicit solution

$$x(t) = x(t, t_0, x_0) = x_0 \frac{t_0 + 1}{t + 1}.$$

For each $\epsilon > 0$, we choose $\delta = \epsilon$ such that $|x_0| < \delta$ implies that

$$|x(t)| \leq |x_0| < \epsilon \quad \text{for all } t \geq t_0 > 0.$$

This shows that the trivial solution is uniformly stable. Meanwhile,

$$\lim_{t \rightarrow +\infty} \|x(t, t_0, x_0)\| = 0,$$

so the trivial solution is asymptotically stable.

Example 5.1.3 Show that the trivial solution of differential equation

$$\frac{dx}{dt} = -x^3$$

is asymptotically stable but not exponentially stable.

Solution We easily obtain the solution of the differential equation

$$x(t) = x(t, t_0, x_0) = \frac{x_0}{\sqrt{1 + 2x_0^2(t - t_0)}}$$

satisfying the initial condition $x(t_0) = x_0$. When $t > t_0$, $|x(t)| \leq |x_0|$ holds. For each $\epsilon > 0$, we choose $\delta = \epsilon$ such that $|x_0| < \delta$ implies that $|x(t)| < \epsilon$ for all $t \geq t_0$. This shows that the trivial solution is uniformly stable.

For any $\alpha > 0$, we have

$$\begin{aligned} \lim_{t \rightarrow +\infty} x(t, t_0, x_0) e^{\alpha(t-t_0)} &= \lim_{t \rightarrow +\infty} \frac{x_0 e^{\alpha(t-t_0)}}{\sqrt{1 + 2x_0^2(t - t_0)}} \\ &= \lim_{t \rightarrow +\infty} \frac{x_0 e^{\alpha(t-t_0)}}{\sqrt{2x_0^2(t - t_0)}} = \infty, \end{aligned}$$

which shows that $|x(t, t_0, x_0)| < \epsilon e^{-\alpha(t-t_0)}$ does not hold. Thus the trivial solution is not exponentially stable.

Exercise 5.1

1. Show that every solution of the differential equation

$$\frac{dx}{dt} = p(t)x$$

is of the form $x(t) = x(t_0) \exp\left[\int_{t_0}^t p(s) ds\right]$ and its trivial solution $x(t) \equiv 0$ is asymptotically stable if and only if

$$\int_{t_0}^t p(s) ds \rightarrow -\infty \text{ as } t \rightarrow +\infty.$$

2. Show that every solution of the differential equation $x'(t) = 2t$ is of the form

$$x(t) = x(t_0) - t_0^2 + t^2,$$

and hence it is stable but not bounded.

3. Show that the problem of stability of the solution $\varphi(t, t_0, x_0)$ of the system

$$\frac{dx}{dt} = f(t, x)$$

can always be reduced to the problem of stability of the trivial solution of the differential system

$$\frac{dy}{dt} = g(t, y),$$

where $y = x - \varphi(t)$ and $g(t, y) = f(t, y + \varphi(t)) - f(t, \varphi(t))$.

5.2 Stability of Linear Systems

In this section, we investigate the stability of the linear system

$$\frac{dx}{dt} = \mathbf{A}(t)x + \mathbf{f}(t), \quad (5.2.1)$$

where $\mathbf{A}(t)$ is an $(n \times n)$ matrix, $\mathbf{f}(t)$ is a column vector and $\mathbf{A}(t), \mathbf{f}(t)$ are continuous in the interval $J = [t_0, \infty)$.

Firstly, we study the stability of the associated homogeneous system

$$\frac{dx}{dt} = \mathbf{A}(t)x, \quad (5.2.2)$$

and give the following results.

Theorem 5.2.1 All solutions of the linear differential system (5.2.2) are stable if and only if they are bounded.

Proof Let $\Phi(t)$ be a fundamental matrix of the differential system (5.2.2). If all solutions of (5.2.2) are bounded, then there exists a constant M such that

$$\|\Phi(t)\| \leq M \quad \text{for all } t \geq t_0.$$

For any given $\epsilon > 0$, we choose $\delta = \epsilon (M \|\Phi^{-1}(t_0)\|)^{-1}$ such that $\|\Delta x_0\| < \delta$ implies that

$$\begin{aligned} \|\mathbf{x}(t, t_0, \mathbf{x}_0 + \Delta \mathbf{x}_0) - \mathbf{x}(t, t_0, \mathbf{x}_0)\| &= \|\Phi(t)\Phi^{-1}(t_0)\Delta \mathbf{x}_0\| \\ &\leq M \|\Phi^{-1}(t_0)\| \|\Delta \mathbf{x}_0\| < \epsilon. \end{aligned}$$

This shows all solution of (5.2.2) are stable.

Conversely, if all solutions of (5.2.2) are stable, then especially the trivial solution is stable. Therefore, for any given $\epsilon > 0$, there is a $\delta = \delta(\epsilon) > 0$ such that $\|\Delta \mathbf{x}_0\| < \delta$ implies that $\|\mathbf{x}(t, t_0, \mathbf{x}_0)\| < \epsilon$ for all $t \geq t_0$. Together with

$$\mathbf{x}(t, t_0, \Delta \mathbf{x}_0) = \Phi(t)\Phi^{-1}(t_0)\Delta \mathbf{x}_0$$

we have

$$\|\mathbf{x}(t, t_0, \Delta \mathbf{x}_0)\| = \|\Phi(t)\Phi^{-1}(t_0)\Delta \mathbf{x}_0\| < \epsilon.$$

Now let $\Delta \mathbf{x}_0$ be a vector $(\delta/2)\mathbf{e}_j$, where $\mathbf{e}_j = (0, \dots, 0, 1, 0, \dots, 0)^T$, then we have

$$\|\Phi(t)\Phi^{-1}(t_0)\Delta x_0\| = \|\varphi_j(t)\| \frac{\delta}{2} < \varepsilon,$$

where $\varphi_j(t)$ is the j th column of $\Phi(t)\Phi^{-1}(t_0)$. Thus, it follows that

$$\|\Phi(t)\Phi^{-1}(t_0)\| = \max_{1 \leq j \leq n} \|\varphi_j(t)\| < \frac{2\varepsilon}{\delta}.$$

Hence, for any solution $x(t, t_0, x_0)$ of the linear differential system (5.2.2), we get

$$\|x(t, t_0, x_0)\| = \|\Phi(t)\Phi^{-1}(t_0)x_0\| < \frac{2\varepsilon}{\delta} \|x_0\|$$

which shows that all solutions of (5.2.2) are bounded.

Corollary 5.2.1 If $A(t) = A$ is a constant matrix, the real part of the multiple eigenvalues of the matrix A are negative and the real parts of the simple eigenvalues of the matrix A are nonpositive, then all solutions of the differential system

$$\frac{dx}{dt} = Ax$$

are stable.

The following result gives necessary and sufficient conditions so that all solutions of (5.2.2) are asymptotically stable.

Theorem 5.2.2 Let $\Phi(t)$ be a fundamental matrix of the differential system (5.2.2). Then all solutions of the differential system (5.2.2) are asymptotically stable if and only if

$$\|\Phi(t)\| \rightarrow 0 \quad \text{as } t \rightarrow \infty. \quad (5.2.3)$$

Proof Every solution $x(t, t_0, x_0)$ of the differential system (5.2.2) can be written as

$$x(t, t_0, x_0) = \Phi(t)\Phi^{-1}(t_0)x_0.$$

Because $\Phi(t)$ is continuous, the condition (5.2.3) implies that there is a constant N such that $\|\Phi(t)\| \leq N$ for all $t \geq t_0$. So

$$\|x(t, t_0, x_0)\| \leq N \|\Phi^{-1}(t_0)\| \|x_0\|$$

and therefore every solution of (5.2.2) is bounded. From Theorem 5.2.1 it follows that every solution of (5.2.2) is stable. Furthermore, since

$$\begin{aligned} \|x(t, t_0, x_0 + \Delta x_0) - x(t, t_0, x_0)\| &= \|\Phi(t)\Phi^{-1}(t_0)\Delta x_0\| \\ &\leq \|\Phi(t)\| \|\Phi^{-1}(t_0)\Delta x_0\| \rightarrow 0 \quad \text{as } t \rightarrow \infty, \end{aligned}$$

it follows that every solution of (5.2.2) is asymptotically stable.

Conversely, if all solutions of (5.2.3) are asymptotically stable, then, especially, the trivial solution $x(t, t_0, x_0) \equiv 0$ is asymptotically stable. Thus

$$\|x(t, t_0, x_0)\| = \|\Phi(t)\| \|\Phi^{-1}(t_0)x_0\| \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

This implies that

$$\|\Phi(t)\| \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

That is, the condition (5.2.3) holds.

Corollary 5.2.2 If $A(t) = A$ is a constant matrix and the real parts of the eigenvalues of the matrix A are negative, then all solutions of the linear differential system

$$\frac{dx}{dt} = Ax$$

are asymptotically stable.

The next result gives necessary and sufficient conditions so that all solutions of the linear differential system (5.2.2) are uniformly stable.

Theorem 5.2.3 Let $\Phi(t)$ be a fundamental matrix of the differential system (5.2.2). Then all solutions of the differential system (5.2.2) are uniformly stable if and only if the condition

$$\|\Phi(t)\Phi^{-1}(s)\| \leq K, \quad t_0 \leq s \leq t < \infty \quad (5.2.4)$$

holds, where K is a positive constant.

Proof Let $u(t) = u(t, t_0, u(t_0))$ be a solution of the differential system (5.2.2). Then for any $t_1 \geq t_0$, we have

$$u(t) = \Phi(t)\Phi^{-1}(t_1)u(t_1).$$

If $v(t) = \Phi(t)\Phi^{-1}(t_1)v(t_1)$ is any other solution, and the condition (5.2.4) holds, then we have

$$\|v(t) - u(t)\| \leq \|\Phi(t)\Phi^{-1}(t_1)\| \|v(t_1) - u(t_1)\| \leq K \|v(t_1) - u(t_1)\|$$

for all $t_0 \leq t_1 \leq t < \infty$.

Hence, if for each $\epsilon > 0$, then $t_1 \geq t_0$ and $\|v(t_1) - u(t_1)\| \leq \delta(\epsilon) = \frac{\epsilon}{K} > 0$ imply that

$$\|v(t) - u(t)\| \leq \epsilon.$$

This shows that the solution $u(t)$ is uniformly stable.

Conversely, if all solutions of the differential system (5.2.2) are uniformly stable, then, especially, the trivial solution $u(t, t_0, 0) \equiv 0$ is uniformly stable. Hence, for any given $\epsilon > 0$, there exists a $\delta = \delta(\epsilon) > 0$ such that $t_1 \geq t_0$ and $\|v(t_1)\| < \delta$ imply that

$$\|v(t)\| \leq \epsilon \quad \text{for all } t \geq t_1.$$

Thus, we have

$$\|\Phi(t)\Phi^{-1}(t_1)v(t_1)\| < \epsilon \quad \text{for all } t \geq t_1.$$

The rest of the proof is the same as the one of Theorem 5.2.1.

Finally, we provide a result about the stability for the linear nonhomogeneous differential system (5.2.1).

Theorem 5.2.4 If the trivial solution of the homogeneous system (5.2.2) is stable, then every solution of the nonhomogeneous system (5.2.1) is also stable. A corresponding theorem holds for asymptotic stability and instability.

Since the definition of stability deals with the difference $\varphi(t) = x_1(t) - x_2(t)$ of two solutions to the nonhomogeneous system (5.2.1) and further the difference is a solution to the corresponding homogeneous equation, the conclusion follows immediately.

The result shows that when dealing with stability problems for the linear differential system (5.2.1), it is sufficient to investigate the stability of the trivial solution of the corresponding homogeneous system (5.2.2).

Exercise 5.2

1. Show that Corollary 5.2.1 holds.

2. Give the proof of Corollary 5.2.2.
3. In the differential system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t)$$

let $\mathbf{A}(t)$ and $\mathbf{f}(t)$ be continuous in $[t_0, \infty)$. Show the following:

- (1) If all solutions are bounded in $[t_0, \infty)$, then they are stable.
- (2) If all solutions are stable and one is bounded, then all solutions are bounded in $[t_0, \infty)$.
4. Uniform stability does not imply asymptotic stability. Give an example to illustrate it.
5. Give an example which shows that asymptotic stability does not imply uniform stability.

5.3 The Method of Lyapunov

The Russian mathematician and engineer A.M. Lyapunov (1857—1918) introduced in his dissertation of 1892 two methods for dealing with stability problems. While the first method is of a special nature, his second method, that is, Lyapunov's direct method has developed into an extraordinarily helpful tool. The method uses a generalized energy function to study the stability of the solutions of the differential systems. The function is called Lyapunov function and generally denoted by V . The main advantage of this method is that the stability can be studied without any prior knowledge of the solutions.

In the section, we consider the following real autonomous differential system

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}), \tag{5.3.1}$$

where $\mathbf{f}(\mathbf{x})$ is continuous in the open set $D \subset R^n$, $\mathbf{0} \in D$, and $\mathbf{f}(\mathbf{0}) = \mathbf{0}$. In this section, we always assume that $\mathbf{f}$ satisfies the local Lipschitz condition with respect to $\mathbf{x}$ in D . Therefore the solution $\mathbf{x}(t)$ of (5.3.1) with the initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$ is uniquely determined and will be denoted by $\mathbf{x}(t, t_0, \mathbf{x}_0)$. In addition, the functions, vectors, and matrices that appear have real-valued components.

Let $V(\mathbf{x})$ be a scalar continuously differentiable function defined in D , that is, $V \in C^1(D)$ and $V(\mathbf{0}) = 0$. In the following, we shall give some definitions.

Definition 5.3.1 $V(\mathbf{x})$ is called positive definite in D if and only if $V(\mathbf{x}) > 0$ for $\mathbf{x} \neq \mathbf{0}, \mathbf{x} \in D$.

Definition 5.3.2 $V(\mathbf{x})$ is called positive semidefinite in D if $V(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in D$.

Definition 5.3.3 $V(\mathbf{x})$ is called negative definite (negative semidefinite) in D if and only if $-V(\mathbf{x})$ is positive definite (positive semidefinite) in D .

Definition 5.3.4 $V(\mathbf{x})$ is called indefinite in D if for any neighborhood $U(0, \delta) \subset D$,

$V(\mathbf{x})$ has both positive values and negative values in $U(0, \delta)$.

Example 5.3.1 The function $V(x_1, x_2) = 2x_1^2 + 3x_2^2$ is positive definite in $D = \mathbb{R}^2$.

Example 5.3.2 The function $V(x_1, x_2, x_3) = x_1^2 + x_2^2$ is positive semidefinite in $D = \mathbb{R}^3$ since it vanishes on the x_3 -axis.

Example 5.3.3 The function $V(x_1, x_2, x_3) = 2x_1^2 + (x_2 + 2x_3)^2$ is positive semidefinite in $D = \mathbb{R}^3$ because it vanishes not only at the origin but also on the line $x_1 = 0, x_2 + 2x_3 = 0$.

Example 5.3.4 The function $V(x_1, x_2) = x_1^2 + x_2^2 + x_1^4 - x_2^4$ is positive definite in the domain $D_1 = \{(x_1, x_2) \mid x_1^2 + x_2^2 < \frac{1}{2}\} \subset \mathbb{R}^2$ but not positive definite in $D_2 = \{(x_1, x_2) \mid x_1^2 + x_2^2 < 2\} \subset \mathbb{R}^2$.

Example 5.3.5 The function $V(x_1, x_2) = x_1^2 - x_2^2$ is indefinite in $D = \mathbb{R}^2$.

Given a real-valued function $V \in C^1(D)$, we use the chain rule to obtain

$$\begin{aligned} \frac{dV(\mathbf{x})}{dt} &= \dot{V}(\mathbf{x}) = \left(\text{grad } V(\mathbf{x}), \frac{d\mathbf{x}}{dt} \right) \\ &= (\text{grad } V(\mathbf{x}), \mathbf{f}(\mathbf{x})) \\ &= \frac{\partial V}{\partial x_1} f_1(\mathbf{x}) + \frac{\partial V}{\partial x_2} f_2(\mathbf{x}) + \cdots + \frac{\partial V}{\partial x_n} f_n(\mathbf{x}) \\ &= \sum_{i=1}^n \frac{\partial V}{\partial x_i} f_i(\mathbf{x}). \end{aligned} \quad (5.3.2)$$

Thus, the derivative of $V(\mathbf{x})$ with respect to t along the solution $\mathbf{x}(t)$ of the autonomous system (5.3.1) is known, although we do not have the explicit solution. It can be used to obtain information about the behavior of V along a solution without prior knowledge of the solution. In the “direct method”, the idea is exploited.

Next we shall provide three theorems regarding the stability, asymptotic stability and instability of the trivial solution of the differential system (5.3.1).

Theorem 5.3.1 Given a neighborhood $S_\rho = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| < \rho\} \subset D$, if there exists a positive definite scalar function $V(\mathbf{x}) \in C^1(S_\rho)$ such that $\dot{V}(\mathbf{x}) \leq 0$ in S_ρ , then the trivial solution of the differential system (5.3.1) is stable.

Proof Let $\epsilon > 0$ be chosen so small that the closed ball $B_\epsilon = \{\mathbf{x} \mid \|\mathbf{x}\| \leq \epsilon\}$ lies in S_ρ . We take

$$m = \min_{\|\mathbf{x}\| = \epsilon} V(\mathbf{x}) > 0$$

and then choose a δ with $0 < \delta < \epsilon$ such that

$$V(\mathbf{x}) < m \quad \text{for } \|\mathbf{x}\| < \delta.$$

If $\mathbf{x}(t, t_0, \mathbf{x}_0)$ is a solution of (5.3.1) with $\|\mathbf{x}_0\| < \delta$, then by $\dot{V}(\mathbf{x}) \leq 0$, we have

$$V(\mathbf{x}(t)) \leq V(\mathbf{x}_0) < m \quad \text{for all } t \geq t_0.$$

Since $V(\mathbf{x})$ only takes on values $\geq m$ on the sphere $\|\mathbf{x}\| = \epsilon$, it follows that $\|\mathbf{x}\| < \epsilon$ for $t \geq t_0$. This shows that the zero solution is stable.

Theorem 5.3.2 If there exists a positive definite scalar function $V(\mathbf{x}) \in C^1(S_\rho)$ such

that $\dot{V}(\mathbf{x})$ is negative definite in S_ρ , then the trivial solution of the differential system (5.3.1) is asymptotically stable.

Proof Since all the conditions of Theorem 5.3.1 are satisfied, the trivial solution of (5.3.1) is stable. It suffices to prove

$$\lim_{t \rightarrow +\infty} \|\mathbf{x}(t)\| = 0.$$

If $\mathbf{x}(t)$ is a solution as defined in Theorem 5.3.1, then $\lim_{t \rightarrow +\infty} V(\mathbf{x}(t)) = \alpha < m$. We first show that $\alpha = 0$. Let us assume that this is not the case. Then the set

$$\Omega = \{\mathbf{x} \in B_\epsilon : \alpha \leq V(\mathbf{x}) \leq m\}$$

is a compact subset of $B_\epsilon \setminus \{0\}$ and

$$\max\{\dot{V}(\mathbf{x}) : \mathbf{x} \in \Omega\} = -\beta < 0.$$

Since the solution $\mathbf{x}(t)$ stays in Ω , we have

$$\frac{dV(\mathbf{x}(t))}{dt} \leq -\beta, t \geq t_0. \quad (5.3.3)$$

Integrating both sides of (5.3.3) from t_0 to t , we get

$$V(\mathbf{x}(t)) \leq V(\mathbf{x}_0) - \beta(t - t_0).$$

The last inequality shows that

$$\lim_{t \rightarrow +\infty} V(\mathbf{x}(t)) = -\infty$$

which contradicts with $V(\mathbf{x}(t)) \geq \alpha > 0$. Thus the trivial solution is asymptotically stable.

Theorem 5.3.3 If there exists a scalar function $V(\mathbf{x}) \in C^1(S_\rho)$ such that $\dot{V}(\mathbf{x})$ is positive definite in S_ρ but $V(\mathbf{x})$ is not negative semidefinite, then the trivial solution of the differential system (5.3.1) is unstable.

Proof We assume that the trivial solution of the differential system (5.3.1) is stable. That is, for any $\epsilon > 0$, there exists a $\delta > 0$, such that $\|\mathbf{x}_0\| < \delta$ implies $\|\mathbf{x}(t, t_0, \mathbf{x}_0)\| < \epsilon$ for all $t \geq t_0$. Since $\dot{V}(\mathbf{x})$ is positive definite in S_ρ , $V(\mathbf{x}(t, t_0, \mathbf{x}_0))$ is an increasing function with respect to t . Because $V(\mathbf{x})$ is not negative semidefinite, there exists a $\mathbf{x}_0$ with $\|\mathbf{x}_0\| < \delta$ such that $V(\mathbf{x}_0) > 0$. Hence, we have

$$V(\mathbf{x}(t, t_0, \mathbf{x}_0)) > V(\mathbf{x}_0) > 0, t \geq t_0.$$

Since $V(\mathbf{x})$ is continuous and $V(\mathbf{0}) = 0$, there exists a $\eta > 0$ ($\eta < \epsilon$) such that

$$\|\mathbf{x}(t, t_0, \mathbf{x}_0)\| \geq \eta, t \geq t_0.$$

From $\dot{V}(\mathbf{x}) > 0$, there exists a positive constant $\beta > 0$ such that $\eta \leq \|\mathbf{x}\| \leq \epsilon$ implies that

$$\dot{V}(\mathbf{x}(t, t_0, \mathbf{x}_0)) \geq \beta, t \geq t_0. \quad (5.3.4)$$

Integrating both sides of the inequality (5.3.4), we have

$$\begin{aligned} V(\mathbf{x}(t, t_0, \mathbf{x}_0)) &= V(\mathbf{x}_0) + \int_{t_0}^t \dot{V}(\mathbf{x}(t, t_0, \mathbf{x}_0)) dt \\ &\geq V(\mathbf{x}_0) + \beta(t - t_0), \end{aligned}$$

which shows that

$$\lim_{t \rightarrow +\infty} V(\mathbf{x}(t, t_0, \mathbf{x}_0)) = +\infty.$$

This contradicts with the fact that $V(\mathbf{x})$ is continuous and $\|\mathbf{x}(t, t_0, \mathbf{x}_0)\| < \epsilon$. Hence, the trivial solution of (5.3.1) is unstable.

Now we shall provide some examples.

Example 5.3.6 Discuss the stability of the trivial solution of the differential equation

$$x''(t) + x'(t) + x = 0.$$

Solution Making substitutions $x_1 = x, x_2 = x'$, we obtain the equivalent system

$$\begin{cases} \frac{dx_1}{dt} = x_2, \\ \frac{dx_2}{dt} = -x_1 - x_2. \end{cases}$$

We choose

$$V(x_1, x_2) = 3x_1^2 + 2x_1x_2 + 2x_2^2.$$

Direct computation yields

$$\dot{V}(x_1, x_2) = -2(x_1^2 + x_2^2).$$

Since $V(x_1, x_2)$ is positive definite in R^2 and $\dot{V}(x_1, x_2)$ is negative definite in R^2 , by Theorem 5.3.2, we get that the trivial solution is asymptotically stable.

Example 5.3.7 Determine the stability of the trivial solution of the following differential system

$$\begin{cases} \frac{dx_1}{dt} = x_2 + x_1(a^2 - x_1^2 - x_2^2), \\ \frac{dx_2}{dt} = -x_1 + x_2(a^2 - x_1^2 - x_2^2). \end{cases}$$

Solution We consider the positive definite function $V(x_1, x_2) = x_1^2 + x_2^2$ in $D = R^2$. A simple computation gives

$$\dot{V}(x_1, x_2) = -2(x_1^2 + x_2^2)(x_1^2 + x_2^2 - a^2).$$

When $a = 0$, $\dot{V}(x_1, x_2)$ is negative definite and hence the trivial solution is asymptotically stable. When $a \neq 0$, $\dot{V}(x_1, x_2)$ is positive definite in the region $x_1^2 + x_2^2 < a^2$. Therefore, the trivial solution of this differential system is unstable.

Example 5.3.8 Determine the stability of the trivial solution of the differential system

$$\begin{cases} \frac{dx_1}{dt} = x_2, \\ \frac{dx_2}{dt} = -f(x_1) - g(x_2), \end{cases}$$

where $f(x_1)$ and $g(x_2)$ are continuous and satisfy the following conditions

$$(1) f(0) = 0, x_1 f(x_1) > 0 (x_1 \neq 0),$$

$$(2) \quad g(0) = 0, \quad x_2 g(x_2) > 0 (x_2 \neq 0).$$

Solution We choose

$$V(x_1, x_2) = \frac{1}{2}x_2^2 + \int_0^{x_1} f(x_1) dx_1.$$

The first condition implies that $V(x_1, x_2)$ is positive definite. A simple computation gives

$$\dot{V}(x_1, x_2) = -x_2 g(x_2).$$

From the second condition, it follows that $\dot{V}(x_1, x_2)$ is negative semidefinite. Therefore the trivial solution of the differential system is stable.

Exercise 5.3

1. Determine the stability of the trivial solution of the differential system

$$\begin{cases} \frac{dx_1}{dt} = -x_1 x_2^4, \\ \frac{dx_2}{dt} = x_2 x_1^4. \end{cases}$$

2. For each of the following systems construct a Lyapunov function of the form $c_1 x_1^2 + c_2 x_2^2$ to determine the trivial solution is stable, asymptotically stable or unstable.

$$(1) \quad \begin{cases} \frac{dx_1}{dt} = -x_1 + e^{x_1} x_2, \\ \frac{dx_2}{dt} = -e^{x_1} x_1 - x_2. \end{cases}$$

$$(2) \quad \begin{cases} \frac{dx_1}{dt} = -x_1^3 + x_1^2 x_2, \\ \frac{dx_2}{dt} = -x_1^3 - x_1^2 x_2. \end{cases}$$

$$(3) \quad \begin{cases} \frac{dx_1}{dt} = 2x_1 x_2 + x_1^3, \\ \frac{dx_2}{dt} = -x_1^2 + x_2^5. \end{cases}$$

$$(4) \quad \begin{cases} \frac{dx_1}{dt} = x_1^3 - x_2, \\ \frac{dx_2}{dt} = x_1 + x_2^3. \end{cases}$$

3. Consider the equation of undamped simple pendulum $x'' + \omega^2 \sin x = 0$, $-\frac{\pi}{2} \leq x \leq$

$\frac{\pi}{2}$. By means of an appropriate Lyapunov function, prove that its trivial solution is stable.

4. Prove by means of a Lyapunov function that the trivial solution of the Van der Pol equation

$$x'' + \mu(x^2 - 1)x' + x = 0$$

is asymptotically stable when $\mu > 0$.

5. For the following system construct a Lyapunov function of the form $ax^2 + by^2 + cz^2$ to determine the stability of the trivial solution

$$\begin{cases} \frac{dx}{dt} = 2y + yz - x^3, \\ \frac{dy}{dt} = -x + xz - y^3, \\ \frac{dz}{dt} = xy - z^3. \end{cases}$$

5.4 Basic Concepts of Two-Dimensional Autonomous Systems

The differential system

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(t, \mathbf{x}), t \in R, \mathbf{x} \in R^n$$

is called **autonomous** if the function $\mathbf{f}(t, \mathbf{x})$ does not depend explicitly on the independent variable t . Otherwise, we call it **non-autonomous**. Thus an autonomous system has the form

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}).$$

In this section, we consider the following two-dimensional autonomous system

$$\begin{cases} \frac{dx}{dt} = P(x, y), \\ \frac{dy}{dt} = Q(x, y). \end{cases} \quad (5.4.1)$$

Throughout, we shall assume that the functions $P(x, y)$ and $Q(x, y)$ are continuous in the region $D \subset R^2$ and satisfy the conditions of existence and uniqueness of solutions to the initial value problem.

The main interest in studying (5.4.1) is twofold:

- I. The systems govern a large number of dynamic processes in applied sciences.
- II. From the geometrical point of view, we can illustrate the qualitative behavior of its solutions.

For the two-dimensional autonomous system (5.4.1), we give the following definitions.

Definition 5.4.1 A solution of $(x(t), y(t))$ of the two dimensional system (5.4.1) may be regarded as a parametrization of a solution curve. The curve with t as the parameter is said to be a **trajectory** or an orbit of (5.4.1), and the xy plane is called the **phase plane**.

Definition 5.4.2 Any two trajectories are either disjoint or identical. Each point of D belongs to exactly one trajectory. The collection of all trajectories of a system is said to be the **phase portrait** of the system.

Definition 5.4.3 Any point $(x_0, y_0) \in D$ at which both $P(x, y)$ and $Q(x, y)$ vanish simultaneously, that is, $P(x_0, y_0) = Q(x_0, y_0) = 0$, is called a **critical point** of the system (5.4.1). A critical point is also referred to as a point of equilibrium or singular point. Further, a critical point is called **isolated** if there exists no other critical point in some neighborhood of (x_0, y_0) .

In fact, if (x_0, y_0) is a critical point of (5.4.1), then $x(t) = x_0, y(t) = y_0$ is a solution of (5.4.1). The corresponding orbit is the singleton $\{(x_0, y_0)\}$.

Definition 5.4.4 If $(x(t), y(t))$ is a solution of (5.4.1) and $x(t + T) = x(t), y(t + T) = y(t)$ for some $T > 0$, then the solution $[x(t), y(t)]$ is called **periodic** with period T .

Obviously, a nonconstant, continuous, periodic function has a smallest period $\omega > 0$, also known as the minimal period.

Example 5.4.1 For the differential system

$$\begin{cases} \frac{dx}{dt} = ax + by, \\ \frac{dy}{dt} = cx + dy, ad - bc \neq 0, \end{cases} \quad (5.4.2)$$

there exists only one critical point $(0, 0)$ in $D = \mathbb{R}^2$.

Example 5.4.2 For the differential system

$$\begin{cases} \frac{dx}{dt} = -y, \\ \frac{dy}{dt} = x, \end{cases} \quad (5.4.3)$$

the particular solution $x = \cos t, y = \sin t$ is a cylindrical spiral in three dimensional space (t, x, y) which passes through the point $(0, 1, 0)$. In addition, for any constant c , $x = \cos(t + c), y = \sin(t + c)$ is also a solution of (5.4.3) whose integral curve or solution curve passes through the point $(-c, 1, 0)$. However, they represent the same trajectory, that is, the circle $x^2 + y^2 = 1$ in phase plane $\mathbb{R}^2$.

For the autonomous differential system (5.4.1), the following properties are fundamental.

Theorem 5.4.1 If $(x(t), y(t))$ is a solution of the differential system (5.4.1), then for any constant c , $(x(t + c), y(t + c))$ is also a solution of the differential system (5.4.1).

Proof Since

$$\begin{aligned} \frac{dx(t + c)}{dt} &= \frac{dx(t + c)}{d(t + c)} = P(x(t + c), y(t + c)), \\ \frac{dy(t + c)}{dt} &= \frac{dy(t + c)}{d(t + c)} = Q(x(t + c), y(t + c)) \end{aligned}$$

it follows that $(x(t + c), y(t + c))$ is also a solution of (5.4.1).

Obviously, any one trajectory of (5.4.1) corresponds to infinite solution of (5.4.1). But the above property does not usually hold for nonautonomous differential system. For example, for the differential system

$$\begin{cases} \frac{dx}{dt} = x, \\ \frac{dy}{dt} = tx, \end{cases} \quad (5.4.4)$$

$x(t) = e^t, y(t) = te^t - e^t$ is a solution of (5.4.4), but $y'(t+c) = (t+c)e^{t+c} \neq tx(t+c)$ unless $c = 0$.

Theorem 5.4.2 Through each point $(x_0, y_0) \in D$ there exists one and only one trajectory of the differential system (5.4.1).

Proof On the contrary, we shall assume that there are two different trajectories $(x_1(t), y_1(t))$ and $(x_2(t), y_2(t))$ passing through (x_0, y_0) . That is, $x_1(t_0) = x_0 = x_2(t_1)$ and $y_1(t_0) = y_0 = y_2(t_1)$. It follows that $t_0 \neq t_1$ by the unique of solutions of the initial value problems. By Theorem 5.4.1, $x^*(t) = x_1(t - t_1 + t_0), y^*(t) = y_1(t - t_1 + t_0)$ is also a solution of the differential system (5.4.1). Notice $x^*(t_1) = x_1(t_0) = x_0 = x_2(t_1)$ and $y^*(t_1) = y_1(t_0) = y_0 = y_2(t_1)$. Therefore, by the unique of solutions of the initial value problems, we have

$$x^*(t) = x_2(t) \quad \text{and} \quad y^*(t) = y_2(t).$$

Thus, $(x_1(t), y_1(t))$ and $(x_2(t), y_2(t))$ are the same trajectory.

Let us denote

$$x(t, x_0, y_0) = x(t, 0, x_0, y_0), y(t, x_0, y_0) = y(t, 0, x_0, y_0).$$

Theorem 5.4.3 If $(x(t, x_0, y_0), y(t, x_0, y_0))$ is a solution of the differential system (5.4.1), then it satisfies that

$$\begin{aligned} x(t_2, x(t_1, x_0, y_0), y(t_1, x_0, y_0)) &= x(t_1 + t_2, x_0, y_0), \\ y(t_2, x(t_1, x_0, y_0), y(t_1, x_0, y_0)) &= y(t_1 + t_2, x_0, y_0). \end{aligned}$$

Proof By Theorem 5.4.1, $(x(t + t_1, x_0, y_0), y(t + t_1, x_0, y_0))$ is also a solution of the differential system. Further, it is obvious that the solutions $(x(t, x_1, y_1), y(t, x_1, y_1))$ and $(x(t + t_1, x_0, y_0), y(t + t_1, x_0, y_0))$ have same initial condition $(x_1, y_1) = (x(t_1, x_0, y_0), y(t_1, x_0, y_0))$ at time $t = 0$. Hence, from the uniqueness of solution of the initial value problems, we find that

$$x(t + t_1, x_0, y_0) = x(t, x_1, y_1)$$

and

$$y(t + t_1, x_0, y_0) = y(t, x_1, y_1).$$

We choose $t = t_2$ in last two identities and obtain the desired conclusion.

From the geometrical point of view, Theorem 5.4.3 shows that if a trajectory starting from $P_0(x_0, y_0)$ arrives at the point $P_1(x_1, y_1) = (x(t_1, x_0, y_0), y(t_1, x_0, y_0))$ after the time t_1 and further reaches the point $P_2 = (x(t_2, x_1, y_1), y(t_2, x_1, y_1))$ again after the time t_2 , then the trajectory starting from $P_0(x_0, y_0)$ also arrives at P_2 after the time $(t_1 + t_2)$.

If the initial point (x_0, y_0) is not a critical point, then the associated trajectory is a curve in the phase plane along which the point $(x(t), y(t))$ moves as t increases. We can exhibit qualitatively the behavior of solutions of the differential system (5.4.1) by

sketching its phase portrait (a collection of its critical points and typical nondegenerate trajectories).

In fact, under rather general hypotheses it can be shown that there are five possibilities for a trajectory of the two dimensional autonomous system (5.4.1). The five possibilities are these:

1. critical point (x_0, y_0) .
2. $(x(t), y(t))$ approaches a critical point as $t \rightarrow \infty$.
3. $(x(t), y(t))$ is unbounded with increasing t .
4. $(x(t), y(t))$ is a periodic solution with a closed trajectory.
5. $(x(t), y(t))$ spirals toward a closed trajectory as $t \rightarrow \infty$.

As a consequence, the qualitative nature of the phase portrait of an autonomous system is determined largely by the locations of its critical points and by the behavior of its trajectory near its critical points.

Exercise 5.4

1. Find the critical points of the given autonomous system

$$\begin{cases} \frac{dx_1}{dt} = x_2, \\ \frac{dx_2}{dt} = -a \sin x_1, \end{cases}$$

where $(x_1, x_2) \in D = \mathbb{R}^2$.

2. Let $(x(t), y(t))$ be a solution of the differential system (5.4.1). Prove if $x(t_0 + \omega) = x(t_0)$, $y(t_0 + \omega) = y(t_0)$ for some $\omega > 0$, then $(x(t), y(t))$ is a periodic solution of period ω , that is, $x(t + \omega) = x(t)$, $y(t + \omega) = y(t)$ for all $t \geq t_0$.

3. Let $(x(t), y(t))$ be a solution of the differential system (5.4.1). Prove the following:

(1) If $(x(t), y(t))$ is periodic of period ω , then the trajectory of this solution is a closed curve in the xy -plane.

(2) If a trajectory of $(x(t), y(t))$ is a closed curve containing no critical points of the differential system (5.4.1), then the solution is periodic.

4. Prove that all solutions of the following differential equations (systems) are periodic.

(1) $x'' + a^2x + bx^3 = 0$, $b > 0$.

(2) $x'' + e^{x^2} - 1 = 0$.

(3)
$$\begin{cases} \frac{dx}{dt} = x_2(1 + x_1^2 + x_2^2), \\ \frac{dy}{dt} = -2x_1(1 + x_1^2 + x_2^2). \end{cases}$$

5. Consider the Van der Pol equation

$$x'' + \mu(x^2 - 1)x' + x = 0$$

which is equivalent to the system

$$\begin{cases} \frac{dx_1}{dt} = x_2, \\ \frac{dx_2}{dt} = \mu(1 - x_1^2)x_2 - x_1, \end{cases}$$

where $x_1 = x, x_2 = x'$. Prove that $(0,0)$ is the only critical point of the system.

5.5 Critical Points of Two-Dimensional Linear Autonomous Systems

In this section, we consider the following two dimensional linear autonomous systems

$$\begin{cases} \frac{dx}{dt} = ax + by, \\ \frac{dy}{dt} = cx + dy, \end{cases} \quad (5.5.1)$$

with constant coefficient matrix $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. We assume that $q = \det \mathbf{A} = ad - bc \neq 0$. This implies that the origin $(0,0)$ is a unique critical point in $D = \mathbb{R}^2$.

In order to investigate the behavior of its trajectories near the critical point $(0,0)$, we need to change the differential system into the normal form. We firstly make use of a result from linear algebra without proof. It says that for constant coefficient matrix $\mathbf{A}$ there exists a nonsingular matrix $\mathbf{T}$ such that $\mathbf{J} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T}$ has the Jordan normal form.

Thus by using the nonsingular linear transformation $\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{T} \begin{pmatrix} u \\ v \end{pmatrix}$, the differential system (5.5.1) changes into

$$\frac{d}{dt} \begin{pmatrix} u \\ v \end{pmatrix} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T} \begin{pmatrix} u \\ v \end{pmatrix} = \mathbf{J} \begin{pmatrix} u \\ v \end{pmatrix}. \quad (5.5.2)$$

The four possibilities of Jordan normal form $\mathbf{J}$ are the following

$$\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \begin{bmatrix} \lambda_1 & 0 \\ 1 & \lambda_1 \end{bmatrix}, \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_1 \end{bmatrix}, \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix},$$

where λ_1 and λ_2 are two distinct real eigenvalues of $\mathbf{A}$, that is, two distinct real roots of the characteristic equation

$$L(\lambda) \triangleq \begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} = \lambda^2 + p\lambda + q = 0.$$

Here $p = -(a + d), q = |\mathbf{A}| = ad - bc$, $\alpha \pm \beta i$ is a pair of complex conjugate roots of $L(\lambda) = 0$. The phase portrait of the differential system (5.5.2) depends on whether the two nonzero eigenvalues λ_1 and λ_2 of $\mathbf{A}$ are

- (1) real and distinct with the same sign;
- (2) real and distinct with opposite signs;

- (3) real and equal;
- (4) complex conjugates with nonzero real part;
- (5) pure imaginary numbers.

For this, there are five different cases which must be studied separately.

Case 1 $q > 0, p^2 - 4q > 0$. λ_1 and λ_2 are real and distinct with same sign.

In this case, the differential system (5.5.2) takes the form

$$\begin{cases} \frac{du}{dt} = \lambda_1 u, \\ \frac{dv}{dt} = \lambda_2 v, \end{cases} \quad (5.5.3)$$

whose general solution is

$$u = c_1 e^{\lambda_1 t}, v = c_2 e^{\lambda_2 t}. \quad (5.5.4)$$

From the stability theory, the solution (5.5.4) implies that when $p > 0$, that is, $\lambda_1 < 0$, $\lambda_2 < 0$, the critical point $(0,0)$ is asymptotically stable; when $p < 0$, that is, $\lambda_1 > 0$, $\lambda_2 > 0$, the critical point $(0,0)$ is unstable.

Further, we study the behavior of its trajectories near the critical point $(0,0)$. From the general solution (5.5.4), it follows that all trajectories of the system (5.5.3) can be written as

$$v = c |u|^{\frac{\lambda_2}{\lambda_1}} \quad \text{and} \quad u = 0.$$

If $u = 0$, then it lies on the v -axis, whereas if $v = 0$, this trajectory lies on the u -axis. The trajectories $v = c |u|^{\frac{\lambda_2}{\lambda_1}}$ are tangent at $(0,0)$ to the u -axis if $\frac{\lambda_2}{\lambda_1} > 1$, to the v -axis if

$0 < \frac{\lambda_2}{\lambda_1} < 1$. Here the critical point $(0,0)$ is called a **node**, as shown in Figure 5.5.1.

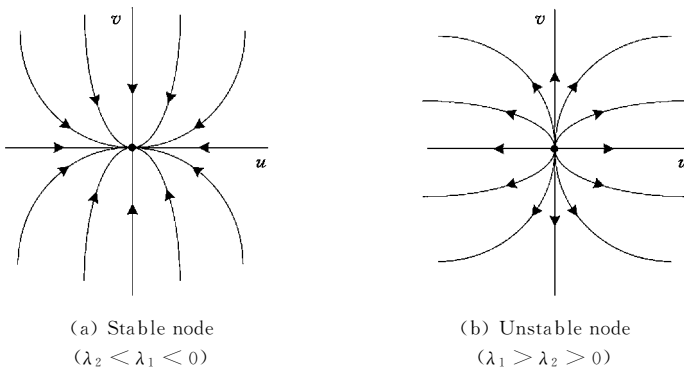


Figure 5.5.1

Case 2 $q > 0, p^2 - 4q = 0$. $\lambda_1 = \lambda_2$.

If the differential system (5.5.2) takes the form

$$\begin{cases} \frac{du}{dt} = \lambda_1 u, \\ \frac{dv}{dt} = \lambda_1 v, \end{cases} \quad (5.5.5)$$

then all trajectories of this system (5.5.5) can be easily written as

$$v = cu \quad \text{and} \quad u = 0.$$

That is, the trajectories are half-lines.

If $\lambda_1 < 0$, both $u(t)$ and $v(t)$ tend to 0 as $t \rightarrow +\infty$ and hence the critical point $(0,0)$ of the system (5.5.5) is asymptotically stable. If $\lambda_1 > 0$, all solutions tend to ∞ as $t \rightarrow +\infty$ and hence the critical point $(0,0)$ of the system (5.5.5) is unstable. Here the critical point $(0,0)$ is called **proper (star-shaped) node**, as shown in Figure 5.5.2.

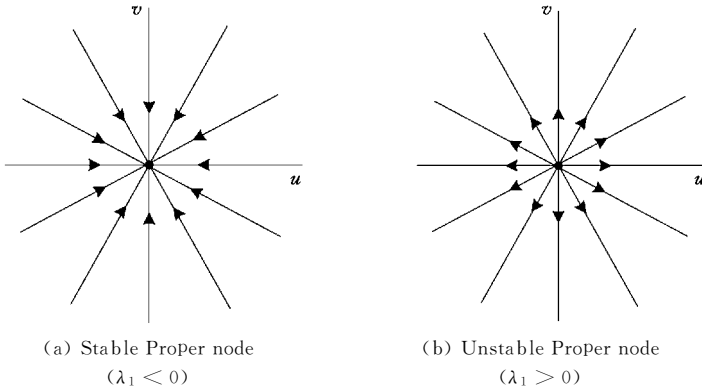


Figure 5.5.2

But if the differential system (5.5.2) takes the form

$$\begin{cases} \frac{du}{dt} = \lambda_1 u, \\ \frac{dv}{dt} = u + \lambda_1 v, \end{cases} \quad (5.5.6)$$

then the general solution of the system is

$$u = c_1 e^{\lambda_1 t}, v = (c_1 t + c_2) e^{\lambda_1 t}.$$

Thus all trajectories of the system (5.5.6) can be written as

$$v = cu + \frac{u}{\lambda_1} \ln |u| \quad \text{and} \quad u = 0.$$

Further, we have

$$\lim_{u \rightarrow 0} v(u) = 0$$

and

$$\lim_{u \rightarrow 0} \frac{dv}{du} = \begin{cases} +\infty, & \lambda_1 < 0, \\ -\infty, & \lambda_1 > 0. \end{cases}$$

Hence each trajectory of the differential system is tangent at $(0,0)$ to the v -axis. Here the

critical point $(0,0)$ is called an **improper node**. If $\lambda < 0$, then we see that this node is a sink, but it is a source if $\lambda > 0$, as shown in Figure 5.5.3.

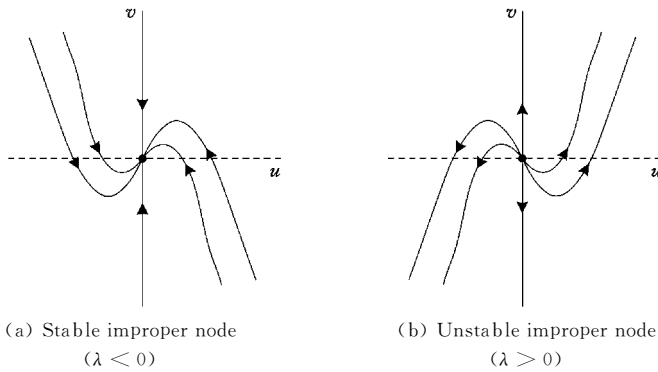


Figure 5.5.3

Case 3 $q > 0, p^2 - 4q < 0$. λ_1 and λ_2 are complex conjugates.

In this case, the differential system (5.5.2) takes the form

$$\begin{cases} \frac{du}{dt} = \alpha u + \beta v, \\ \frac{dv}{dt} = -\beta u + \alpha v, \alpha \neq 0. \end{cases} \quad (5.5.7)$$

Making substitutions $u = r \cos \theta, v = r \sin \theta$, the system (5.5.7) is changed into

$$\begin{cases} \frac{dr}{dt} = \alpha r, \\ \frac{d\theta}{dt} = -\beta, \end{cases} \quad (5.5.8)$$

whose general solution is

$$r = c_1 e^{\alpha t}, \theta = -\beta t + c_2, c_1 > 0. \quad (5.5.9)$$

If $\alpha < 0$, then the trajectories are spirals that approach the origin. In this case, the critical point $(0,0)$ is asymptotically stable, and is called a **stable focus**.

If $\alpha > 0$, then all trajectories of (5.5.8) spiral away from the origin $(0,0)$ as $t \rightarrow +\infty$. In this case, the critical point $(0,0)$ is unstable, and is called an **unstable focus** (see Figure 5.5.4).

Case 4 $q > 0, p = 0$. λ_1 and λ_2 are pure imaginary numbers. In this case, the differential system (5.5.2) takes the form

$$\begin{cases} \frac{du}{dt} = \beta v, \\ \frac{dv}{dt} = -\beta u, \end{cases} \quad (5.5.10)$$

When $\alpha = 0$ in the general solution (5.5.9), it follows that the general solution of the differential system (5.5.10) is

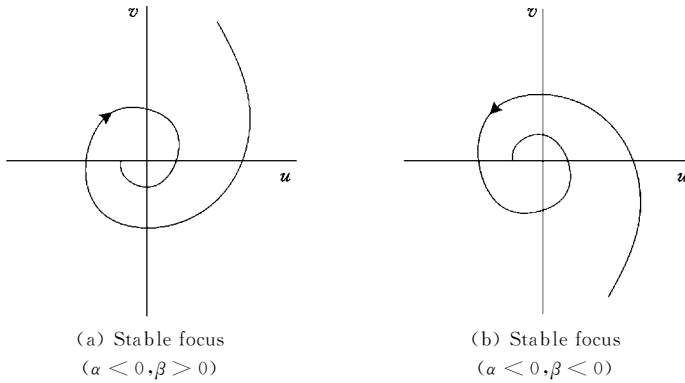


Figure 5.5.4

$$r(t) = c_1, \theta(t) = -\beta t + c_2, c_1 > 0.$$

Thus, the trajectories are closed curves, that is, circles around the origin. The phase portrait of the system (5.5.10) has the form described in Figure 5.5.5. In this case, the critical point $(0,0)$ is stable but not asymptotically stable, and is named a **center**, as shown in Figure 5.5.5.

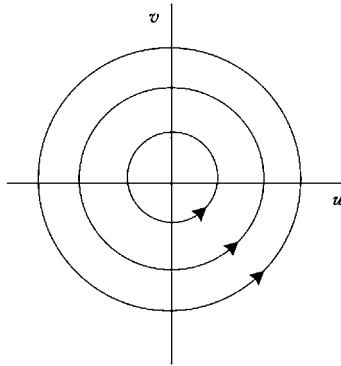
Stable center ($\alpha = 0, \beta < 0$)

Figure 5.5.5

Case 5 $q < 0$, λ_1 and λ_2 are real with opposite signs.

Here the situation is the same as in Case 1. But $\lambda_1 \lambda_2 < 0$, from the system (5.5.3), it is obvious that $u=0$ and $v=0$ are the trajectories of the system (5.5.3). If $\lambda_1 < 0 < \lambda_2$, the general solution of the form (5.5.4) implies that $u \rightarrow 0, v \rightarrow \infty$ as $t \rightarrow \infty$. The nonlinear trajectories

$$v = c \left| u \right|^{\frac{\lambda_2}{\lambda_1}}$$

resemble hyperbolas. This type of critical point is called a **saddle point**. Obviously, the saddle points illustrated in Figure 5.5.6 are unstable.

We summarize the above analysis in the following theorem.

Theorem 5.5.1 For the differential system (5.5.1), let $p = -\text{tr} \mathbf{A}$, $q = \det \mathbf{A}$. Then the

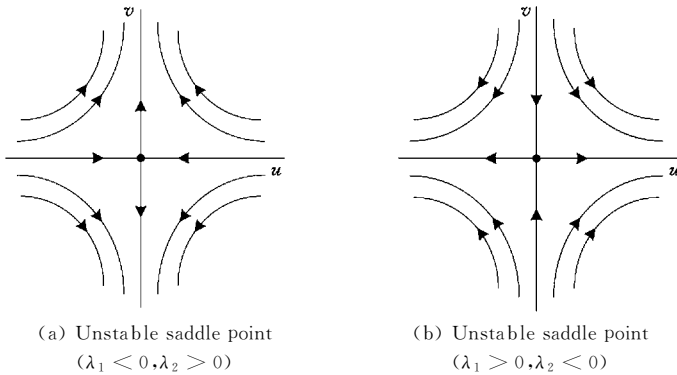


Figure 5.5.6

behavior of its trajectories near the critical point $(0,0)$ is as follows:

- (1) if $q > 0$, $p^2 - 4q > 0$, then the critical point $(0,0)$ is a node;
- (2) if $q > 0$, $p^2 - 4q = 0$, then the critical point $(0,0)$ is a proper (star-shaped) node or an improper node;
- (3) if $q > 0$, $p^2 - 4q < 0$, then the critical point $(0,0)$ is a focus;
- (4) if $q > 0$, $p = 0$, then the critical point $(0,0)$ is a center;
- (5) if $q < 0$, then the critical point $(0,0)$ is a saddle point.

In addition, in cases (1), (2) and (4), the stability of the critical point $(0,0)$ is determined by the sign of p , that is, when $p > 0$, $(0,0)$ is stable, and when $p < 0$, $(0,0)$ is unstable.

Remark 5.5.1 Actually, $u = 0$ contains three trajectories:

- (1) $u = 0, v > 0$; (2) $u = 0, v < 0$; (3) $u = 0, v = 0$.

For simplicity, we say that $u = 0$ is a trajectory of the differential system (5.5.2).

Remark 5.5.2 If the differential system (5.5.1) and (5.5.2) are coupled by the linear transformation $\begin{pmatrix} x \\ y \end{pmatrix} = T \begin{pmatrix} u \\ v \end{pmatrix}$, then their phase portraits are also coupled by the same affine mapping $\begin{pmatrix} x \\ y \end{pmatrix} = T \begin{pmatrix} u \\ v \end{pmatrix}$ of R^2 which transforms straight lines into straight lines, closed orbits into closed orbits and etc. Although affine distortion appears, but the affine mapping preserves the characteristic features such as the behavior as $t \rightarrow \infty$, and does not change the types of the critical point $(0,0)$. In this way, we obtain an insight into the global properties of the differential system (5.5.1) with $\det A \neq 0$.

Remark 5.5.3 If $(x(t), y(t))$ is a solution of the differential system (5.5.1), then the pair of functions $(x(-t), y(-t))$ is a solution of a related equation, in which the constant coefficient matrix A is replaced by $-A$. The functions $(x(t), y(t))$ and $(x(-t), y(-t))$ have the same trajectories, only the direction of arrows is reversed. This suits all possible cases.

Exercise 5.5

1. Find all critical points of each of the following differential systems and determine

whether they are stable or unstable.

$$(1) \begin{cases} \frac{dx}{dt} = x + 4y, \\ \frac{dy}{dt} = x + y - x^2. \end{cases}$$

$$(2) \begin{cases} \frac{dx}{dt} = 8x - y^2, \\ \frac{dy}{dt} = y - x^2. \end{cases}$$

2. Determine the type of stability of the critical point (0,0) of each of the following linear systems and sketch the phase portraits.

$$(1) \begin{cases} \frac{dx}{dt} = x + 2y, \\ \frac{dy}{dt} = x - 2y. \end{cases}$$

$$(2) \begin{cases} \frac{dx}{dt} = 2x - y, \\ \frac{dy}{dt} = 2x + y. \end{cases}$$

$$(3) \begin{cases} \frac{dx}{dt} = x - y, \\ \frac{dy}{dt} = 2x - y. \end{cases}$$

$$(4) \begin{cases} \frac{dx}{dt} = 3x + y, \\ \frac{dy}{dt} = -x + y. \end{cases}$$

$$(5) \begin{cases} \frac{dx}{dt} = -x - y, \\ \frac{dy}{dt} = x - y. \end{cases}$$

$$(6) \begin{cases} \frac{dx}{dt} = x + y, \\ \frac{dy}{dt} = 3x - y. \end{cases}$$

$$(7) \begin{cases} \frac{dx}{dt} = 7x + y, \\ \frac{dy}{dt} = -3x + 4y. \end{cases}$$

$$(8) \begin{cases} \frac{dx}{dt} = y, \\ \frac{dy}{dt} = 2x - y. \end{cases}$$

3. Each of the following systems has a single critical point (x_0, y_0) . Apply Theorem (5.5.1) to classify this critical point as to type and stability and construct the phase portrait for the given system.

$$(1) \begin{cases} \frac{dx}{dt} = x - 2y, \\ \frac{dy}{dt} = 3x - 4y - 2. \end{cases}$$

$$(2) \begin{cases} \frac{dx}{dt} = x + y - 7, \\ \frac{dy}{dt} = 3x - y - 5. \end{cases}$$

$$(3) \begin{cases} \frac{dx}{dt} = x - 2y + 1, \\ \frac{dy}{dt} = x + 3y - 9. \end{cases}$$

$$(4) \begin{cases} \frac{dx}{dt} = x - 5y - 5, \\ \frac{dy}{dt} = x - y - 3. \end{cases}$$

5.6 Limit Cycles of Two-Dimensional Autonomous Systems

In order to sketch the phase portrait, we study the limit cycles of the following two dimensional autonomous systems

$$\begin{cases} \frac{dx}{dt} = P(x, y), \\ \frac{dy}{dt} = Q(x, y). \end{cases} \quad (5.6.1)$$

where $P(x, y)$ and $Q(x, y)$ are continuous in $D = \mathbb{R}^2$ and satisfy the conditions of the existence and uniqueness of solution of the initial value problems.

We begin this section with the following definitions.

Definition 5.6.1 Let Γ be a closed trajectory of the differential system (5.6.1). If there exists a $\delta > 0$, such that each nonclosed trajectory of the differential system (5.5.1) starting from both sides of the neighborhood $U(\Gamma, \delta)$ approaches spirally the closed trajectory Γ either as $t \rightarrow +\infty$ or as $t \rightarrow -\infty$, then the closed trajectory Γ is named a limit cycle of the differential system.

Obviously, it implies that a limit cycle of the differential system (5.6.1) is an isolated closed trajectory.

Definition 5.6.2 Suppose that there exists one side of Γ (either inside or outside) satisfying the requirements of Definition (5.6.1). In this case, Γ is called a unilateral limit cycle.

Definition 5.6.3 Let Γ be a limit cycle of the differential system (5.6.1). If there exists a $\delta > 0$ such that each trajectory of the system (5.6.1) starting from the neighborhood $U(\Gamma, \delta)$ approaches spirally Γ as $t \rightarrow +\infty$ ($t \rightarrow -\infty$), then Γ is said to be stable (unstable). If each trajectory of the system (5.6.1) starting from the unilateral neighborhood $U(\Gamma, \delta)$ approaches spirally the limit cycle Γ as $t \rightarrow +\infty$ ($t \rightarrow -\infty$), then Γ is called one-sided stable (unstable). Sometimes, if one side of Γ is stable and the other side of Γ is unstable, then the limit cycle is named semi-stable.

The stability of a limit cycle is displayed in Figure 5.6.1.

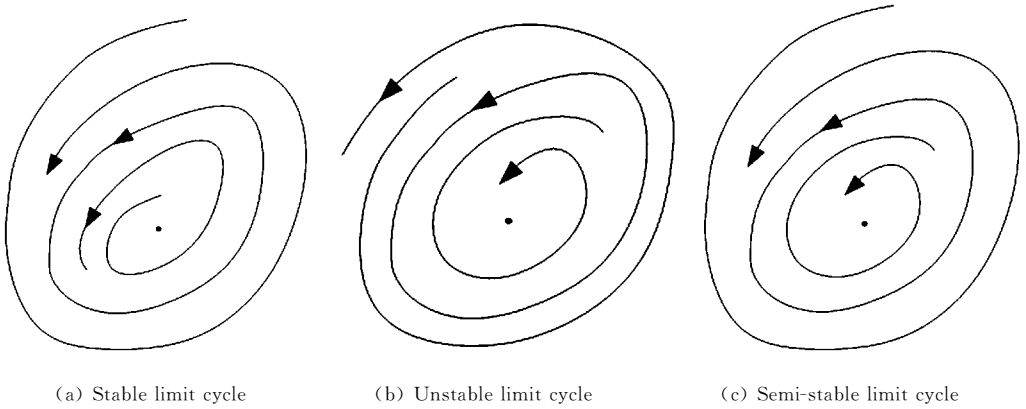


Figure 5.6.1

Example 5.6.1 Consider the nonlinear differential system

$$\begin{cases} \frac{dx}{dt} = -y - x(x^2 + y^2 - 1), \\ \frac{dy}{dt} = x - y(x^2 + y^2 - 1). \end{cases} \quad (5.6.2)$$

Since the term $x^2 + y^2$ appears in both the equations, we introduce polar coordinates (r, θ) , where

$$x = r \cos \theta, y = r \sin \theta.$$

It follows that the equivalent differential system is

$$\begin{cases} \frac{dr}{dt} = -r(r^2 - 1), \\ \frac{d\theta}{dt} = -1. \end{cases} \quad (5.6.3)$$

We can solve easily the solution

$$\begin{cases} r(t) = \frac{r_0}{[r_0^2 + (1 - r_0^2)e^{-2t}]^{\frac{1}{2}}}, \\ \theta(t) = t + \theta_0, \end{cases} \quad (5.6.4)$$

where $r_0 = r(0)$ and $\theta_0 = \theta(0)$.

In addition, it is obvious that the two particular solutions are

$$\begin{cases} r=0, \\ \theta=\theta_0+t \end{cases} \quad \text{and} \quad \begin{cases} r=1, \\ \theta=\theta_0+t, \end{cases}$$

which correspond to the critical point $(0,0)$ and closed trajectory $x^2 + y^2 = 1$ respectively.

If $r_0 \neq 1$, the trajectories defined by the formula (5.6.4) are not closed but rather have a spiral behavior. If $r_0 < 1$, the trajectories are spirals lying inside the circle $x^2 + y^2 = 1$. They approach this circle as $t \rightarrow +\infty$, while they approach the only critical point $(0,0)$ as $t \rightarrow -\infty$. If $r_0 > 1$, the trajectories are spirals lying outside the cycle $r = 1$. These outer trajectories also approach this cycle as $t \rightarrow +\infty$, while $r(t) \rightarrow \infty$ as $t \rightarrow \frac{1}{2} \ln \left(1 - \frac{1}{r_0^2} \right)$. The phase portrait is displayed in Figure 5.6.2. This shows that $x^2 + y^2 = 1$ is a limit cycle of the system (5.6.2).

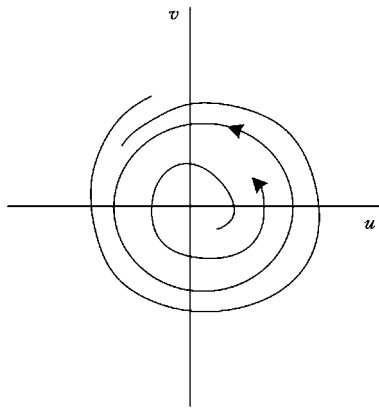


Figure 5.6.2

In the following, we give a result which provides sufficient conditions for the existence of closed trajectories of the differential system (5.6.1).

Theorem 5.6.1 (Poincare-Bendixson Theorem)

Suppose that a solution $(x(t), y(t))$ of a limit cycle of the differential system (5.6.1) remains in a bounded region D_0 of the xy -plane which contains no critical points of the system (5.6.1). Then its trajectory must spiral into a simple closed curve, that is, the trajectory of a periodic solution of the system (5.6.1).

Remark 5.6.1 This theorem shows that it exists a closed trajectory of the system (5.6.1) in D_0 , but this closed trajectory is not always a limit cycle of the system (5.6.1). The result has been proved that if $P(x, y)$ and $Q(x, y)$ are analytical in D_0 , then these closed trajectories in D_0 are limit cycles of the differential system.

Remark 5.6.2 Poincare-Bendixson Theorem is not generalized simply from two dimensional autonomous systems (5.6.1) to three dimensional and above autonomous systems.

Here, we shall not give the proof of the important theorem. One example is displayed

to explain the significance of this result.

Example 5.6.2 Consider the following differential system

$$\begin{cases} \frac{dx}{dt} = y, \\ \frac{dy}{dt} = -x + (1 - 2x^2 - 3y^2)y. \end{cases} \quad (5.6.5)$$

For any given solution $(x(t), y(t))$ of the differential system (5.6.5), we notice that

$$\begin{aligned} \frac{d}{dt}(x^2(t) + y^2(t)) &= 2x(t) \frac{dx}{dt} + 2y \frac{dy}{dt} \\ &= 2(1 - 2x^2 - 3y^2)y^2. \end{aligned}$$

Because the quantity $(1 - 2x^2 - 3y^2)$ is positive for $x^2 + y^2 < \frac{1}{3}$ and negative for $x^2 + y^2 > \frac{1}{2}$,

the function $x^2(t) + y^2(t)$ is increasing if $x^2 + y^2 < \frac{1}{3}$ and decreasing if $x^2 + y^2 > \frac{1}{2}$. Hence if

a solution $(x(t), y(t))$ starts in the annulus

$$\frac{1}{3} < x^2 + y^2 < \frac{1}{2}$$

at $t = t_0$, then this trajectory will remain in the annulus for all $t \geq t_0$. Further, since this annulus contains no critical points of the system (5.6.5), it follows that the trajectory of this solution $(x(t), y(t))$ must spiral into a simple closed trajectory which is a nontrivial periodic solution the system (5.6.5).

The following results provide sufficient conditions for the non-existence of closed trajectories, and hence, especially, limit cycles of the differential system (5.6.1).

Theorem 5.6.2 (Bendixson's Theorem) Supposed that the functions $P(x, y)$ and $Q(x, y)$ have continuous partial derivatives in the simply connected region D . If

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$$

has the same sign throughout the region D , then the differential system (5.6.1) has no closed trajectories in D .

Proof Let Ω be a region in D which is bounded by a closed trajectory Γ of the differential system (5.6.1). By Green's Theorem, we have

$$\oint_{\Gamma} P(x, y) dy - Q(x, y) dx = \iint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy. \quad (5.6.6)$$

Since Γ is a closed trajectory of the system (5.6.1), along the closed trajectory Γ , it follows that

$$\begin{aligned} P(x, y) dy - Q(x, y) dx &= \left(P \frac{dy}{dt} - Q \frac{dx}{dt} \right) dt \\ &= (PQ - QP) dt \\ &= 0. \end{aligned}$$

Hence, from the identity (5.6.6), we have

$$\iint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy = 0.$$

But this double integral can be zero only if its integrand changes sign. This is a contradiction with the condition. The proof is completed.

Theorem 5.6.3 (Dulac's Theorem) Supposed that the functions $P(x, y)$ and $Q(x, y)$ have continuous partial derivatives in the simply connected region D . If there exists a continuously differentiable function $B(x, y)$ such that

$$\frac{\partial(BP)}{\partial x} + \frac{\partial(BQ)}{\partial y}$$

has the same sign throughout the region D , then the differential system (5.6.1) does not exist a closed trajectory in D .

Proof Replacing $P(x, y)$ and $Q(x, y)$ in Theorem 5.6.2 by $B(x, y)P(x, y)$ and $B(x, y)Q(x, y)$ respectively, we easily prove this theorem.

Remark 5.6.3 Obviously, Bendixson's Theorem is a special case of Dulac's Theorem with $B(x, y) \equiv 1$. But Bendixson's Theorem only depends on $P(x, y)$ and $Q(x, y)$, does not use the other auxiliary function $B(x, y)$ in determining the non-existence of a closed trajectory of the differential system (5.6.1). Thus, it exists still unique advantages.

Example 5.6.3 Consider the nonlinear differential system

$$\begin{cases} \frac{dx}{dt} = x(x^2 + y^2 - 2x - 3) - y, \\ \frac{dy}{dt} = y(x^2 + y^2 - 2x - 3) + x. \end{cases} \quad (5.6.7)$$

Here, $P(x, y) = x(x^2 + y^2 - 2x - 3) - y$, $Q(x, y) = y(x^2 + y^2 - 2x - 3) + x$.

Since

$$\begin{aligned} \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} &= 4x^2 + 4y^2 - 6x - 6 \\ &= 4 \left[\left(x - \frac{3}{4} \right)^2 + y^2 - \frac{33}{16} \right], \end{aligned}$$

we have

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} < 0$$

in the disc D of radius $\sqrt{\frac{33}{16}}$ centered at $\left(\frac{3}{4}, 0 \right)$. Thus, from Bendixson's Theorem, it follows that the differential system (5.6.7) does not exist closed trajectory in this disc D .

Example 5.6.4 Consider the differential system

$$\begin{cases} \frac{dx}{dt} = y^2 \\ \frac{dy}{dt} = -x - \beta y - y^3, \end{cases} \quad (x, y) \in \mathbb{R}^2 \quad (5.6.8)$$

Determine the nonexistence of closed trajectories of the differential system (5.6.8) in the

following two cases: (1) $\beta = 0$; (2) $\beta \neq 0$.

Solution. (1) when $\beta = 0$, the system (5.6.8) takes the form

$$\begin{cases} \frac{dx}{dt} = y^2, \\ \frac{dy}{dt} = -x - y^3. \end{cases}$$

Since

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -3y^2 \leq 0 \quad \text{in } R^2$$

and it is not identical zero in any sub-region of R^2 , Bendixson's Theorem implies that the differential system (5.6.8) with $\beta = 0$ has no closed trajectory in R^2 .

(2) when $\beta \neq 0$, we choose $B(x, y) = e^{ax+by}$, where a and b are two undetermined constants.

Since

$$\frac{\partial(BP)}{\partial x} + \frac{\partial(BQ)}{\partial y} = e^{ax+by} [-\beta - bx - \beta by + (a-3)y^2 - by^3],$$

we take $a = 3, b = 0$, that is, $B(x, y) = e^{3x}$, and further obtain

$$\frac{\partial(BP)}{\partial x} + \frac{\partial(BQ)}{\partial y} = -\beta e^{3x},$$

which implies that

$$\frac{\partial(BP)}{\partial x} + \frac{\partial(BQ)}{\partial y}$$

has the same sign throughout the region R^2 . By Dulac's Theorem, the differential system (5.6.8) with $\beta \neq 0$ has no closed trajectory in R^2 .

Exercise 5.6

1. Determine the existence of closed trajectory of the following differential system

$$\begin{cases} \frac{dx}{dt} = x(y-1), \\ \frac{dy}{dt} = x+y-2y^2. \end{cases}$$

2. Prove that a linear autonomous system has no an isolated closed trajectory.
3. Find all limit cycles of each of the following differential systems.

$$(1) \begin{cases} \frac{dx}{dt} = y + x(x^2 + y^2)^{\frac{1}{2}}(x^2 + y^2 - 3), \\ \frac{dy}{dt} = -x + y(x^2 + y^2)^{\frac{1}{2}}(x^2 + y^2 - 3). \end{cases}$$

$$(2) \begin{cases} \frac{dx}{dt} = -y, \\ \frac{dy}{dt} = x + y(x^2 + y^2 - 1). \end{cases}$$

$$(3) \begin{cases} \frac{dx}{dt} = -y - x(x^2 + y^2 - 9)^2, \\ \frac{dy}{dt} = x - y(x^2 + y^2 - 9)^2. \end{cases}$$

4. Show that the following nonlinear differential system has no nontrivial periodic solutions in the given region.

$$\begin{cases} \frac{dx}{dt} = x - xy^2 + y^3, \\ \frac{dy}{dt} = 3y - x^2y + x^3, \end{cases} D = \{(x, y) : x^2 + y^2 < 4\}.$$

5. Prove the existence of a nontrivial periodic solution of the following differential system

$$\begin{cases} \frac{dx}{dt} = 2x - 2y - x(x^2 + y^2), \\ \frac{dy}{dt} = 2x + 2y - y(x^2 + y^2). \end{cases}$$

6. For the differential system (5.6.1), if the functions $P(x, y)$, $Q(x, y)$, $B(x, y)$ and $F(x, y)$ have continuously partial derivatives in the simply connected region D and

$$\frac{\partial(BP)}{\partial x} + \frac{\partial(BQ)}{\partial y} + B \left(P \frac{\partial F}{\partial x} + Q \frac{\partial F}{\partial y} \right)$$

has the same sign throughout the region D , then the differential system (5.6.1) does not exist closed trajectory in D .

Chapter 6 Numerical Methods for Ordinary Differential Equations

In this chapter, we mainly consider the numerical methods for ordinary differential equations. We will focus on Euler's Method, the improved Euler Method and the RungeKutta Method which are used for solving the scalar ordinary differential equations. For the application of these methods, we mainly use Maple2020 for our implementation tools.

6.1 Introduction to Numerical Methods

Many questions in science and technology can be used as definite solutions of ordinary differential equations. In this chapter, we mainly consider the initial value problems of ordinary differential equations.

Many physical systems are related to time. From satellite orbit operation to simple pendulum motion and from chemical reaction to species competition, all of these systems are changing over time. Ordinary differential equation is a mathematical language to describe continuous change. Solution of ordinary differential equation aims to get the differentiable function $y(t)$ which satisfies the given equation. The main task of this chapter is to study its numerical methods.

Consider the initial value problem of the first order ordinary differential equation

$$\begin{cases} \frac{dy}{dx} = f(x, y), \\ y(x_0) = y_0, x \in [x_0, b] \end{cases} \quad (6.1.1)$$

where f is continuous and satisfies the Lipschitz condition with respect to y in the domain $D = \{(x, y) \mid a \leq x \leq b, y \in R\}$, that is, there is a real number $L > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \leq L |y_1 - y_2|, \forall y_1, y_2 \in R \quad (6.1.2)$$

and L is called the Lipschitz constant of f .

Theorem 6.1.1 Suppose that f is continuous in the domain $D = \{(x, y) \mid a \leq x \leq b, y \in R\}$ and f satisfies the Lipschitz condition with respect to y . Then for any $x_0 \in [a, b]$, $y_0 \in R$, the ordinary differential equation (6.1.1) has the unique continuous differentiable solution $y(x), x \in [a, b]$.

Theorem 6.1.2 If f is continuous in the domain D and satisfies Lipschitz condition with respect to y , further, the initial value problem $y'(x) = f(x, y), y(x_0) = s$ has the

solution $y(x,s)$, then

$$|y(x,s_1) - y(x,s_2)| \leqslant e^{L|x-x_0|} |s_1 - s_2|.$$

For the numerical methods, we should find the series of discrete points

$$x_1 < x_2 < \cdots < x_n < x_{n+1} < \cdots$$

on which the approximation of solution $y(x)$ is $y_1, y_2, \cdots, y_n, y_{n+1}, \cdots$.

The distance between two adjacent nodes $h_n = x_{n+1} - x_n$ is called step size. In the following, unless otherwise specified, we always assume $h_i = h (i=0,1,\cdots)$ as constant. In this case, the node point can be specified as $x_n = x_0 + nh, n=0,1,2,\cdots$

In this chapter, we should first discretize the ordinary differential equation (6.1.1), then establish the recurrence formula of numerical solution. One kind of method has to compute y_{n+1} which should also know the former point solution y_n . This kind of method is called one step method. The other kind of method should know the former k points of $y_n, y_{n-1}, \cdots, y_{n-k+1}$, which is also called k step method. Secondly, we will investigate the error estimate between numerical solution y_n and the exact solution $y(x_n)$, which is related to the stability of recurrence formula.

6.2 Numerical Approximation: Euler’s Method

As we all know, in the xoy plane, the graph of solution $y = y(x)$ with ordinary differential equation (6.1.1) is called the integral curve. Each tangent slope of point (x,y) on the integral curve is the value of function $f(x,y)$.

Based on the above geometrical explanation, we start from the initial point $P_0(x_0,y_0)$. First we carry the point $P_0(x_0,y_0)$ forward to $P_1(x_1,y_1)$ according to the direction of the directional field in this point. Then carry the point $P_1(x_1,y_1)$ forward to point $P_2(x_2,y_2)$ based on the directional field. Look forward to making a broken line $\overrightarrow{P_1 P_2 P_3 \cdots}$ which can be seen in the Figure 6.2.1.

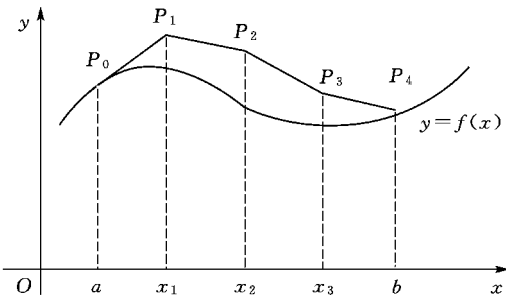


Figure 6.2.1

In general, if we have obtained the point of the polyline P_n , then we can carry the next point forward from $P_n(x_n,y_n)$ to $P_{n+1}(x_{n+1},y_{n+1})$ according to the directional field. Obviously, the two points have the relationship

$$\frac{y_{n+1} - y_n}{x_{n+1} - x_n} = f(x_n, y_n),$$

which can also be written as

$$y_{n+1} = y_n + hf(x_n, y_n). \quad (6.2.1)$$

This kind of method is called Euler's method. Actually, this is the mean difference to approximate the derivative of the ordinary differential equation (6.1.1)

$$\frac{y(x_{n+1}) - y(x_n)}{h} \approx y'(x_n) = f(x_n, y(x_n)).$$

If the initial value y_0 is known, then using (6.2.1) we can get

$$\begin{cases} y_1 = y_0 + hf(x_0, y_0), \\ y_2 = y_1 + hf(x_1, y_1), \\ \vdots \end{cases}$$

From the above formula, we can get all the solutions of y_n for the original ordinary differential equations.

Example 6.2.1 Solve the initial value problem

$$\begin{cases} y' = y - \frac{2x}{y}, 0 < x < 1, \\ y(0) = 1. \end{cases}$$

Solution Obviously, we find that it is a Bernoulli equation. By using the elementary integral method, we get the particular solution of this problem

$$y(x) = \sqrt{2x+1}.$$

Next, we try to use numerical method to solve this problem.

Here, we use Euler's method. Its recurrence formulation is

$$y_{n+1} = y_n + h \left(y_n - \frac{2x_n}{y_n} \right).$$

Set the step $h = 0.1$, with $x_0 = 0, y_0 = 1, f(x, y) = y - \frac{2x}{y}$, the iterative formula is

$$y_1 = y_0 + h \left(y_0 - \frac{2x_0}{y_0} \right) = 1 + 0.1(1 - 0) = 1.1000$$

$$y_2 = y_1 + h \left(y_1 - \frac{2x_1}{y_1} \right) = 1.1 + 0.1 \left(1.1 - \frac{0.2}{1.1} \right) = 1.1918$$

$$y_3 = y_2 + h \left(y_2 - \frac{2x_2}{y_2} \right) = 1.1918 + 0.1 \left(1.1918 - \frac{0.4}{1.1918} \right) = 1.2774$$

$$y_4 = y_3 + h \left(y_3 - \frac{2x_3}{y_3} \right) = 1.2774 + 0.1 \left(1.2774 - \frac{0.6}{1.2774} \right) = 1.3582$$

$$y_5 = y_4 + h \left(y_4 - \frac{2x_4}{y_4} \right) = 1.3582 + 0.1 \left(1.3582 - \frac{0.8}{1.3582} \right) = 1.4351$$

$\vdots$

Because we need to compute y_{10} , we just use Maple to help us. First, we write a Maple

```
program for Euler's method
Euler:= proc(f,a,b,y0,n)
    description "Solves the ODE  $y'(t) = f(t,y)$  using the Euler's method.
        Input parameters: f is a function of two variables  $f(t,y)$ ; a and b are
        the endpoints of the interval of integration  $[a,b]$ ; n is the number of
        subintervals of  $[a,b]$ ;  $y_0$  is the initial condition";
local fm, h, i, t, w, wp; # Local variables
h:=evalf[(b-a)/n];
t:= Array([seq(a + i * h, i=0..n)]);
w := Array(1..n+1);
w[1] := y0;
# This loop implements the difference equation for Euler's method
for i from 1 to n do
    fm := apply(f, t[i], w[i]);
    w[i+1] := w[i] + h * fm;
od;
Array([seq([t[i],w[i]],i=1..n+1)]): # Output
end;
```

Then we input the initial condition and initial parameters for example 6.2.1.

```
[> f := (u, v) -> v - 2 * u/v;
[> t0 := 0: tf := 1: y0 := 1.0: n := 10:
[> with(plots):
[> z := Euler(f, t0, tf, y0, n);
[> z;
```

From Table 6.2.1, the first column is the value of x_n , the second column is the value of y_n . Compare the results with the data we got before, we see that they fit very well. Next, we plot the figures of numerical solutions with $h=0.2$, $h=0.1$, $h=0.05$.

Table 6.2.1 The solution of ealer's Method for Example 6.2.1 with $h=0.1$

0.	1.0
0.1000000000	1.100000000
0.2000000000	1.191818182
0.3000000000	1.277437834
0.4000000000	1.358212600
0.5000000000	1.435132919
0.6000000000	1.508966254
0.7000000000	1.580338238
0.8000000000	1.649783431
0.9000000000	1.717779348
⋮	⋮

The results can be seen from Figure 6.2.2.

```
[> t0 := 0; tf := 1; y0 := 1.0; n := 5;
[> z1 := Euler(f, t0, tf, y0, n);
[> a := listplot(z1, style = pointline, color = green);
[> t0 := 0; tf := 1; y0 := 1.0; n := 10;
[> z2 := Euler(f, t0, tf, y0, n);
[> b := listplot(z2, style = pointline, color = red);
[> t0 := 0; tf := 1; y0 := 1.0; n := 20;
[> z3 := Euler(f, t0, tf, y0, n);
[> c := listplot(z3, style = pointline, color = blue);
[> d := plot(sqrt(1 + 2 * x), x = 0 .. 1, color = black);
[> display([a, b, c, d], title = "Graphs of Euler approximations with step sizes h=0.2,
h=0.1, h=0.05", legend = ["h=0.2", "h=0.1", "h=0.05", "exact"]);
```

From Figure 6.2.2, we see that with the increasing of grid points, the approximation effect is getting better and better. In fact, using Maple2020, we can get the same figure with much more simpler ways:

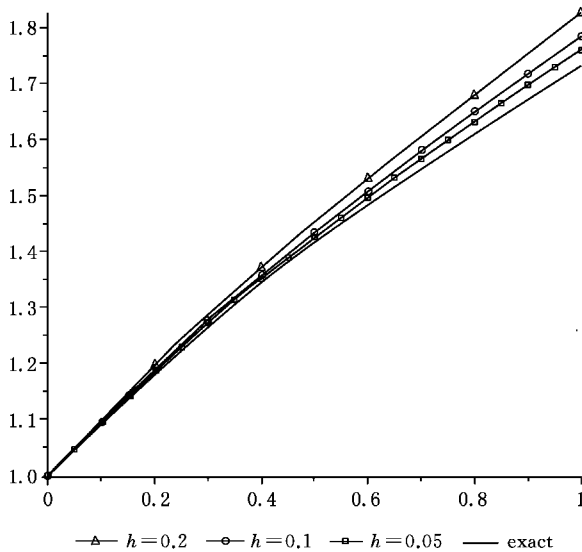


Figure 6.2.2 Graphs of Euler approximations with step sizes

```
[> with(Student[NumericalAnalysis]):
[> Euler(diff(y(x), x) = y(x) - 2 * x/y(x), y(0) = 1, x = 1, output = plot)
```

Here, we load the Student Package and use its resources “NumericalAnalysis”. Then, we can use the built in function Euler to get the above graphs of Figure 6.2.3 (the blue line is the Euler solutions and the red line is exact solution).

Until now, we also want to know the error estimate of Euler’s method. This time, we can use the Euler function which is defined in Student Package [NumericalAnalysis].

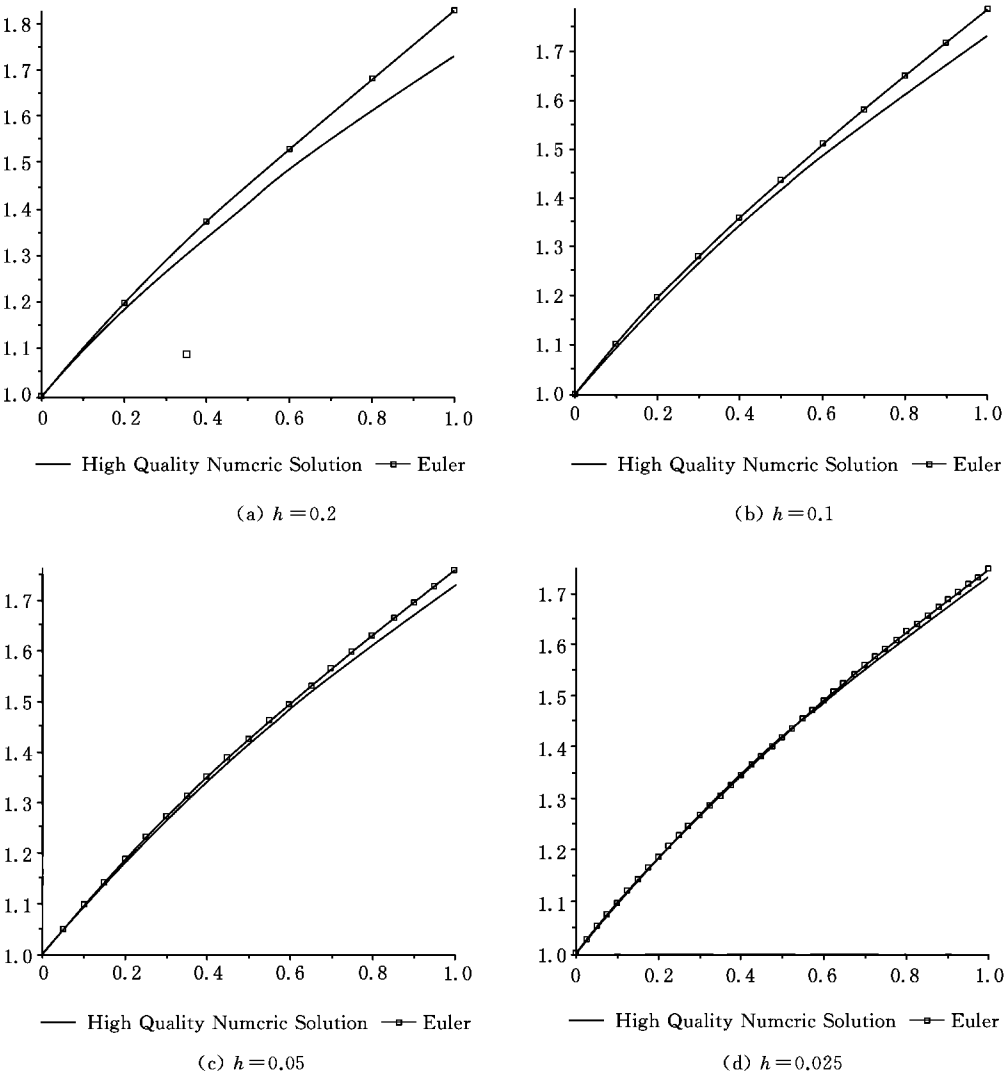


Figure 6.2.3

```
[> Euler(diff(y(x), x)=y(x)-2 * x/y(x), y(0)=1, x=1, 'numsteps'=10, 'digits'=5,
output=information);
```

From Table 6.2.2, we see that the number step $h = 0.1$. The second column is the exact solutions of $y(x_n)$, the third column is the numerical solutions of Euler's method and the last column is the error which can be defined as $\text{error} = |y_n - y(x_n)|$. From these data, we see that the error between exact solution and Euler's method is not very good.

Table 6.2.2 The error Euler's method for Example 6.2.1 with $h=0.1$

x	[Maple's numeric solution]	[Euler]	[Error]
0.	1.	1.	0.
0.10000	1.0954	1.1000	0.0046000
0.20000	1.1832	1.1918	0.0086182
0.30000	1.2649	1.2774	0.012538
0.40000	1.3416	1.3582	0.016613
0.50000	1.4142	1.4351	0.020933
0.60000	1.4832	1.5090	0.025766
0.70000	1.5492	1.5803	0.031138
0.80000	1.6125	1.6498	0.037283
$\vdots$	$\vdots$	$\vdots$	$\vdots$

`[>Euler(diff(y(x), x) = y(x) - 2 * x/y(x), y(0) = 1, x = 1, 'numsteps'=40, 'digits'=5, output=information)`

If we decrease the number step to $h=0.025$ (see Table 6.2.3), we find that the error between exact solution and Euler's method has been improved. But from the increasing of x_n , the error is still getting bigger and bigger which coincides with the above Figure 6.2.3 (d).

Table 6.2.3 The error of Euler's method for Example 6.2.1 $h=0.05$

x	[Maple's numeric solution]	[Euler]	[Error]
0.	1.	1.	0.
0.025000	1.0247	1.0250	0.00030000
0.050000	1.0488	1.0494	0.00060549
0.075000	1.0724	1.0733	0.00085832
0.10000	1.0954	1.0966	0.0011957
0.12500	1.1180	1.1195	0.0014511
0.15000	1.1402	1.1419	0.0016543
0.17500	1.1619	1.1638	0.0019324
0.20000	1.1832	1.1854	0.0022099
$\vdots$	$\vdots$	$\vdots$	$\vdots$

If users want to learn more details of Euler function in Student Package [Numerical Analysis], one can use the EulerTutor() function which can be seen from Figure 6.2.4.

`[>EulerTutor(diff(y(x), x) = y(x) - 2 * x/y(x), y(0) = 1, x = 1)`

Example 6.2.2 Using explicit Euler's method to solve the initial value problem

$$\begin{cases} \frac{dy}{dx} = -y + x + 1, 0 < x < 1, \\ y(0) = 1. \end{cases}$$

Solution We see that it is a first-order linear ordinary differential equation. One can easily get its general solution

$$y(x) = x + Ce^{-x}.$$

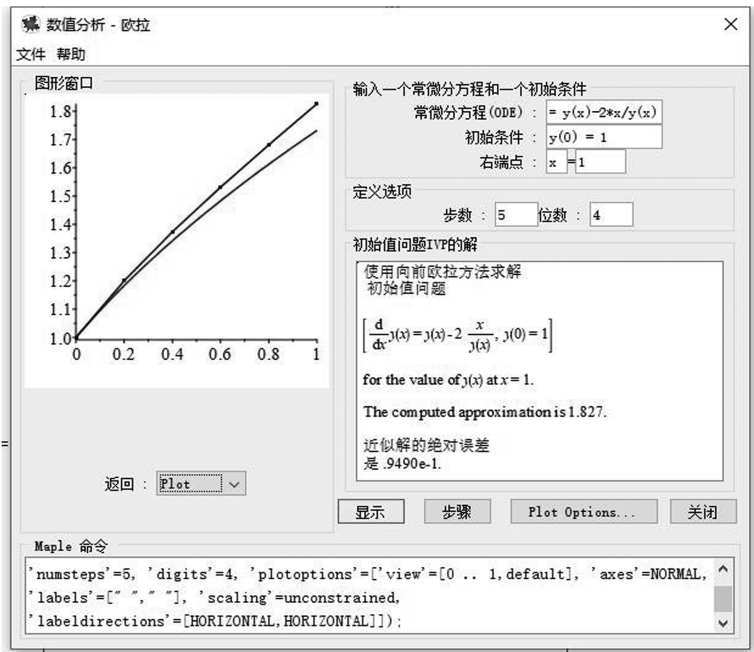


Figure 6.2.4 Numerical Analysis of Eler’s method

Using the initial condition $y(0) = 1$, then $C = 1$. So the exact solution of this problem is $y(x) = x + e^{-x}$.

Next, we try to use Euler’s method to solve this problem. The iterative formula of this problem is

$$y_{n+1} = y_n + h(-y_n + x_n + 1).$$

Let the step $h = 0.1$, with $x_0 = 0, y_0 = 1, f(x, y) = -y + x + 1$, the iterative formula is

$$y_1 = y_0 + h(-y_0 + x_0 + 1) = 1 + 0.1(-1 + 0 + 1) = 1$$

$$y_2 = y_1 + h(-y_1 + x_1 + 1) = 1 + 0.1(-1 + 0.1 + 1) = 1.01$$

$$y_3 = y_2 + h(-y_2 + x_2 + 1) = 1.01 + 0.1(-1.01 + 0.2 + 1) = 1.029$$

$$y_4 = y_3 + h(-y_3 + x_3 + 1) = 1.029 + 0.1(-1.029 + 0.3 + 1) = 1.056 \ 1$$

$$y_5 = y_4 + h(-y_4 + x_4 + 1) = 1.056 \ 1 + 0.1(-1.056 \ 1 + 0.4 + 1) = 1.090 \ 49$$

⋮

Then we input the initial condition and initial parameters for Example 6.2.2.

```
[> f := (u, v) -> -v+u+1;
[> t0 := 0: tf := 1: y0 := 1.0: n := 10:
[> with(plots)
[> z := Euler(f, t0, tf, y0, n);
[> z;
```

As shown in Table 6.2.4 The first column is the values of x_n , the second column is the values of y_n , from which we see that it coincides with the above computation perfectly. Next, we plot the

figures of numerical solutions with $h = 0.2, h = 0.1, h = 0.05$. The results can be seen from Figure 6.2.5.

Table 6.2.4 The solution of Euler’s method for Example 6.2.2 with $h = 0.1$

0.	1.0
0.1000000000	1.0
0.2000000000	1.010000000
0.3000000000	1.029000000
0.4000000000	1.056100000
0.5000000000	1.090490000
0.6000000000	1.131441000
0.7000000000	1.178296900
0.8000000000	1.230467210
0.9000000000	1.287420489
$\vdots$	$\vdots$

11×2 Array

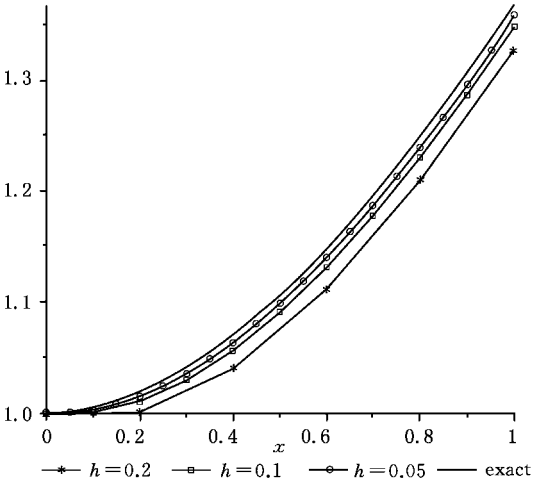


Figure 6.2.5 Graphs of Euler approximations with step sizes

Then, we use Maple2020 to get a better show of this problem

```
[> with(Student[NumericalAnalysis]):
```

```
[> Euler(diff(y(x), x) = -y(x)+x+1, y(0)=1, x=1, numsteps=5, output=plot)
```

From Figure 6.2.6, the solution of Euler’s method is getting better and better with the increasing of mesh points, which show that the algorithm is convergent.

```
[> Euler(diff(y(x), x) = -y(x)+x+1, y(0) = 1, x = 1, 'numsteps' = 10,
```

```
'digits' = 5, output = information):
```

```
[> Euler(diff(y(x), x) = -y(x)+x+1, y(0) = 1, x = 1, 'numsteps' = 40,
```

```
'digits' = 5, output = information):
```

From Table 6.2.5,6.2.6, we see that the accuracy has been improved greatly with the

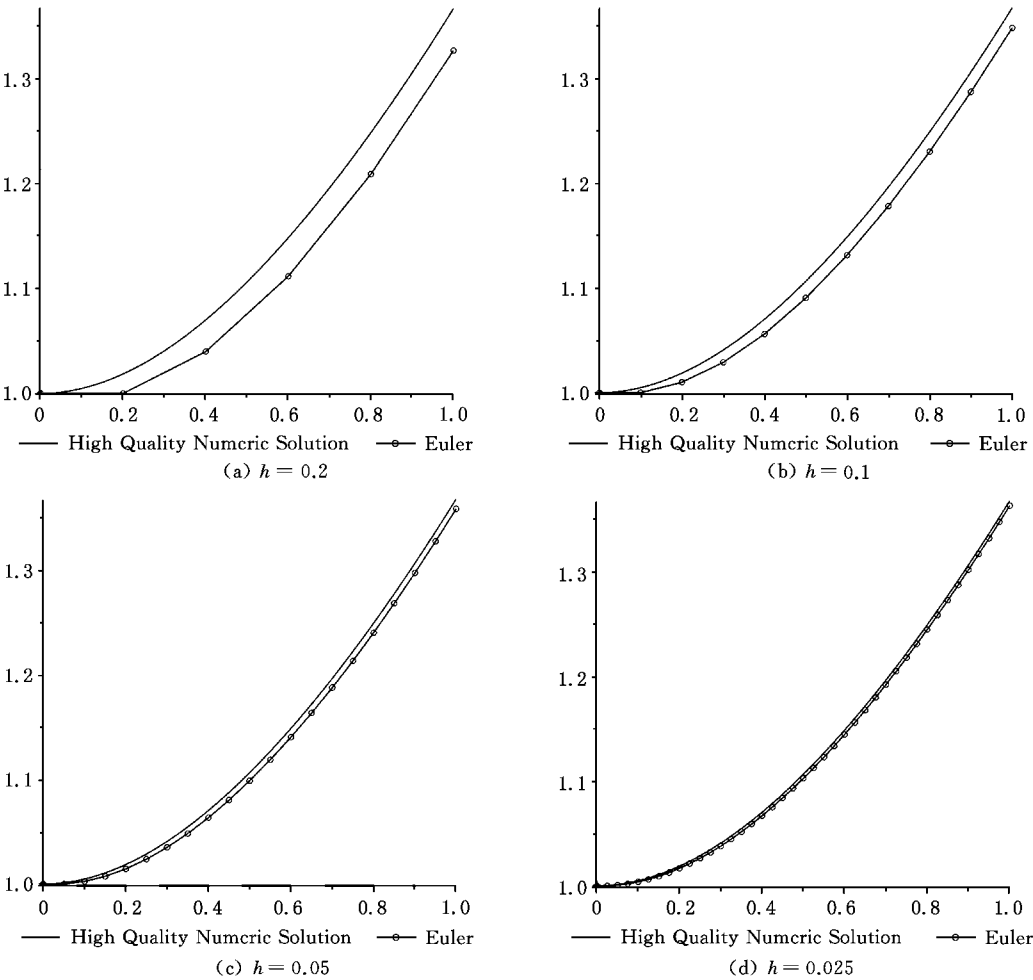


Figure 6.2.6 Euler and exact solution for Exanple 6.2.2

mesh refinement.

Table 6.2.5 The error of Euler’s method for Example 6.2.2 with $h=0.1$

x	[Maple’s numeric solution]	[Euler]	[Error]
0.	1.	1.	0.
0.10000	1.0048	1.	0.0048000
0.20000	1.0187	1.0100	0.0087000
0.30000	1.0408	1.0290	0.011800
0.40000	1.0703	1.0561	0.014200
0.50000	1.1065	1.0905	0.016010
0.60000	1.1488	0.1314	0.017359
0.70000	1.1966	1.1783	0.018303
0.80000	1.2493	1.2305	0.018833
$\vdots$	$\vdots$	$\vdots$	$\vdots$

$h = 0.1$

Table 6.2.6 The error of Euler’s method for Example 6.2.2 with $h=0.025$

x	[Maple’s numeric solution]	[Euler]	[Error]
0.	1.	1.	0.
0.025000	1.0003	1.	0.00030000
0.050000	1.0012	1.0006	0.00057500
0.075000	1.0027	1.0019	0.00084062
0.10000	1.0048	1.0037	0.0011121
0.12500	1.0075	1.0061	0.0014043
0.15000	1.0107	1.0091	0.0016317
0.17500	1.0145	1.0126	0.0019084
0.20000	1.0187	1.0167	0.0020482
$\vdots$	$\vdots$	$\vdots$	$\vdots$

$h=0.025$

Example 6.2.3 Solve the initial value problem

$$\begin{cases} \frac{dy}{dx} = \frac{1}{3}y(8 - y), 0 < x < 5, \\ y(0) = 1. \end{cases}$$

Solution This equation is called logistic population equation. Its general form is

$$\frac{dy}{dx} = \frac{1}{n}y(m - y), y(0) = 1.$$

Here, we choose $n=3, m=8$. Using variable separation method, we can easily get its exact solution

$$y(x) = \frac{8}{1 + 7e^{\frac{8x}{3}}}.$$

Now we use Euler’s method to solve this problem $y_{n+1} = y_n + \frac{h}{3}y_n(8 - y_n)$. Here, we use the parameters $h = 1, x_0 = 0, y_0 = 1$, then we get the iterative formulas

$$\begin{aligned} y_1 &= y_0 + \frac{h}{3}y_0(8 - y_0) = 1 + \frac{1}{3}(8 - 1) = 3.33 \\ y_2 &= y_1 + \frac{h}{3}y_1(8 - y_1) = 3.33 + \frac{1}{3}(3.33)(8 - 3.33) = 8.51 \\ y_3 &= y_2 + \frac{h}{3}y_2(8 - y_2) = 8.51 + \frac{1}{3}(8.51)(8 - 8.51) = 7.06 \\ y_4 &= y_3 + \frac{h}{3}y_3(8 - y_3) = 7.06 + \frac{1}{3}(7.06)(8 - 7.06) = 9.27 \\ y_5 &= y_4 + \frac{h}{3}y_4(8 - y_4) = 9.27 + \frac{1}{3}(9.27)(8 - 9.27) = 5.35 \end{aligned}$$

Note that the result of each calculation feeds into the next one. The resulting table of approximation values is can be seen in Table 6.2.7.

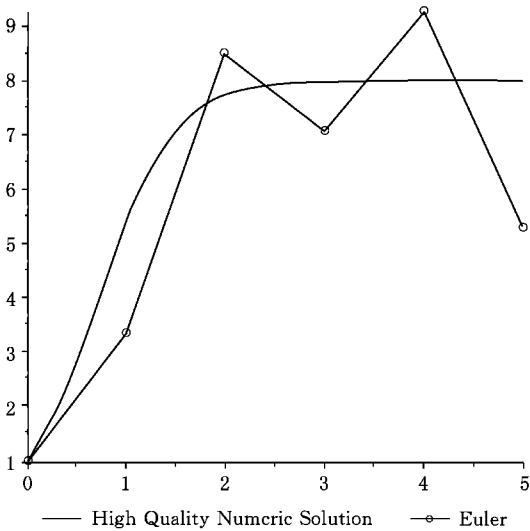
Table 6.2.7 The solution of Euler’s method for Example 6.2.3 with $h=1$

x_n	0	1	2	3	4	5
y_n	1	3.33	8.51	7.06	9.27	5.35

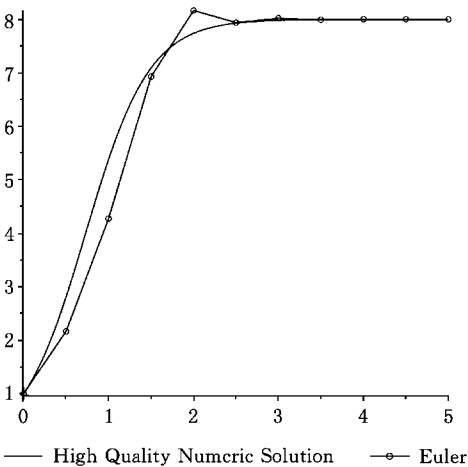
Now, we use Maple2020 to show the solutions when $h = 1, 0 < x < 5$

```
[>with(Student[NumericalAnalysis]):  
[>Euler(diff(y(x), x) = 1/3 * y(x) * (8 - y(x)), y(0) = 1, x = 5, numsteps = 5,  
output = plot)
```

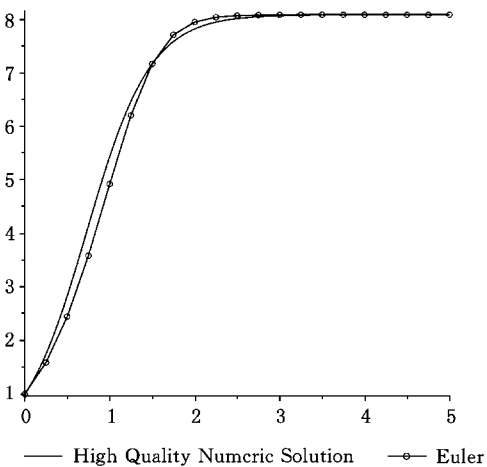
From Figure 6.2.7, it can be seen that Euler’s method behaves really bad, for there are oscillations in the solution. But we can improve our solution by mesh refinement.



(a) $h = 1$



(b) $h = 0.5$



(c) $h = 0.25$

Figure 6.2.7 Graphs of Euler approximation for Example 6.2.3 with $h = 1, 0.5, 0.25$

Then, we give a global view of this problem which can be seen from Figure 6.2.8.

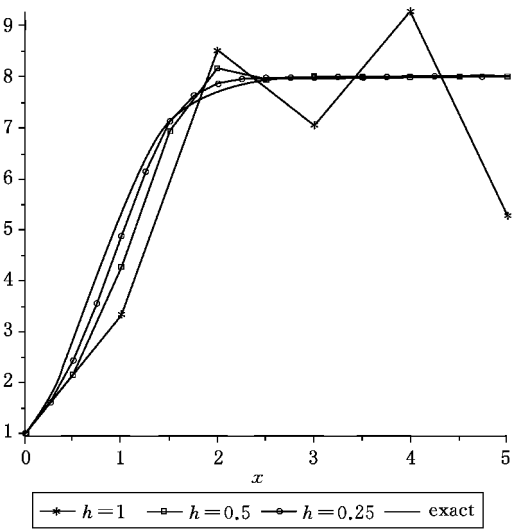


Figure 6.2.8 Graphs of Euler approximations with step sizes

Finally, we give an error analysis of this question:

```
[> Euler(diff(y(x), x) = 1/3 * y(x) * (8 - y(x)), y(0) = 1, x = 5, 'numsteps' = 5, 'digits' = 5, output = information);
```

From Table 6.2.8, it can be seen that the error is unacceptable and the solution is not convergent to the exact solution. But, if we use numsteps=20, all these problems can be corrected, see Table 6.2.9:

Table 6.2.8 The error of Euler's method for Example 6.2.3 with $h=1$

x	[Maple's numeric solution]	[Euler]	[Error]
0.	1.	1.	0.
1.	5.3822	3.3333	2.0489
2.	7.7385	8.5185	0.78002
3.	7.9813	7.0462	0.93512
4.	7.9987	9.2864	1.2877
5.	7.9999	5.3043	2.6956

We end this example with the directional curve shown in the following:

```
[> with(DEtools);
[> dfieldplot(diff(y(t), t) = 1/3 * y(t) * (8 - y(t)), y(t), t = 0 .. 5, y = 0 .. 12, axes = boxed, color = black);
```

Otherwise, we can use the following command:

```
[> solc := dsolve([diff(y(t), t) = 1/3 * y(t) * (8 - y(t)), y(0) = y0], y(t));
```

$$\text{solc} := y(t) = -\frac{8y_0}{e^{\frac{8t}{3}}y_0 - 8e^{\frac{8t}{3}} - y_0}$$

Table 6.2.9 The error of Euler’s method for Example 6.2.3 with $h=0.25$

x	[Maple’s numeric solution]	[Euler]	[Error]
0.	1.	1.	0.
0.25000	1.7414	1.5833	0.15807
0.50000	2.8118	2.4300	0.38182
0.75000	4.1082	3.5579	0.5530
1.	5.3822	4.8749	0.50726
1.2500	6.4014	6.1445	0.25692
1.5000	7.0909	7.0946	0.0036811
1.7500	7.5059	7.6299	0.12398
2.	7.7385	7.8652	0.12671
2.2500	7.8636	7.9536	0.089956
2.5000	7.9294	7.9843	0.054939
0.7500	7.9636	7.9948	0.031159
3.	7.9813	7.9983	0.016951
3.2500	7.9904	7.9994	0.0090167
3.5000	7.9951	7.9998	0.0047055
3.7500	7.9975	7.9999	0.0024352
4.	7.9987	8.0000	0.0012784
4.2500	7.9993	8.0000	0.00069280
4.5000	7.9997	8.0000	0.00029760
4.7500	7.9998	8.0000	0.00019920
5.	7.9999	8.0000	0.000099733

```
[>yvals := seq(i, i = 1 .. 10);  
[>toplot := seq(subs(y = y0, rhs(solc)), y0 = yvals);  
[>sols := plot([toplot], t = 0 .. 5, color = black);  
[>display(pvf, sols, color = black, view = [0 .. 5, 0 .. 12]);
```

The results can be shown in Figure 6.2.9 and Figure 6.2.10.

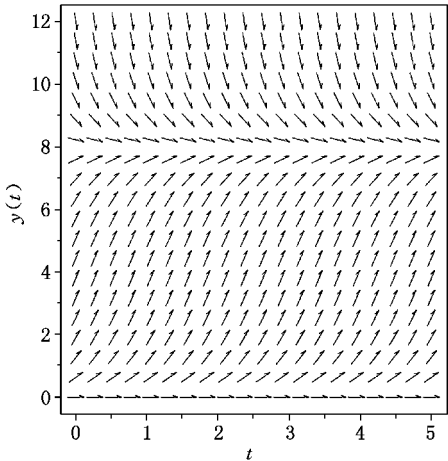


Figure 6.2.9 Direction field of Example 6.2.3

6.3 An Improvement in Euler’s Method

In the previous section, we see that Euler’s method is easy and simple, but its

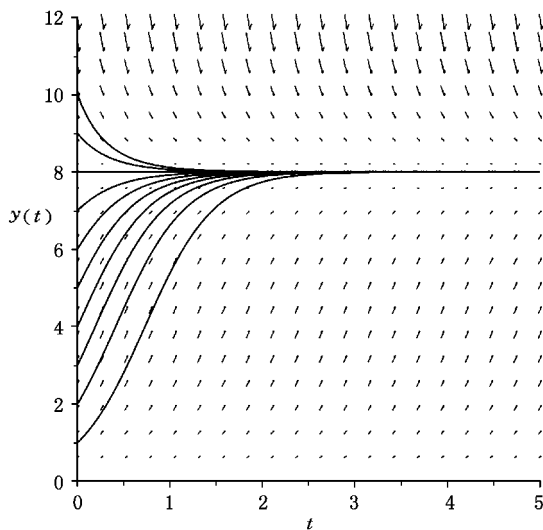


Figure 6.2.10 Direction field with ten solutions of Example 6.2.3

accuracy is not very good. So we want to make some changes to Euler's method to improve its efficiency. Specifically, first, we use Euler's method to get a preliminary approximation of $\bar{y}_{n+1}$ which is called predictor value. Though the accuracy of $\bar{y}_{n+1}$ may be very bad, we then use a corrector formula to get a much better approximation. This kind of predictor-corrector system is usually called improved Euler's method (or modified Euler's method)

$$\begin{cases} \bar{y}_{n+1} = y_n + hf(x_n, y_n), \\ y_{n+1} = y_n + \frac{h}{2}[f(x_n, y_n) + f(x_{n+1}, \bar{y}_{n+1})]. \end{cases} \quad (6.3.1)$$

We can also rewrite the above formula as the next average form

$$\begin{cases} y_p = y_n + hf(x_n, y_n), \\ y_c = y_n + hf(x_{n+1}, y_p), \\ y_{n+1} = \frac{1}{2}(y_p + y_c). \end{cases} \quad (6.3.2)$$

Example 6.3.1 Use Modified Euler's Method to solve the initial value problem

$$\begin{cases} \frac{dy}{dx} = x + y, 0 < x < 1 \\ y(0) = 1. \end{cases} \quad (6.3.3)$$

Solution This equation is a first-order linear ordinary differential equation. The exact solution is

$$y(x) = 2e^x - x - 1.$$

Here $f(x, y) = x + y$. In (6.3.1), the predictor-corrector formula for the modified Euler's method is

$$\bar{y}_{n+1} = y_n + h(x_n + y_n)$$

$$y_{n+1} = y_n + \frac{h}{2} [(x_n + y_n) + (x_{n+1} + \bar{y}_{n+1})]$$

Set step size $h = 0.1$, then $x_n = 0 + hn = 0.1n$. Substituting these data to (6.3.1), we can obtain

$$y_{n+1} = 1.105y_n + 0.0105n + 0.005, \quad y_0 = 1.$$

Then we have

$$\begin{aligned} n = 0, y_1 &= 1.110 \ 0 \\ n = 1, y_2 &= 1.242 \ 1 \\ n = 2, y_3 &= 1.398 \ 5 \\ &\vdots \end{aligned}$$

In fact, we can also use Maple2020 to calculate this modified Euler's method. The program code is as follows.

```
Imp_Euler:=proc(f,a,b,y0,n)
    description "Solves the ODE y'(t) = f(t, y) using the Modified Euler's method.
        Input parameters: f is a function of two variables f(t, y); a and b are
        the endpoints of the interval of integration [a, b]; n is the number of
        subintervals of [a, b]; y0 is the initial condition";
    local fm,h,i,t,w,wp,u,k1,k2; # Local variables
    h:=evalf((b-a)/n);
    t:= Array([seq(a + i * h, i=0..n)]);
    w := Array(1..n+1);
    u := Array(1..n+1);
    w[1] := y0;
    # This loop implements the difference equation for modified Euler's method
    for i from 1 to n do
        k1 := apply(f,t[i],w[i]);
        u[i+1] := w[i] + h * k1;
        k2 := apply(f,t[i+1],u[i+1]);
        w[i+1] := w[i] + h * 1/2 * (k1 + k2);
    od;
    Array([seq([t[i],w[i]],i=1..n+1)]): # Output
end;
```

Then we input the initial condition and initial parameters for Example 6.3.1.

```
[> f := (u, v) -> u + v;
> t0 := 0; tf := 1; y0 := 1.0; n := 10;
> with(plots):
> z := Imp_Euler(f, t0, tf, y0, n);
> z;
```

From Table 6.3.1, we see that the first column is mesh point value, and the second

column is the solution of modified Euler’s method which coincides with the data we got before. Next, we plot the figures with $h = 0.2, h = 0.1, h = 0.05$. The solutions of modified Euler’s method can be found in Figure 6.3.1. From this figure, we see that the accuracy of modified Euler’s method has been improved greatly, even when $h = 0.2$. This shows the capability of the modified Euler’s method for solving ordinary differential equations.

Table 6.3.1 Solutions of modified Euler’s method for Example 6.3.1 with $h = 0.1$

0.	1.0
0.1000000000	1.110000000
0.2000000000	1.242050000
0.3000000000	1.398465250
0.4000000000	1.581804101
0.5000000000	1.794893532
0.6000000000	2.040857353
0.7000000000	2.323147375
0.8000000000	2.645577849
0.9000000000	3.012363523
1.000000000	3.428161693

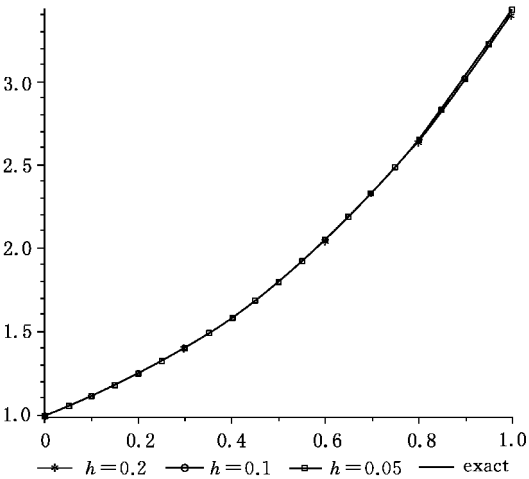


Figure 6.3.1 Graphs of modified Euler approximations with step sizes

Then, we try to use Maple to compare the accuracy of Euler’s method and modified Euler’s method. This time, we use command “InitialValueProblem” in the Student Package [NumericalAnalysis].

```
[>with(Student[NumericalAnalysis]);  
[>DE := diff(y(x), x) = x + y(x);  
[>InitialValueProblem(DE, y(0) = 1, x = 1, method = rungekutta, submethod =  
meuler, comparewith = [[euler]], output = information, digits = 5, numsteps = 10);
```

From Table 6.3.2, we can find that the accuracy of modified Euler’s method is much

better than the one of Euler’s method. Combined with the previous figures of these two methods, the numerical capability of modified Euler’s method is obviously better than the one of Euler’s method.

Table 6.3.2 Errors of modified Euler’s method and Euler’s method for Example 6.3.1 with $h=0.1$

x	[Maple’s numeric solution]	[R-K Mod.Euler]	[Error]	[Euler]	[Error]
0.	1.	1.	0.	1.	0.
0.10000	1.1103	1.1100	0.00030000	1.1000	0.010300
0.20000	1.2428	1.2421	0.00075000	1.2200	0.022800
0.30000	1.3997	1.3985	0.0012347	1.3620	0.03770
0.40000	1.5836	1.5818	0.0017959	1.5282	0.055400
0.50000	1.7974	1.7948	0.0025065	1.7210	0.076380
0.60000	2.0442	2.0409	0.0033426	1.9431	0.10108
0.70000	2.3275	2.3231	0.0043526	2.1974	0.13007
0.80000	2.6511	2.6456	0.0055222	2.4872	0.16392
0.90000	3.0192	3.0124	0.0068365	2.8159	0.20330
1.	3.4366	3.4282	0.0084383	3.1875	0.24912

Example 6.3.2 Use modified Euler’s Method to solve the initial value problem

$$\begin{cases} \frac{dy}{dx} = \frac{1}{x}(y^2 + y), 1 \leq x \leq 3 \\ y(1) = -2. \end{cases}$$

(6.3.4)

Here, we use $h = 0.5$, compare the numerical solution with exact solution.

Solution First, we can use variable separation method to get the general solution of (6.3.4), which can be written as $y = Cx(y + 1)$. Then using the initial condition $y(1) = -2$, we can get $C = 2$. So we can obtain the exact solution is

$$y = \frac{2x}{1 - 2x}.$$

In connection with $y(x_n) = \frac{2x_n}{1 - 2x_n}, n = 0, 1, 2, \dots$, modified Euler’s method gives

$$y_{n+1} = y_n + \frac{h}{2}\{f(x_n, y_n) + f(x_{n+1}, y_n + hf(x_n, y_n))\}$$
$$y_0 = y(1) = -2.$$

Using $h = 0.5, x_n = x_0 + hn = 1 + 0.5n, f(x, y) = \frac{1}{x}(y^2 + y)$, we have

$$y_{n+1} = \frac{0.125y_n^4 + (0.75 + 0.25n)y_n^3 + (2.125 + 1.5n + 0.25n^2)y_n^2 + 0.25(3 + n)^2(2 + n)y_n}{(1 + 0.5n)^2(3 + n)}$$

For $y_0 = -2$, so we have

$$n = 0, y_1 = -1.500\ 0$$
$$n = 1, y_2 = -1.335\ 9$$
$$n = 2, y_3 = -1.252\ 5$$
$$n = 3, y_4 = -1.202\ 1$$

Next, we use the Maple2020 code in the previous section to compute the solution of modified Euler's method which can be seen in Figure 6.3.2.

```

1 Imp_Euler:=proc(f,a,b,y0,n)
2   description "Solves the ODE  $y'(t) = f(t, y)$  using the Modified Euler's method.
3   Input parameters: f is a function of two variables f(t, y); a and b are
4   the endpoints of the interval of integration [a, b]; n is the number of
5   subintervals of [a, b]; y0 is the initial condition";
6
7   local fm,h,i,t,w,wp,u,k1,k2; # Local variables
8   h:=evalf((b-a)/n);
9   t:= Array([seq(a + i*h, i=0..n)]);
10  w := Array(1..n+1);
11  u := Array(1..n+1);
12  w[1] := y0;
13  # This loop implements the difference equation for improved Euler's method
14  for i from 1 to n do
15      k1 := apply(f,t[i],w[i]);
16      u[i+1] := w[i] + h*k1;
17      k2 := apply(f,t[i+1],u[i+1]);
18      w[i+1] := w[i] + h*1/2*(k1 + k2);
19  od:
20  Array([seq([t[i],w[i]],i=1..n+1)]): # Output
21 end;
```

Figure 6.3.2 Maple2020 code for modified Euler method

```

[> f := (u, v) -> (v^2 + v)/u;
[> t0 := 1: tf := 3: y0 := -2.0: n := 4:
[> z1 := Imp_Euler(f, t0, tf, y0, n);
```

From Table 6.3.3, we see that the solution perfectly coincides with the solution we got before. Then, we use the Maple2020 code to compute the solutions of Euler's method which can be seen in Figure 6.3.3.

Table 6.3.3 Solution of Example 6.3.2 with M-Euler method

1.	-2.0
1.500000000	-1.500000000
2.000000000	-1.335937500
2.500000000	-1.252458658
3.000000000	-1.202087254

```

1 Euler_my:=proc(f,a,b,y0,n)
2   description "Solves the ODE  $y'(t) = f(t,y)$  using the Euler's method.
3   Input parameters: f is a function of two variables f(t,y); a and b are
4   the endpoints of the interval of integration [a,b]; n is the number of subintervals of
5   [a,b]; y0 is the initial condition";
6   local fm,h,i,t,w,wp; # Local variables
7   h:=evalf((b-a)/n);
8   t:= Array([seq(a + i*h, i=0..n)]);
9   w := Array(1..n+1);
10  w[1] := y0;
11  # This loop implements the difference equation for Euler's method
12  for i from 1 to n do
13      fm := apply(f,t[i],w[i]);
14      w[i+1] := w[i] + h*fm;
15  od:
16  Array([seq([t[i],w[i]],i=1..n+1)]): # Output
17 end;
```

Figure 6.3.3 Maple2020 code for Euler's method

```
[> z2 := Euler_my(f, t0, tf, y0, n);
```

The data above in Table 6.3.4 is the solution of Euler’s method, we see that Euler’s method converges to the totally wrong solution which can be seen in Figure 6.3.4.

Table 6.3.4 Solution of Example 6.3.2 with Euler’s method

1.	−2.0
1.500000000	−1.000000000
2.000000000	−1.000000000
2.500000000	−1.000000000
3.000000000	−1.000000000

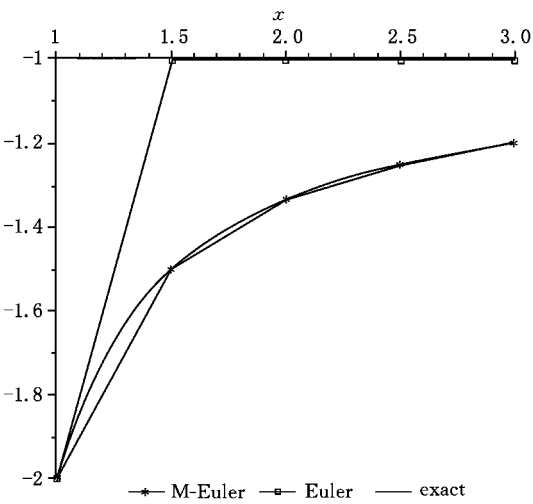


Figure 6.3.4 Graphs of Modified Euler vs Euler with step sizes $h=0.5$

In the Figure 6.3.4, we clearly see that modified Euler’s method converges to the exact solution while Euler’s method does not. From this phenomenon, we find that in this example, modified Euler’s method is much more effective than Euler’s method.

6.4 The Runge-Kutta Method

The Runge-Kutta method was first proposed by German scientists C. Runge and M. W. Kutta in the late 19th century. Later, it was improved to varying degrees.

The Runge-Kutta method is still widely used in the recent years.

6.4.1 Second-Order Runge-Kutta Method

In the previous sections, we point out that using Taylor series method, one can get the one step method of (6.1.1). Suppose $y(t)$ is the solution of initial value problem 6.1.1. Using Taylor series, we can get

$$\begin{aligned}
 y(t+h) = & y(t) + hy'(t) + \frac{1}{2} h^2 y''(t) + \frac{1}{3!} h^3 y'''(t) + \cdots \\
 & + \frac{1}{p!} h^p y^{(p)}(t) + \frac{1}{(p+1)!} h^{p+1} y^{(p+1)}(\xi), t < \xi < t+h \quad (6.4.1)
 \end{aligned}$$

From (6.1.1), we see that

$$y'(t) = f(t, y(t)),$$

Substituting this into (6.4.1), we get

$$\begin{aligned}
 y(t+h) = & y(t) + hf(t, y(t)) + \frac{1}{2} h^2 f'(t, y(t)) + \\
 & \frac{1}{3!} h^3 f''(t, y(t)) + \cdots + \frac{1}{p!} h^p f^{(p-1)}(t, y(t)) + \\
 & \frac{1}{(p+1)!} h^{p+1} y^{(p+1)}(\xi).
 \end{aligned}$$

Set

$$\Phi(t, y, h) = f(t, y(t)) + \frac{1}{2} h f'(t, y(t)) + \cdots + \frac{1}{p!} h^{p-1} f^{(p-1)}(t, y(t)).$$

Then, we get

$$y(t_{n+1}) \simeq y(t_n) + h\Phi(t_n, y(t_n), h).$$

So we obtain the one step method

$$\begin{cases} y_{n+1} = y_n + h\Phi(t_n, y_n, h), n = 0, 1, \dots, N-1, \\ y_0 = \eta. \end{cases}$$

However, it is unrealistic to expand several terms with Taylor series to improve the accuracy because it is necessary to calculate the values of several partial derivatives. For many problems, this is very complicated and the workload is heavy. From the modified Euler's method, we see that if we set $\Phi(t, y, h)$ to be the linear combination of two different function values, it does improve the accuracy compared with Euler's method. We now use this thought to consider one step method discussed above.

In the Taylor series of $y(t+h)$, we set $p=2$, then

$$\begin{aligned}
 y(t+h) = & y(t) + hf(t, y(t)) + \frac{1}{2} h^2 f'(t, y(t)) + O(h^3) \\
 = & y(t) + h\Phi(t, y, h) + O(h^3),
 \end{aligned}$$

where

$$\Phi(t, y, h) = f(t, y(t)) + \frac{1}{2} h f'(t, y(t)).$$

Because

$$f'(t, y(t)) = \frac{d}{dt} f(t, y) = f'_t(t, y) + f'_y(t, y) y'(t)$$

and

$$y'(t) = f(t, y),$$

the above formula can be written as

$$y(t+h) = y(t) + h \left\{ f(t, y) + \frac{h}{2} [f'_t(t, y) + f'_y(t, y) f(t, y)] \right\} + O(h^3) \quad (6.4.2)$$

On the other hand, set

$$\Phi(t, y, h) = c_1 K_1 + c_2 K_2, \quad (6.4.3)$$

where $K_1 = f(t, y)$, $K_2 = f(t + a_2 h, y + b_{21} h K_1)$, the parameters c_1, c_2, a_2, b_{21} are the undetermined coefficients. We expand K_2 as

$$K_2 = f(t, y) + a_2 h f'_t(t, y) + b_{21} h f(t, y) f'_y(t, y) + O(h^2).$$

Then, (6.4.3) can be written as

$$\begin{aligned} \Phi(t, y, h) &= (c_1 + c_2) f(t, y) \\ &\quad + c_2 h [a_2 f'_t(t, y) + b_{21} f'_y(t, y) f(t, y)] + O(h^2) \\ y(t+h) &= y(t) + h \Phi(t, y, h) + O(h^3) \\ &= y(t) + h \{ (c_1 + c_2) f(t, y) + c_2 h [a_2 f'_t(t, y) + b_{21} f'_y(t, y) f(t, y)] \} \\ &\quad + O(h^3). \end{aligned} \quad (6.4.4)$$

Comparing (6.4.2) with (6.4.4), we find that

$$\begin{cases} c_1 + c_2 = 1, \\ c_2 a_2 = \frac{1}{2}, \\ c_2 b_{21} = \frac{1}{2}. \end{cases} \quad (6.4.5)$$

The system (6.4.5) has four unknown variables c_1, c_2, a_2, b_{21} , but only three equations, so we can choose one as free variable

$$\begin{cases} b_{21} = a_2, \\ c_1 = \frac{2a_2 - 1}{2a_2}, \\ c_2 = \frac{1}{2a_2}. \end{cases}$$

Substituting the above parameters into (6.4.3) and (6.4.4), we get the second-order Runge-Kutta method

$$y_{n+1} = y_n + h(c_1 K_1 + c_2 K_2), n = 0, 1, 2, \dots, N-1$$

where

$$\begin{aligned} K_1 &= f(t_n, y_n), \\ K_2 &= f(t_n + a_2 h, y_n + a_2 h K_1), \\ c_1 &= \frac{2a_2 - 1}{2a_2}, c_2 = \frac{1}{2a_2}, y_0 = \eta \end{aligned} \quad (6.4.6)$$

Particularly, if we set $a_2 = 1$, then $c_1 = c_2 = \frac{1}{2}$, then (6.4.6) becomes

$$\begin{aligned} y_{n+1} &= y_n + \frac{h}{2} (K_1 + K_2), n = 0, 1, 2, \dots, N-1 \\ K_1 &= f(t_n, y_n), \end{aligned}$$

$$K_2 = f(t_n + h, y_n + hK_1),$$

$$y_0 = \eta.$$

This is the modified Euler's method. If we set $a_2 = \frac{1}{2}$, then $c_1 = 0, c_2 = 1$, so we can obtain

$$y_{n+1} = y_n + hK_2, n = 0, 1, \dots, N - 1$$

$$K_1 = f(t_n, y_n),$$

$$K_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{1}{2}hK_1\right),$$

$$y_0 = \eta \quad (6.4.7)$$

We call formula (6.4.7) Midpoint method.

If we set $a_2 = \frac{2}{3}$, then $c_1 = \frac{1}{4}, c_2 = \frac{3}{4}$, so that we get Heun method

$$y_{n+1} = y_n + \frac{h}{4}(K_1 + 3K_2), n = 0, 1, 2, \dots, N - 1$$

$$K_1 = f(t_n, y_n),$$

$$K_2 = f\left(t_n + \frac{2}{3}h, y_n + \frac{2}{3}hK_1\right),$$

$$y_0 = \eta. \quad (6.4.8)$$

Example 6.4.1 Use Midpoint method and Heun method to solve the initial value problem

$$\begin{cases} y' = -y + t^2 + 1, 0 \leq t \leq 1, \\ y(0) = 1, \end{cases}$$

then compare with the exact solution.

Solution Set $h = 0.1, t_n = 0.1n, y_0 = 1$, and $f(t, y) = -y + t^2 + 1$. Using Midpoint method, we can get

$$K_1 = f(t_0, y_0) = f(0, 1) = 0,$$

$$K_2 = f\left(t_0 + \frac{h}{2}, y_0 + \frac{h}{2}K_1\right) = f(0.05, 1) = 0.0025,$$

$$y_1 = y_0 + hK_2 = 1 + 0.1(0.0025) = 1.00025,$$

Then, $y_2, y_3, \dots$ can be computed in the same way.

Next, if we use Heun method, then

$$K_1 = f(t_0, y_0) = f(0, 1) = 0,$$

$$K_2 = f\left(t_0 + \frac{2}{3}h, y_0 + \frac{2}{3}hK_1\right) = f(0.0667, 1) = 0.0044,$$

$$y_1 = y_0 + \frac{1}{4}h(K_1 + 3K_2) = 1 + 0.025(3)(0.0044) = 1.00033,$$

Then, $y_2, y_3, \dots$ can be computed in the same way.

Again, we use Maple2020 to verify our results above. First, we give the Maple code for Midpoint method and Heun method which can be seen from Figure 6.4.1 and 6.4.2.

[> f := (u, v) -> -v + u^2 + 1;

```
[> t0 := 0: tf := 1: y0 := 1.0: n := 10:
[> z1 := midp_Euler(f, t0, tf, y0, n):
[> z2 := Heun(f, t0, tf, y0, n):
```

```
1 midp_Euler:=proc(f,a,b,y0,n)
2   description "Solves the ODE  $y'(t) = f(t, y)$  using the midpoint method.
3   Input parameters: f is a function of two variables f(t, y); a and b are
4   the endpoints of the interval of integration [a, b]; n is the number of
5   subintervals of [a, b]; y0 is the initial condition";
6
7   local fm,h,i,t,w,wp,u,v,k1,k2; # Local variables
8   h:=evalf((b-a)/n);
9   t:= Array([seq(a + i*h, i=0..n)]);
10  w := Array(1..n+1);
11  u := Array(1..n+1);
12  v := Array(1..n+1);
13  w[1] := y0;
14  # This loop implements the difference equation for midpoint Euler's method
15  for i from 1 to n do
16      k1 := apply(f,t[i],w[i]);
17      u[i+1] := w[i] + 0.5*h*k1;
18      v[i+1] := t[i] + 0.5*h;
19      k2 := apply(f,v[i+1],u[i+1]);
20      w[i+1] := w[i] + h*k2;
21  od:
22  Array([seq([t[i],w[i]],i=1..n+1)]): # Output
23  end;
```

Figure 6.4.1 Maple2020 code for Mid-point Euler's method

```
1 Heun:=proc(f,a,b,y0,n)
2   description "Solves the ODE  $y'(t) = f(t, y)$  using the 2-order heun method.
3   Input parameters: f is a function of two variables f(t, y); a and b are
4   the endpoints of the interval of integration [a, b]; n is the number of
5   subintervals of [a, b]; y0 is the initial condition";
6
7   local fm,h,i,t,w,wp,u,v,k1,k2; # Local variables
8   h:=evalf((b-a)/n);
9   t:= Array([seq(a + i*h, i=0..n)]);
10  w := Array(1..n+1);
11  u := Array(1..n+1);
12  v := Array(1..n+1);
13  w[1] := y0;
14  # This loop implements the difference equation for Heun method
15  for i from 1 to n do
16      k1 := apply(f,t[i],w[i]);
17      u[i+1] := w[i] + 2./3*h*k1;
18      v[i+1] := t[i] + 2./3*h;
19      k2 := apply(f,v[i+1],u[i+1]);
20      w[i+1] := w[i] + 0.25*h*(k1+3*k2);
21  od:
22  Array([seq([t[i],w[i]],i=1..n+1)]): # Output
23  end;
```

Figure 6.4.2 Maple2020 code for Second-order Heun method

From Table 6.4.1 and Table 6.4.2, we can see that the solutions are both convergent to the exact solutions which can be seen clearly in Figure 6.4.1 and Figure 6.4.2.

Table 6.4.1 Solution of Midpoint Euler method

Midpoint method	
0.	1.0
0.1000000000	1.000250000
0.2000000000	1.002426250
0.3000000000	1.008245756
0.4000000000	1.019262409
0.5000000000	1.036882480
0.6000000000	1.062378644
0.7000000000	1.096902673
0.8000000000	1.141496919
0.9000000000	1.197104712
⋮	⋮

Table 6.4.2 Solution of Heun method

Heun method	
0.	1.0
0.1000000000	1.000333333
0.2000000000	1.002585000
0.3000000000	1.008472758
0.4000000000	1.019551179
0.5000000000	1.037227150
0.6000000000	1.062773904
0.7000000000	1.097343716
0.8000000000	1.141979396
0.9000000000	1.197624687
⋮	⋮

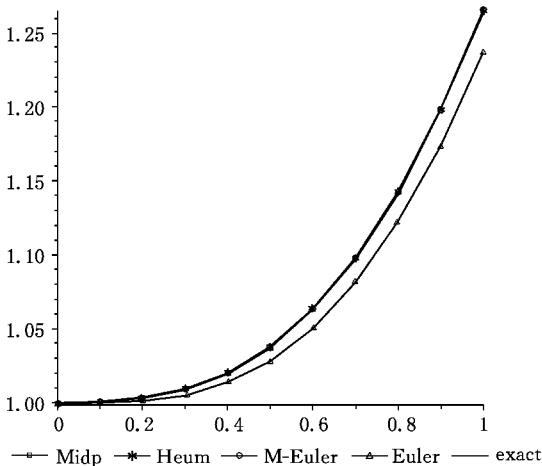


Figure 6.4.3 Graphs of Midpoint, Heun, imp_Euler, Euler with step sizes $h=0.1$

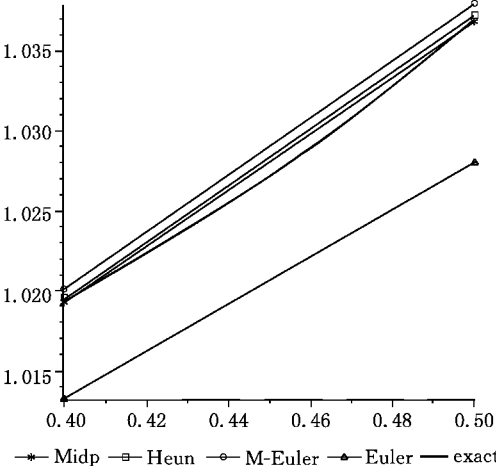


Figure 6.4.4 Graphs of Midpoint, Heun, imp_Euler, Euler with step sizes $h=0.1$ (closer view)

Figure 6.4.3 shows that except Euler’s method behaves poorly in the computation of this example, modified Euler’s method, Heun method and Midpoint method all behave very well. But, if we give close view of Figure 6.4.3 which can be seen in Figure 6.4.4, we see that Midpoint method is best, and Heun method is better than modified Euler’s method but worse than Midpoint method in this example.

6.4.2 Third-Order and Fourth-Order Runge-Kutta Method

First, we give the formula of third-order Heun and Rung-Kutta method.

Third-order Heun method

$$\begin{cases} y_{n+1} = y_n + \frac{h}{4}(K_1 + 3K_3), n = 0, 1, \dots, N-1, \\ K_1 = f(t_n, y_n), \\ K_2 = f\left(t_n + \frac{1}{3}h, y_n + \frac{1}{3}hK_1\right), \\ K_3 = f\left(t_n + \frac{2}{3}h, y_n + \frac{2}{3}hK_2\right), \\ y_0 = \eta. \end{cases}$$

Third-order Runge-Kutta method

$$y_{n+1} = y_n + \frac{h}{9}(2K_1 + 3K_2 + 4K_3), n = 0, 1, \dots, N-1,$$

where

$$\begin{aligned} K_1 &= f(t_n, y_n), \\ K_2 &= f\left(t_n + \frac{1}{2}h, y_n + \frac{1}{2}hK_1\right), \\ K_3 &= f\left(t_n + \frac{3}{4}h, y_n + \frac{3}{4}hK_2\right), \\ y_0 &= \eta. \end{aligned}$$

The above two methods are third-order explicit methods. Their local discretization errors are $O(h^4)$. In application, the most commonly used one step method is fourth-order Runge-Kutta method

$$\begin{aligned} y_{n+1} &= y_n + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4), n = 0, 1, \dots, N-1, \\ K_1 &= f(t_n, y_n), \\ K_2 &= f\left(t_n + \frac{1}{2}h, y_n + \frac{1}{2}hK_1\right), \\ K_3 &= f\left(t_n + \frac{1}{2}h, y_n + \frac{1}{2}hK_2\right), \\ K_4 &= f(t_n + h, y_n + hK_3), \\ y_0 &= \eta. \end{aligned}$$

Since this method has the earliest application, it is also called the classical Runge-Kutta method. Its local discretization error is $O(h^5)$. Next, we give the Maple2020 code for fourth-order Runge-Kutta method, which can be seen from Figure 6.4.5.

Example 6.4.2 Use classical fourth-order Runge-Kutta method to solve initial value problem

$$\begin{cases} y' = -y + t^2 + 1, 0 \leq t \leq 1, \\ y(0) = 1. \end{cases}$$

Solution We set $h = 0.1, N = 10, t_n = 0.1n, y_0 = 1$. Here $f(t, y) = -y + t^2 + 1$.

```

1 RK4:=proc(f,a,b,y0,n)
2   description "Solves the ODE y'(t) = f(t, y) using the 4-order Runge-Kutta method.
3   Input parameters: f is a function of two variables f(t, y); a and b are
4   the endpoints of the interval of integration [a, b]; n is the number of
5   subintervals of [a, b]; y0 is the initial condition";
6
7   local fm,h,i,t,w,wp,u,v,k1,k2,k3,k4; # Local variables
8   h:=evalf((b-a)/n);
9   t:= Array([seq(a + i*h, i=0..n)]);
10  w := Array(1..n+1);
11  u := Array(1..n+1);
12  v := Array(1..n+1);
13  w[1] := y0;
14  # This loop implements the difference equation for 4-order Runge-Kutta method
15  for i from 1 to n do
16      k1 := apply(f,t[i],w[i]);
17      u[i+1] := w[i] + 0.5*h*k1;
18      v[i+1] := t[i] + 0.5*h;
19      k2 := apply(f,v[i+1],u[i+1]);
20      u[i+1] := w[i] + 0.5*h*k2;
21      v[i+1] := t[i] + 0.5*h;
22      k3 := apply(f,v[i+1],u[i+1]);
23      u[i+1] := w[i] + h*k3;
24      v[i+1] := t[i] + h;
25      k4 := apply(f,v[i+1],u[i+1]);
26      w[i+1] := w[i] + 1.0/6.0*h*(k1+2*k2+2*k3+k4);
27  od:
28  Array([seq([t[i],w[i]],i=1..n+1)]): # Output
29  end;

```

Figure 6.4.5 Maple2020 code for fourth-order Runge-Kutta method

We can obtain

$$\begin{aligned}
 K_1 &= f(t_0, y_0) = f(0, 1) = 0, \\
 K_2 &= f\left(t_0 + \frac{h}{2}, y_0 + \frac{h}{2}K_1\right) = f(0.05, 1) = 0.002\ 5, \\
 K_3 &= f\left(t_0 + \frac{h}{2}, y_0 + \frac{h}{2}K_2\right) = f(0.05, 1.000\ 125) = 0.002\ 375, \\
 K_4 &= f(t_1, y_0 + hK_3) = f(0.1, 1.000\ 237\ 5) = 0.009\ 762\ 5, \\
 y_1 &= y_0 + (K_1 + 2K_2 + 2K_3 + K_4)\left(\frac{h}{6}\right) = 1.000\ 325\ 208.
 \end{aligned}$$

Using the Maple2020 code RK4, we get the following table

```

[>f := (u, v) -> -v + u^2 + 1;
[> t0 := 0: tf := 1: y0 := 1.0: n := 10;
[>RK4(f, t0, tf, y0, n);

```

In Table 6.4.3, the first column is t_n , the second column is numerical solution of fourth-order Runge-Kutta method, the third column is the exact solution at t_n , and the fourth column is numerical error. From this table, we see that the error of fourth-order Runge-Kutta method is much more accurate than Midpoint method and modified Euler's method. We can conform this conclusion by using the following command.

```

[>with(Student[NumericalAnalysis]);
[>RungeKutta(diff(y(t), t) = -y(t) + t^2 + 1, y(0) = 1, t = 1, comparewith =
[[rk4]], output = information, digits = 10, numsteps = 10)

```

Table 6.4.3 Solution of Example 6.4.2 with 4th-order Runge-Kutta method

0.	1.0	1.	0.
0.1000000000	1.000325208	1.000325164	4.4×10^{-8}
0.2000000000	1.002538594	1.002538494	1.00×10^{-7}
0.3000000000	1.008363723	1.008363559	1.64×10^{-7}
0.4000000000	1.019360144	1.019359908	2.36×10^{-7}
0.5000000000	1.036938993	1.036938681	3.12×10^{-7}
0.6000000000	1.062377119	1.062376728	3.91×10^{-7}
0.7000000000	1.096829865	1.096829392	4.73×10^{-7}
0.8000000000	1.141342626	1.141342072	5.54×10^{-7}
0.9000000000	1.196861317	1.196860681	6.36×10^{-7}
1.0000000000	1.264241835	1.264241118	7.17×10^{-7}

```
[> RungeKutta(diff(y(t), t) = -y(t) + t^2 + 1, y(0) = 1, t = 1, comparewith =
[[rk4]]], output = plot, digits = 10)
```

Table 6.4.4 Solution of Example 6.4.2 comparison

t	[Maple's numeric solution]	[R-K Midpt.]	[Error]	[R-K 4h Ord.]	[Error]
0.	1.	1.	0.	1.	0.
0.1000000000	1.00082516392808	1.00050000000000	0.000174836071918705	1.00032520833333	$4.44052521419991 \times 10^{-8}$
0.2000000000	1.00253849384404	1.00290250000000	0.000364006155964436	1.00253859402865	$1.00184610385412 \times 10^{-7}$
0.3000000000	1.00836355863656	1.00892676250000	0.000563203863436446	1.00836372340773	$1.64771164712363 \times 10^{-7}$
0.4000000000	1.01935990792872	1.02012872006250	0.000768812133778107	1.01936014391227	$2.35983551810648 \times 10^{-7}$
0.5000000000	1.03693868057473	1.03791649165656	0.000977811081830149	1.03693899255056	$3.11975822953059 \times 10^{-7}$
0.6000000000	1.06237672781195	1.06356442494919	0.00118769713724198	1.06237711900530	$3.91193349491559 \times 10^{-7}$
0.7000000000	1.09682939241718	1.09822580457902	0.00139641216183573	1.09682986475129	$4.72334108048145 \times 10^{-7}$
0.8000000000	1.14134207176556	1.14294435314404	0.00160228137845397	1.14134262608023	$5.54314671719069 \times 10^{-7}$
0.9000000000	1.19686068051880	1.19866463959533	0.00180395907652819	1.19686131675920	$6.36240400586630 \times 10^{-7}$
1.	1.26424111765711	1.26624149883377	0.00200038117665891	1.26424183503644	$7.17379323500467 \times 10^{-7}$

The results can be shown in Figure 6.4.6.

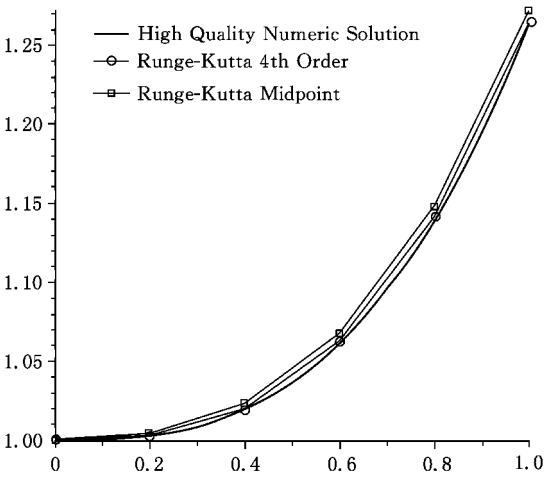


Figure 6.4.6 Comparison between 4th-order Runge-Kutta method and Midpoint method

Appendix Application of Maple2020 for Ordinary Differential Equations

In this appendix, we mainly consider the application of Maple software for ordinary differential equations. Mathematical software is a computer software designed for the calculation and processing of mathematical formulas, functions and data. Using mathematical software for the data processing of math, physics, biology, sociology and engineering technology can replace the complicated manual calculation, and greatly improve the work efficiency and accuracy.

Maple is a very excellent computer mathematics software. Maple software is developed by Waterloo University in Canada and is known as “mathematician’s software” in the fields of mathematics and science. Maple has millions of users around the world and is widely used in engineering, science and education. Users penetrate more than 96% of the world’s major universities and research institutes and more than 81% of the global fortune 500 enterprises. Maple system has built-in advanced modeling and simulation modules, including the world’s most powerful symbolic calculation, infinite precision numerical calculation, innovative Internet connection and powerful 4GL language, with more than 5 000 calculation commands. Maple’s mathematics and analysis functions cover almost all mathematical branches, such as calculus, linear algebra, differential equations, special functions, image and sound processing, statistics, dynamic systems, etc. It is a rare and powerful auxiliary teaching tool.

A1 Introduction for Maple2020

Maple is a cross platform software, which is applicable to various mainstream operating systems, including Windows system, Linux system, MAC system, etc. In this appendix, we focus our attention on Maple2020 which is a very powerful new version of Maple family. After the software installation is completed, the first time you see Maple2020, you will find the following frame.

There are two working interfaces of Maple2020. One is file mode and another is worksheet mode. Generally speaking, both interfaces can be used, but for experienced users, they prefer the worksheet mode that needs to write code.

When you start typing, your typing appears to the right of the prompt; see Figure A1.1 and A1.2.

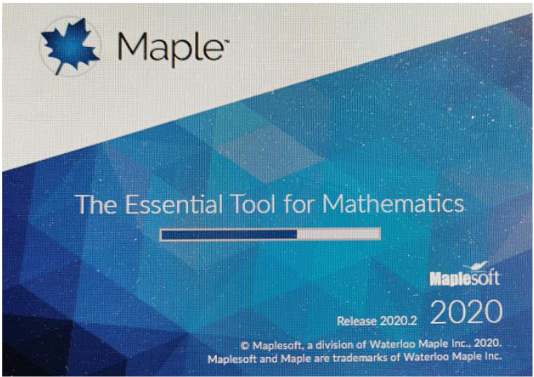


Figure A1.1 Maple2020 startup interface

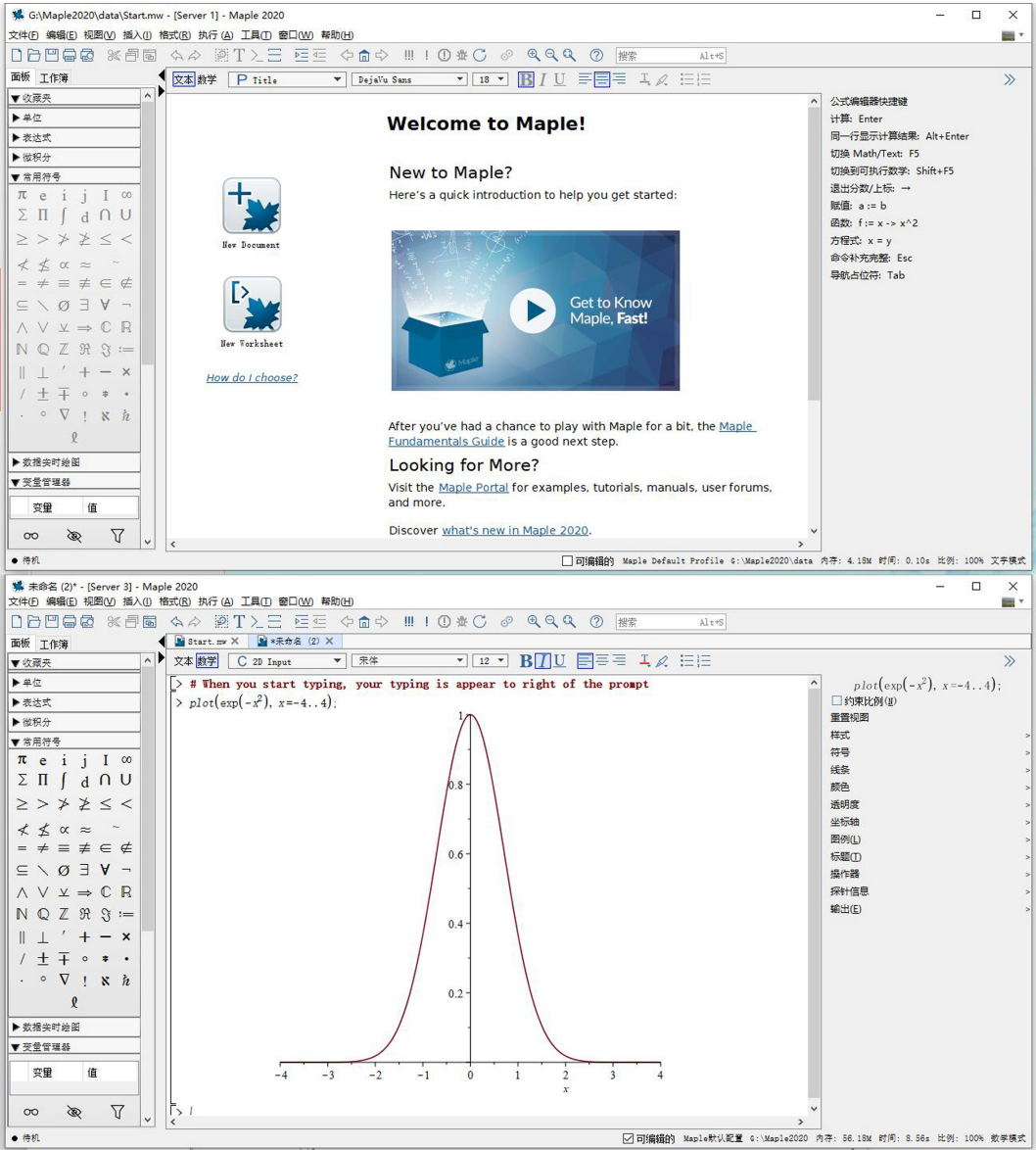


Figure A1.2 Maple2020 main interface of worksheet mode

A2 Basic Operation for Maple2020

Example A2.1 Compute 2^{99} .

Solution In Maple2020, if we type

```
[> 2^99;
```

Then, we will get

```
633825300114114700748351602688
```

By the way, if we use the reverse operation

```
[> ifactor(633825300114114700748351602688);
```

Naturally, we will find that

```
(2)^99
```

We see that Maple can complete complex operations in a very short time.

Again, if we type the following command

```
[> 2.0^99;
```

then Maple will give us a different value

```
6.338253001 10^29
```

If we use floating-point parameters for operation, Maple will represent the return value in scientific enumeration.

Example A2.2 Factor the polynomial $x^4 - 4x^3 - 7x^2 + 22x + 24$.

Solution We type the following command in Maple2020

```
[> f:=x->x^4-4*x^3-7*x^2+22*x+24;
```

```
f:=x->x^4-4*x^3-7*x^2+22*x+24
```

In this way, we have defined a function

$$f(x) = x^4 - 4x^3 - 7x^2 + 22x + 24.$$

Next, we use factor command to give its factorization form.

```
[> factor(f(x));
```

```
(x+1)(x+2)(x-3)(x-4)
```

We can use plot command to graph the figure of the above fourth order polynomial

```
[> plot(f(x), x=-3..5, size=[300,240]);
```

From the Figure A2.1, we see that this fourth order polynomial has four roots

$$x = -2, x = -1, x = 3, x = 4.$$

Example A2.3 Compute the sum $\sum_{k=1}^n k^4$.

Solution In this example, we have to compute the sum

$$1^4 + 2^4 + 3^4 + \cdots + n^4$$

In Maple2020, we can use the command sum to finish the computation

```
[> Sum(k^4, k=1..n)=sum(k^4,k=1..n);
```

$$\sum_{k=1}^n k^4 = \frac{(n+1)^5}{5} - \frac{(n+1)^4}{2} + \frac{(n+1)^3}{3} - \frac{n}{30} - \frac{1}{30}$$

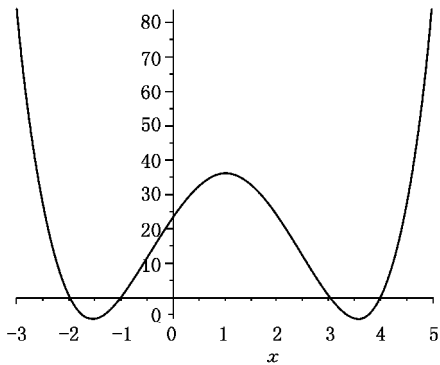


Figure A2.1 Figure of polynomial in Example A2.2

Here, in the above command, `Sum()` is the inert form of function `sum()`. It only gives the formulation of this expression but does not give the result. In order to get the final form, we can use the `simplify()` command.

```
[> simplify(%);
```

%Represents the formula of the previous line

$$\sum_{k=1}^n k^4 = \frac{1}{5}n^5 + \frac{1}{2}n^4 + \frac{1}{3}n^3 - \frac{1}{30}n$$

```
[> factor(rhs(%));
```

$$\frac{n(2n+1)(n+1)(3n^2+3n-1)}{30}$$

If you want to conform your computation, you can use `expand` command

```
[> expand(%);
```

$$\frac{1}{5}n^5 + \frac{1}{2}n^4 + \frac{1}{3}n^3 - \frac{1}{30}n$$

Example A2.4 Here $f(x,y)=x\tan(xy)$, calculate partial derivative $\frac{\partial f}{\partial x} \frac{\partial^2 f}{\partial x \partial y}$.

Solution In Maple, we type the following command

```
[> f:=(x,y)->x*tan(x*y);
```

$$f:=(x,y)\mapsto x\cdot \tan(y\cdot x)$$

```
[> Diff(f(x,y), x)=diff(f(x,y), x);
```

$$\frac{\partial}{\partial x}(x\tan(yx))=\tan(yx)+xy(1+\tan(yx)^2)$$

```
[> Diff(f(x,y), x,y)=diff(f(x,y), x,y);
```

$$\frac{\partial}{\partial x \partial y}(x\tan(yx))=2x(1+\tan(yx)^2)+2x^2y\tan(yx)(1+\tan(yx)^2)$$

We can also use D operator to finish the task

```
[> D[1](f);
```

$$(x,y)\mapsto \tan(y\cdot x)+x\cdot y\cdot (1+\tan(y\cdot x)^2)$$

[> D[1,2](f);

$$(x, y) \mapsto 2 \cdot x \cdot (1 + \tan(y \cdot x)^2) + 2 \cdot x^2 \cdot \tan(y \cdot x) \cdot y \cdot (1 + \tan(y \cdot x)^2)$$

From the result above, we see that D operator's solution is still a mapping.

Example A2.5 Compute the integration $\int \frac{1}{t^6 - 1} dt$.

Solution First, we see that $(t^6 - 1)$ can be factorized as

[> factor(t^6-1);

$$(t - 1)(t + 1)(t^2 + t + 1)(t^2 - t + 1)$$

Next, we can convert the above expression from rational fraction into partial fraction

[> convert(1/(t^6-1), parfrac, t);

$$-\frac{1}{6(t+1)} + \frac{-t-2}{6(t^2+t+1)} + \frac{t-2}{6(t^2-t+1)} + \frac{1}{6(t-1)}$$

Finally, we can use command int to give the partial fraction's integration

[> int(%, t);

$$\begin{aligned} & \frac{\ln(t-1)}{6} - \frac{\ln(t+1)}{6} - \frac{\ln(t^2+t+1)}{12} - \frac{\sqrt{3} \arctan\left(\frac{(2t+1)\sqrt{3}}{3}\right)}{6} \\ & + \frac{1}{12} \ln(t^2-t+1) - \frac{\sqrt{3}}{6} \arctan\left(\frac{(2t-1)\sqrt{3}}{3}\right) \end{aligned}$$

We can also use int to give the exact solution

[> int(1/(t^6-1), t);

$$\begin{aligned} & \frac{\ln(t-1)}{6} - \frac{\ln(t+1)}{6} - \frac{\ln(t^2+t+1)}{12} - \frac{\sqrt{3} \arctan\left(\frac{(2t+1)\sqrt{3}}{3}\right)}{6} \\ & + \frac{1}{12} \ln(t^2-t+1) - \frac{\sqrt{3}}{6} \arctan\left(\frac{(2t-1)\sqrt{3}}{3}\right) \end{aligned}$$

Again, we can use simplify() and diff() to confirm our computation

[> diff(%, t);

$$\begin{aligned} & \frac{1}{6(t-1)} - \frac{1}{6(t+1)} - \frac{2t+1}{12(t^2+t+1)} - \frac{1}{3\left(\frac{(2t+1)^2}{3} + 1\right)} + \\ & \frac{2t-1}{12(t^2-t+1)} - \frac{1}{3\left(\frac{(2t-1)^2}{3} + 1\right)} \end{aligned}$$

[> simplify(%);

$$\frac{1}{t^6 - 1}$$

Example A2.6 Find the eigenvalues and eigenvectors of the matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Solution In Maple, we type the following command

```
[> with(LinearAlgebra):
[> A:=Matirx([[0,1,1],[1,0,1],[1,1,0]]):
[> a:=CharacteristicPolynomial(A, λ);
a:=λ3 - 3λ - 2
[> factor(a);
(λ - 2)(λ + 1)2
```

The eigenvalue $\lambda_1 = 2$ has the corresponding eigenvector $v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

The eigenvalues $\lambda_{2,3} = -1$ has multiplicity 2. The two linearly independent eigenvectors are

$v_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, v_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$. We use Eigenvectors to confirm

our results

```
[> [Eigenvectors(A)];
[[[-1], [-1, -1, 1]],
 [[-1], [0, 1, 1]],
 [[2], [1, 0, 1]]]
```

We see that Maple's results fit well with our above computation.

A3 Maple2020 for Solving Ordinary Differential Equations

In this section, we will review the basic method for solving ordinary differential equations. If we want to solve the differential equation

$$\frac{dy}{dt} = f(t, y)$$

we can use the Maple command: `dsolve(diff(y(t), t) = f(t, y(t)), y(t))`. For more information of `dsolve()`, we can use Maple help, using `? dsolve` see Figure A3.1.

Example A3.1 Solve the ordinary differential equation

$$y' - y^2 \cos t = 0.$$

Solution This equation is a separable equation. It can be written as

$$\frac{dy}{y^2} = \cos t \, dt.$$

Integrating both sides of this equation yields

$$y = \frac{1}{C - \sin t}.$$

We use `dsolve` command to check our computation above.

```
[> sola:=dsolve(diff(y(t), t)=y(t)^2 * cos(t), y(t));
```

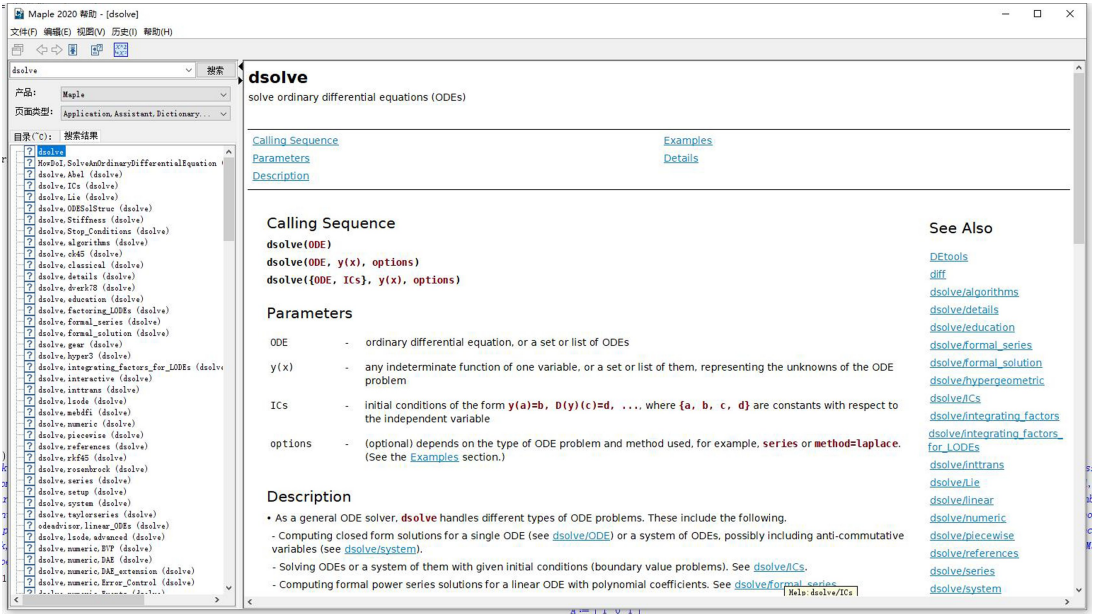


Figure A3.1 Maple2020 help of command dsolve

$$\text{sola} := y(t) = -\frac{1}{\sin(t) - _C1}$$

From the result above, we see that Maple's solution is coincident with our computation.

We then graph several figures with various values of C, using plot

```
[> toplot := seq(subs(_C1=i, rhs(sola)), i=2..10);
toplot := -1/(sin(t) - 2), -1/(sin(t) - 3), -1/(sin(t) - 4), -1/(sin(t) - 5), -1/(sin(t) - 6),
-1/(sin(t) - 7), -1/(sin(t) - 8), -1/(sin(t) - 9), -1/(sin(t) - 10)
[> plot([toplot], t=0..2 * Pi, view=[0..2 * Pi, 0..1], color=black);
```

The results can be shown in Figure A3.2.

Then, we use the following command to graph the direction field of this problem

```
[> dfieldplot(diff(y(t),t)=y(t)^2 * cos(t), y(t), t=0..2 * Pi, y=0..1, axes=boxed,
color=black);
```

If we give the initial conditions $y(0)=0.2, y(0)=0.4, y(0)=0.6$, then we can use the command DEplot to get the direction field as in Figure A3.3 and Figure A3.4.

```
[> ivs := [y(0) = 0.2, y(0) = 0.4, y(0) = 0.6];
[> eq1 := diff(y(t), t) = y(t)^2 * cos(t);
[> DEplot(eq1, y(t), t = 0..2 * Pi, y = 0 .. 2, ivs, arrows = medium, linecolor =
black)
```

Example A3.2 Solve the equation

$$(1 + t^2) \frac{dy}{dt} + 2ty = t^2.$$

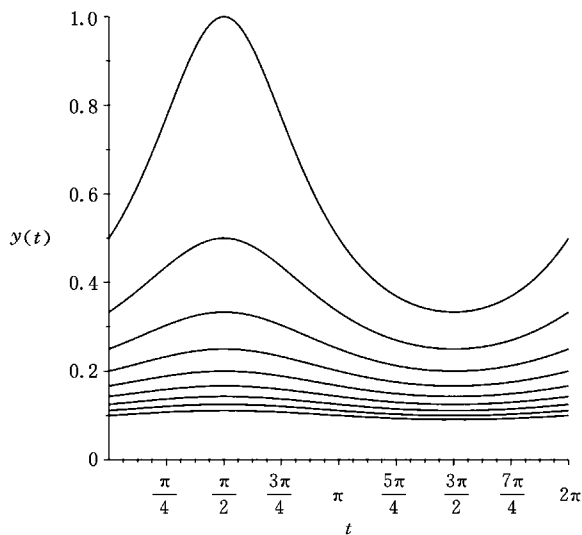


Figure A3.2 Solution of Example A3.1

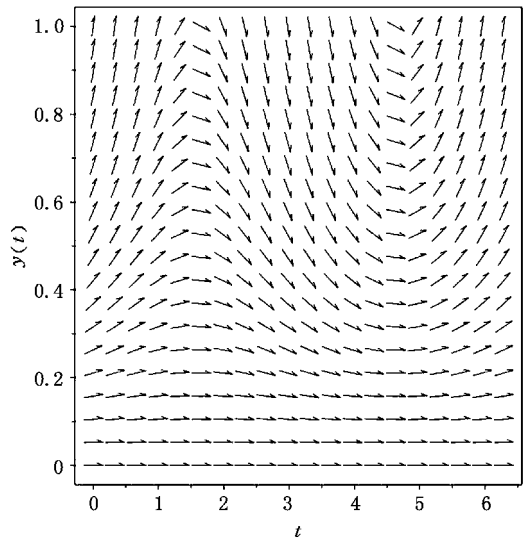


Figure A3.3 Direction field of Example A.3.1

Solution This equation is a first order linear nonhomogeneous equation. We can use integral factor method to solve it. Indeed, we obtain the general solution.

$$y(t) = \frac{t^3}{3(1+t^2)} + \frac{C}{1+t^2}.$$

Then, we use Maple to verify our computation

```
[> eq1 := (t^2 + 1) * diff(y(t), t) + 2 * t * y(t) = t^2;
```

$$\text{eq1} := (t^2 + 1) \left(\frac{d}{dt} y(t) \right) + 2ty(t) = t^2$$

```
[> odeadvisor(eq1);
```

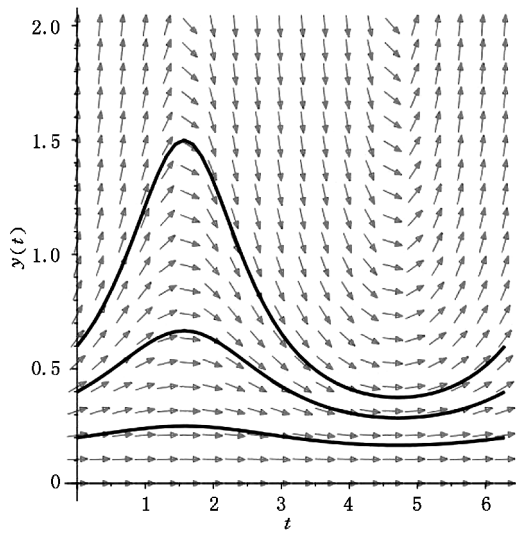


Figure A3.4 Direction field and three solutions of Example A3.1

[_linear]

Here, we use the command `odeadvisor()` to get the classification of the equation. Then we use `dsolve()` to find its exact solution.

[> solb := dsolve(eq1);

$$\text{solb} := y(t) = \frac{\frac{t^3}{3} + _C1}{t^2 + 1}$$

From the formulation of this solution, it does fit quite well with the solution above. Next, we try to plot the direction field of this equation.

[> toplot := seq(subs(_C1 = 1/3 * i, rhs(solb)), i = 1 .. 10);

$$\text{toplot} := \frac{\frac{t^3}{3} + \frac{1}{3}}{t^2 + 1}, \frac{\frac{t^3}{3} + \frac{2}{3}}{t^2 + 1}, \frac{\frac{t^3}{3} + 1}{t^2 + 1}, \frac{\frac{t^3}{3} + \frac{4}{3}}{t^2 + 1}, \frac{\frac{t^3}{3} + \frac{5}{3}}{t^2 + 1}, \frac{\frac{t^3}{3} + 2}{t^2 + 1}, \frac{\frac{t^3}{3} + \frac{7}{3}}{t^2 + 1},$$

$$\frac{\frac{t^3}{3} + \frac{8}{3}}{t^2 + 1}, \frac{\frac{t^3}{3} + 3}{t^2 + 1}, \frac{\frac{t^3}{3} + \frac{10}{3}}{t^2 + 1}$$

[> plot([toplot], t = 0 .. 2 * Pi, view = [0 .. 2 * Pi, 0 .. 2], color = black);

[> with(DEtools):

[> ivs := [y(0) = 0.5, y(0) = 1.0, y(0) = 1.5];

[> DEplot(eq1, y(t), t = 0 .. 2 * Pi, y = 0 .. 2, ivs, arrows = medium, linecolor = black); The results can be shown in Figure A3.5 and Figure A3.6.

Example A3.3 Solve the equation

$$(x^2 + y^2) \frac{dy}{dx} = 2xy.$$

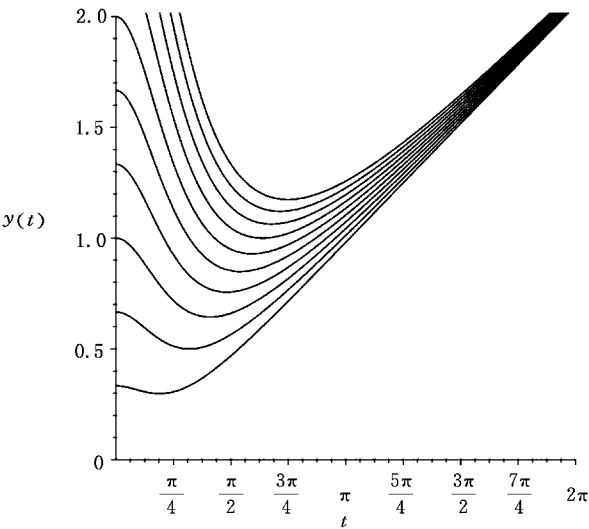


Figure A3.5 Ten solutions of Example A.3.2

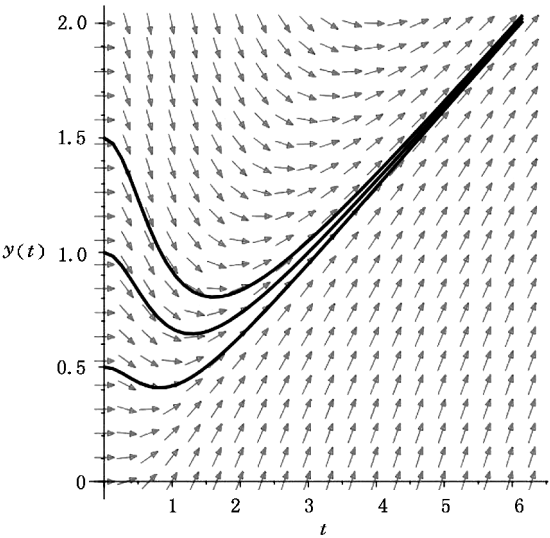


Figure A3.6 Direction field and three solutions of Example A.3.2

Solution This equation is a homogeneous equation. We easily obtain the general solution

$$x^2 - y^2 = Cy.$$

Then, we use Maple to confirm our solution

```
[> eq := (x^2 + y(x)^2) * diff(y(x), x) = 2 * x * y(x);
```

$$\text{eq} := (x^2 + y(x)^2) \left(\frac{d}{dx} y(x) \right) = 2xy(x)$$

```
[> odeadvisor(eq);
```

```
[[_homogeneous, class A], _rational, _dAlembert]
```

We see that Maple can classify the above equation into homogeneous equation.

```
[> sola:=dsolve(eq);
```

$$y(x) = -\frac{-1 + \sqrt{4_C1^2 x^2 + 1}}{2_C1}, y(x) = \frac{1 + \sqrt{4_C1^2 x^2 + 1}}{2_C1}$$

The form of the above solution is different with our computation, but they can be transformed into the same formula.

```
[> sola[1];
```

$$y(x) = -\frac{-1 + \sqrt{4_C1^2 x^2 + 1}}{2_C1}$$

```
[> sola[2];
```

$$y(x) = \frac{1 + \sqrt{4_C1^2 x^2 + 1}}{2_C1}$$

```
[> toplot:= seq(subs(_C1 = i, rhs(sola[1])), i = 1 .. 5);
```

$$\text{toplot} := \frac{1}{2} - \frac{\sqrt{4x^2 + 1}}{2}, \frac{1}{4} - \frac{\sqrt{16x^2 + 1}}{4}, \frac{1}{6} - \frac{\sqrt{36x^2 + 1}}{6}, \frac{1}{8} - \frac{\sqrt{64x^2 + 1}}{8},$$

$$\frac{1}{10} - \frac{\sqrt{100x^2 + 1}}{100}$$

```
[> p1:=plot([toplot], x = -1 .. 1, view = [-1 .. 1, -1 .. 0], scaling =
constrained, color = black);
```

```
[> toplot:= seq(subs(_C1 = i, rhs(sola[2])), i = 1 .. 5);
```

$$\text{toplot} := \frac{1}{2} + \frac{\sqrt{4x^2 + 1}}{2}, \frac{1}{4} + \frac{\sqrt{16x^2 + 1}}{4}, \frac{1}{6} + \frac{\sqrt{36x^2 + 1}}{6}, \frac{1}{8} + \frac{\sqrt{64x^2 + 1}}{8},$$

$$\frac{1}{10} + \frac{\sqrt{100x^2 + 1}}{100}$$

```
[> p2:=plot([toplot], x = -1 .. 1, view = [-1 .. 1, 0 .. 1], scaling = constrained,
color = black);
```

```
[> display([p1,p2]);
```

Next, we use Maple to graph the direction field of this problem

```
[> with(DEtools);
```

```
[> ivs := [y(0) = 0.1, y(0) = 0.2, y(0) = -0.1, y(0) = -0.2];
```

```
[> DEplot(eq, y(x), x = -1 .. 1, y = -1 .. 1, ivs, arrows = smalltwo, linecolor =
black, color = green)
```

The results can be shown in Figure A3.7 and A3.8.

Example A3.4 Find the solution of initial value problem

$$\begin{cases} y'' + 9y = 18\cos(3x) - 30\sin(3x) \\ y(0) = 0, y'(0) = 3. \end{cases}$$

Solution The characteristic equation of the corresponding homogenous equation is

$$\lambda^2 + 9 = 0$$

whose eigenvalues are $\lambda_{1,2} = \pm 3i$. Thus the general solution of the corresponding homogenous equation is

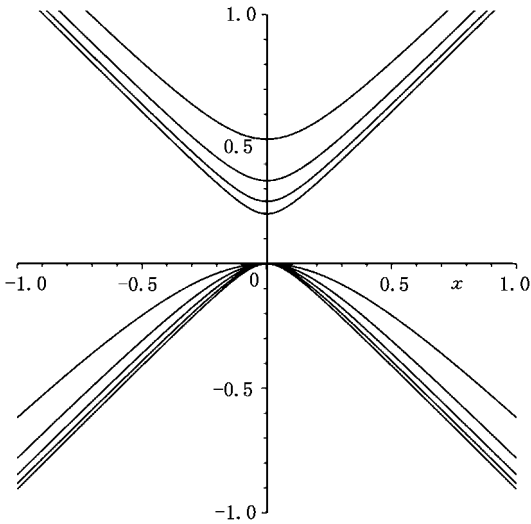


Figure A3.7 Ten solutions of Example A.3.3

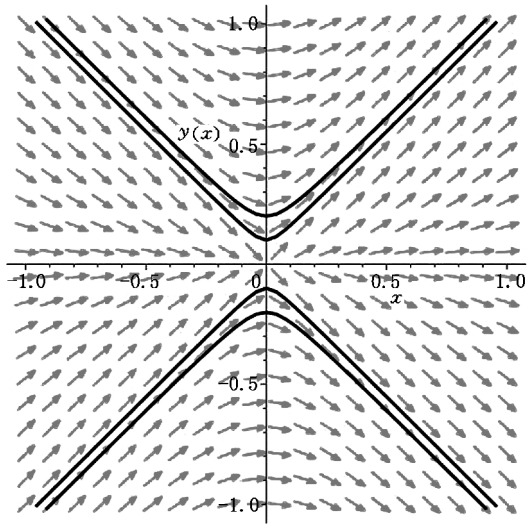


Figure A3.8 Direction field and four solutions of Example A.3.3

$$y = C_1 \cos(3x) + C_2 \sin(3x).$$

Since $+3i$ is a single characteristic root, the particular solution has the formula

$$y_1 = x[A \cos(3x) + B \sin(3x)].$$

Substituting y_1 into the equation, we find $A=5, B=3$. The general solution of the equation is

$$y = C_1 \cos(3x) + C_2 \sin(3x) + 5x \cos(3x) + 3x \sin(3x).$$

Using the initial condition, we obtain

$$y(x) = -\frac{2}{3} \sin(3x) + x[5 \cos(3x) + 3 \sin(3x)].$$

Next, we use Maple to confirm our computation

```
[>yp := x -> x * (a * cos(3 * x) + b * sin(3 * x)):
```

```
[> diff(yp(x), x);
      a cos(3x) + b sin(3x) + x[-3a sin(3x) + 3b cos(3x)]
[> diff(yp(x), x $ 2);
      -6a sin(3x) + 6b cos(3x) + x[-9a cos(3x) - 9b sin(3x)]
[> eqn := diff(yp(x), x $ 2) + 9 * yp(x) = 18 * cos(3 * x) - 30 * sin(3 * x);
[> simplify(eqn);
      -6a sin(3x) + 6b cos(3x) = 18cos(3x) - 30sin(3x)
Equating coefficients readily yields  $a = 5, b = 3$ .
It follows that  $y_p = x(5\cos(3x) + 3\sin(3x))$ ,
      y = C1 cos(3x) + C2 sin(3x) + x[5cos(3x) + 3sin(3x)]
[> diff(y(x), x);
      -3C1 sin(3x) + 3C2 cos(3x) + 5cos(3x) + 3sin(3x) + x[-15sin(3x) + 9cos(3x)]
[> cvals := solve({y(0) = 0, D(y)(0) = 3}, {C1, C2});
      cvals: = {C1 = 0, C2 = - $\frac{2}{3}$ }
[> subs(cvals, y(x));
      - $\frac{2\sin(3x)}{3}$  + x[5cos(3x) + 3sin(3x)]
```

We graph the final solution using plot in Figure A3.9.

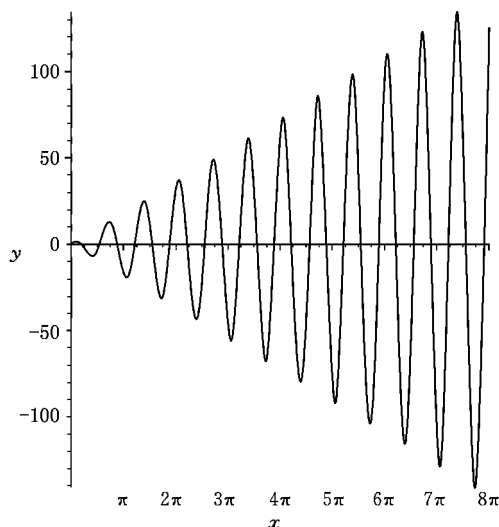


Figure A3.9 Solution of Example A3.4

```
[> plot(subs(cvals, y(x)), x = 0 .. 8 * Pi, color = black);
```

We verify the calculation by using dsolve

```
[> y := 'y';
[> eq := diff(y(x), x $ 2) + 9 * y(x) = 18 * cos(3 * x) - 30 * sin(3 * x);
eq :=  $\frac{d^2}{dx^2}y(x) + 9y(x) = 18\cos(3x) - 30\sin(3x)$ 
```

[>ics := y(0) = 0, D(y)(0) = 3;

ics := y(0) = 0, D(y)(0) = 3

[>dsolve({eq, ics});

$$y(x) = -\frac{2\sin(3x)}{3} + 3x\sin(3x) + 5x\cos(3x).$$

Example A3.5 Solve the system $X' = \begin{pmatrix} -1 & -1 \\ 2 & -3 \end{pmatrix} X$.

Solution Here $A = \begin{pmatrix} -1 & -1 \\ 2 & -3 \end{pmatrix}$. The characteristic equation of matrix A is

$$\det(A - \lambda E) = \begin{vmatrix} -1 - \lambda & -1 \\ 2 & -3 - \lambda \end{vmatrix} = \lambda^2 + 4\lambda + 5 = 0.$$

Then we find the eigenvalues are $\lambda_1 = -2 - i, \lambda_2 = -2 + i$, and the corresponding eigenvectors are $v_1 = \begin{pmatrix} 1 \\ 1 + i \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 1 - i \end{pmatrix}$ respectively. We use Maple to confirm the above results.

[> with(LinearAlgebra):

[>A := Matrix([[-1, -1], [2, -3]]);

$$\begin{bmatrix} -1 & -1 \\ 2 & -3 \end{bmatrix}$$

[> Eigenvectors(A);

$$\begin{bmatrix} -2 + i \\ -2 - i \end{bmatrix}, \begin{bmatrix} \frac{1}{2} + \frac{i}{2} & \frac{1}{2} - \frac{i}{2} \\ 1 & 1 \end{bmatrix}$$

Note that here

$$v_1 = \begin{pmatrix} \frac{1}{2}(1 - i) \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 1 + i \end{pmatrix}, v_2 = \begin{pmatrix} \frac{1}{2}(1 + i) \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 1 - i \end{pmatrix}.$$

For the eigenvector $\begin{pmatrix} 1 \\ 1 - i \end{pmatrix}$, the corresponding complex solution is

$$\begin{aligned} e^{(-2+i)t} \begin{pmatrix} 1 \\ 1 - i \end{pmatrix} &= e^{-2t} (\cos(t) + i\sin(t)) \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} + i \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right\} \\ &= e^{-2t} \left\{ \begin{pmatrix} \cos t \\ \cos t + \sin t \end{pmatrix} + i \begin{pmatrix} \sin t \\ \sin t - \cos t \end{pmatrix} \right\}. \end{aligned}$$

So the two corresponding linearly independent solutions are

$$e^{-2t} \begin{pmatrix} \cos t \\ \cos t + \sin t \end{pmatrix}, e^{-2t} \begin{pmatrix} \sin t \\ \sin t - \cos t \end{pmatrix}.$$

The general solution of the linear systems is

$$X(t) = e^{-2t} \left\{ C_1 \begin{pmatrix} \cos t \\ \cos t + \sin t \end{pmatrix} + C_2 \begin{pmatrix} \sin t \\ \sin t - \cos t \end{pmatrix} \right\}.$$

We verify this results by using matrixDE

```
[> with(DEtools):
```

```
[> matrixDE(A, t);
```

$$\begin{bmatrix} e^{-2t} \sin(t) & e^{-2t} \cos(t) \\ e^{-2t} \sin(t) - e^{-2t} \cos(t) & e^{-2t} \cos(t) + e^{-2t} \sin(t) \end{bmatrix}, [0 \quad 0]$$

From this, we see that the result given by Maple is the same with our computation above.

Then, we use dsolve to get the general solution to this linear systems by Maple.

```
[> sys_ode := diff(x(t), t) = -x(t) - y(t), diff(y(t), t) = 2 * x(t) - 3 * y(t);
```

$$\text{sys_ode} := \frac{d}{dt}x(t) = -x(t) - y(t), \frac{d}{dt}y(t) = 2x(t) - 3y(t).$$

```
[> dsolve([sys_ode]);
```

$$\{x(t) = e^{-2t}(\sin(t)_C1 + \cos(t)_C2),$$

$$y(t) = e^{-2t}(\sin(t)_C1 + \sin(t)_C2 - \cos(t)_C1 + \cos(t)_C2)\}.$$

We see that Maple's solution fits quite well to our calculation above. We use DEplot to graph the direction field of the system along with various solutions.

```
[> ival := seq(-1 + 0.25 * i, i = 0 .. 8);
```

```
[> i1 := seq([x(0) = 1, y(0) = i], i = ival);
```

```
[> i2 := seq([x(0) = i, y(0) = 1], i = ival);
```

```
[> i3 := seq([x(0) = -1, y(0) = i], i = ival);
```

```
[> i4 := seq([x(0) = i, y(0) = -1], i = ival);
```

```
[> DEplot([sys_ode], [x(t), y(t)], t = 0 .. 10, [i1, i2, i3, i4], x = -1..1, y = -1..1,
scene = [x(t), y(t)], scaling = constrained, linecolor = black, thickness = 1,
stepsize = 0.05, arrows = large, color = green).
```

The direction field can be shown in Figure A3.10.

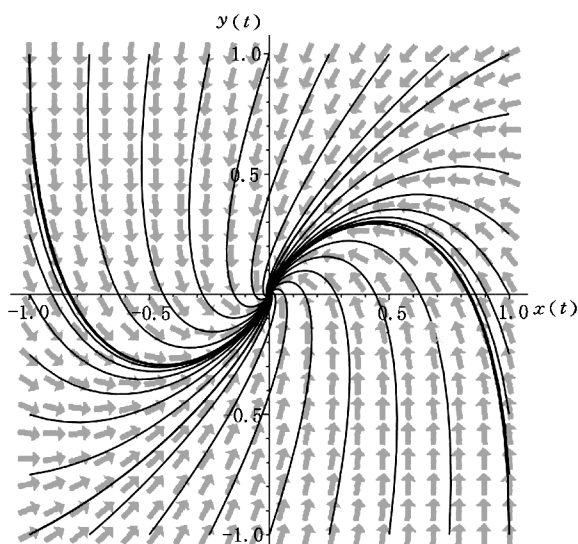


Figure A3.10 Direction field of Example A.3.5

References

- [1] 丁同仁,李承治. 常微分方程教程[M]. 北京:高等教育出版社,2004.
- [2] 东北师范大学微分方程教研室. 常微分方程[M]. 北京:高等教育出版社,2005.
- [3] 陆启韶,彭临平,杨卓琴. 常微分方程与动力系统[M]. 北京:北京航空航天大学出版社,2010.
- [4] 马知恩,周义仓. 常微分方程定性及稳定性方法[M]. 北京:科学出版社,2001.
- [5] 王高雄,周之铭,朱思铭. 常微分方程[M]. 北京:高等教育出版社,2020.
- [6] 伍卓群,李勇. 常微分方程[M]. 北京:高等教育出版社,2004.
- [7] 张芷芬,丁同仁,黄文灶,等. 微分方程定性理论[M]. 北京:科学出版社,1985.
- [8] 周宇虹,罗建书. Ordinary differential equations and their applications-theories and models[M]. 北京:科学出版社,2011.
- [9] ABELL M L, BARSELTON J D. Maple by example[M].3th ed. New York: Elsevier Academic Press,2005.
- [10] EDWARDS C H, PENNEY D E.常微分方程基础:第5版[M]. 北京:机械工业出版社,2005.
- [11] HALE J K. Ordinary differential equations[M]. New York: Krieger Publishing Company, 1980.
- [12] HARTMAN P. Ordinary differential equations[M]. Boston: Birkhäuser,1982.
- [13] PERKO L. Differential equations and dynamical systems[M]. New York: Springer-Verlag,1993.
- [14] WALTER W F. Ordinary differential equations[M]. 北京:世界图书出版公司,1998.